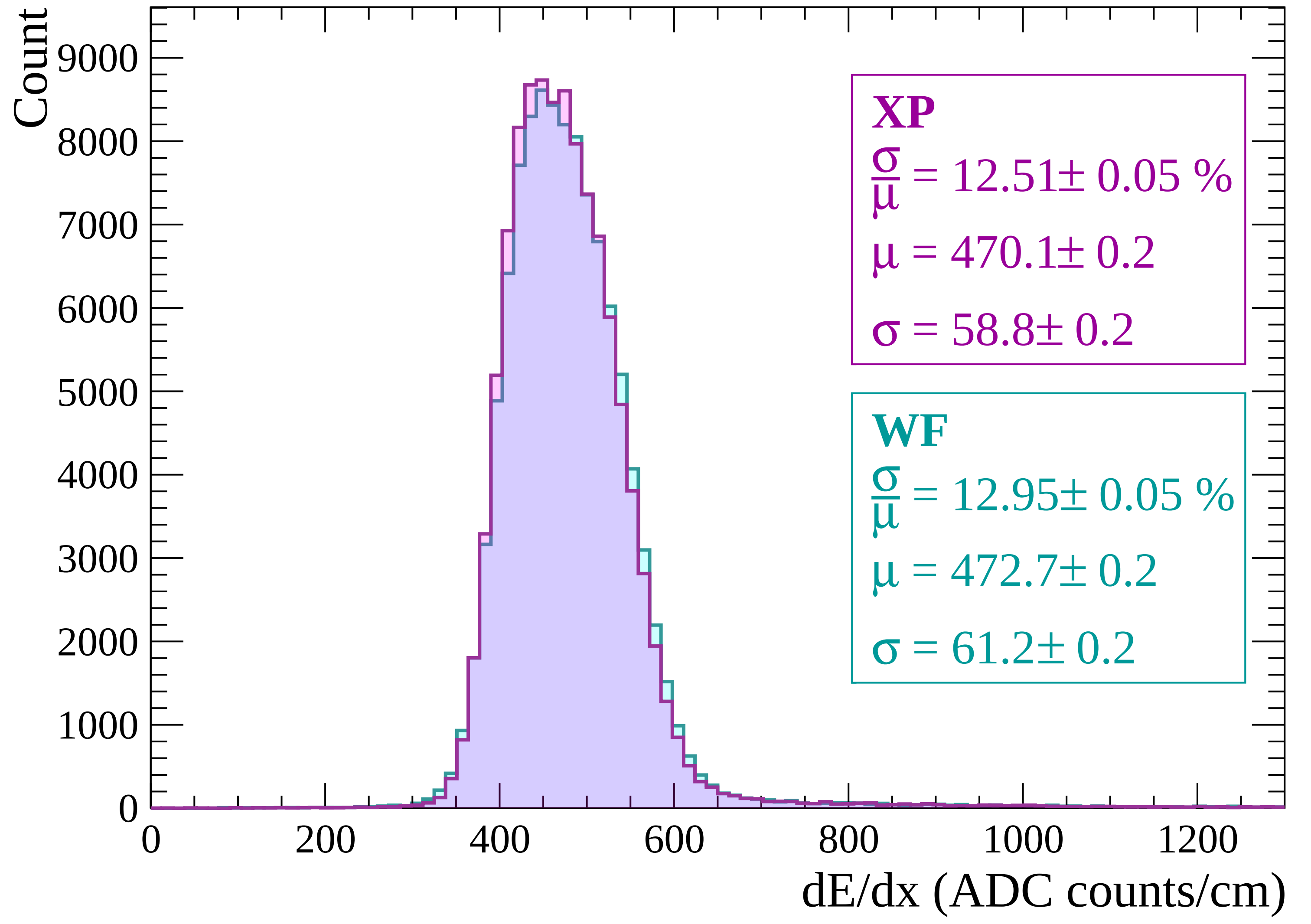
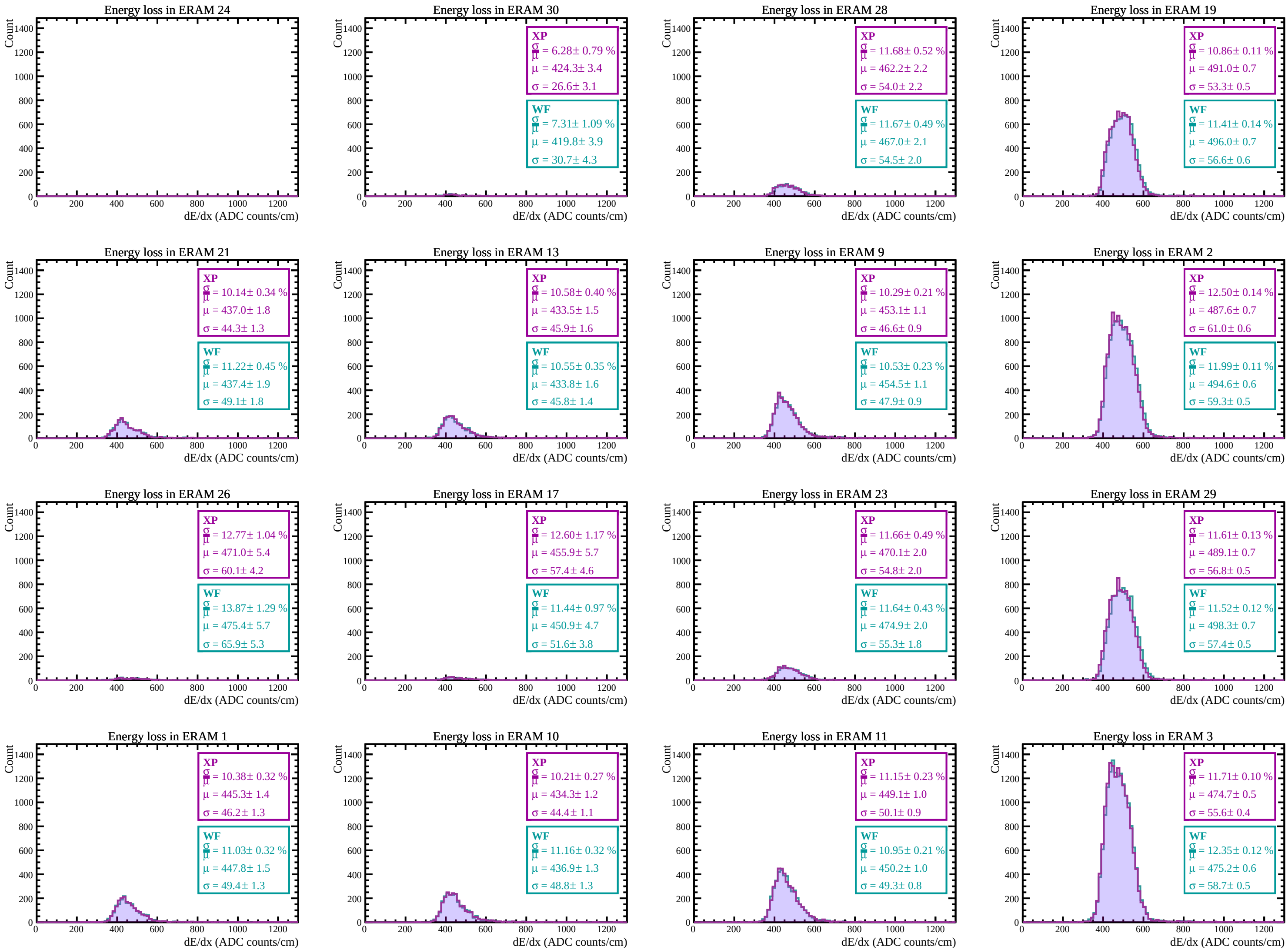
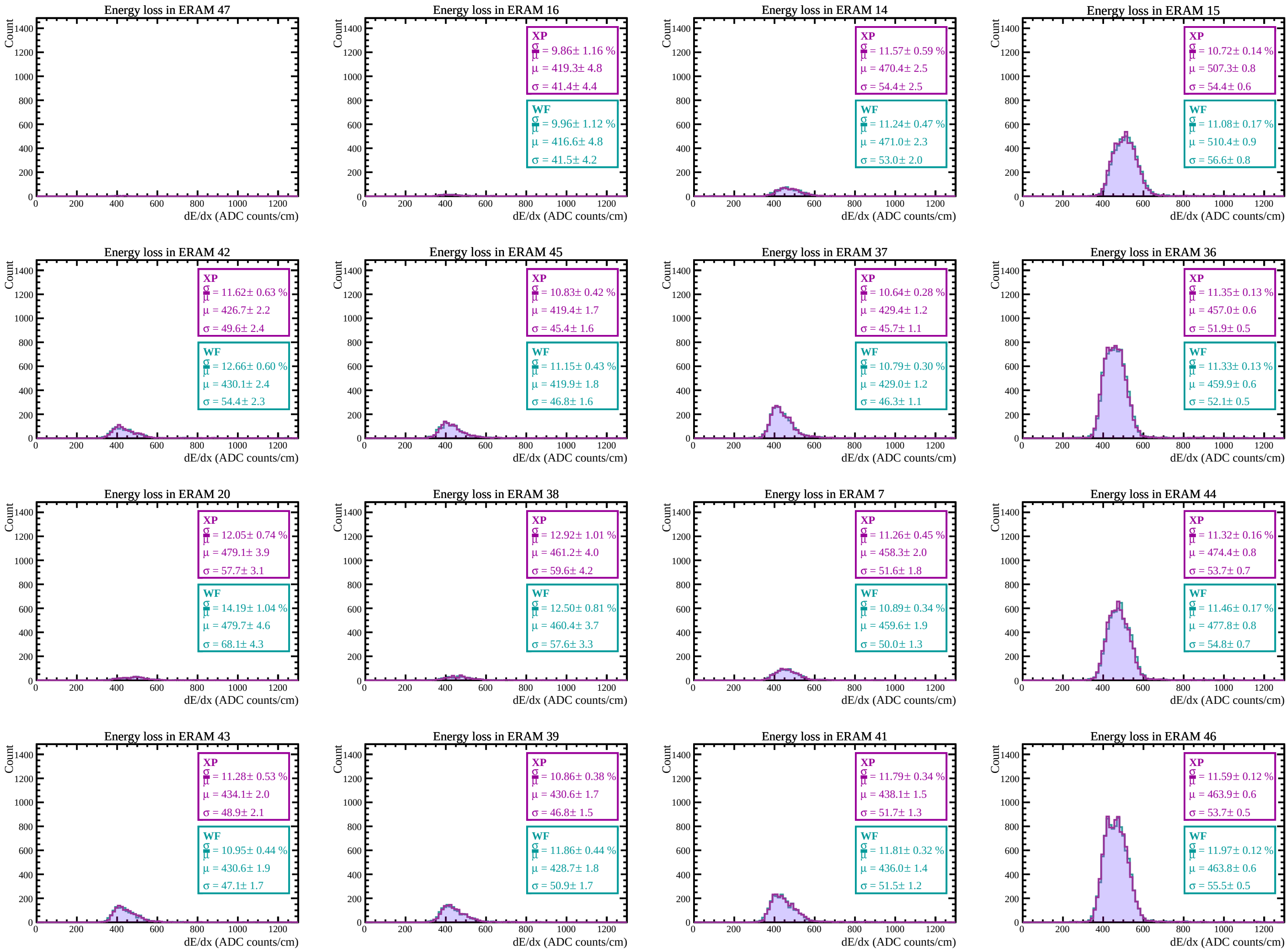
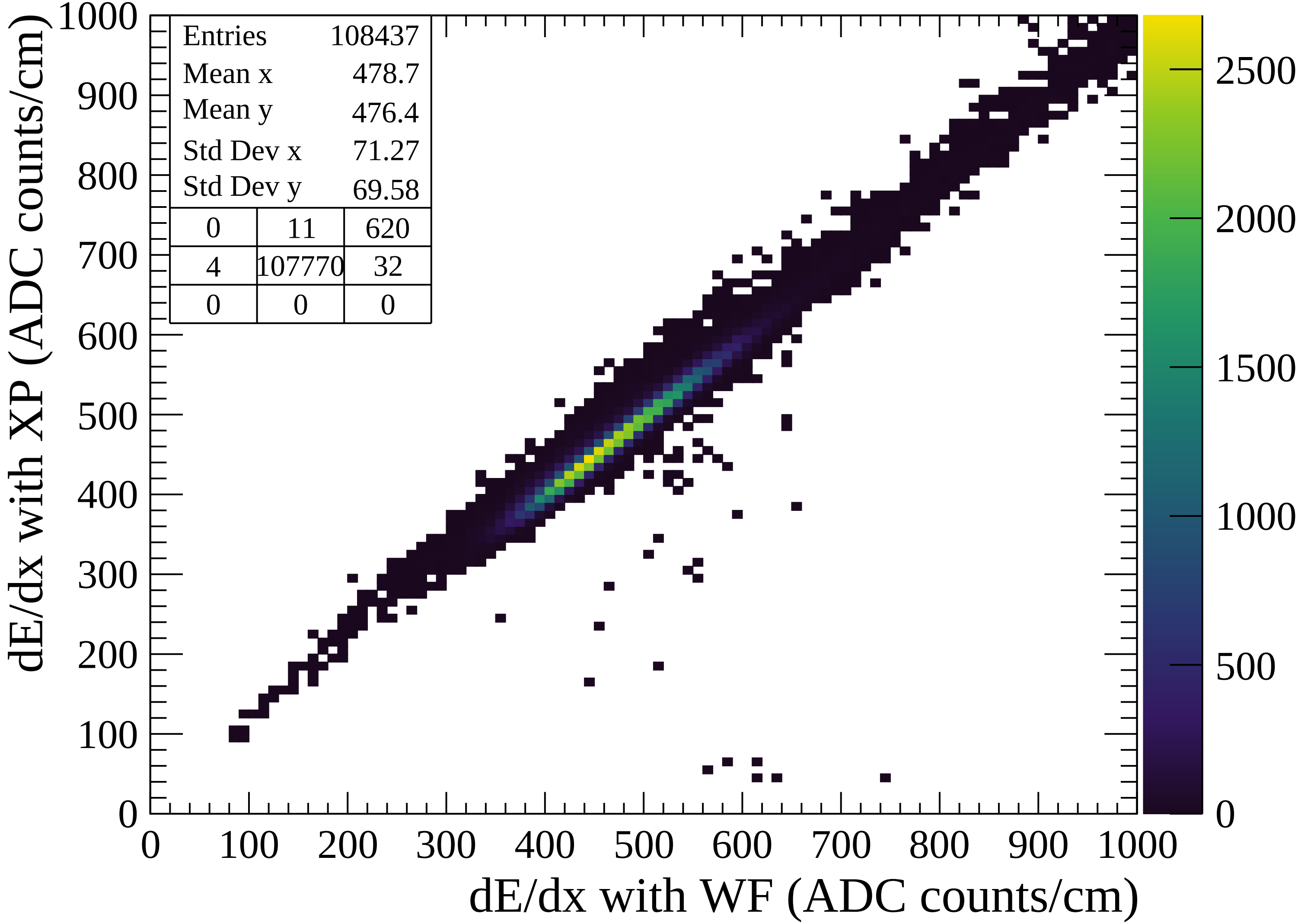
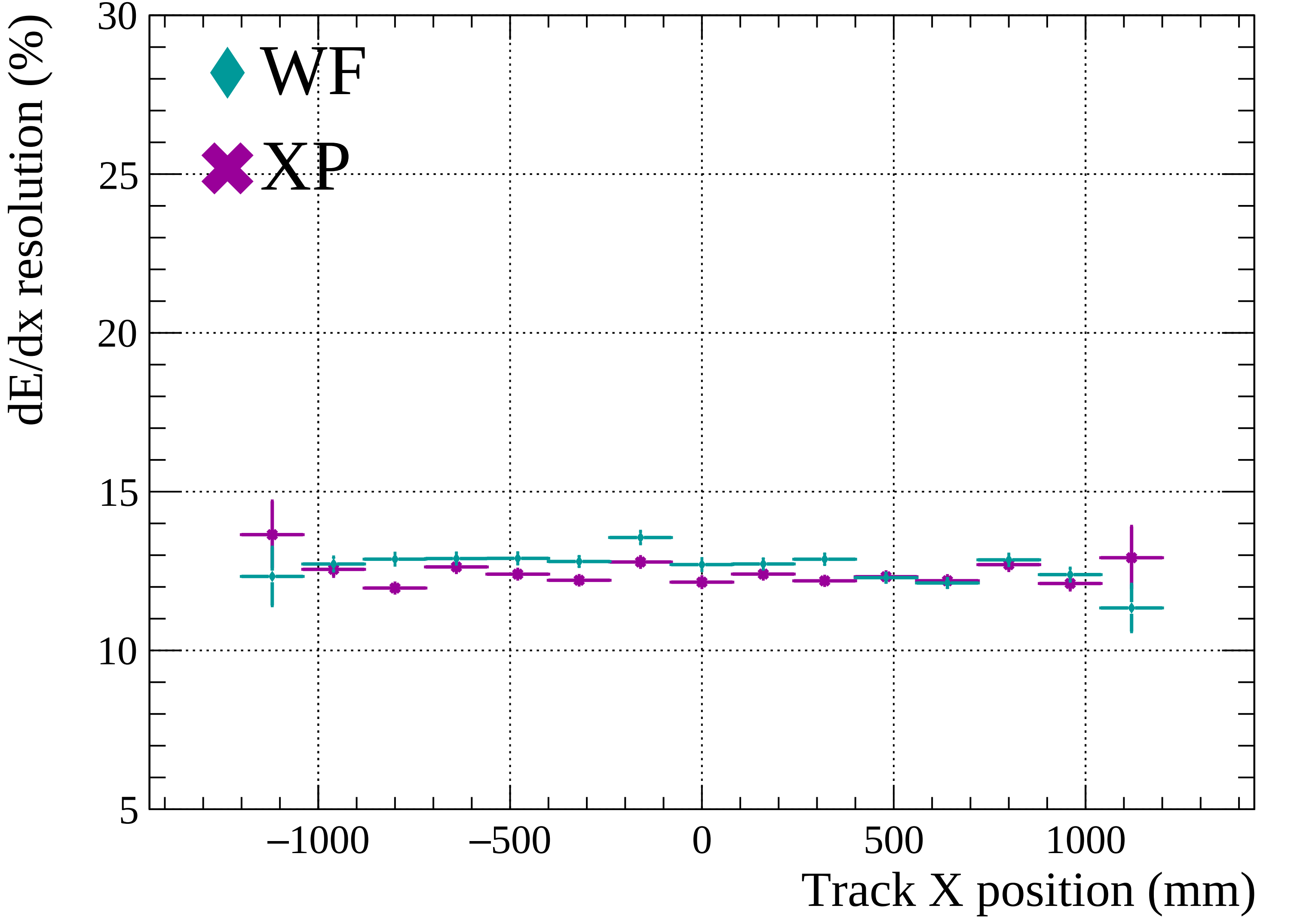


Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.51± 0.05 %
μ = 470.1± 0.2
σ = 58.8± 0.2
WF
$\frac{\sigma}{\mu}$ = 12.95± 0.05 %
μ = 472.7± 0.2
σ = 61.2± 0.2

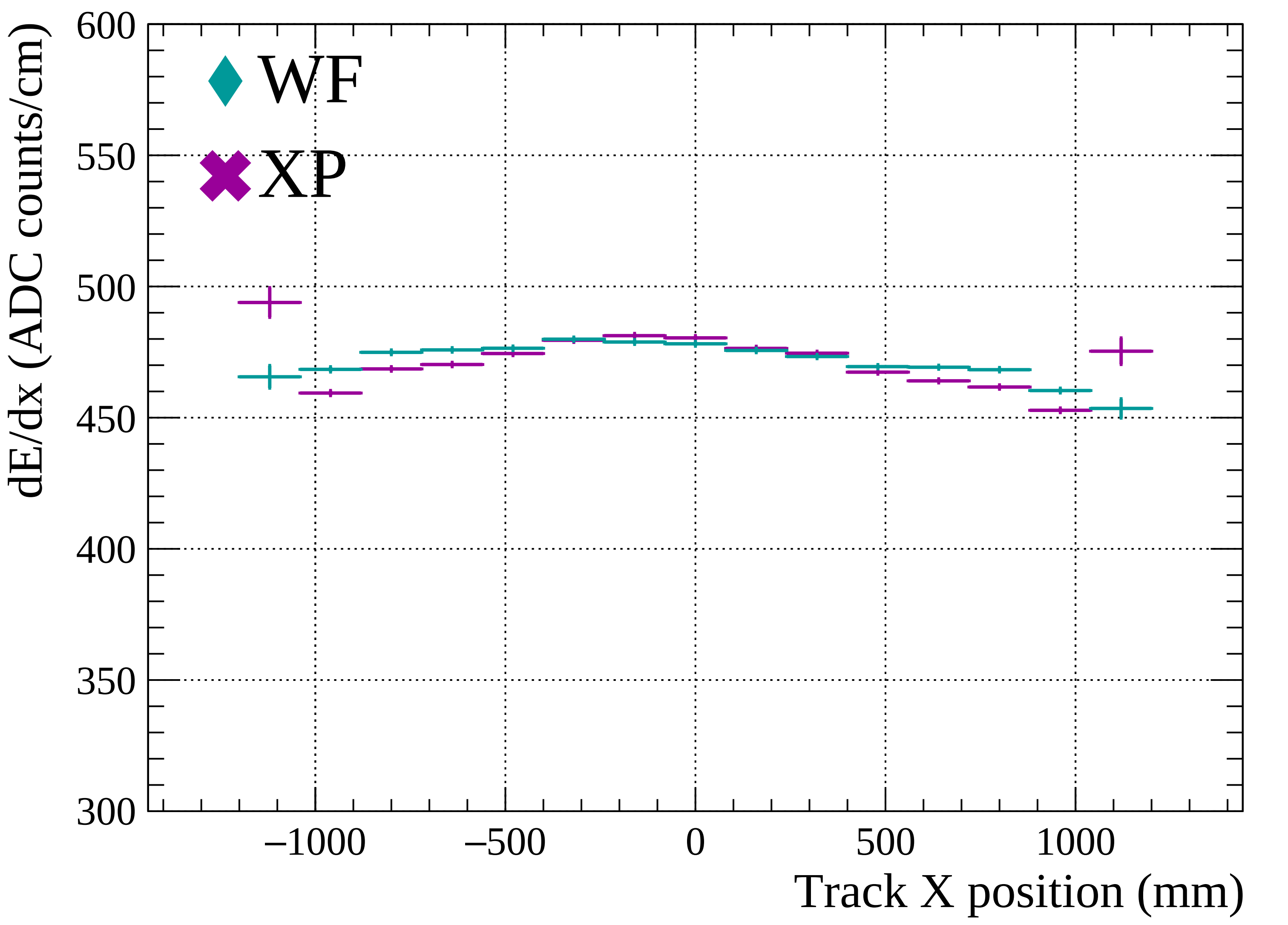

Energy loss in ERAM 24
Energy loss in ERAM 30
XP
σ/μ = 6.28± 0.79 %
μ = 424.3± 3.4
σ = 26.6± 3.1
WF
σ/μ = 7.31± 1.09 %
μ = 419.8± 3.9
σ = 30.7± 4.3
Energy loss in ERAM 28
XP
σ/μ = 11.68± 0.52 %
μ = 462.2± 2.2
σ = 54.0± 2.2
WF
σ/μ = 11.67± 0.49 %
μ = 467.0± 2.1
σ = 54.5± 2.0
Energy loss in ERAM 19
XP
σ/μ = 10.86± 0.11 %
μ = 491.0± 0.7
σ = 53.3± 0.5
WF
σ/μ = 11.41± 0.14 %
μ = 496.0± 0.7
σ = 56.6± 0.6
Energy loss in ERAM 21
XP
σ/μ = 10.14± 0.34 %
μ = 437.0± 1.8
σ = 44.3± 1.3
WF
σ/μ = 11.22± 0.45 %
μ = 437.4± 1.9
σ = 49.1± 1.8
Energy loss in ERAM 13
XP
σ/μ = 10.58± 0.40 %
μ = 433.5± 1.5
σ = 45.9± 1.6
WF
σ/μ = 10.55± 0.35 %
μ = 433.8± 1.6
σ = 45.8± 1.4
Energy loss in ERAM 9
XP
σ/μ = 10.29± 0.21 %
μ = 453.1± 1.1
σ = 46.6± 0.9
WF
σ/μ = 10.53± 0.23 %
μ = 454.5± 1.1
σ = 47.9± 0.9
Energy loss in ERAM 2
XP
σ/μ = 12.50± 0.14 %
μ = 487.6± 0.7
σ = 61.0± 0.6
WF
σ/μ = 11.99± 0.11 %
μ = 494.6± 0.6
σ = 59.3± 0.5
Energy loss in ERAM 26
XP
σ/μ = 12.77± 1.04 %
μ = 471.0± 5.4
σ = 60.1± 4.2
WF
σ/μ = 13.87± 1.29 %
μ = 475.4± 5.7
σ = 65.9± 5.3
Energy loss in ERAM 17
XP
σ/μ = 12.60± 1.17 %
μ = 455.9± 5.7
σ = 57.4± 4.6
WF
σ/μ = 11.44± 0.97 %
μ = 450.9± 4.7
σ = 51.6± 3.8
Energy loss in ERAM 23
XP
σ/μ = 11.66± 0.49 %
μ = 470.1± 2.0
σ = 54.8± 2.0
WF
σ/μ = 11.64± 0.43 %
μ = 474.9± 2.0
σ = 55.3± 1.8
Energy loss in ERAM 29
XP
σ/μ = 11.61± 0.13 %
μ = 489.1± 0.7
σ = 56.8± 0.5
WF
σ/μ = 11.52± 0.12 %
μ = 498.3± 0.7
σ = 57.4± 0.5
Energy loss in ERAM 1
XP
σ/μ = 10.38± 0.32 %
μ = 445.3± 1.4
σ = 46.2± 1.3
WF
σ/μ = 11.03± 0.32 %
μ = 447.8± 1.5
σ = 49.4± 1.3
Energy loss in ERAM 10
XP
σ/μ = 10.21± 0.27 %
μ = 434.3± 1.2
σ = 44.4± 1.1
WF
σ/μ = 11.16± 0.32 %
μ = 436.9± 1.3
σ = 48.8± 1.3
Energy loss in ERAM 11
XP
σ/μ = 11.15± 0.23 %
μ = 449.1± 1.0
σ = 50.1± 0.9
WF
σ/μ = 10.95± 0.21 %
μ = 450.2± 1.0
σ = 49.3± 0.8
Energy loss in ERAM 3
XP
σ/μ = 11.71± 0.10 %
μ = 474.7± 0.5
σ = 55.6± 0.4
WF
σ/μ = 12.35± 0.12 %
μ = 475.2± 0.6
σ = 58.7± 0.5
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 16
XP
σ/μ = 9.86± 1.16 %
μ = 419.3± 4.8
σ = 41.4± 4.4
WF
σ/μ = 9.96± 1.12 %
μ = 416.6± 4.8
σ = 41.5± 4.2
Energy loss in ERAM 14
XP
σ/μ = 11.57± 0.59 %
μ = 470.4± 2.5
σ = 54.4± 2.5
WF
σ/μ = 11.24± 0.47 %
μ = 471.0± 2.3
σ = 53.0± 2.0
Energy loss in ERAM 15
XP
σ/μ = 10.72± 0.14 %
μ = 507.3± 0.8
σ = 54.4± 0.6
WF
σ/μ = 11.08± 0.17 %
μ = 510.4± 0.9
σ = 56.6± 0.8
Energy loss in ERAM 42
XP
σ/μ = 11.62± 0.63 %
μ = 426.7± 2.2
σ = 49.6± 2.4
WF
σ/μ = 12.66± 0.60 %
μ = 430.1± 2.4
σ = 54.4± 2.3
Energy loss in ERAM 45
XP
σ/μ = 10.83± 0.42 %
μ = 419.4± 1.7
σ = 45.4± 1.6
WF
σ/μ = 11.15± 0.43 %
μ = 419.9± 1.8
σ = 46.8± 1.6
Energy loss in ERAM 37
XP
σ/μ = 10.64± 0.28 %
μ = 429.4± 1.2
σ = 45.7± 1.1
WF
σ/μ = 10.79± 0.30 %
μ = 429.0± 1.2
σ = 46.3± 1.1
Energy loss in ERAM 36
XP
σ/μ = 11.35± 0.13 %
μ = 457.0± 0.6
σ = 51.9± 0.5
WF
σ/μ = 11.33± 0.13 %
μ = 459.9± 0.6
σ = 52.1± 0.5
Energy loss in ERAM 20
XP
σ/μ = 12.05± 0.74 %
μ = 479.1± 3.9
σ = 57.7± 3.1
WF
σ/μ = 14.19± 1.04 %
μ = 479.7± 4.6
σ = 68.1± 4.3
Energy loss in ERAM 38
XP
σ/μ = 12.92± 1.01 %
μ = 461.2± 4.0
σ = 59.6± 4.2
WF
σ/μ = 12.50± 0.81 %
μ = 460.4± 3.7
σ = 57.6± 3.3
Energy loss in ERAM 7
XP
σ/μ = 11.26± 0.45 %
μ = 458.3± 2.0
σ = 51.6± 1.8
WF
σ/μ = 10.89± 0.34 %
μ = 459.6± 1.9
σ = 50.0± 1.3
Energy loss in ERAM 44
XP
σ/μ = 11.32± 0.16 %
μ = 474.4± 0.8
σ = 53.7± 0.7
WF
σ/μ = 11.46± 0.17 %
μ = 477.8± 0.8
σ = 54.8± 0.7
Energy loss in ERAM 43
XP
σ/μ = 11.28± 0.53 %
μ = 434.1± 2.0
σ = 48.9± 2.1
WF
σ/μ = 10.95± 0.44 %
μ = 430.6± 1.9
σ = 47.1± 1.7
Energy loss in ERAM 39
XP
σ/μ = 10.86± 0.38 %
μ = 430.6± 1.7
σ = 46.8± 1.5
WF
σ/μ = 11.86± 0.44 %
μ = 428.7± 1.8
σ = 50.9± 1.7
Energy loss in ERAM 41
XP
σ/μ = 11.79± 0.34 %
μ = 438.1± 1.5
σ = 51.7± 1.3
WF
σ/μ = 11.81± 0.32 %
μ = 436.0± 1.4
σ = 51.5± 1.2
Energy loss in ERAM 46
XP
σ/μ = 11.59± 0.12 %
μ = 463.9± 0.6
σ = 53.7± 0.5
WF
σ/μ = 11.97± 0.12 %
μ = 463.8± 0.6
σ = 55.5± 0.5

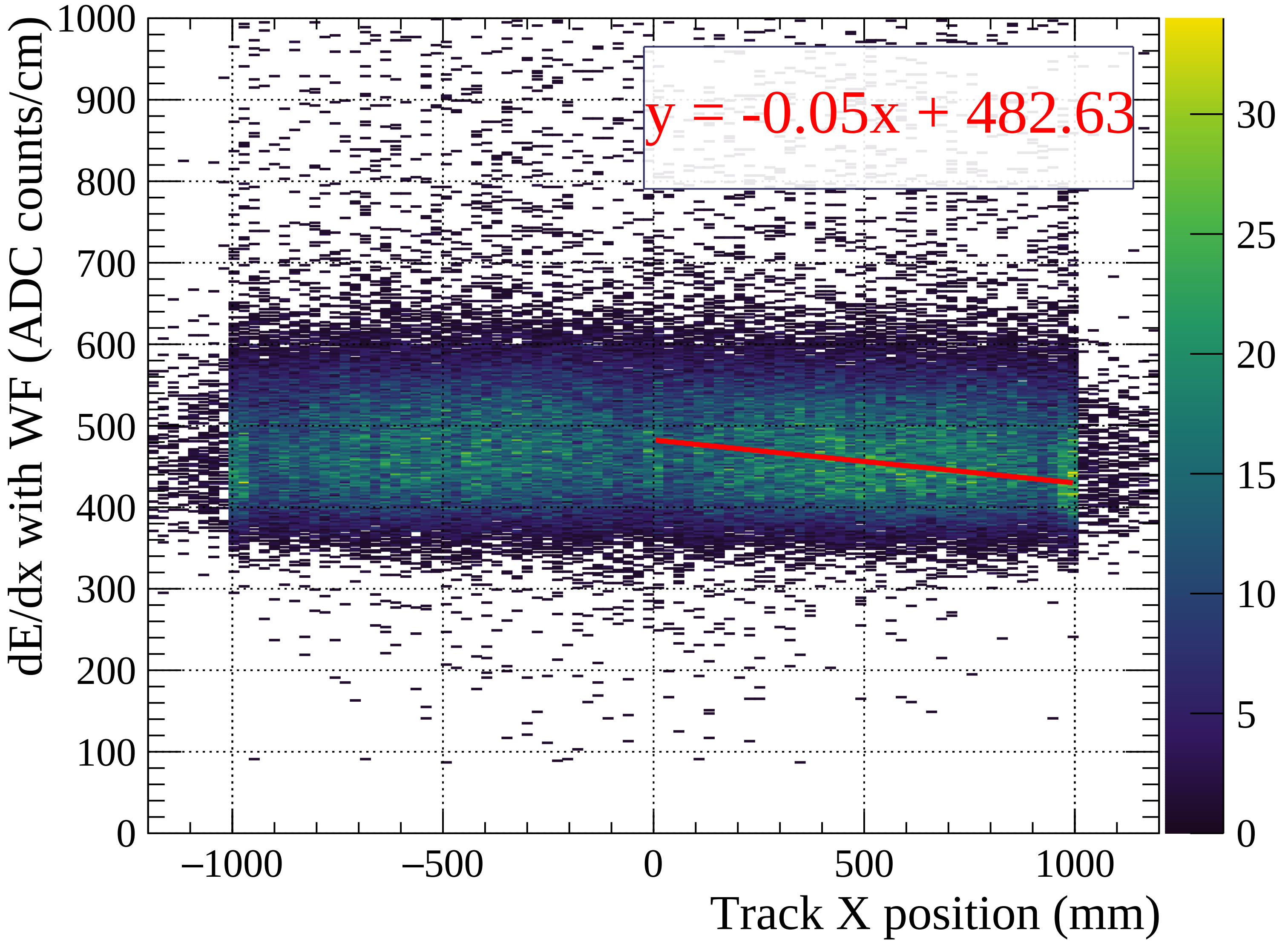

Entries 108437
Mean x 478.7
Mean y 476.4
Std Dev x 71.27
Std Dev y 69.58
0 11 620
4 107770 32
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500



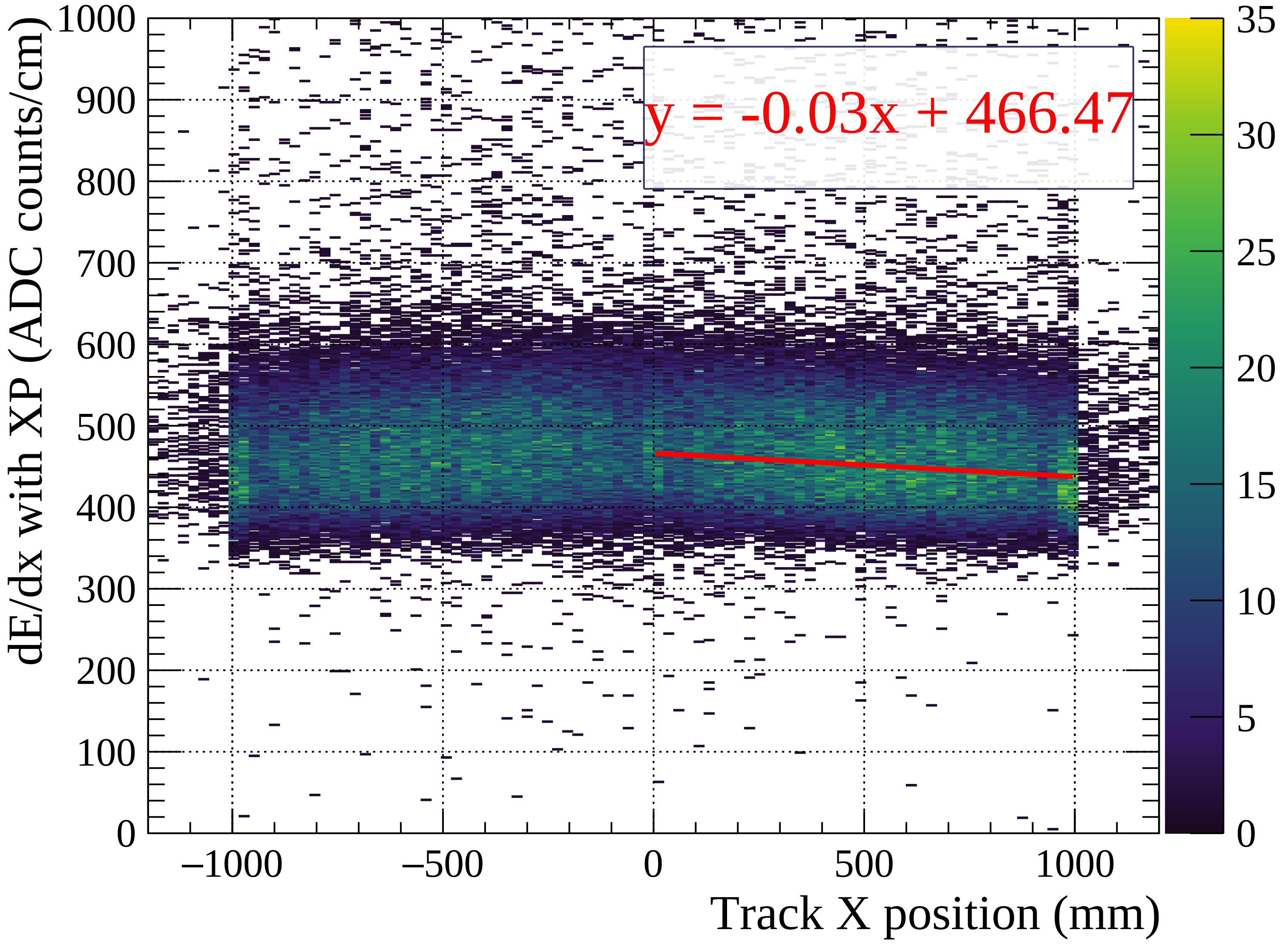
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)



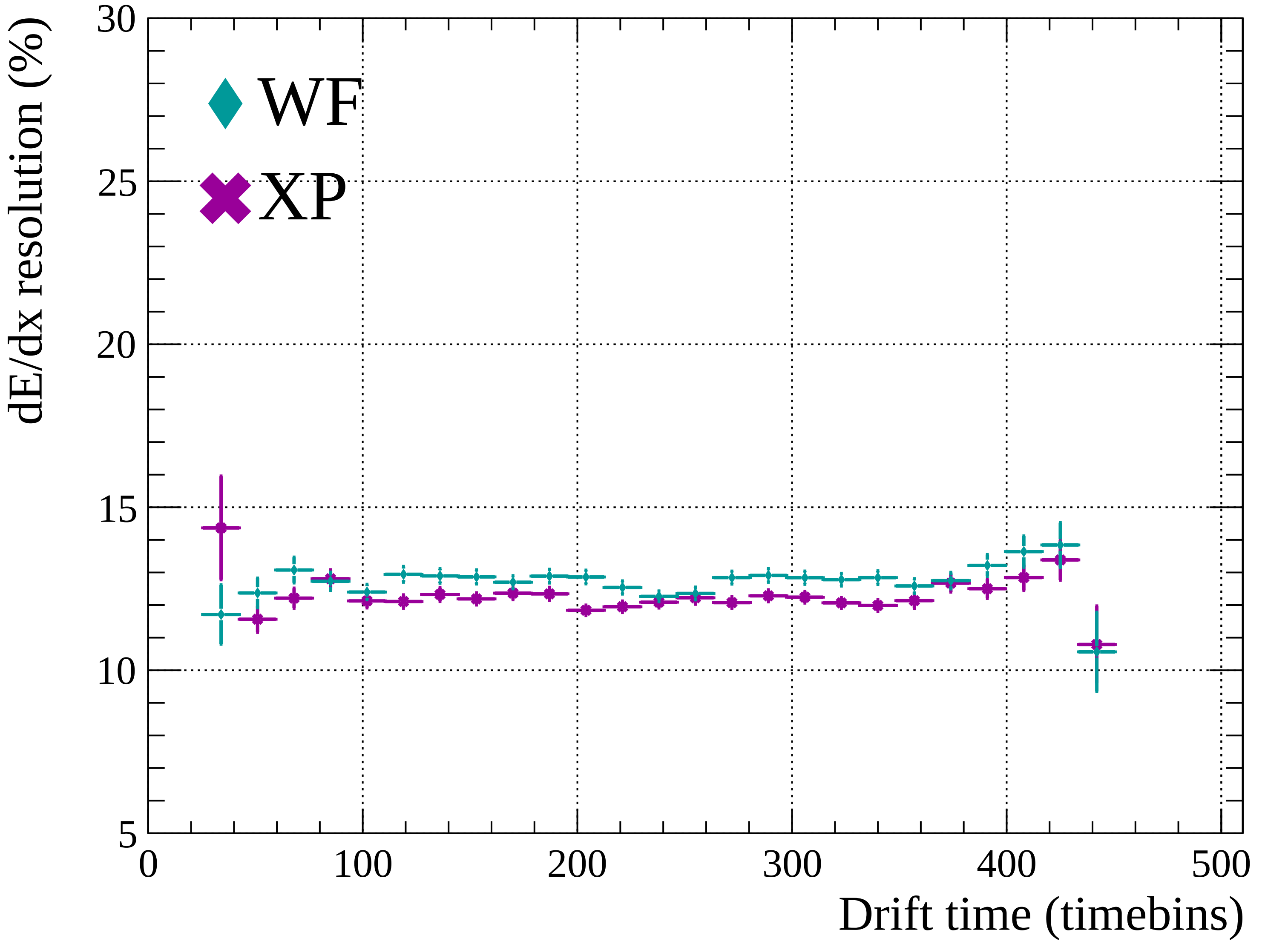
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)

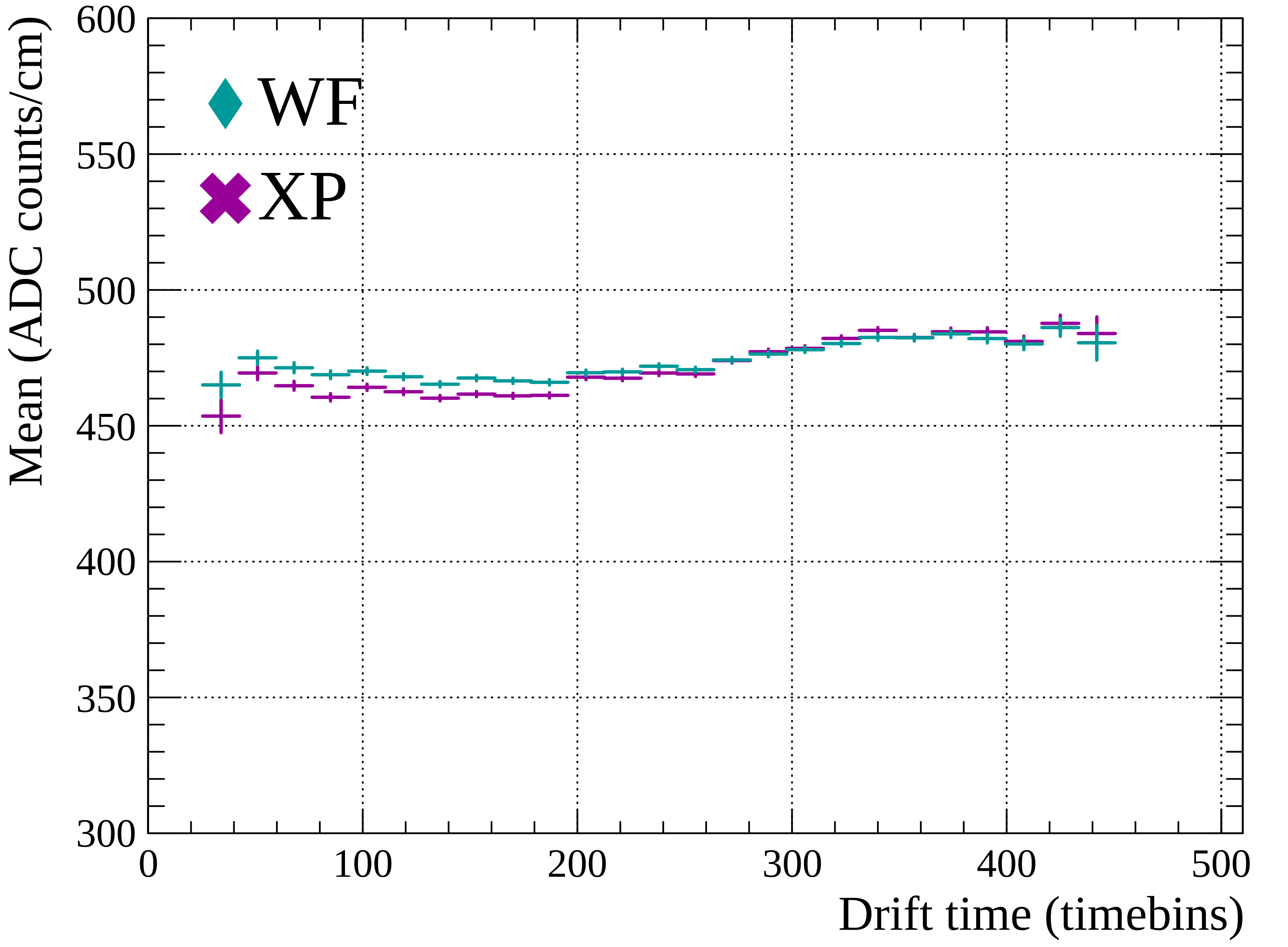


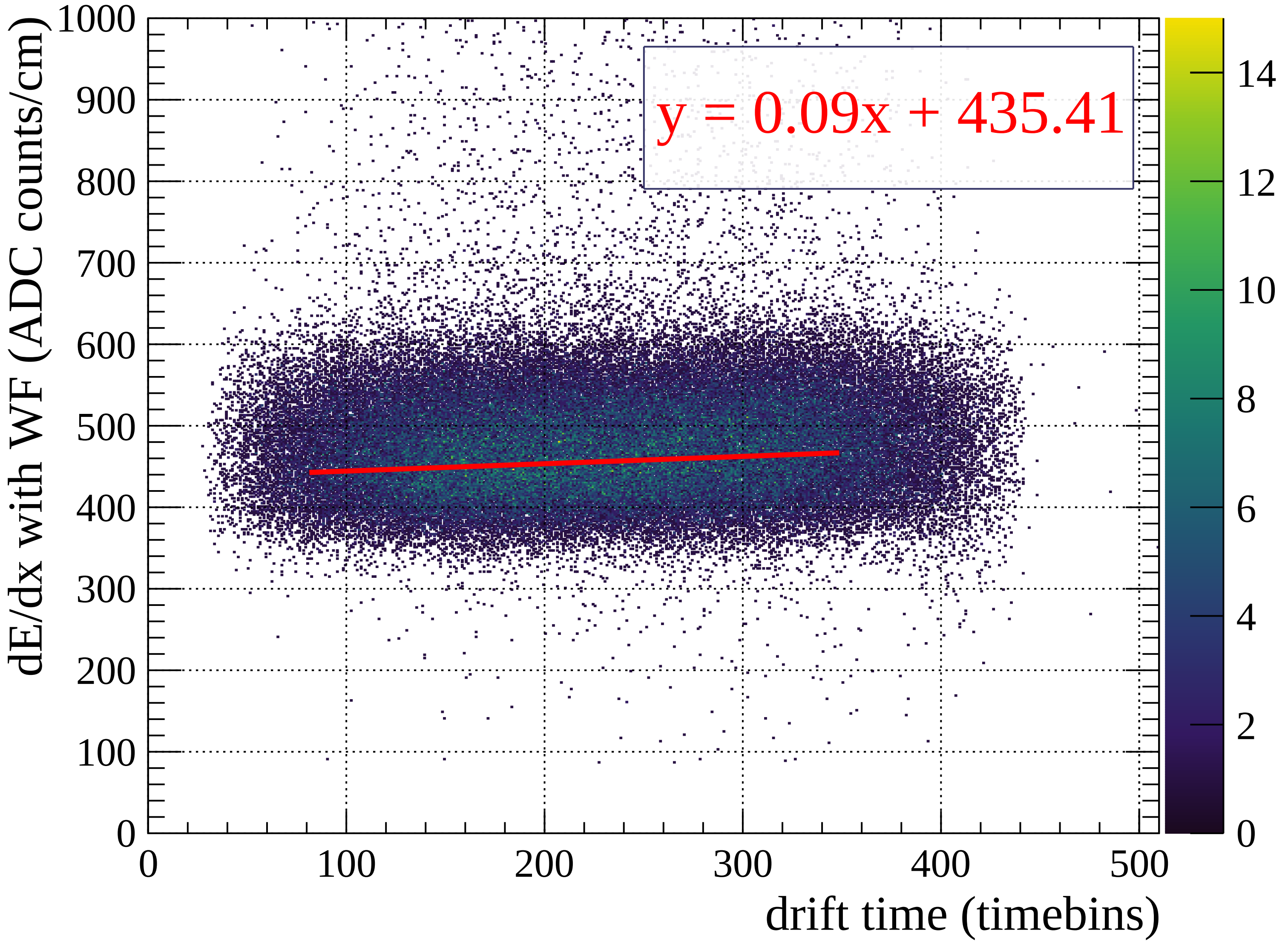
y = -0.05x + 482.63
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
5
10
15
20
25
30

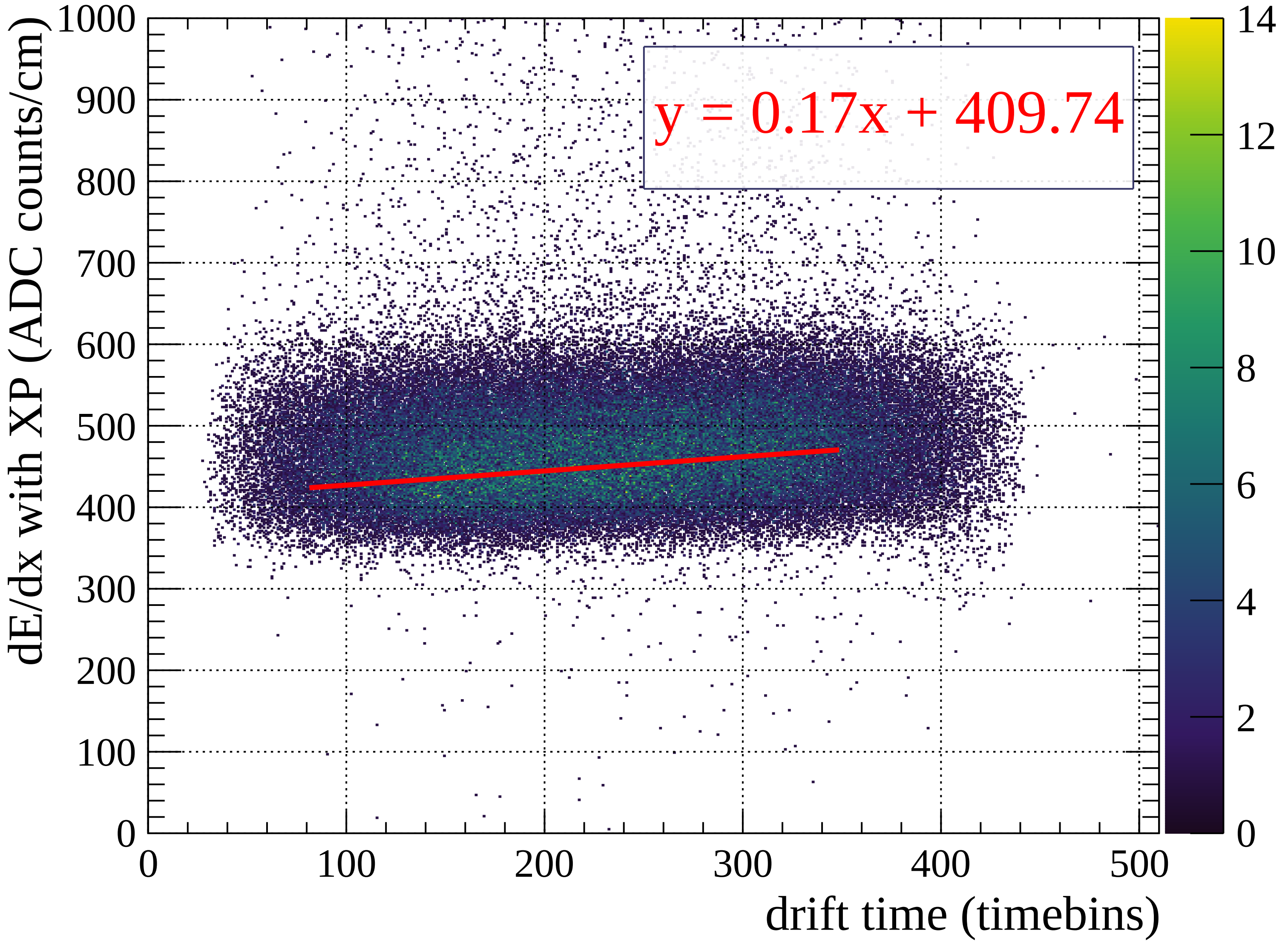


y = -0.03x + 466.47
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
5
10
15
20
25
30
35

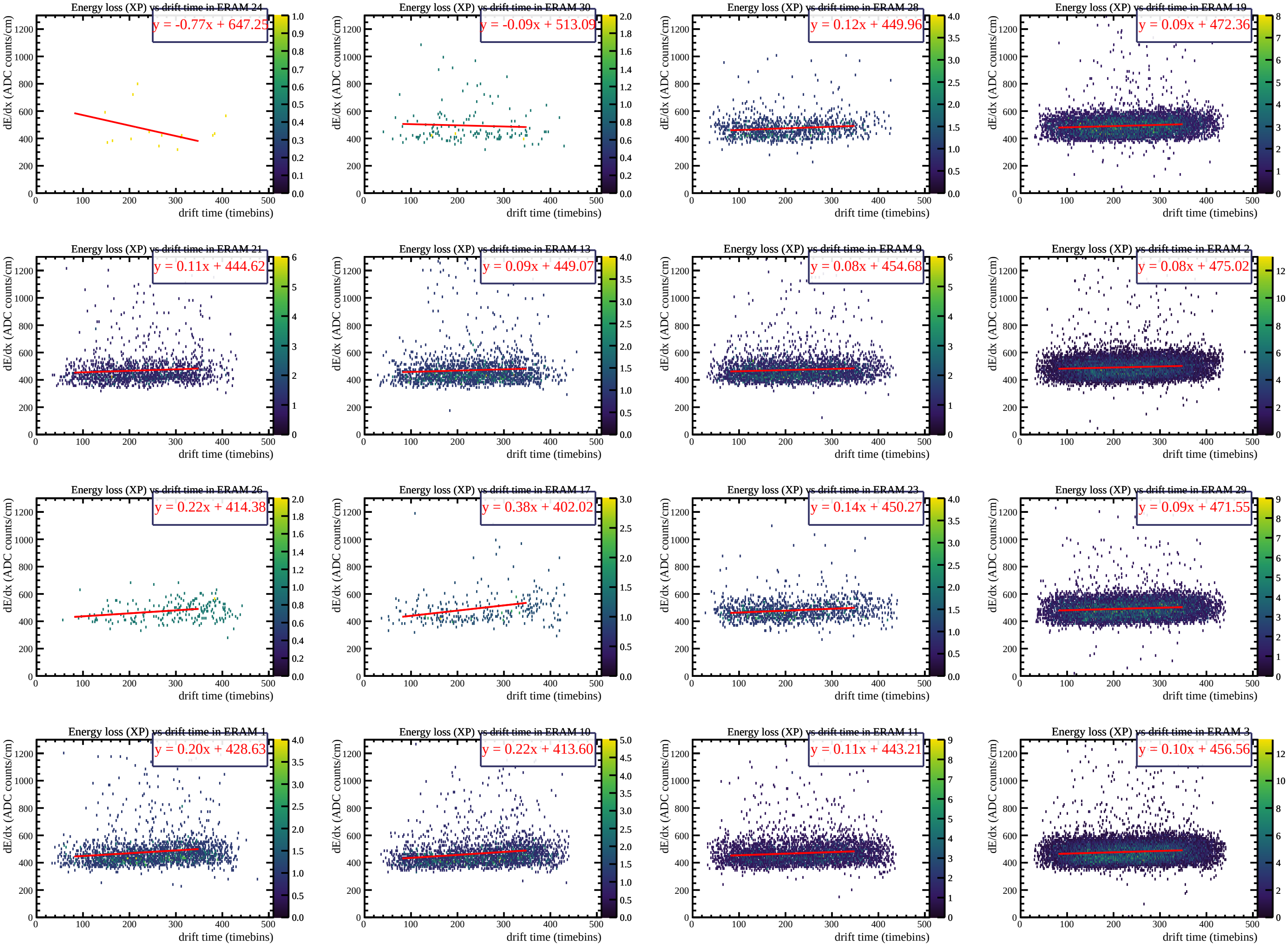

dE/dx resolution (%)
Drift time (timebins)
WF
XP
30
25
20
15
10
5
0
100
200
300
400
500

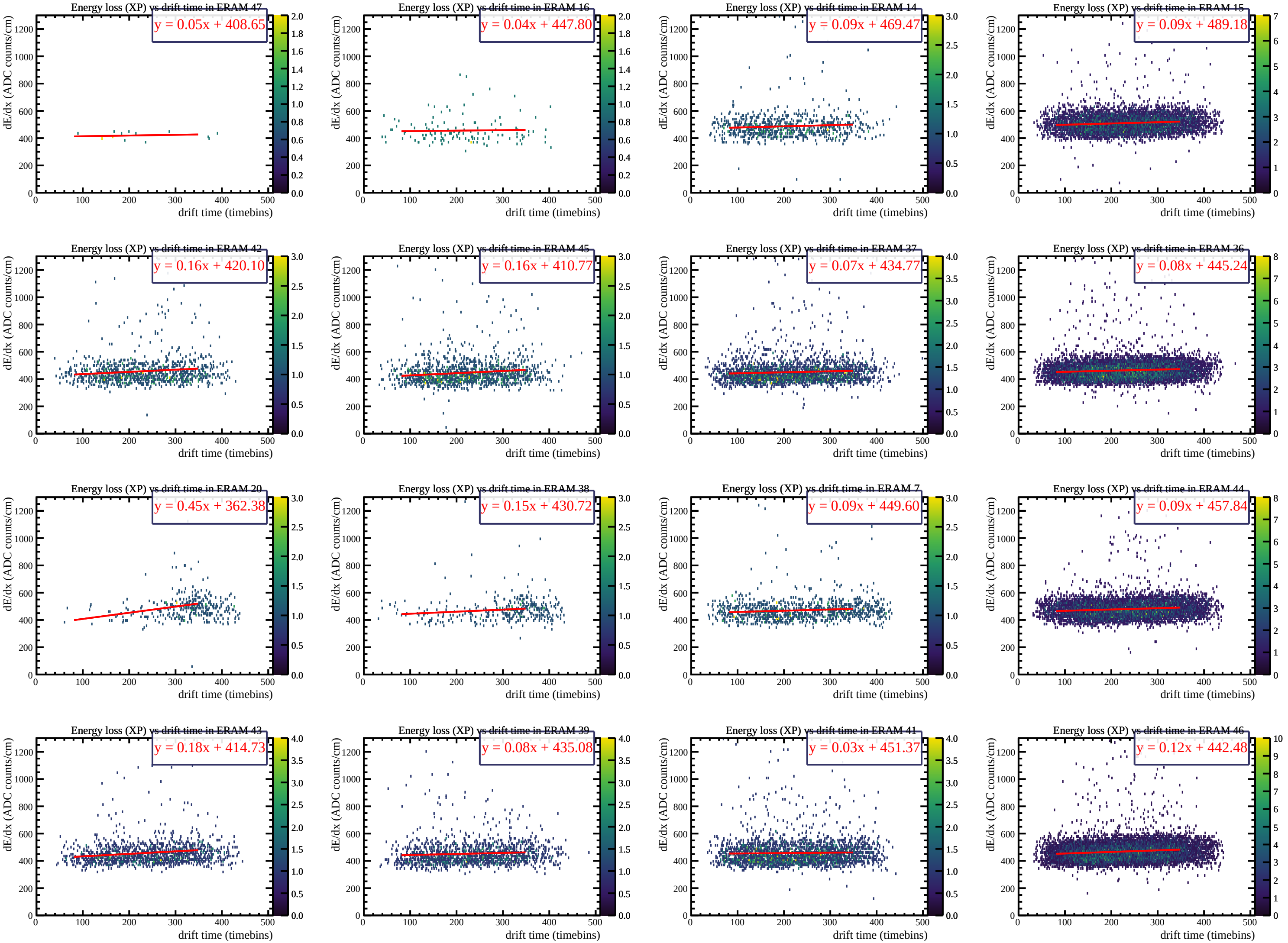

WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

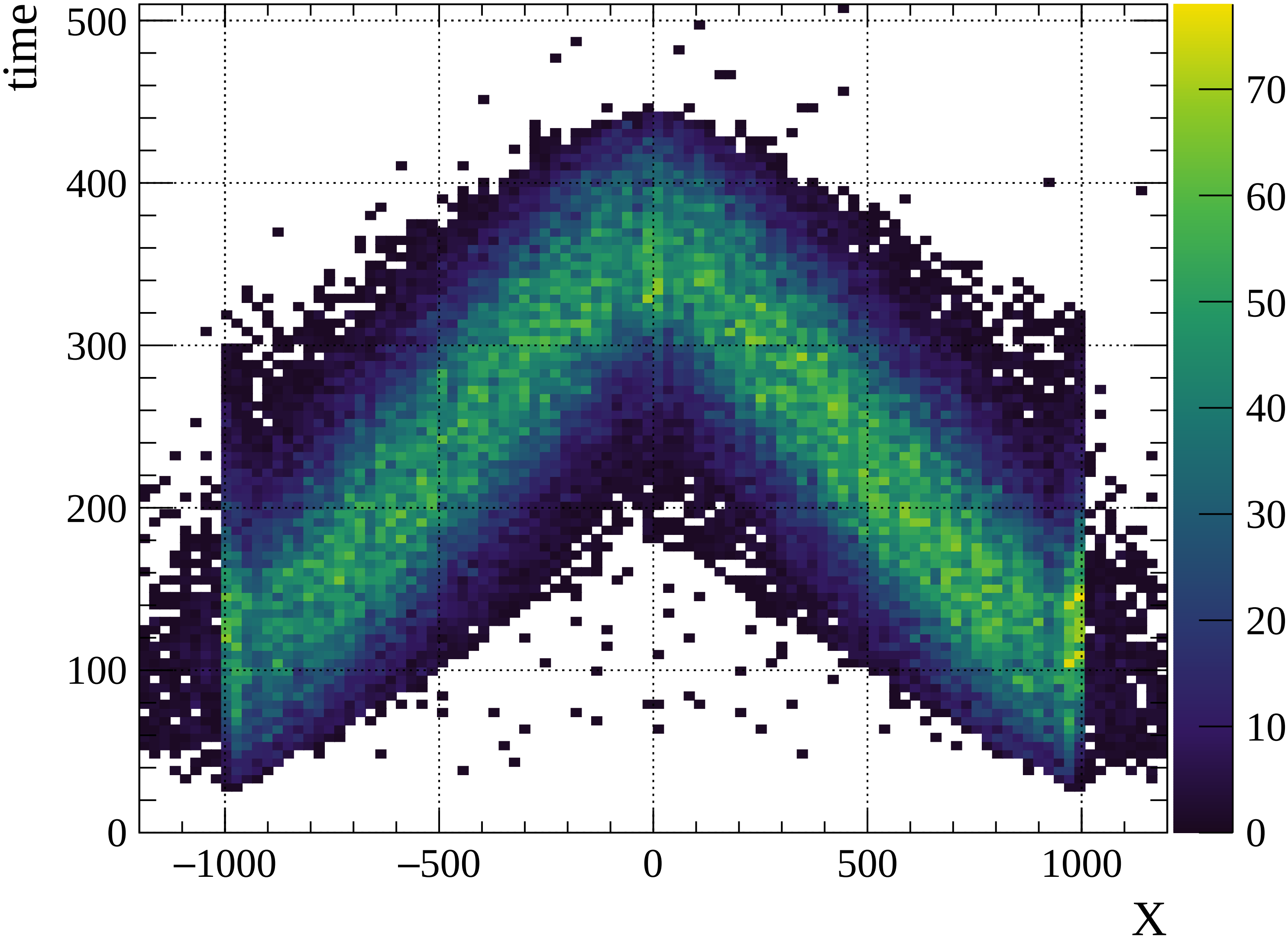

y = 0.09x + 435.41
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

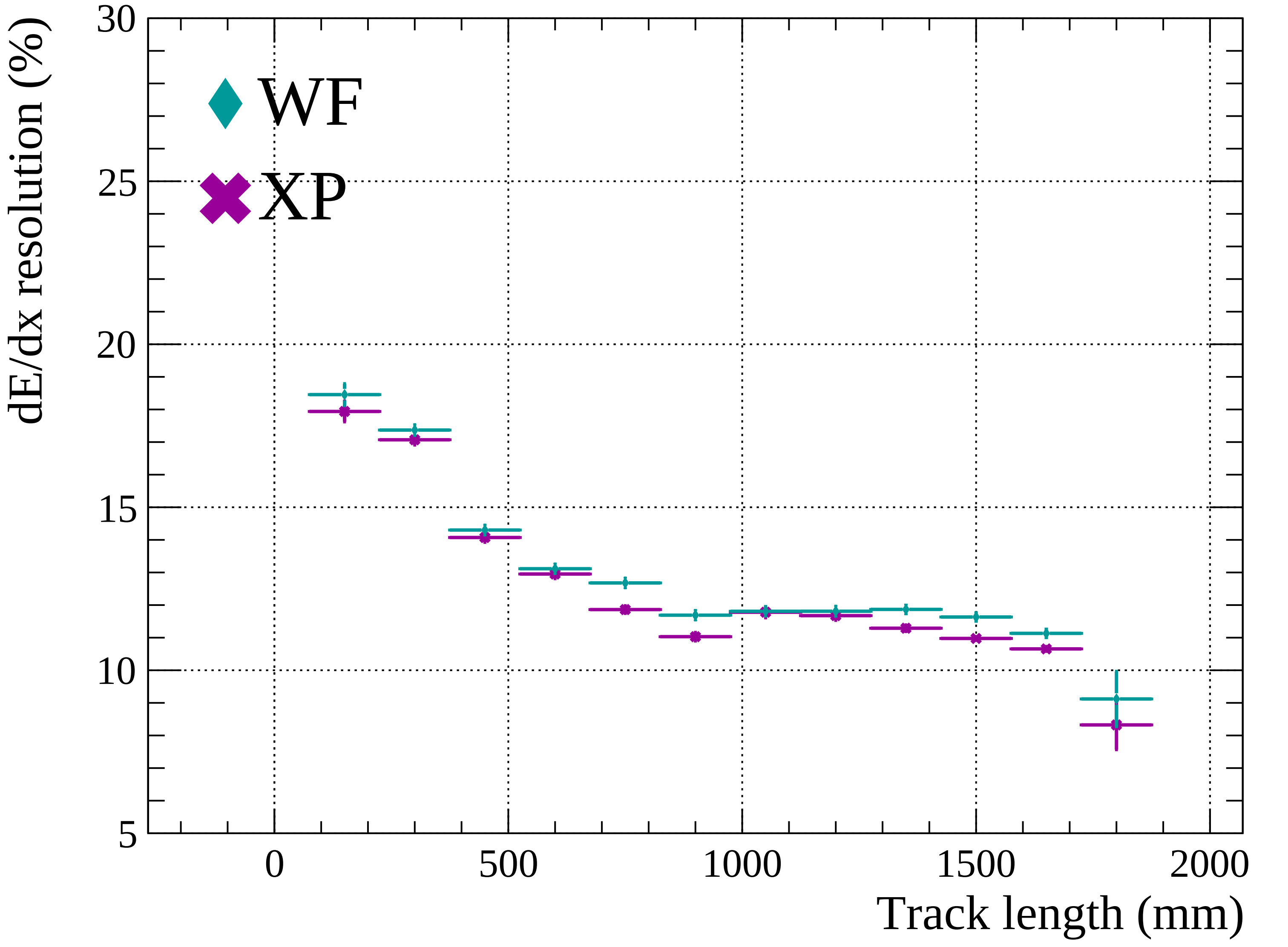


y = 0.17x + 409.74
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

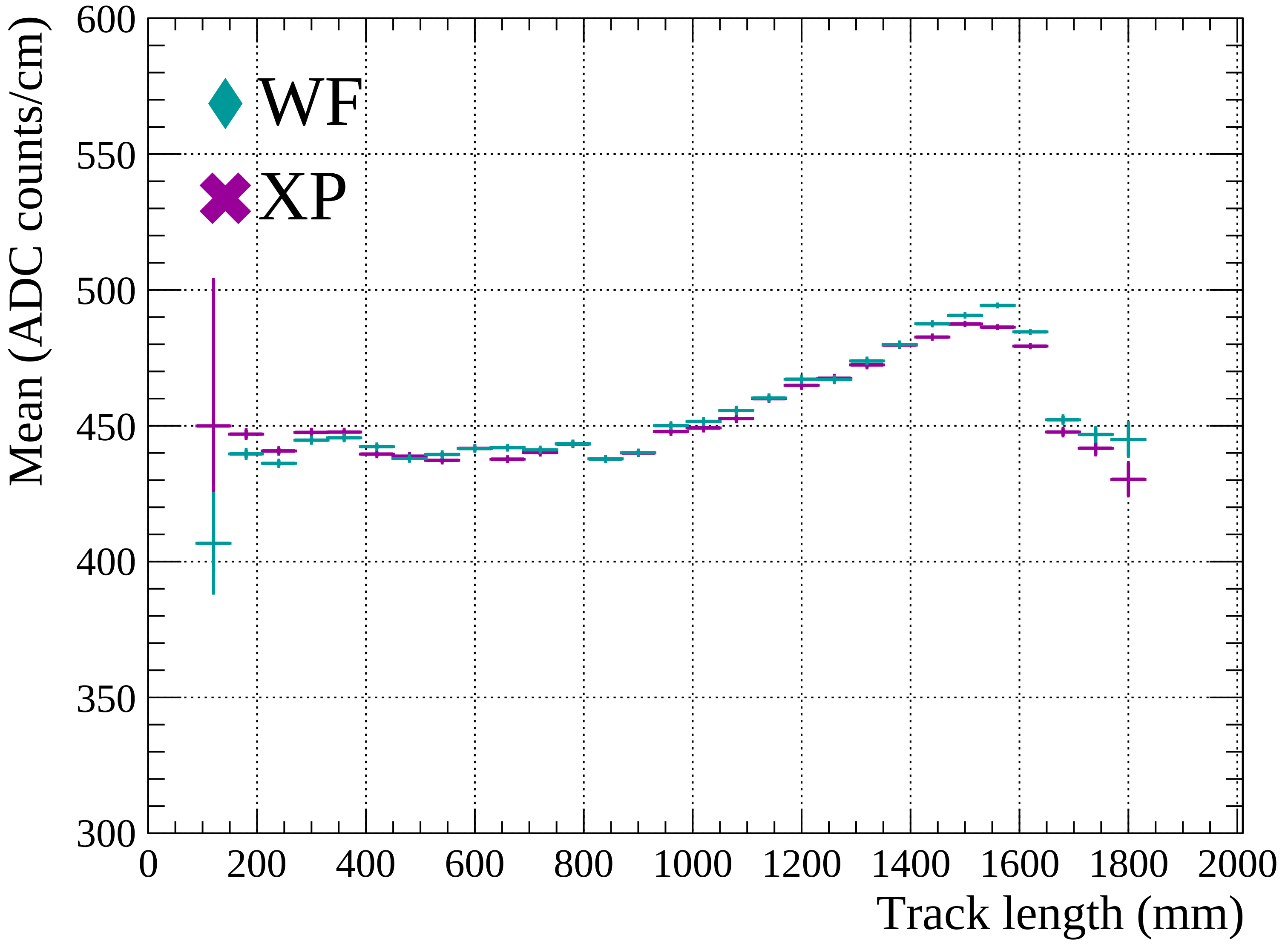

Energy loss (XP) vs drift time in ERAM 24
y = -0.77x + 647.25
Energy loss (XP) vs drift time in ERAM 30
y = -0.09x + 513.09
Energy loss (XP) vs drift time in ERAM 28
y = 0.12x + 449.96
Energy loss (XP) vs drift time in ERAM 19
y = 0.09x + 472.36
Energy loss (XP) vs drift time in ERAM 21
y = 0.11x + 444.62
Energy loss (XP) vs drift time in ERAM 13
y = 0.09x + 449.07
Energy loss (XP) vs drift time in ERAM 9
y = 0.08x + 454.68
Energy loss (XP) vs drift time in ERAM 2
y = 0.08x + 475.02
Energy loss (XP) vs drift time in ERAM 26
y = 0.22x + 414.38
Energy loss (XP) vs drift time in ERAM 17
y = 0.38x + 402.02
Energy loss (XP) vs drift time in ERAM 23
y = 0.14x + 450.27
Energy loss (XP) vs drift time in ERAM 29
y = 0.09x + 471.55
Energy loss (XP) vs drift time in ERAM 1
y = 0.20x + 428.63
Energy loss (XP) vs drift time in ERAM 10
y = 0.22x + 413.60
Energy loss (XP) vs drift time in ERAM 11
y = 0.11x + 443.21
Energy loss (XP) vs drift time in ERAM 3
y = 0.10x + 456.56
dE/dx (ADC counts/cm)
drift time (timebins)

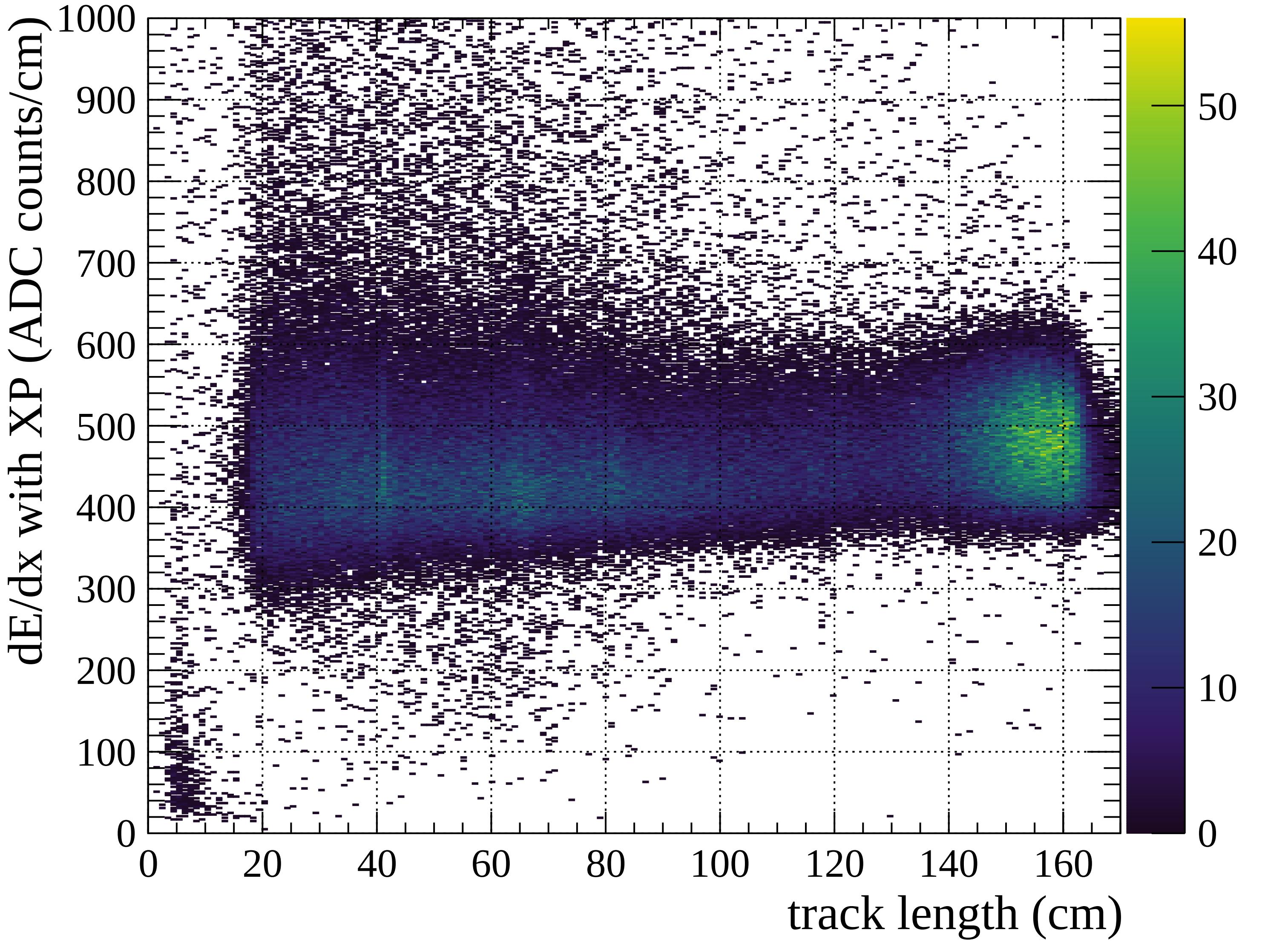

Energy loss (XP) vs drift time in ERAM 47
y = 0.05x + 408.65
Energy loss (XP) vs drift time in ERAM 16
y = 0.04x + 447.80
Energy loss (XP) vs drift time in ERAM 14
y = 0.09x + 469.47
Energy loss (XP) vs drift time in ERAM 15
y = 0.09x + 489.18
Energy loss (XP) vs drift time in ERAM 42
y = 0.16x + 420.10
Energy loss (XP) vs drift time in ERAM 45
y = 0.16x + 410.77
Energy loss (XP) vs drift time in ERAM 37
y = 0.07x + 434.77
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 445.24
Energy loss (XP) vs drift time in ERAM 20
y = 0.45x + 362.38
Energy loss (XP) vs drift time in ERAM 38
y = 0.15x + 430.72
Energy loss (XP) vs drift time in ERAM 7
y = 0.09x + 449.60
Energy loss (XP) vs drift time in ERAM 44
y = 0.09x + 457.84
Energy loss (XP) vs drift time in ERAM 43
y = 0.18x + 414.73
Energy loss (XP) vs drift time in ERAM 39
y = 0.08x + 435.08
Energy loss (XP) vs drift time in ERAM 41
y = 0.03x + 451.37
Energy loss (XP) vs drift time in ERAM 46
y = 0.12x + 442.48
dE/dx (ADC counts/cm)
drift time (timebins)

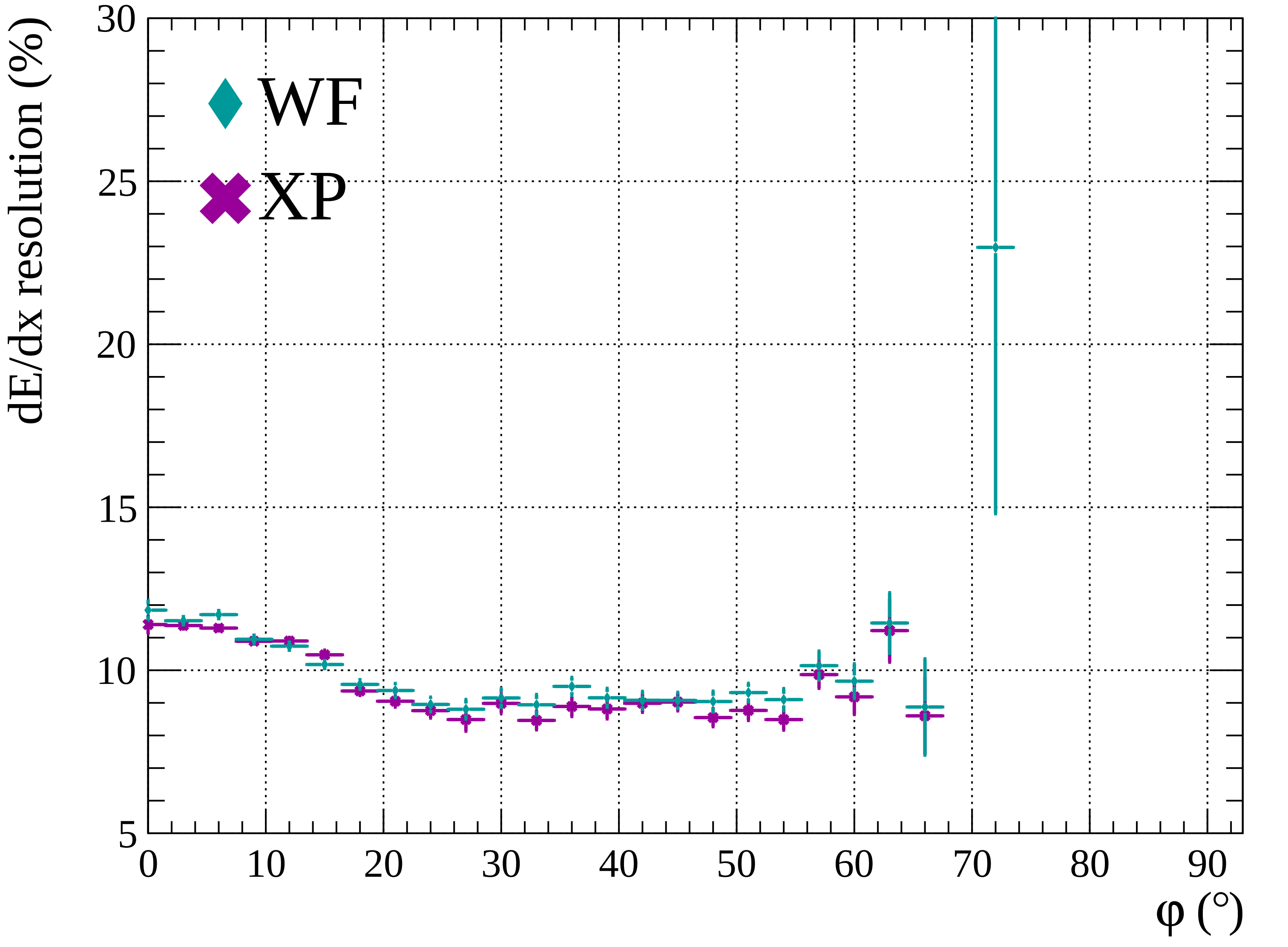

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
70
60
50
40
30
20
10
0



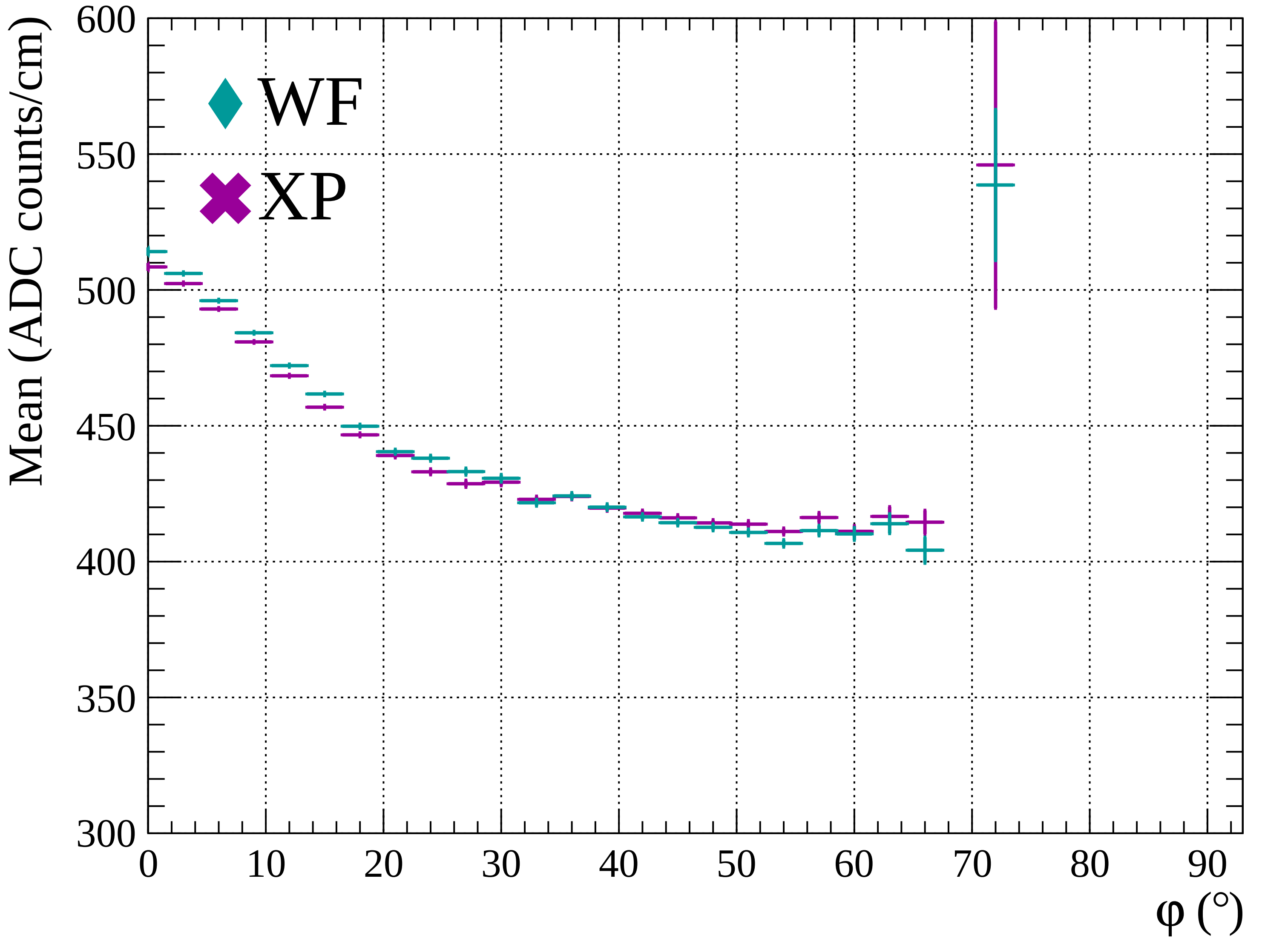
WF
XP
dE/dx resolution (%)
Track length (mm)
30
25
20
15
10
5
0
500
1000
1500
2000


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Track length (mm)

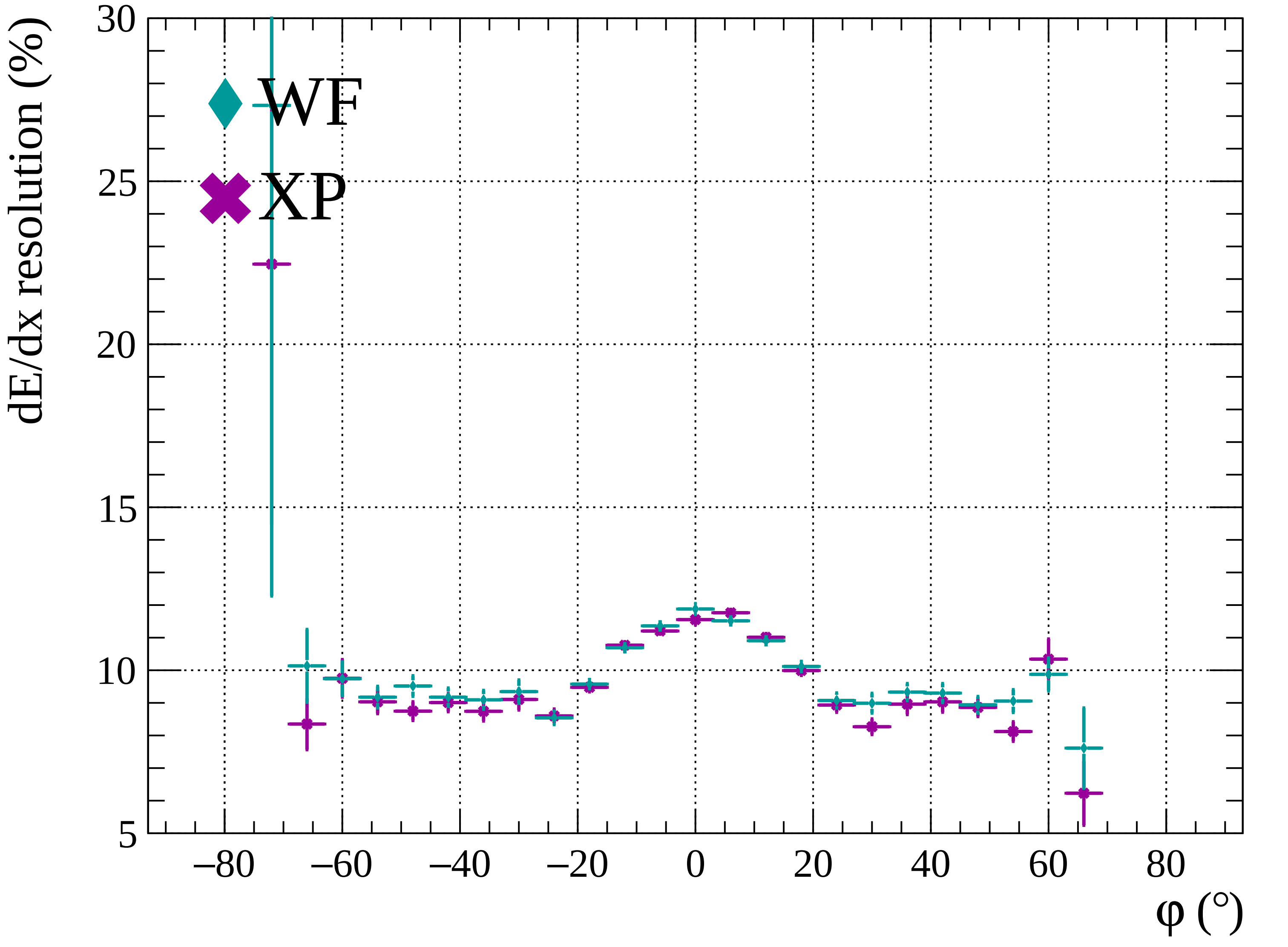

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
50
40
30
20
10
0



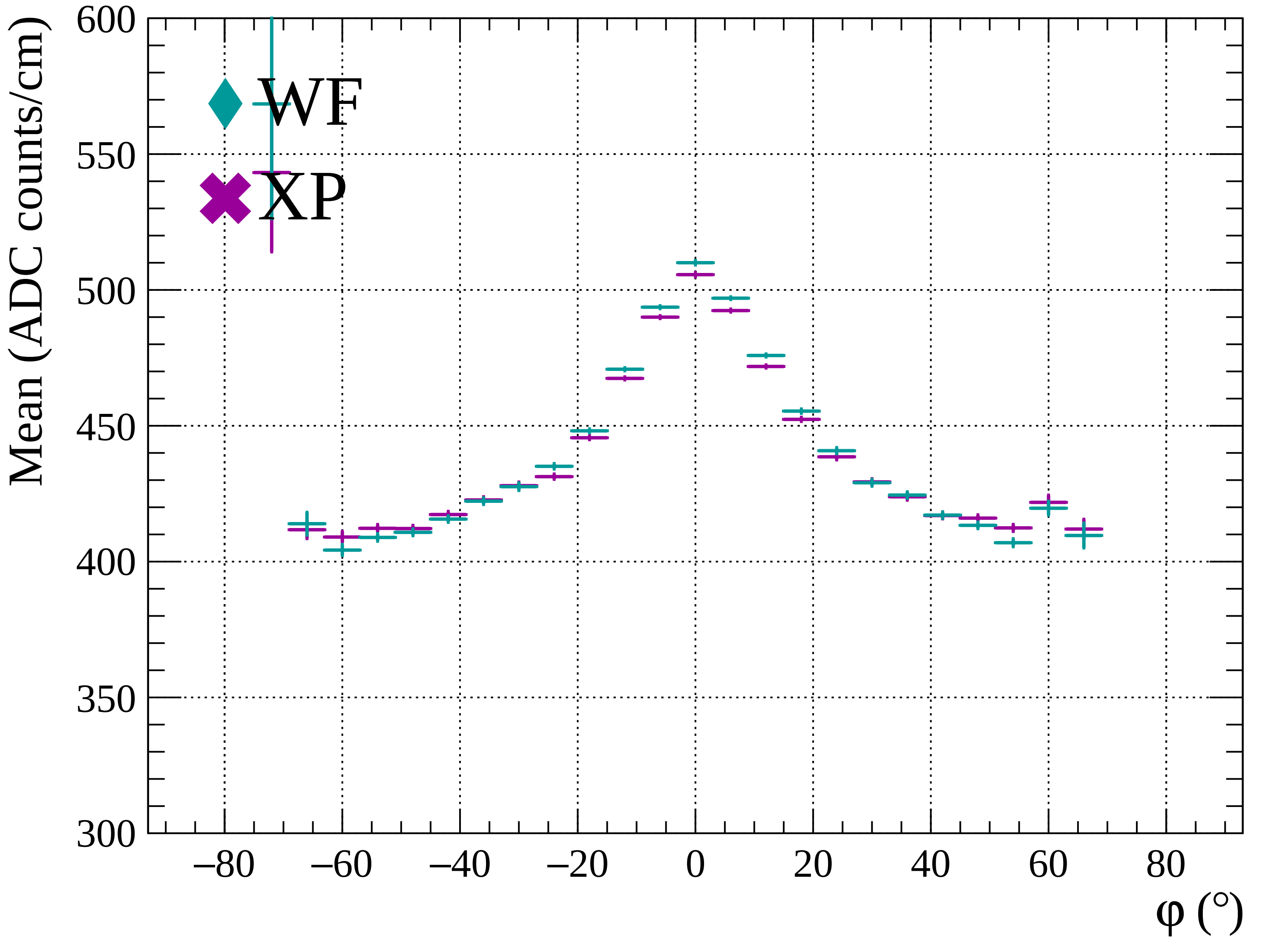
WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



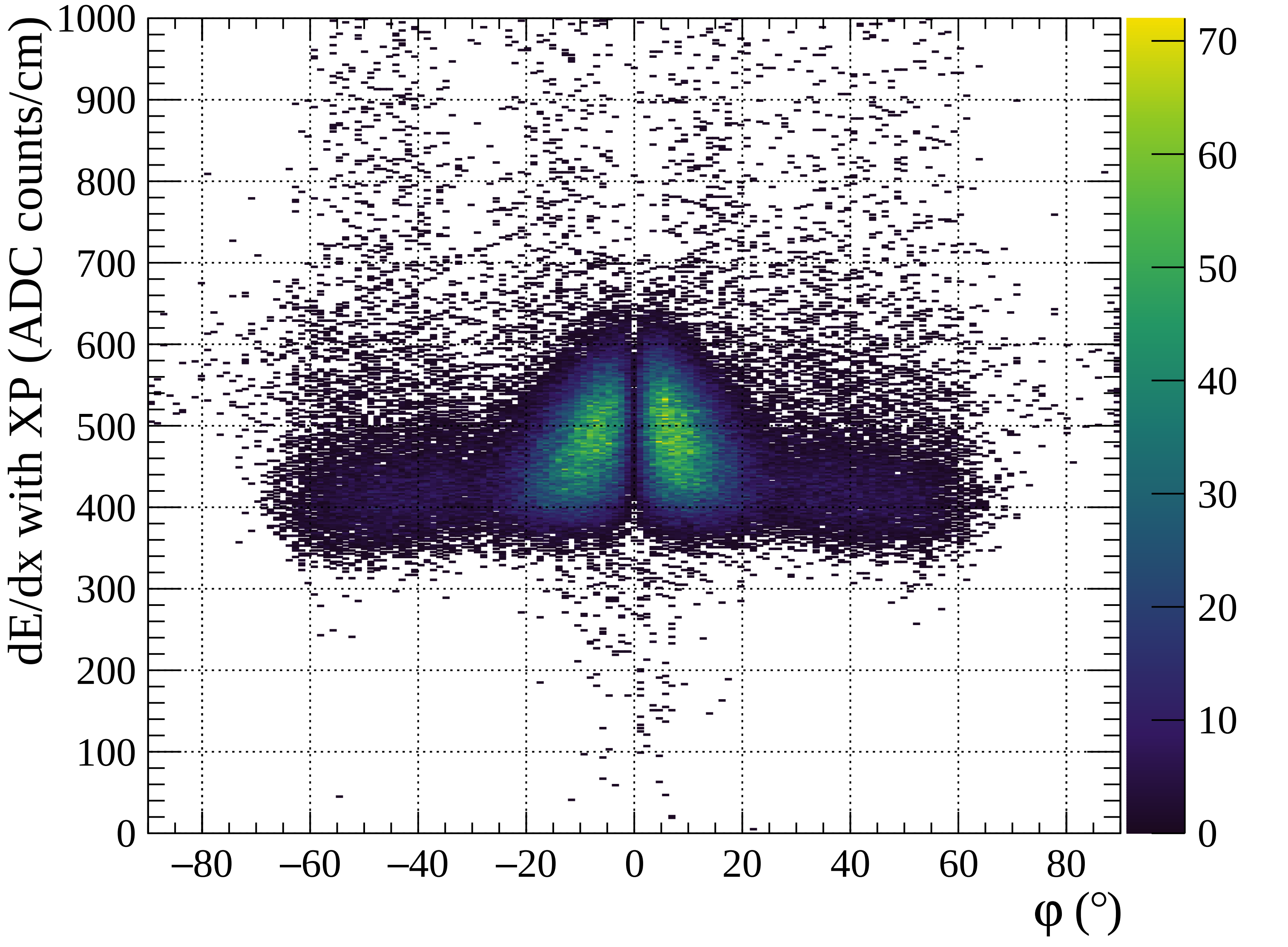
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
φ (°)



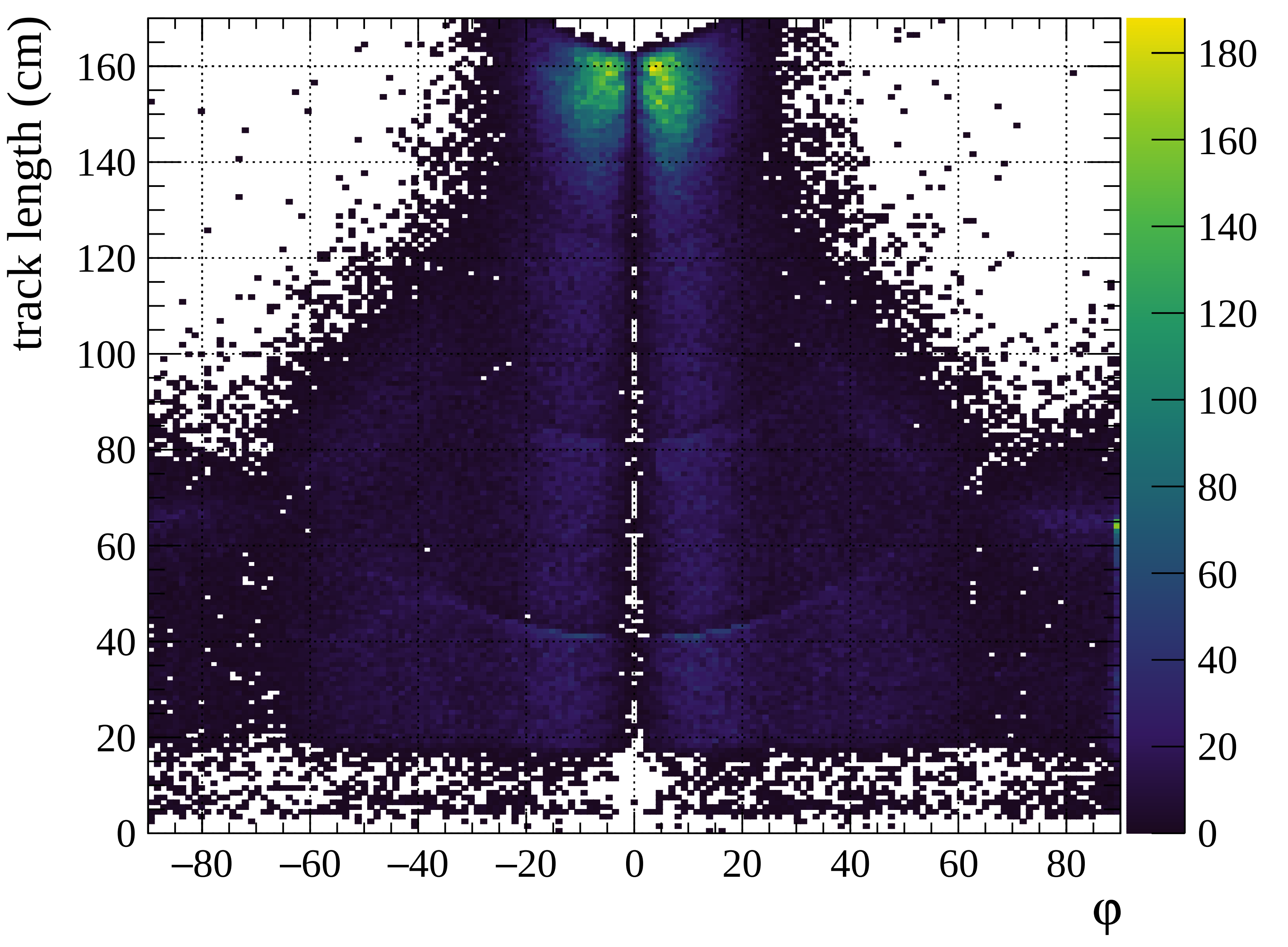
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

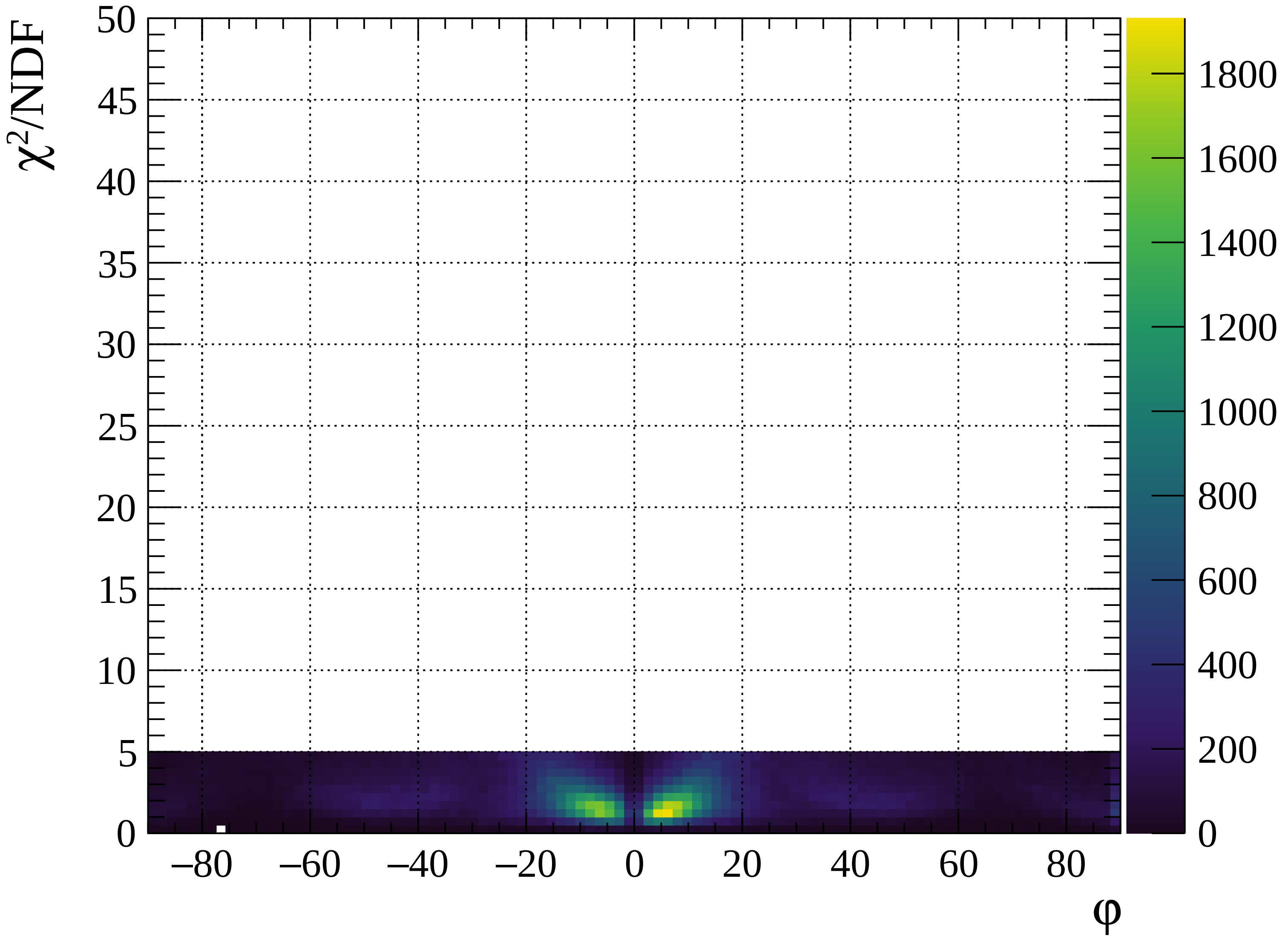


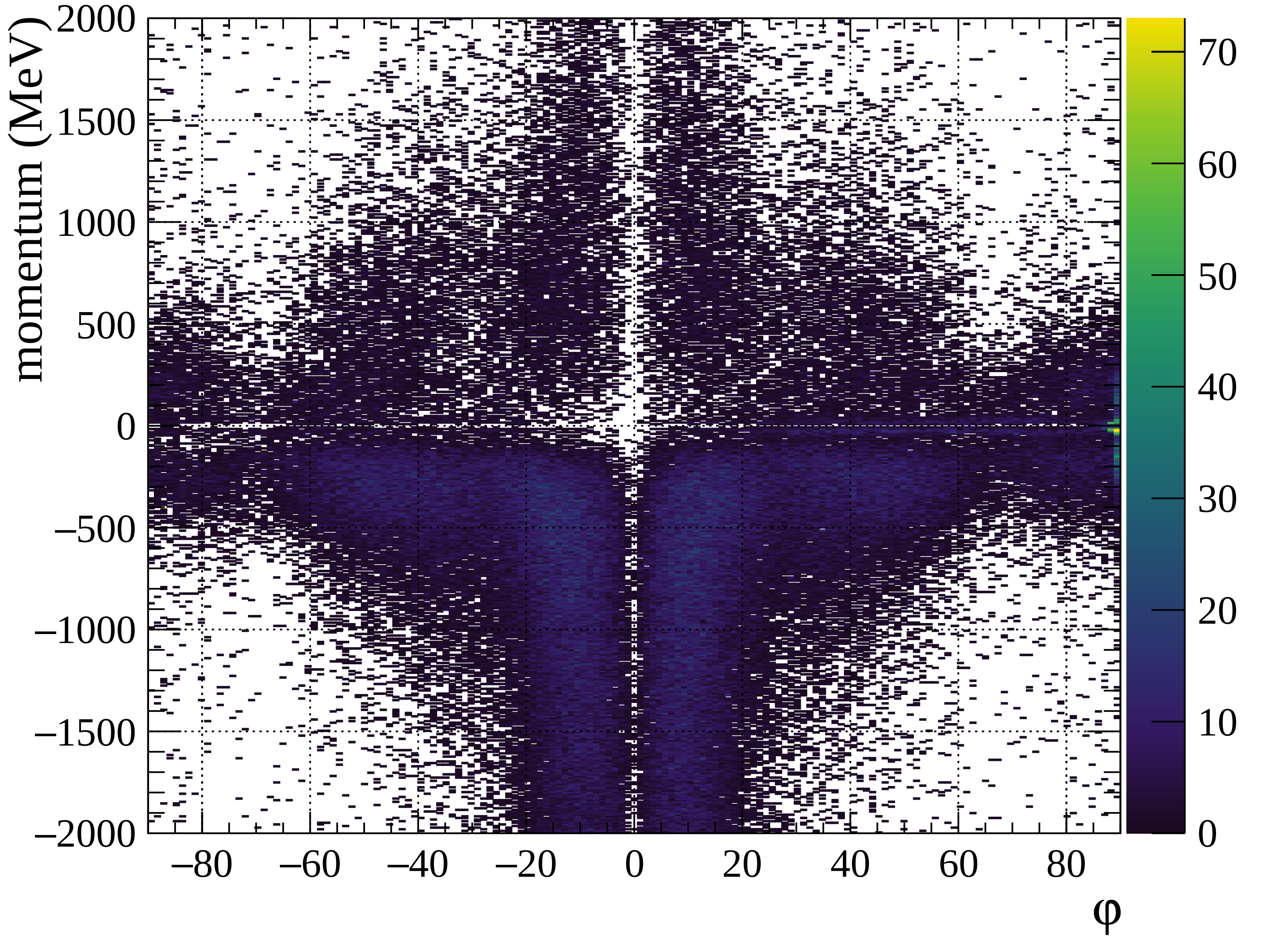
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)
WF
XP



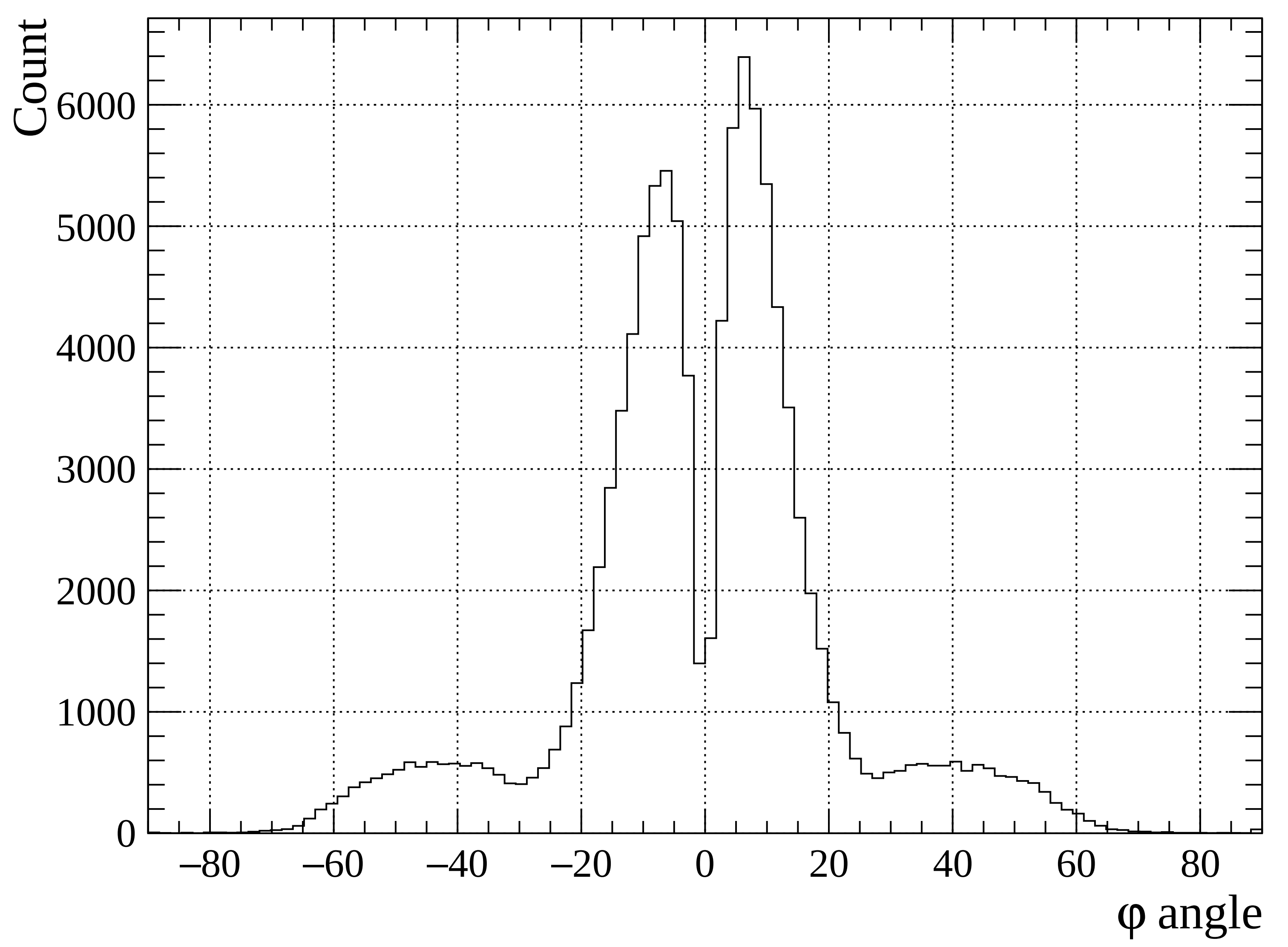
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
70
60
50
40
30
20
10
0

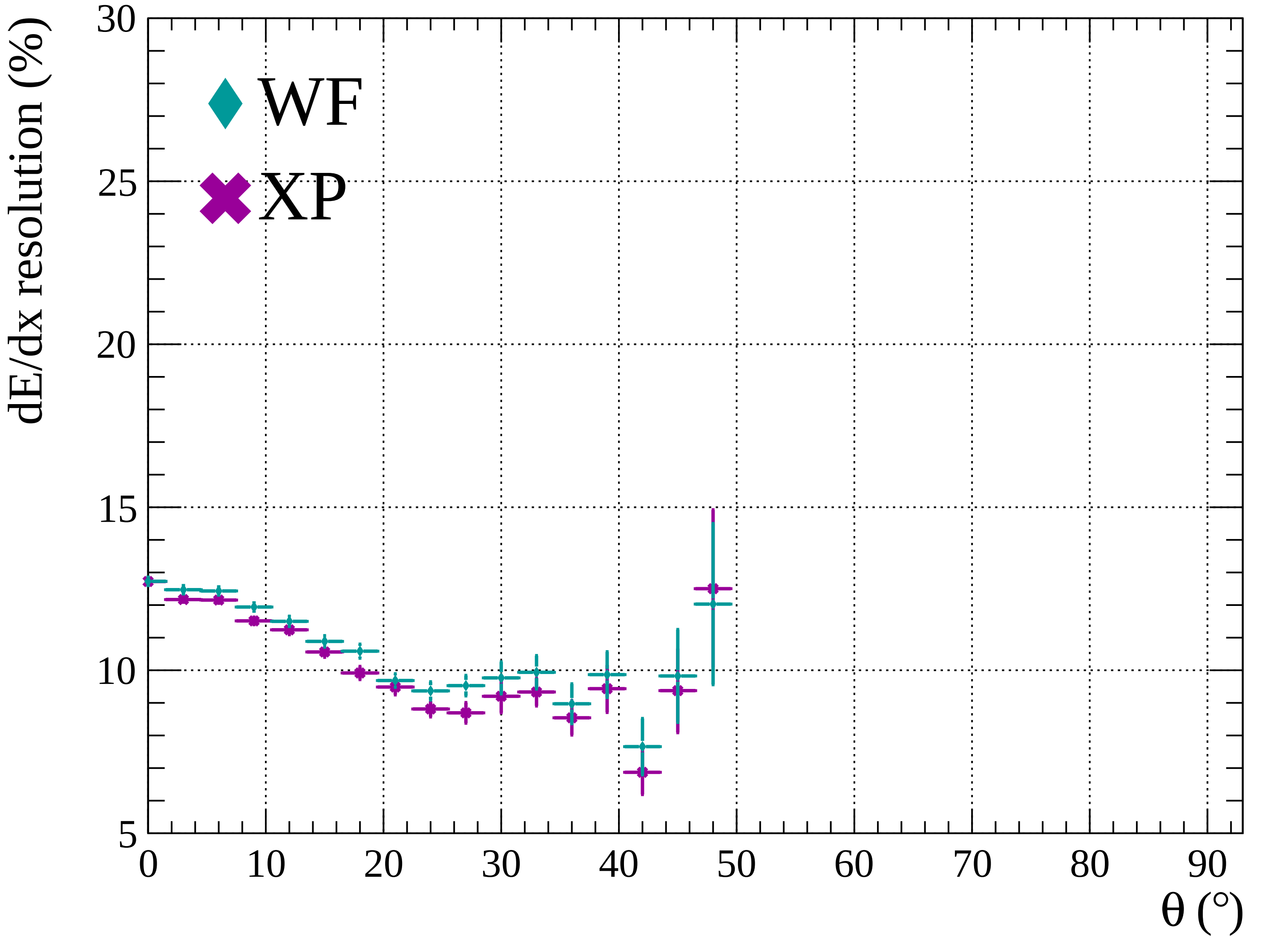

track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
20
40
60
80
100
120
140
160
180


χ²/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
1800
1600
1400
1200
1000
800
600
400
200
0

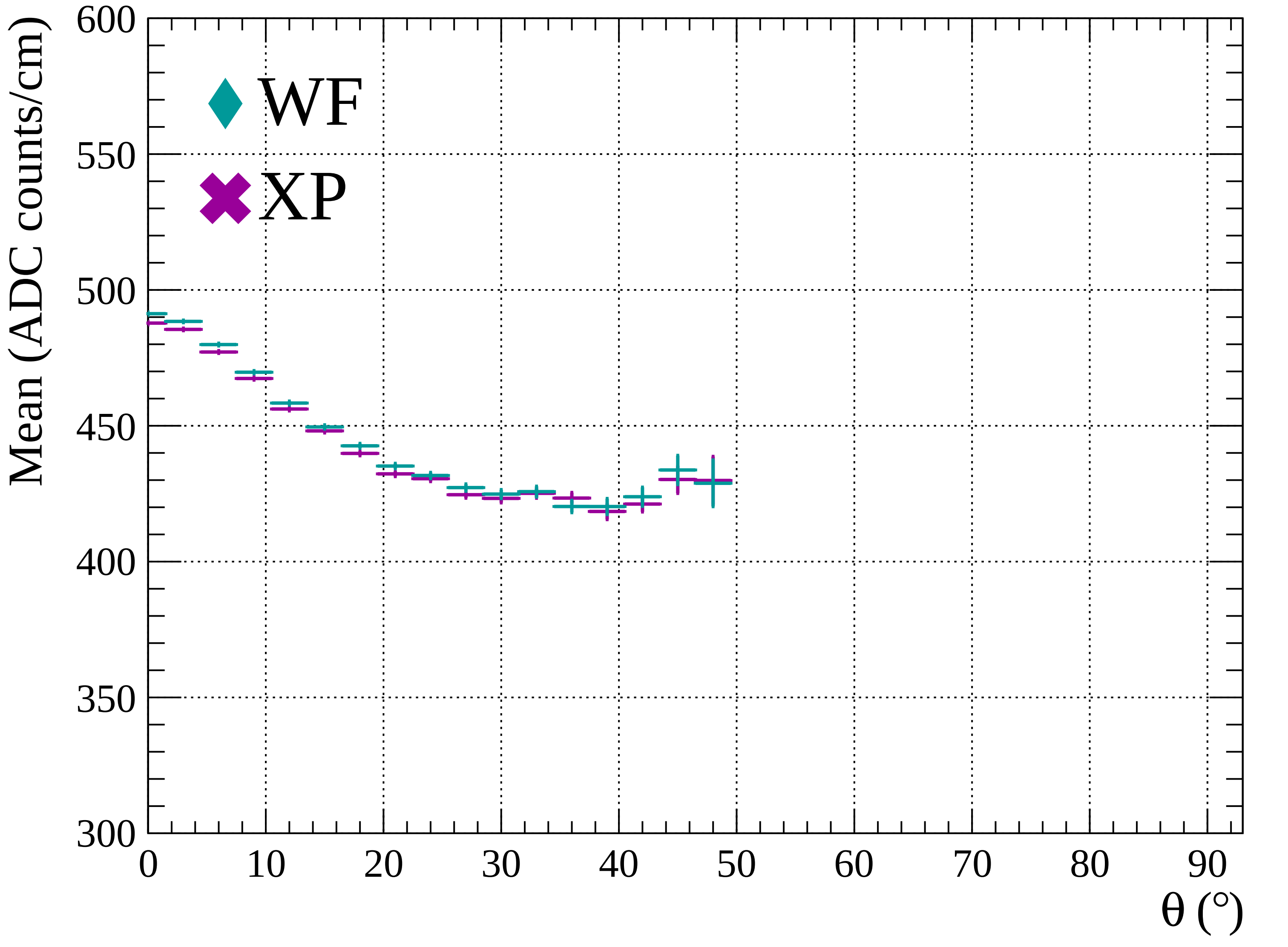


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
70
60
50
40
30
20
10
0

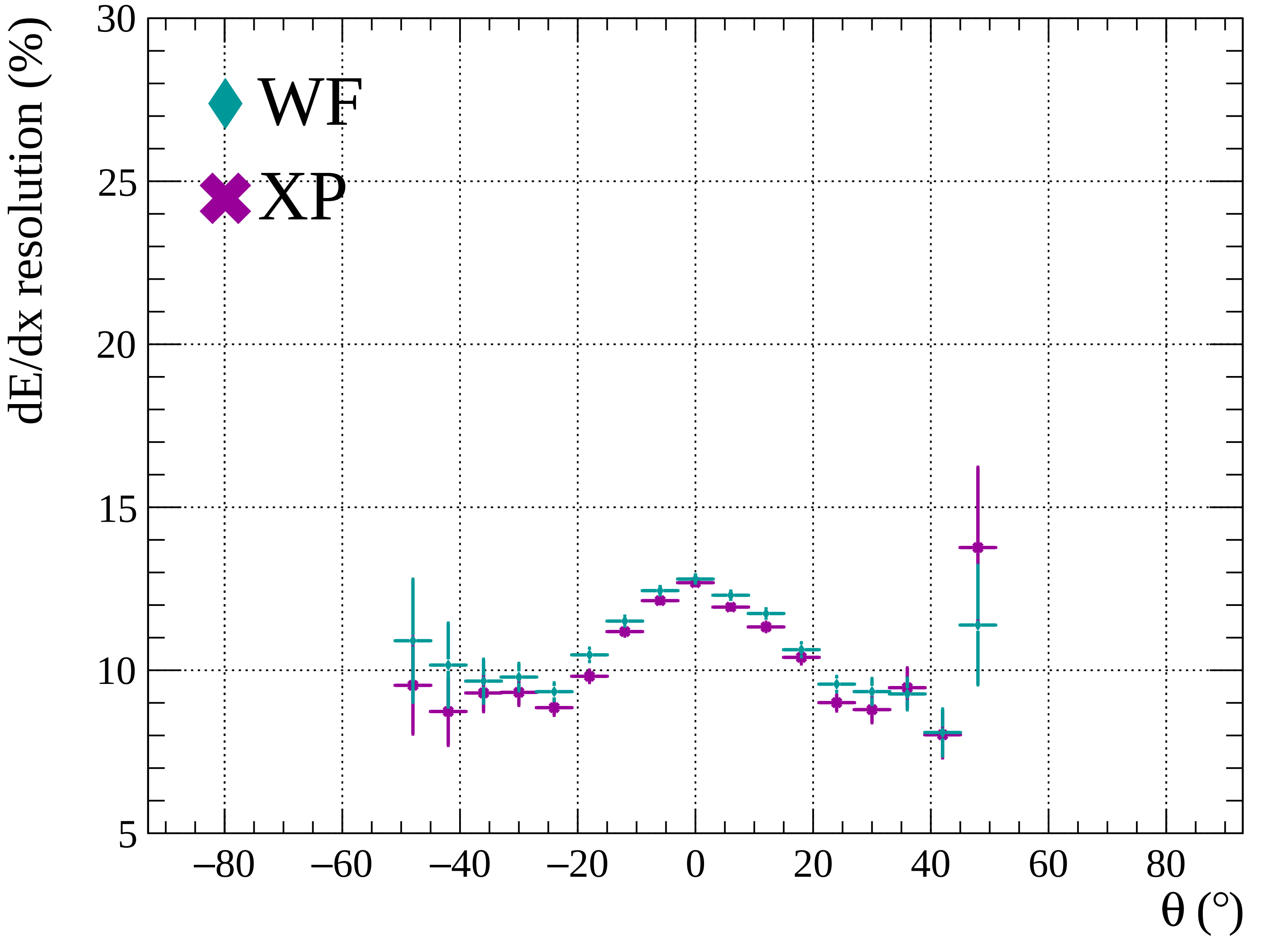

Count
6000
5000
4000
3000
2000
1000
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



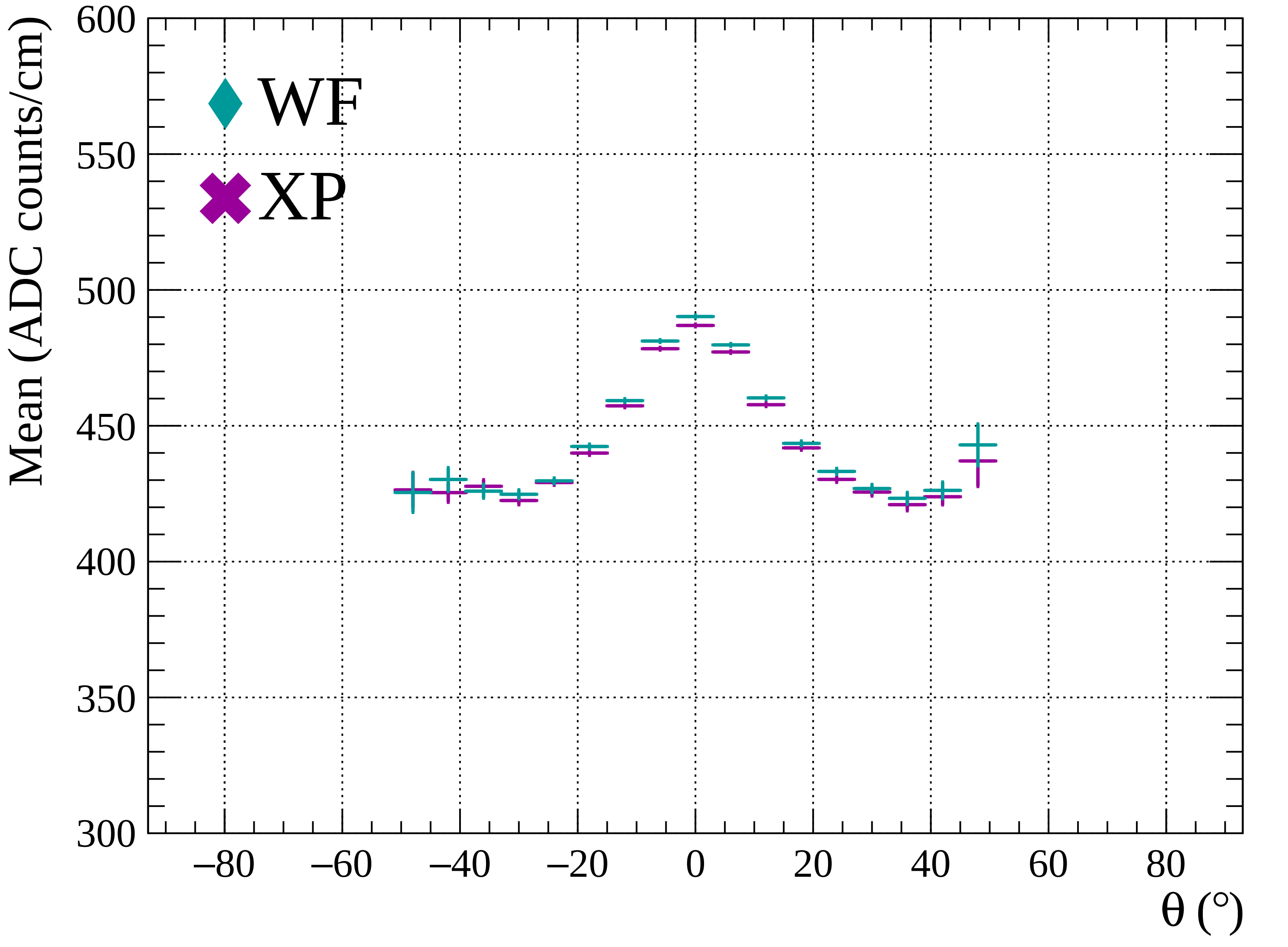
WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



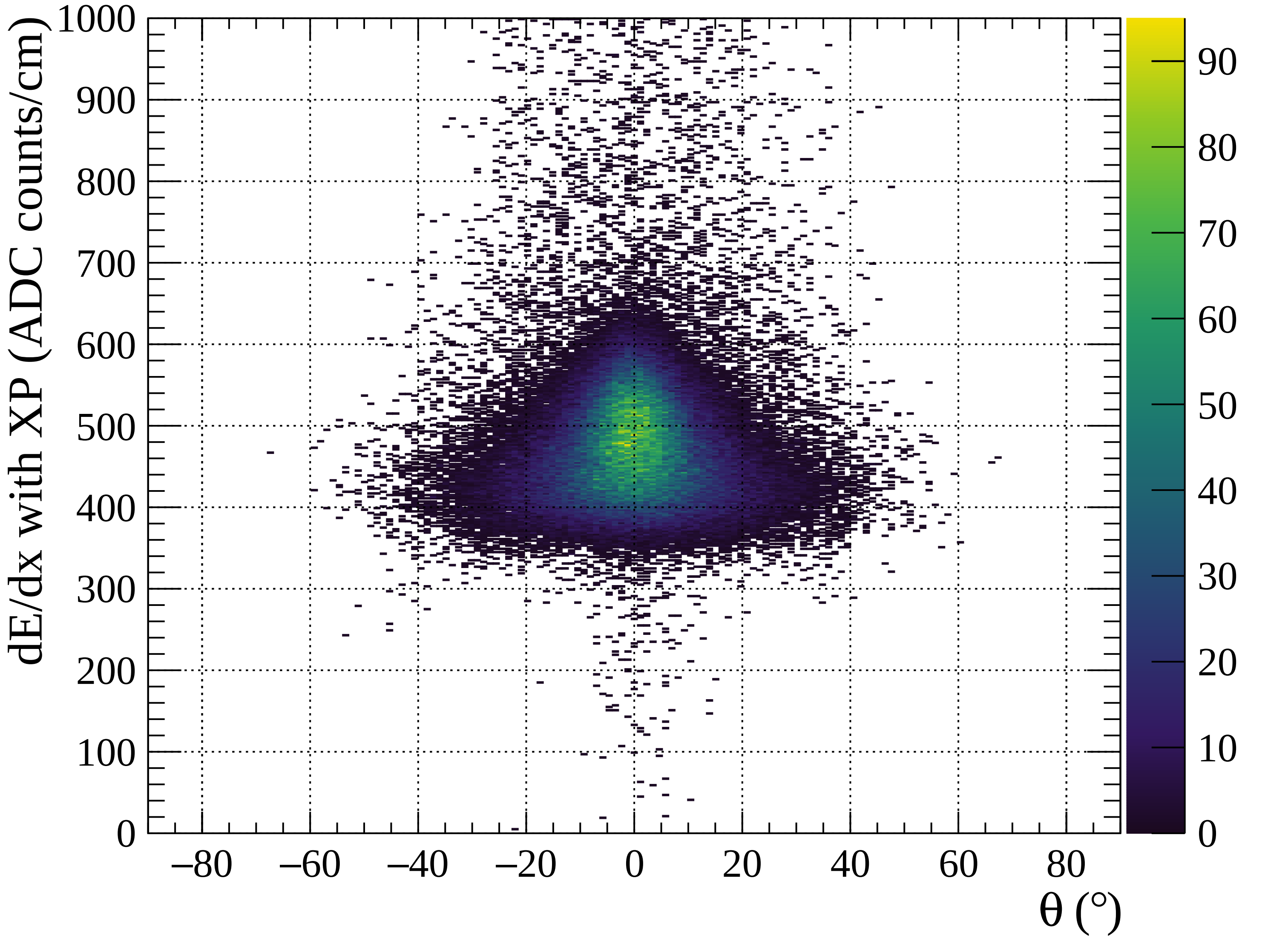
Mean (ADC counts/cm)
θ (°)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

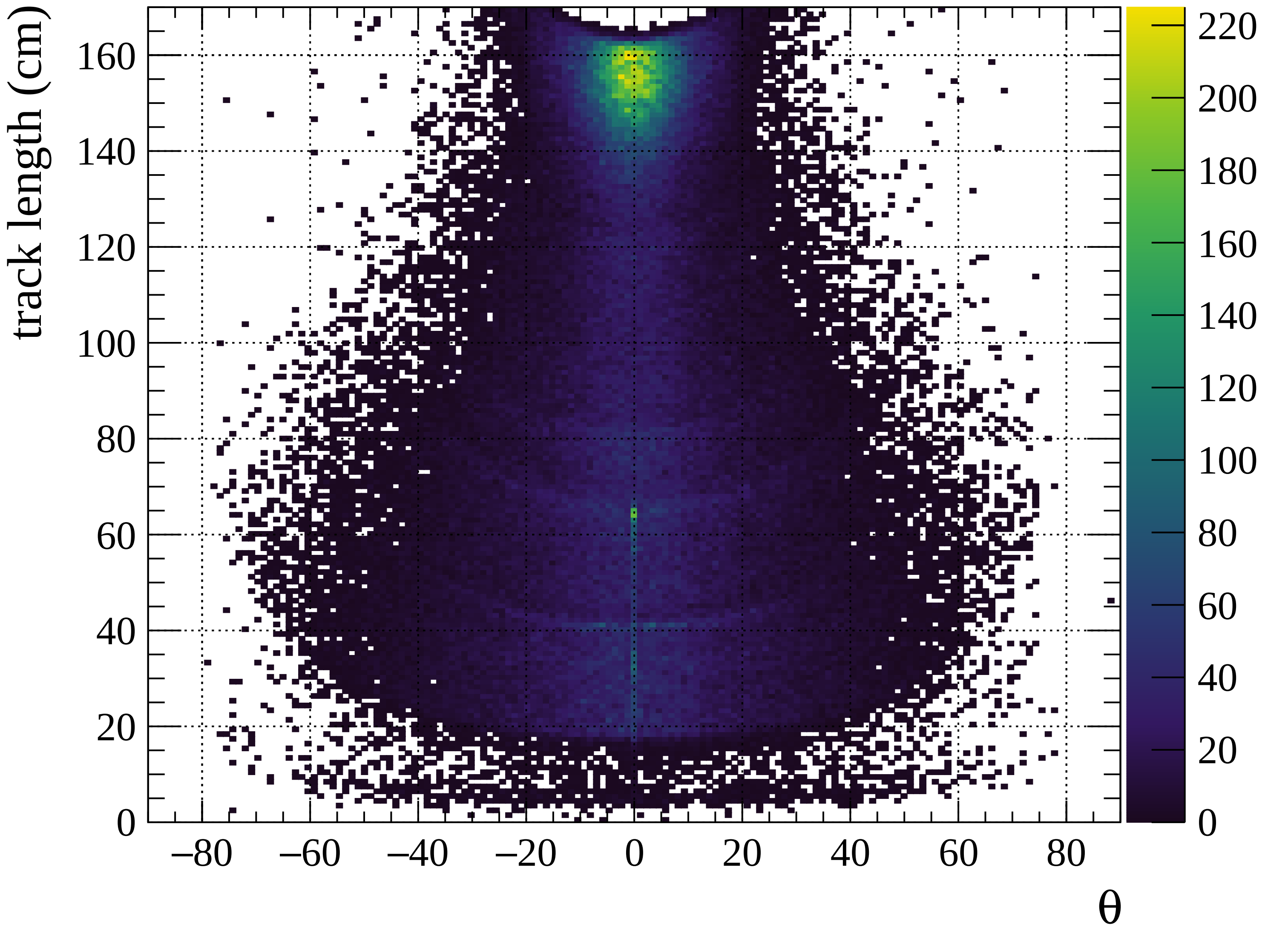


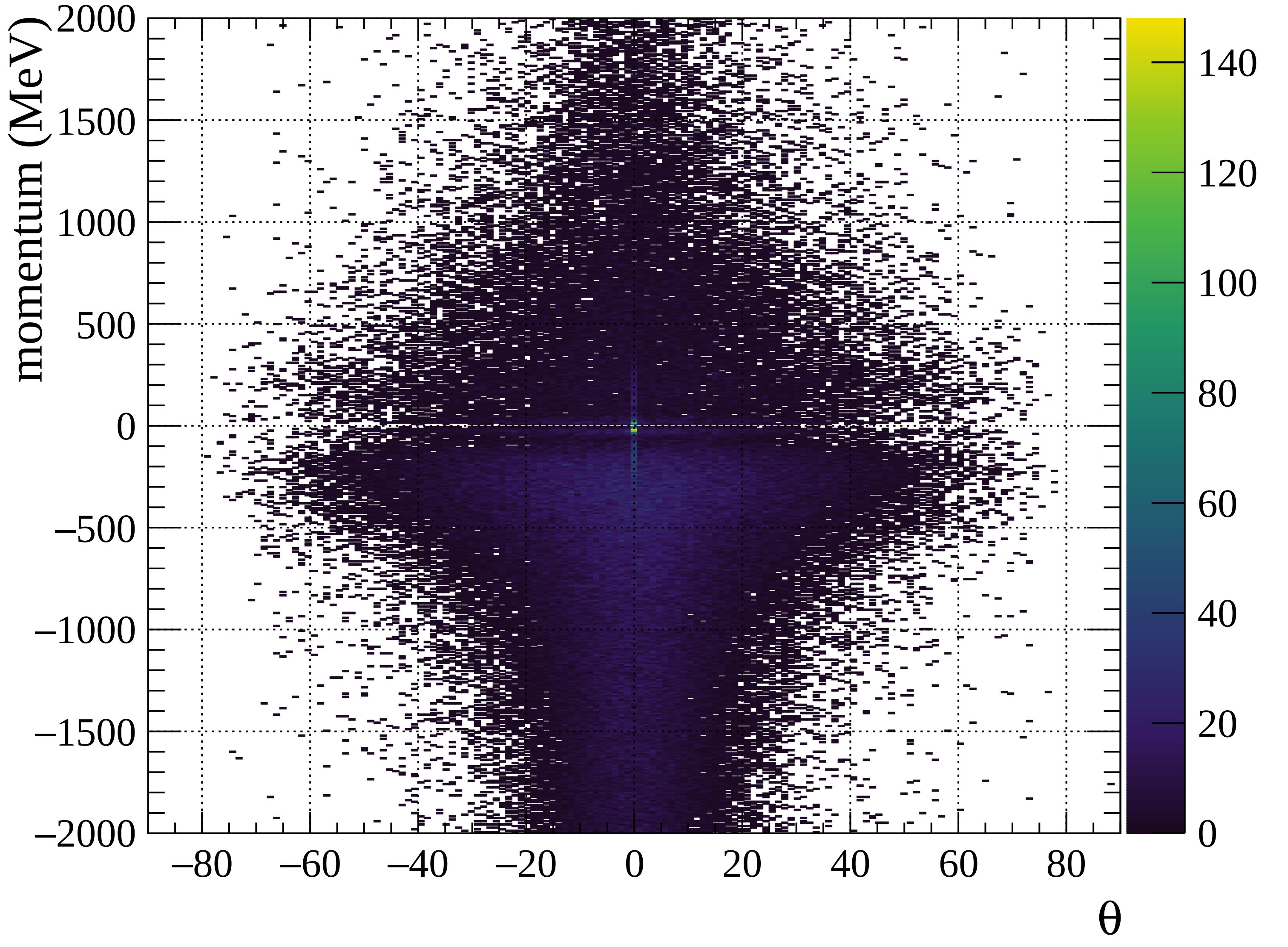
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



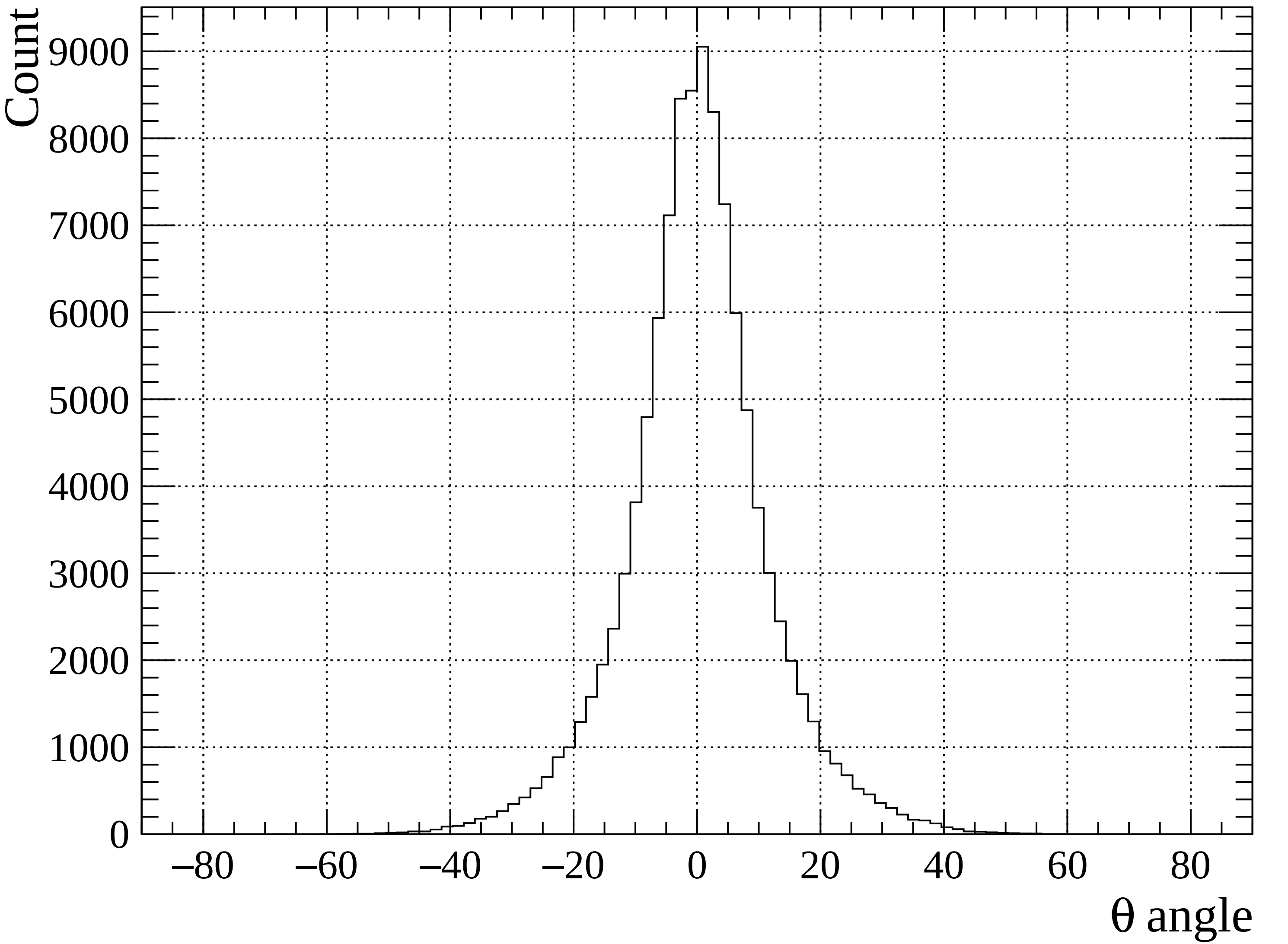
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

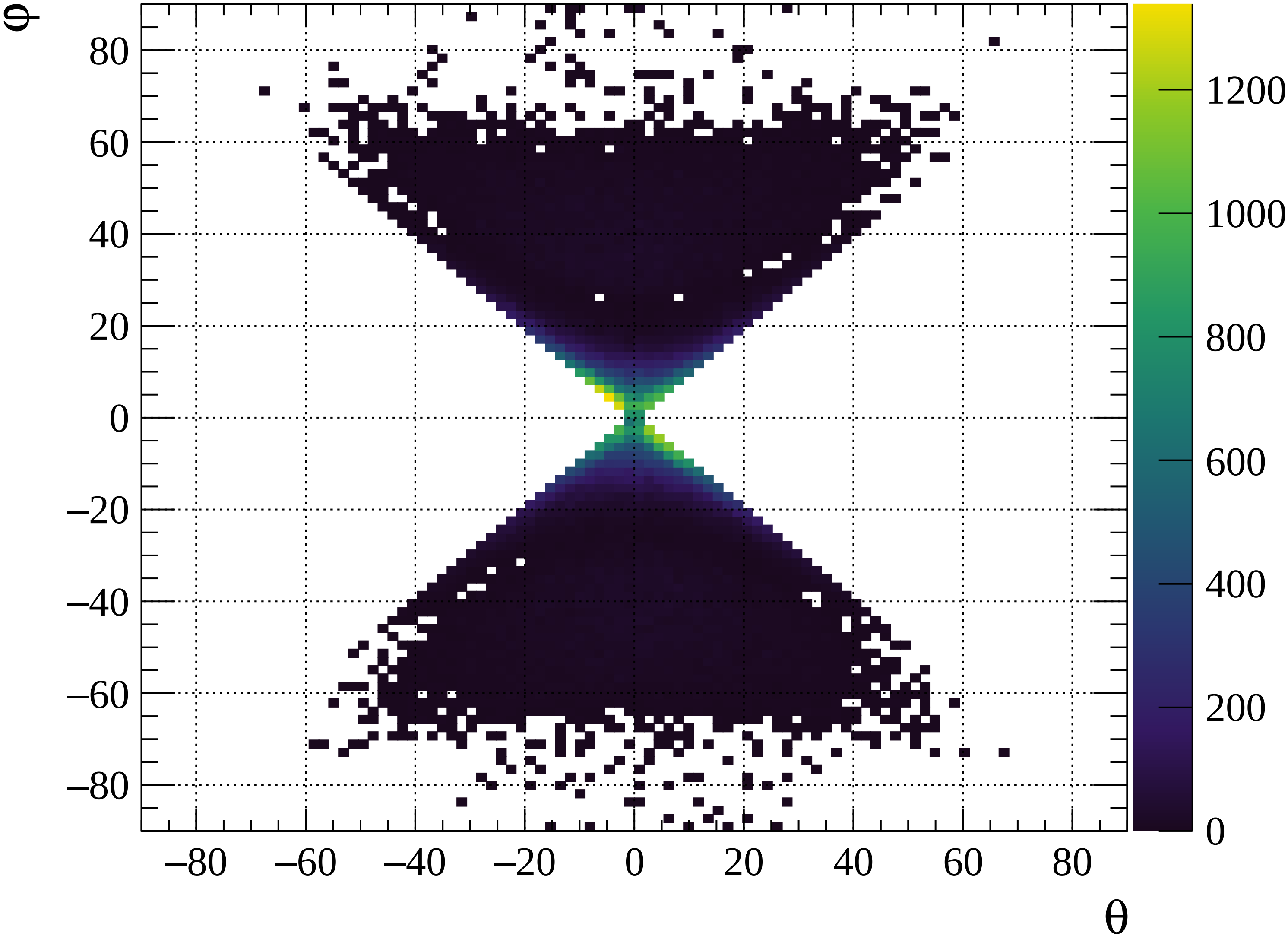

dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
90
80
70
60
50
40
30
20
10
0

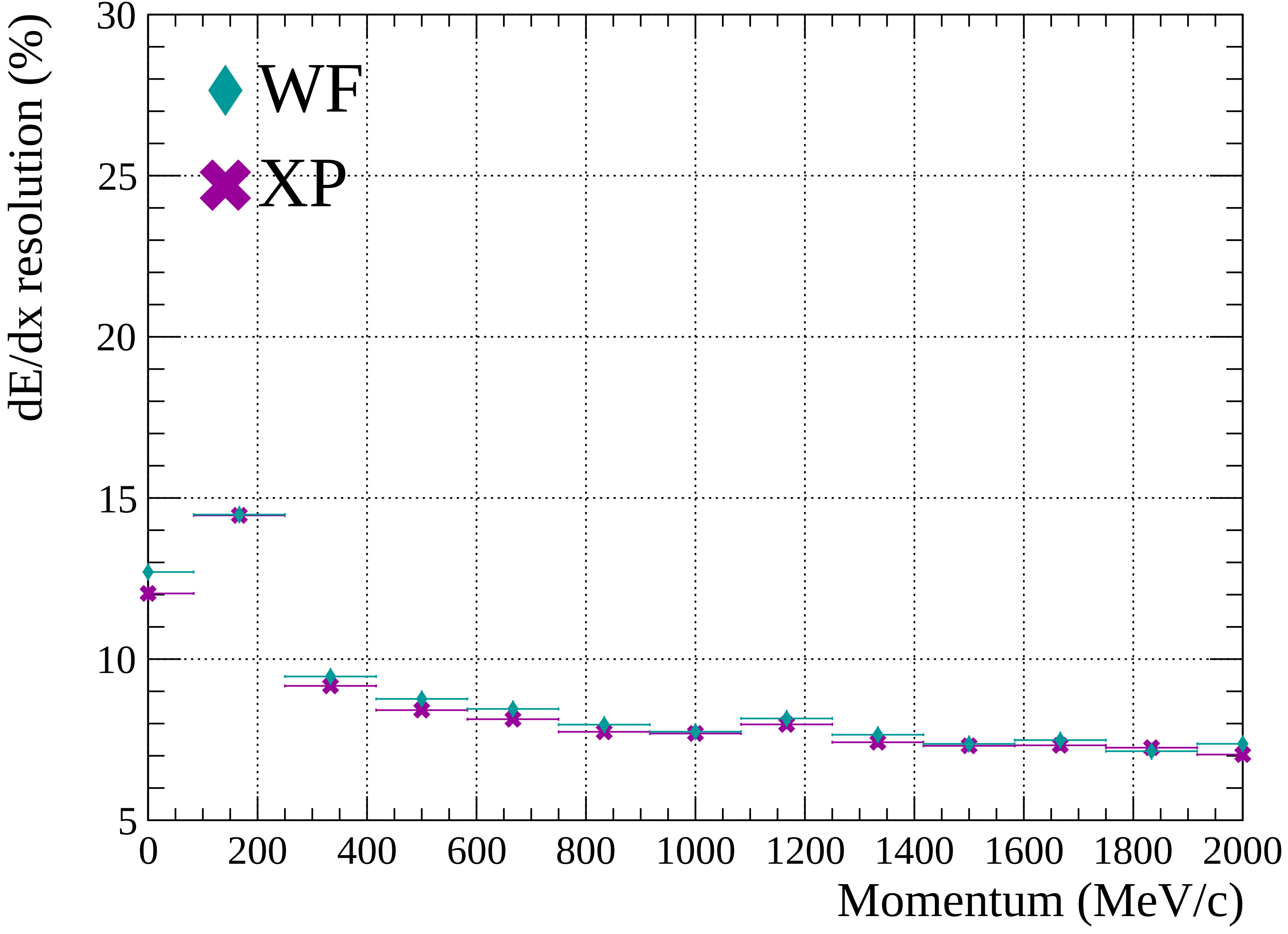

track length (cm)
θ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
20
40
60
80
100
120
140
160
180
200
220



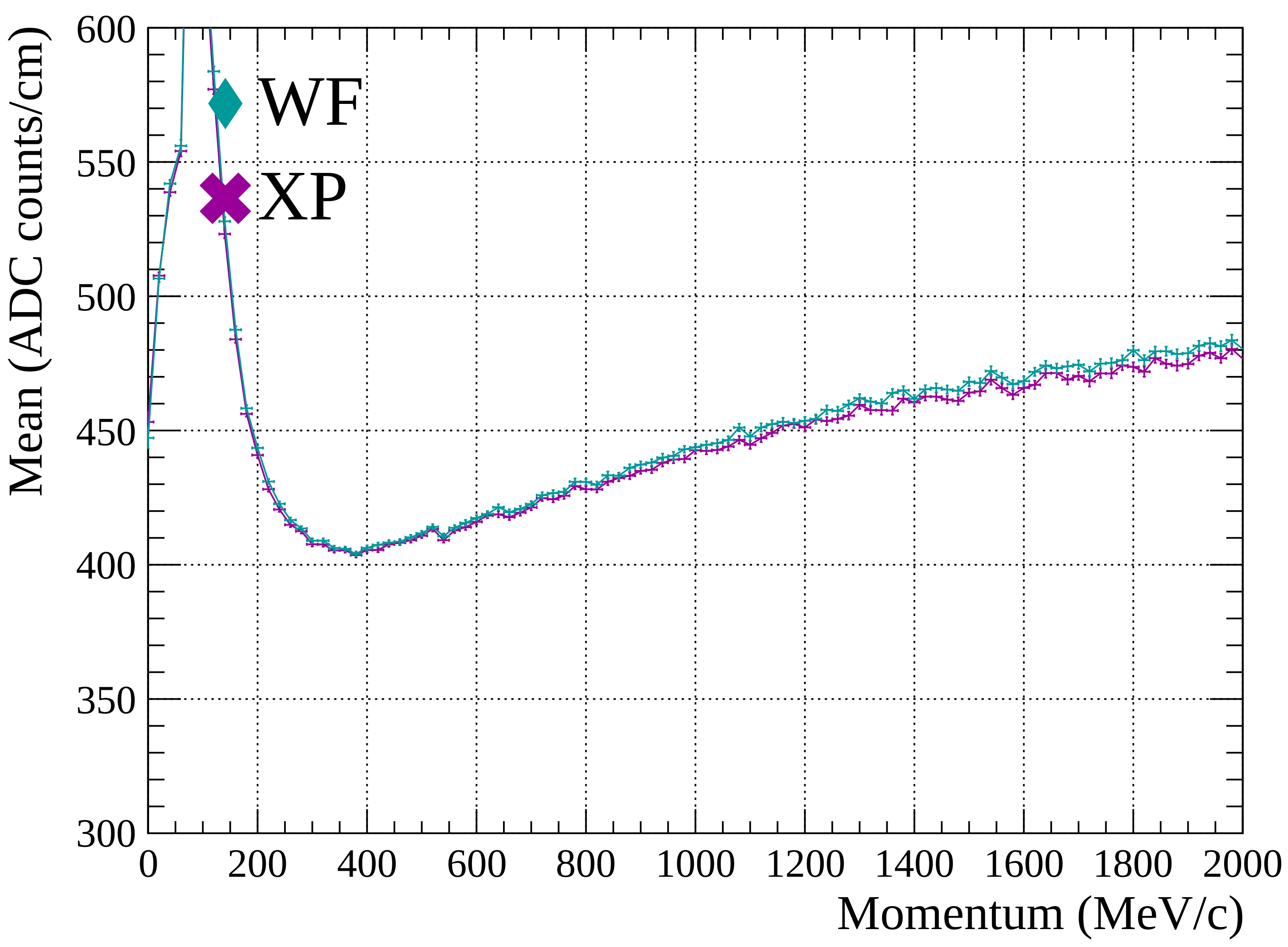
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
140
120
100
80
60
40
20
0


Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

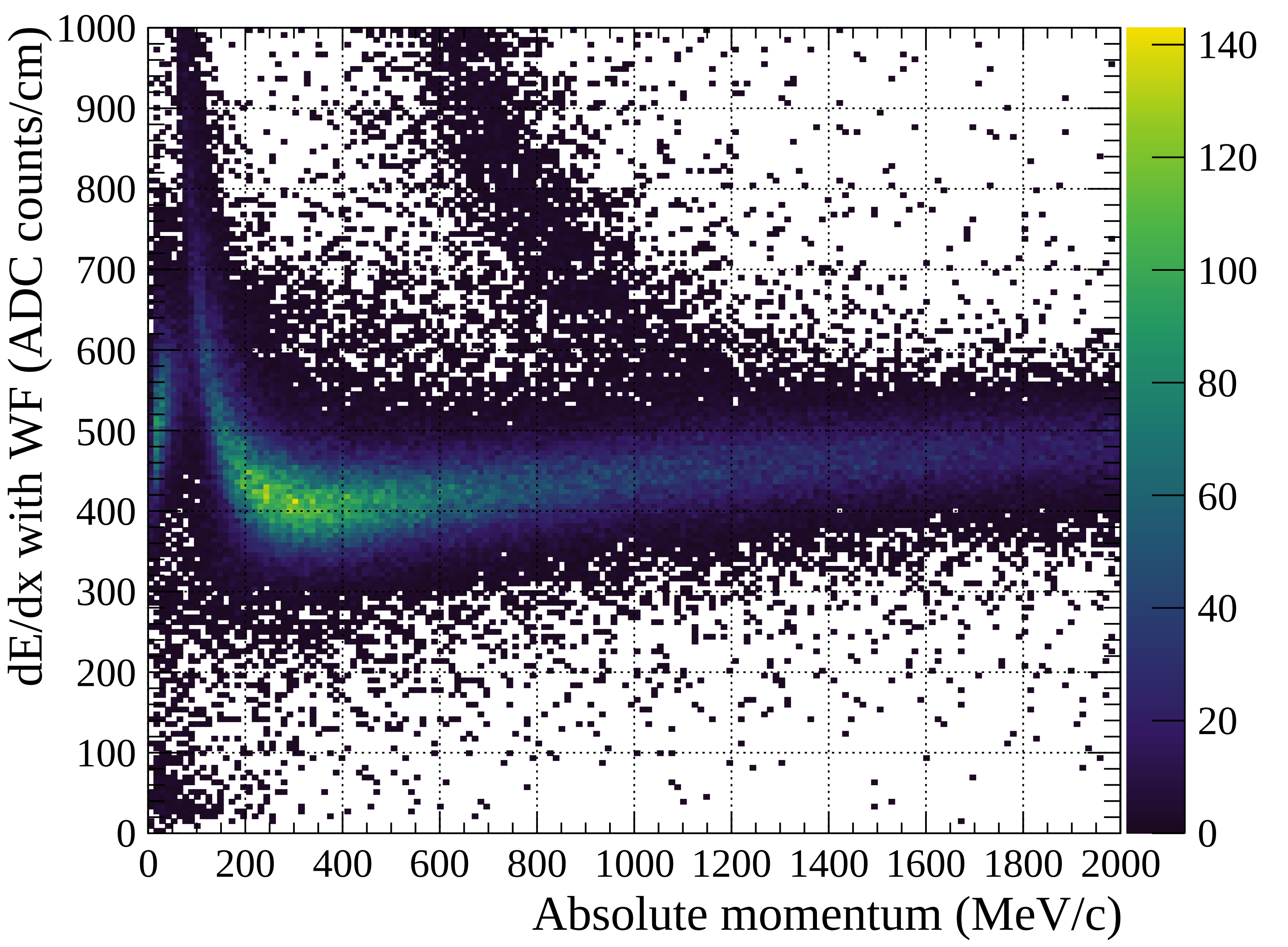

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
1200
1000
800
600
400
200
0



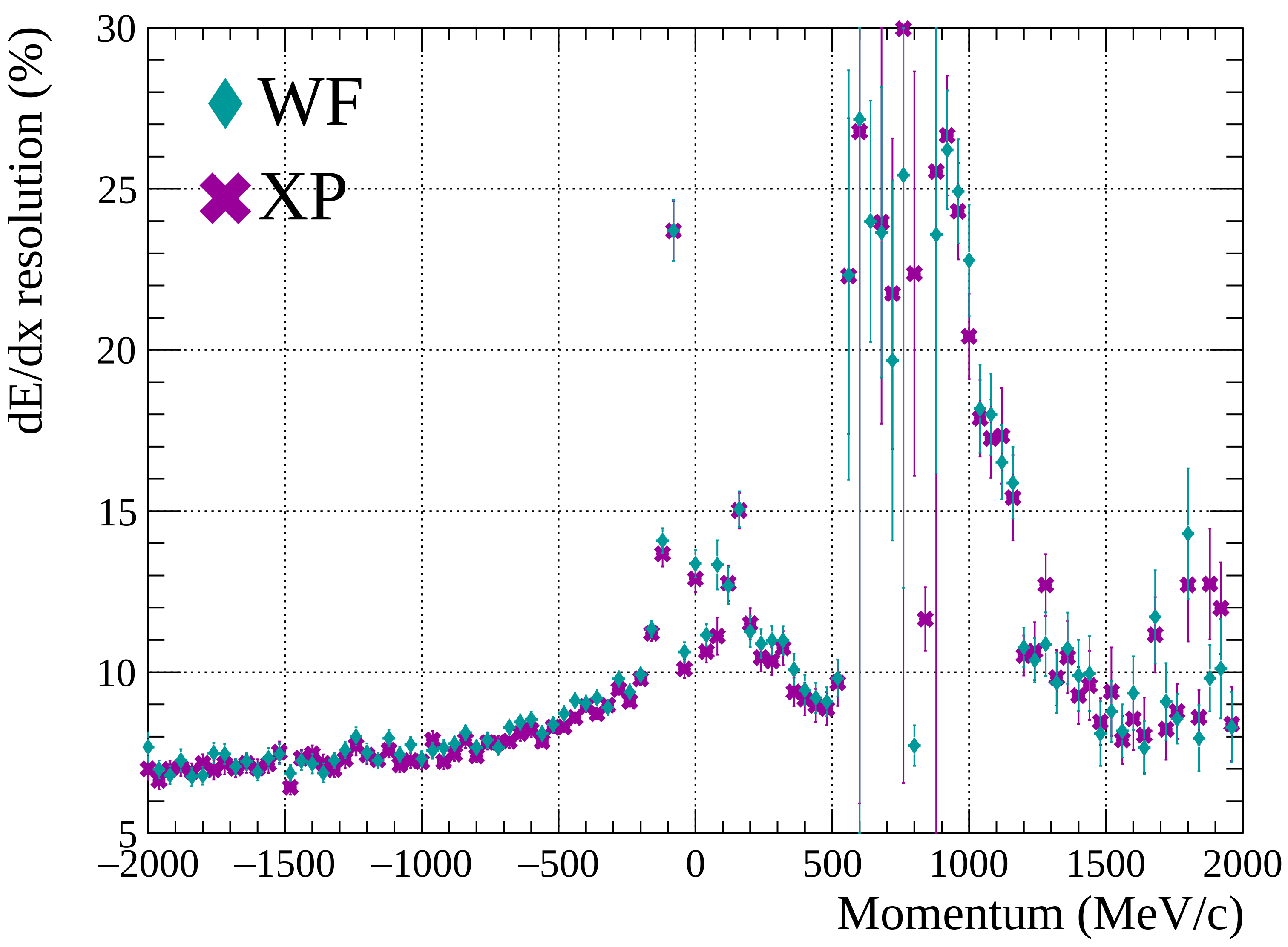
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000



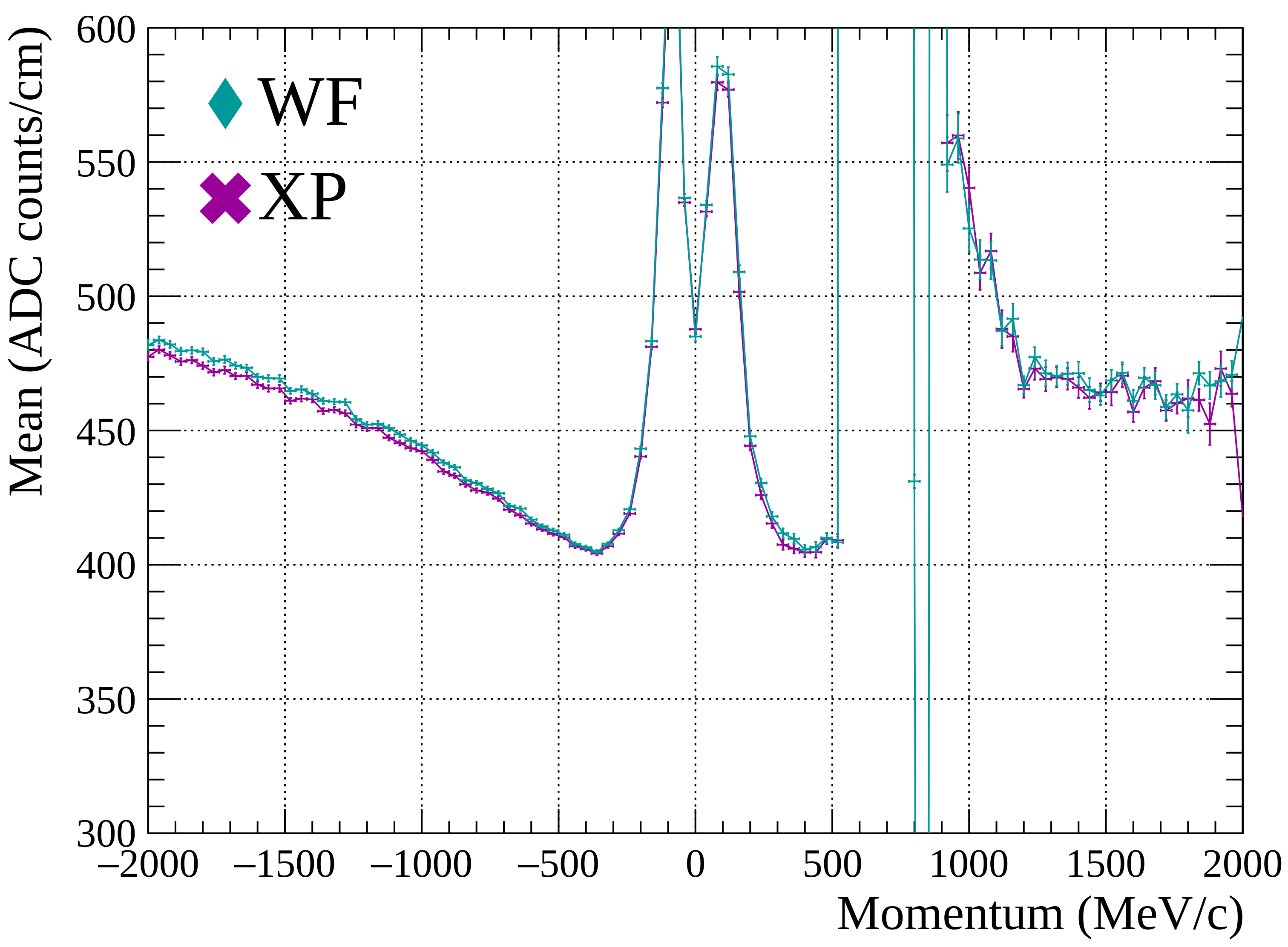
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

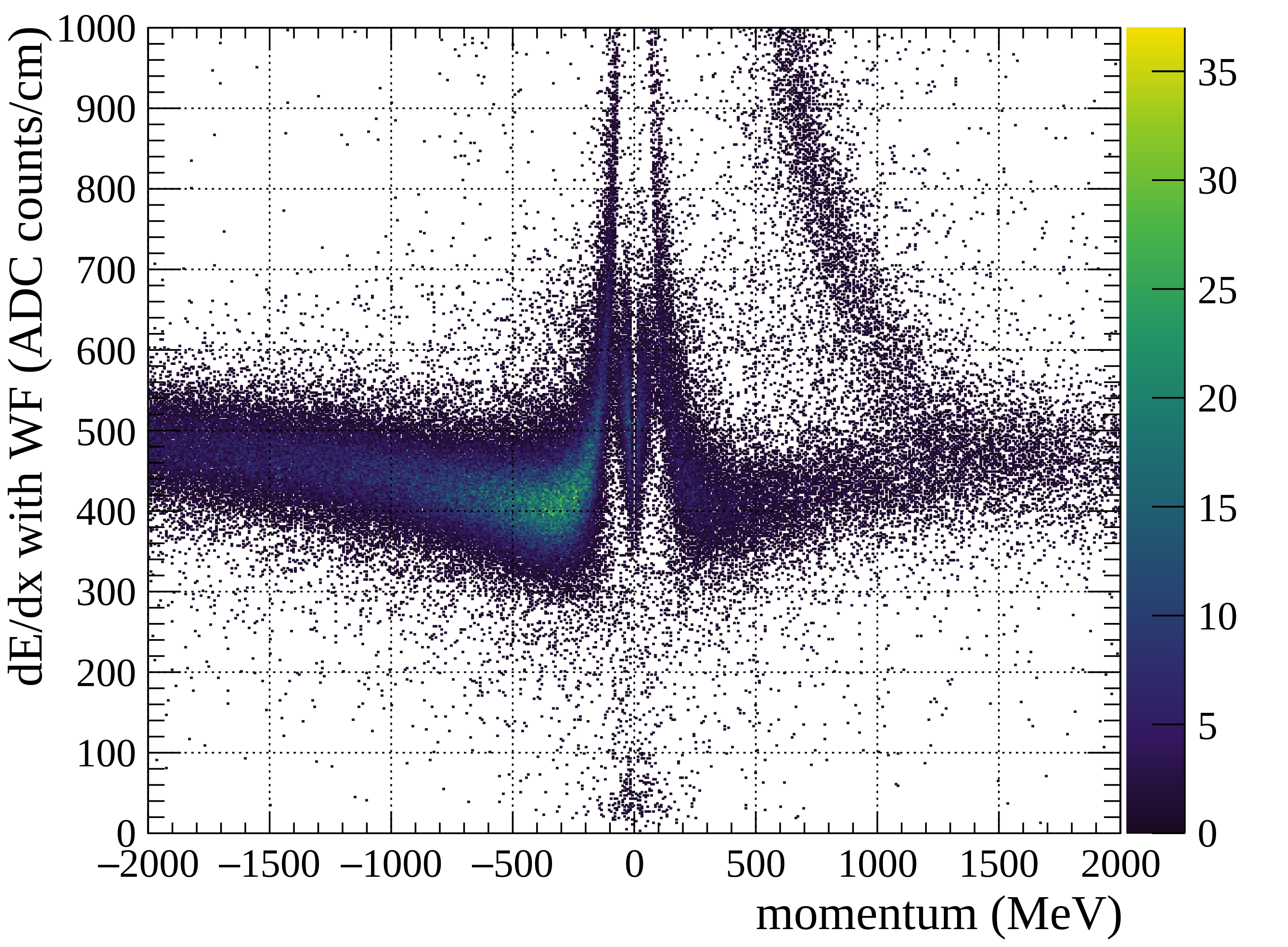


dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
20
40
60
80
100
120
140

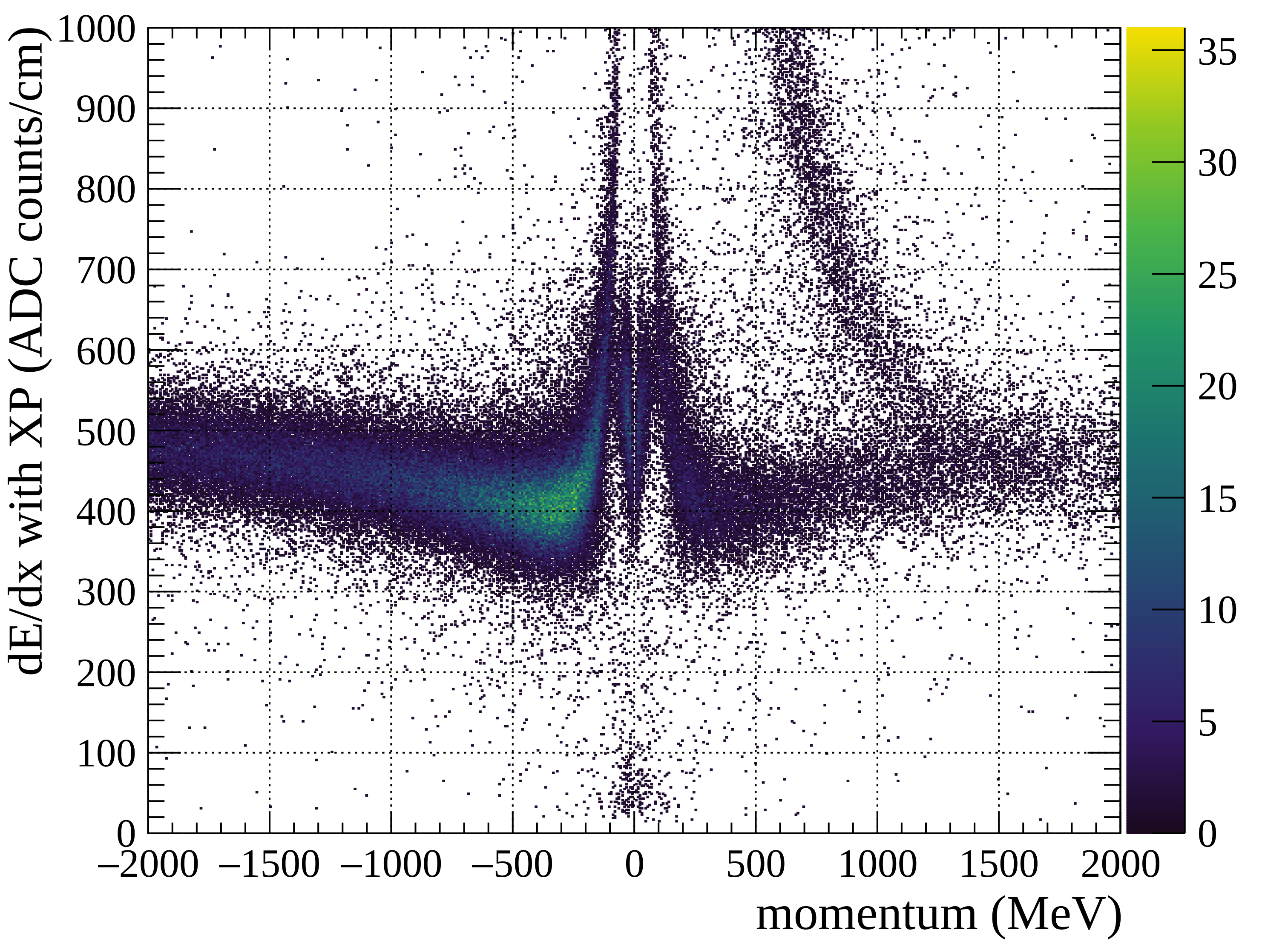


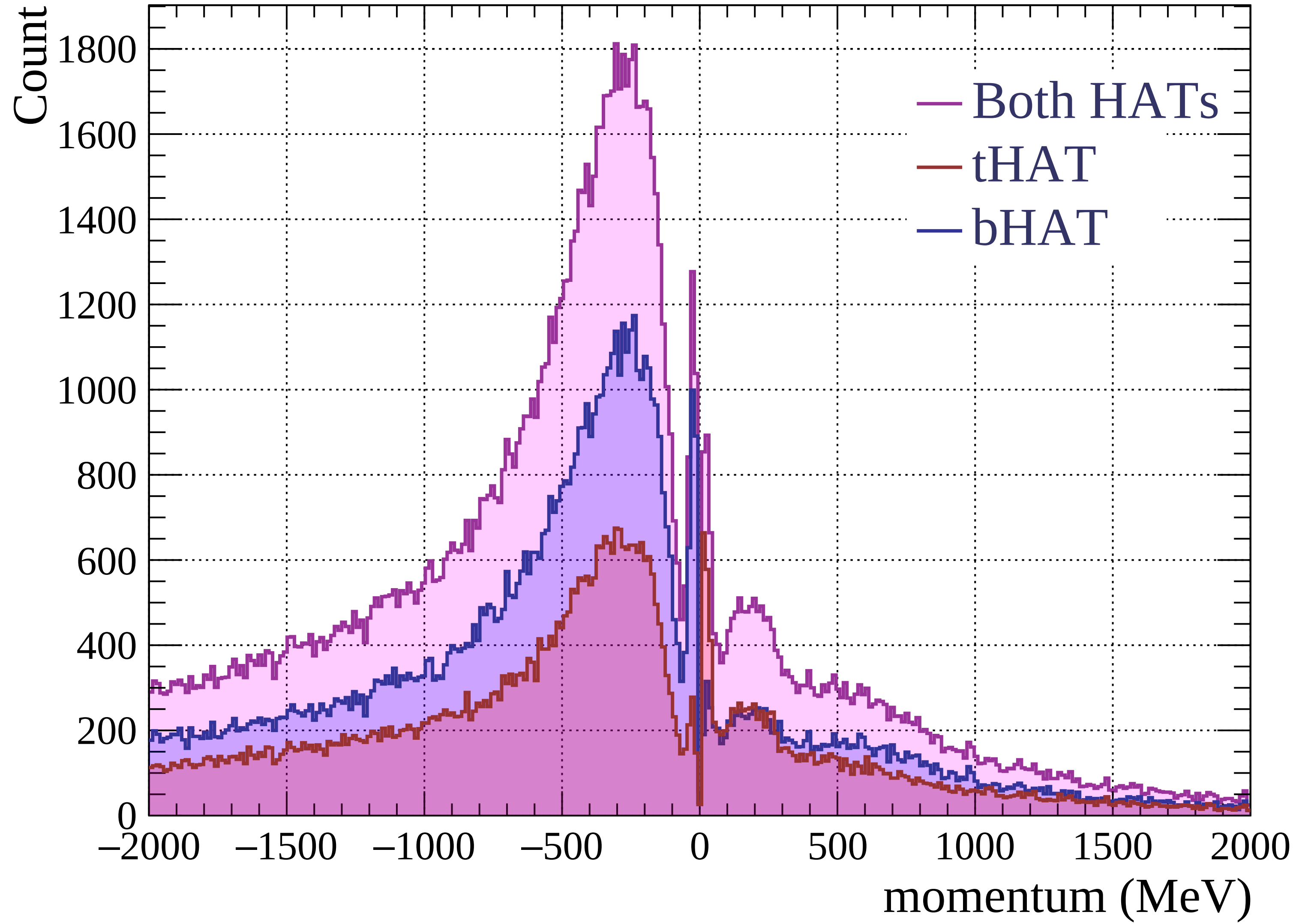
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

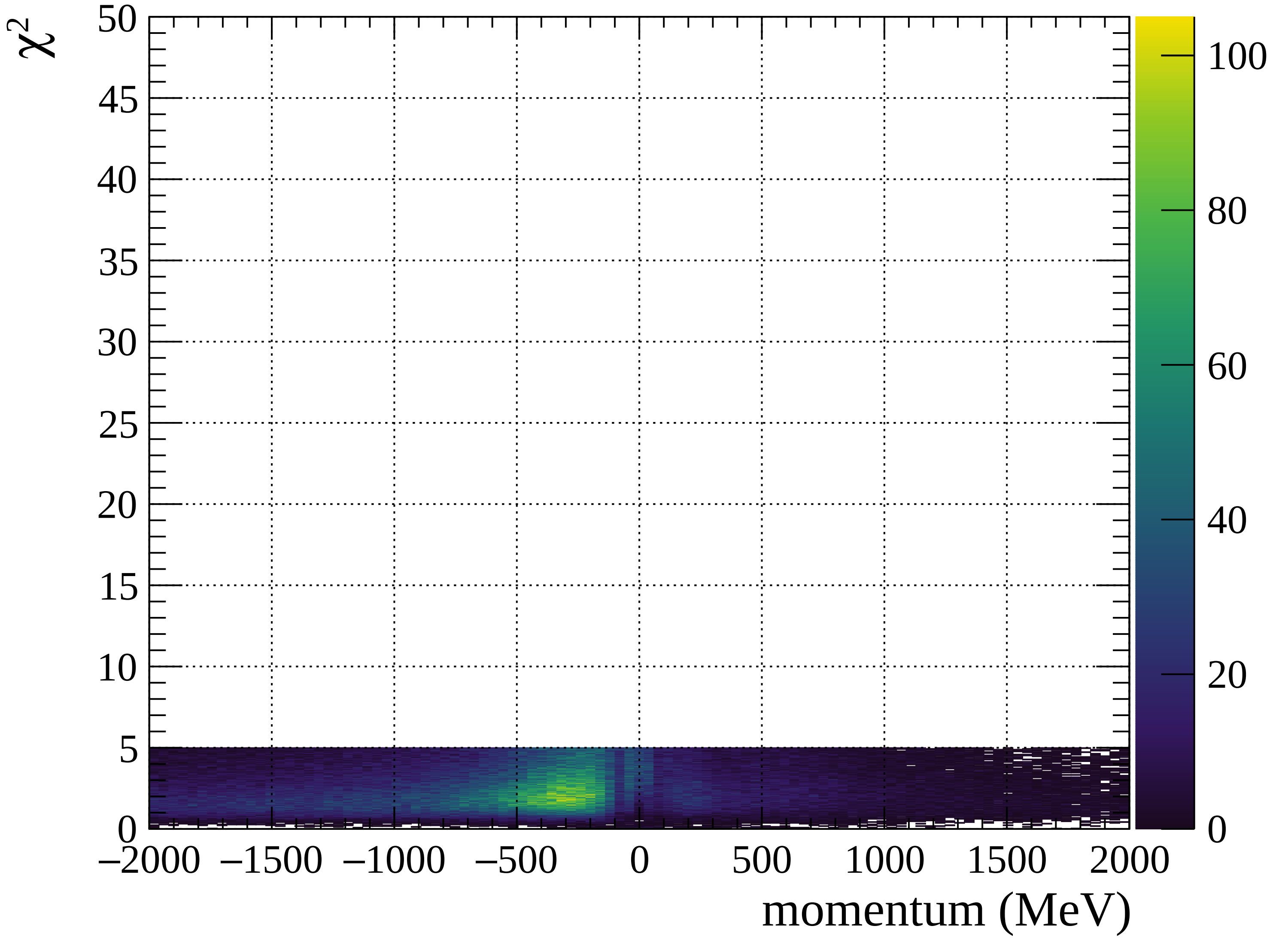

WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000



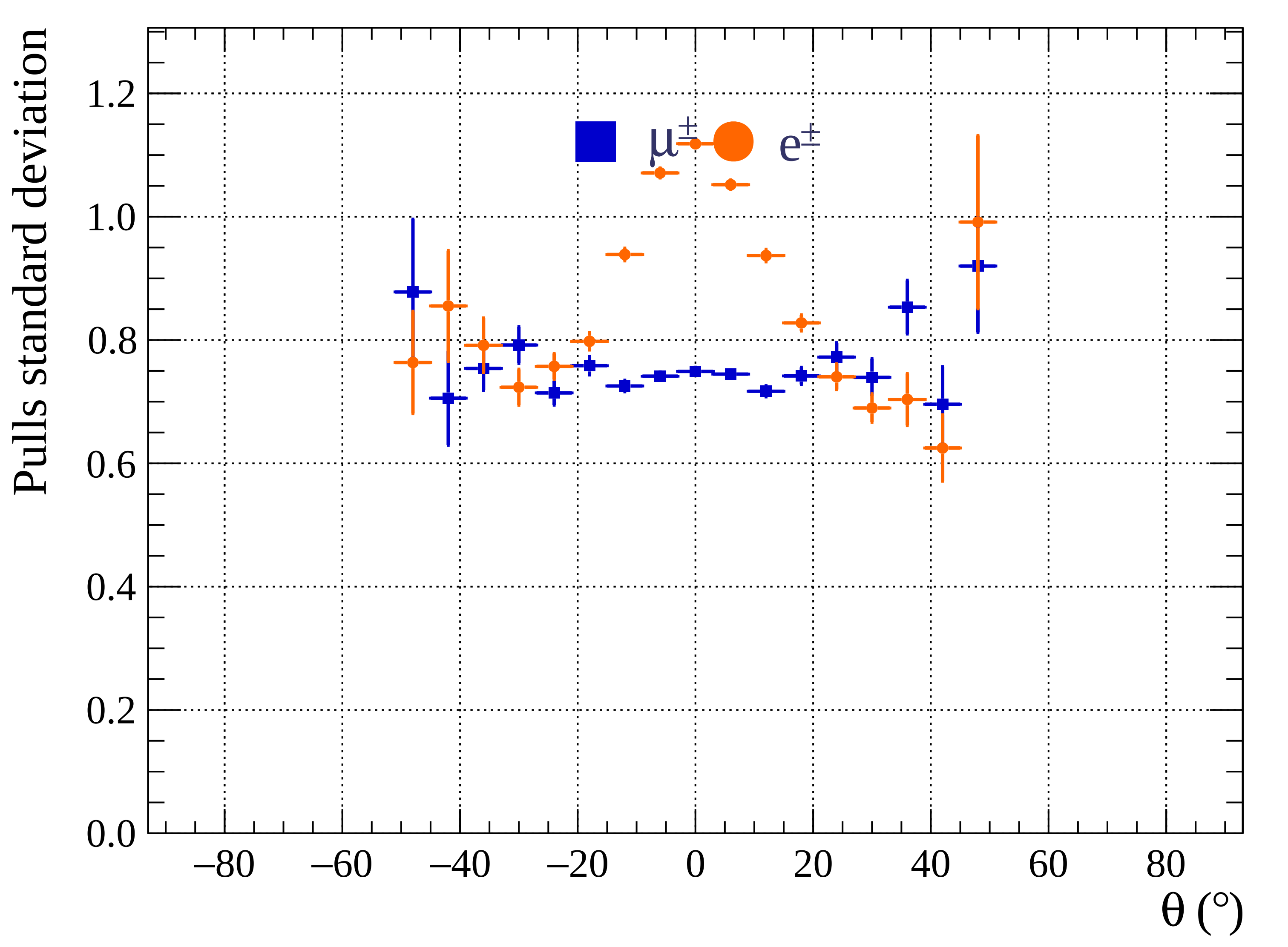
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
35
30
25
20
15
10
5
0


dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
0
5
10
15
20
25
30
35

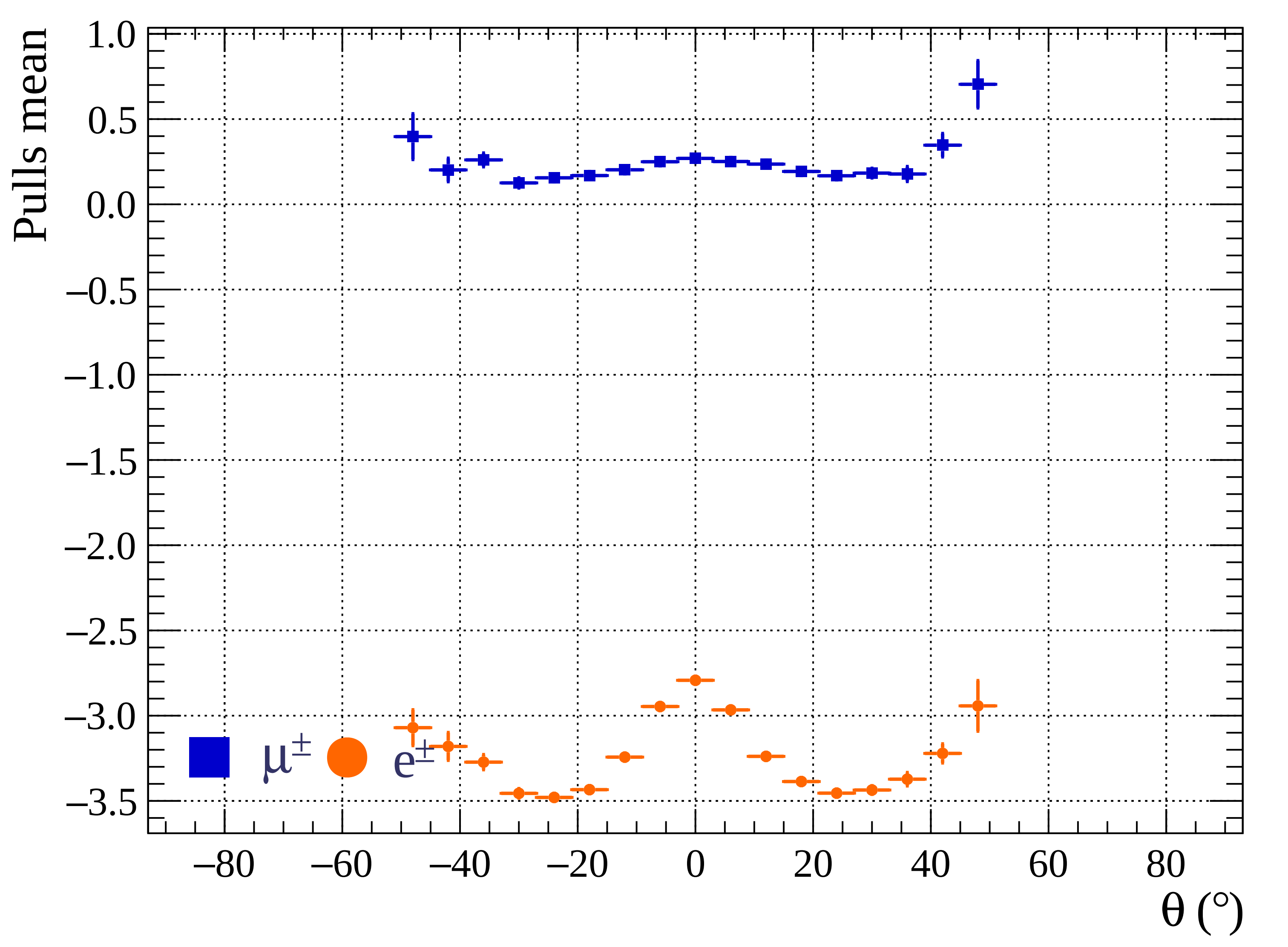

Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
200
400
600
800
1000
1200
1400
1600
1800
−2000
−1500
−1000
−500
0
500
1000
1500
2000



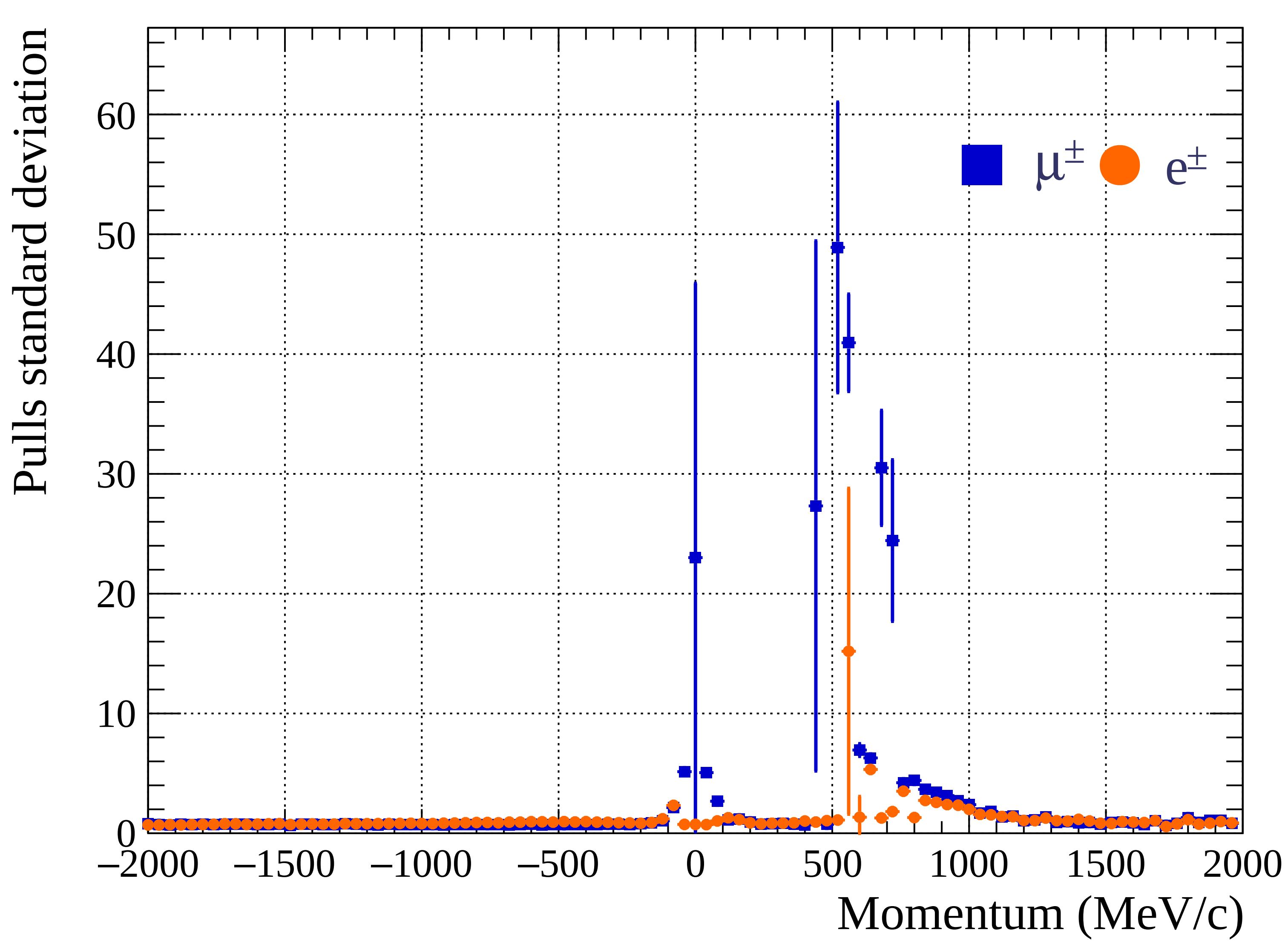
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
100
80
60
40
20
0

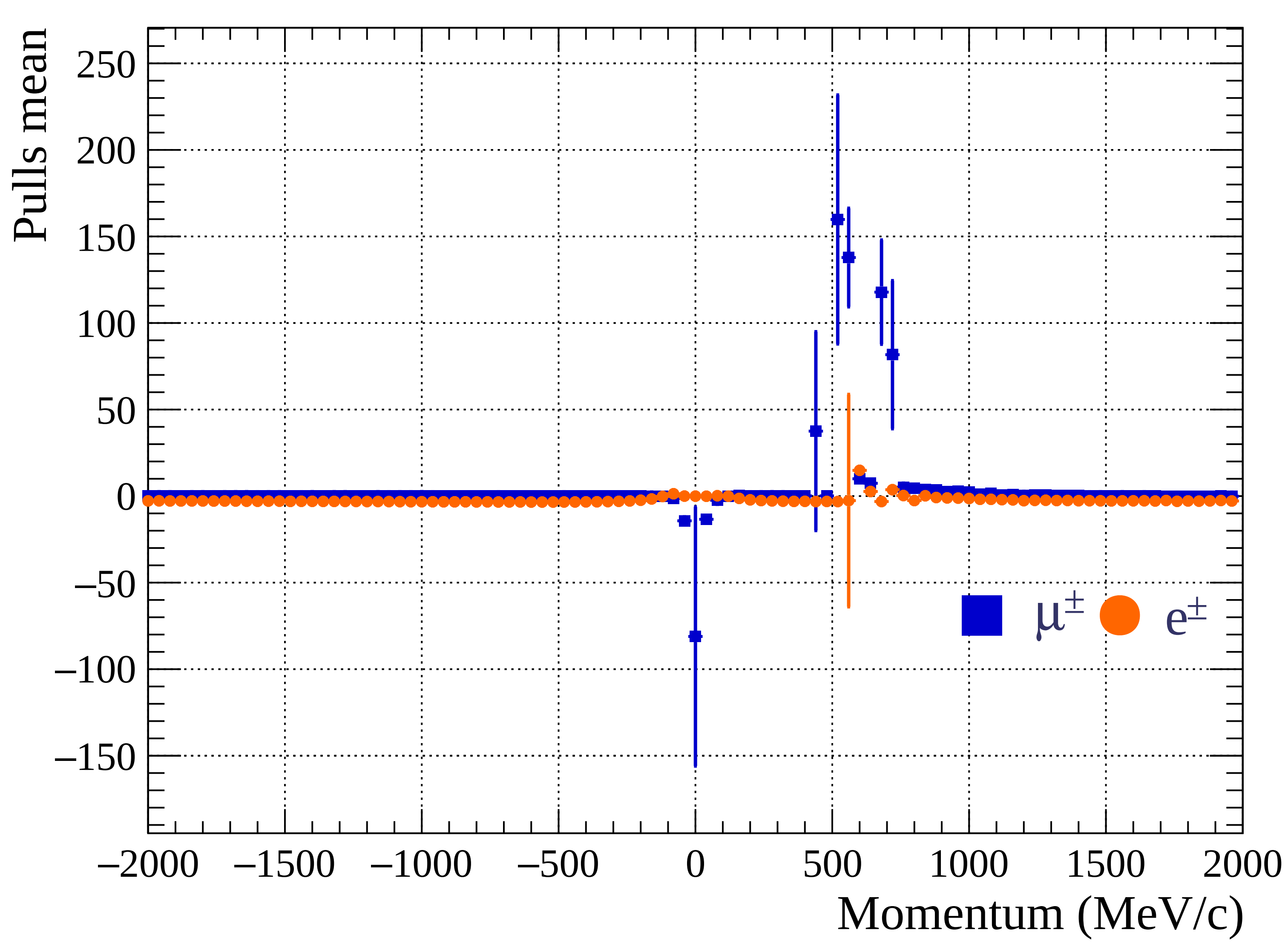


Pulls standard deviation
1.2
1.0
0.8
0.6
0.4
0.2
0.0
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)

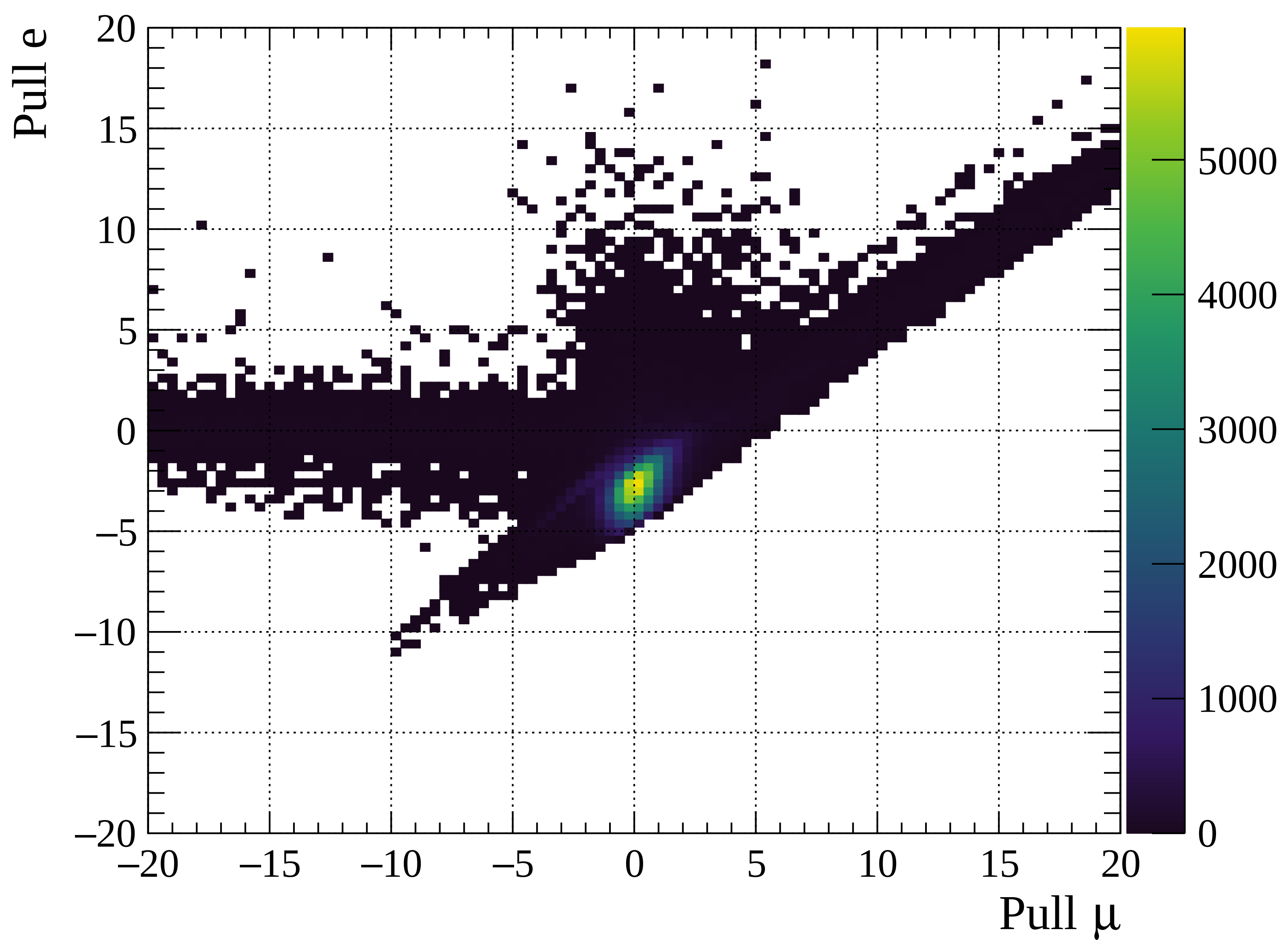


Pulls mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

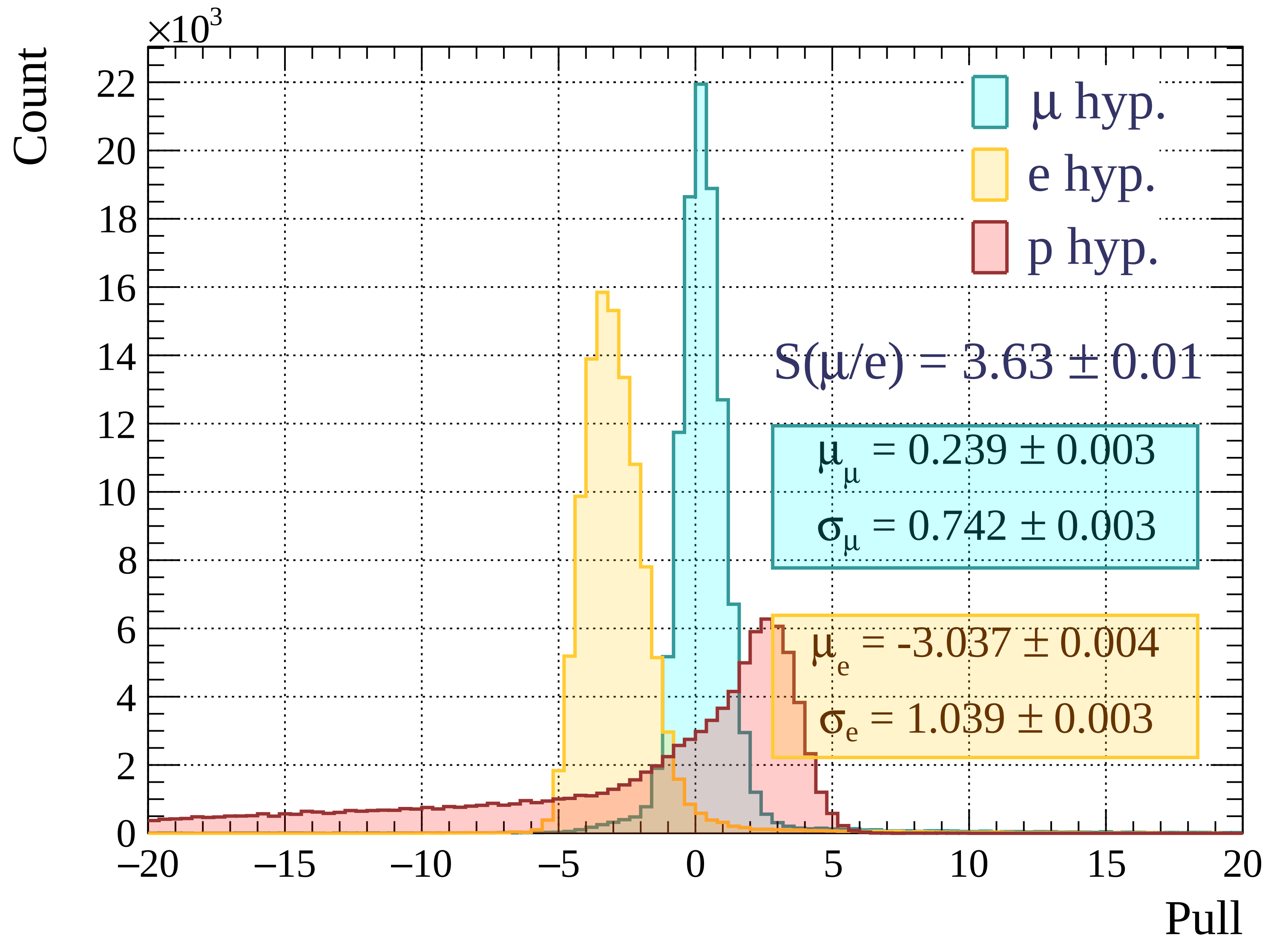

Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
10
20
30
40
50
60
−2000
−1500
−1000
−500
0
500
1000
1500
2000



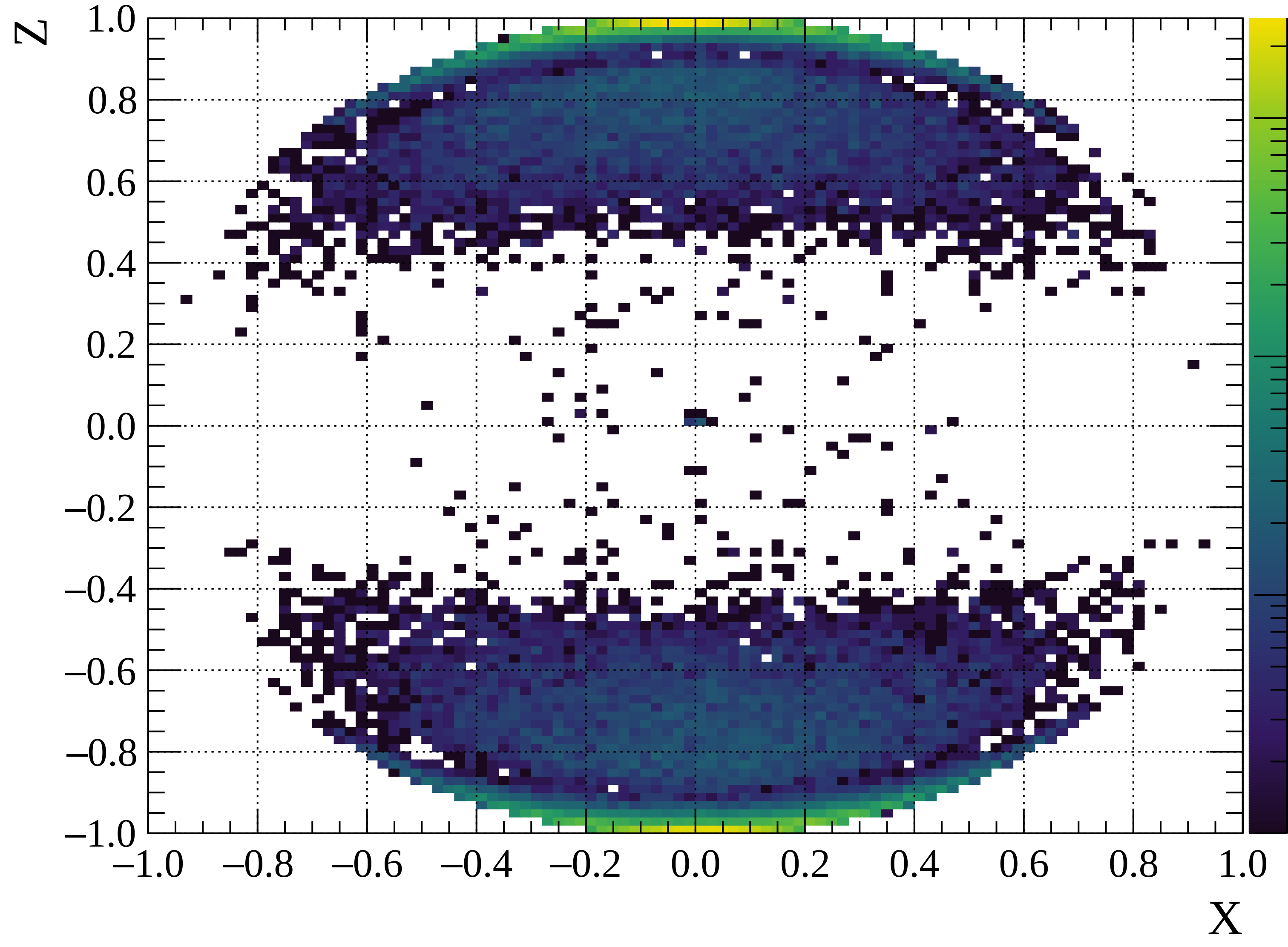
Pulls mean
250
200
150
100
50
0
−50
−100
−150
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±



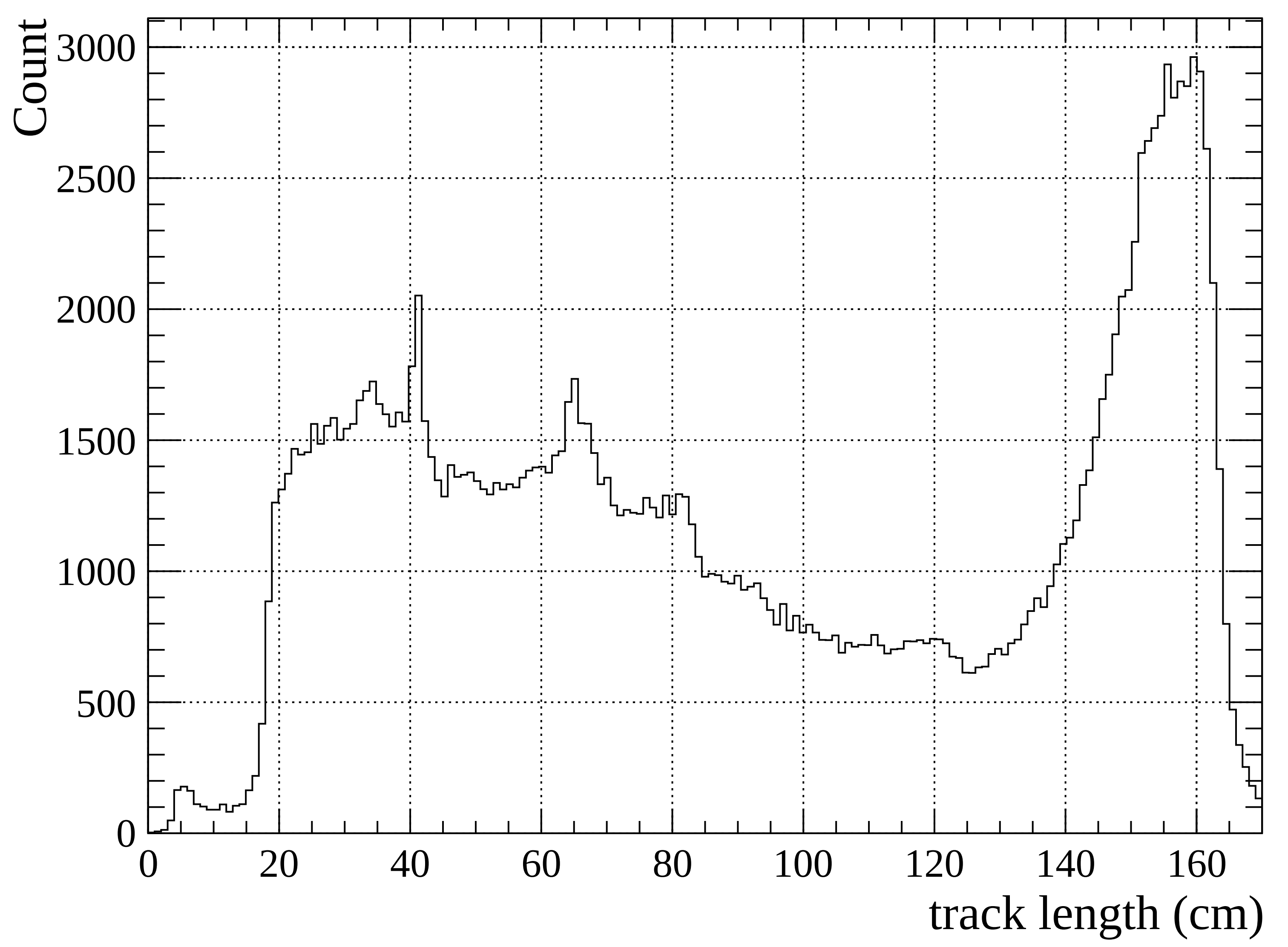
Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
5000
4000
3000
2000
1000
0

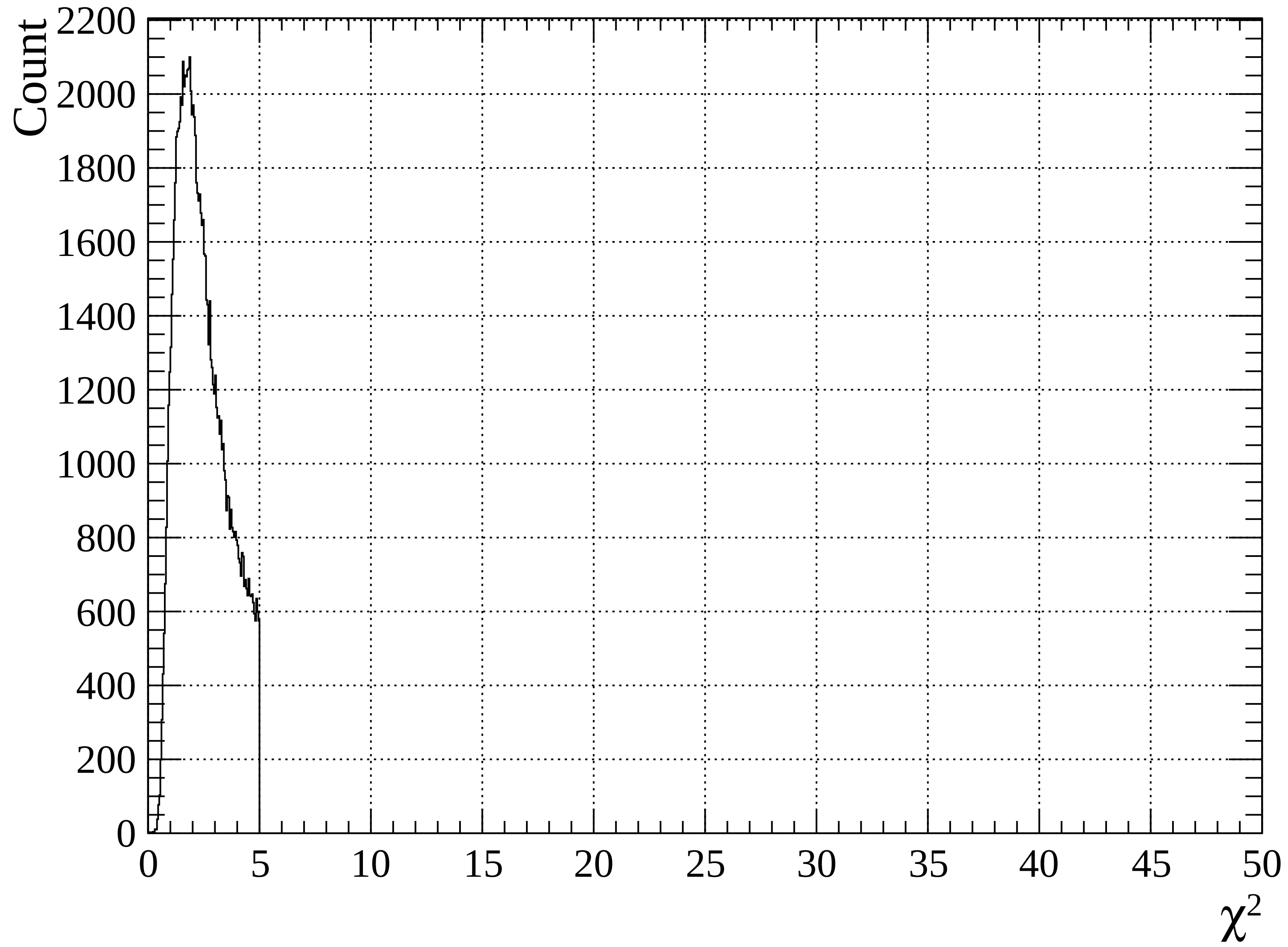


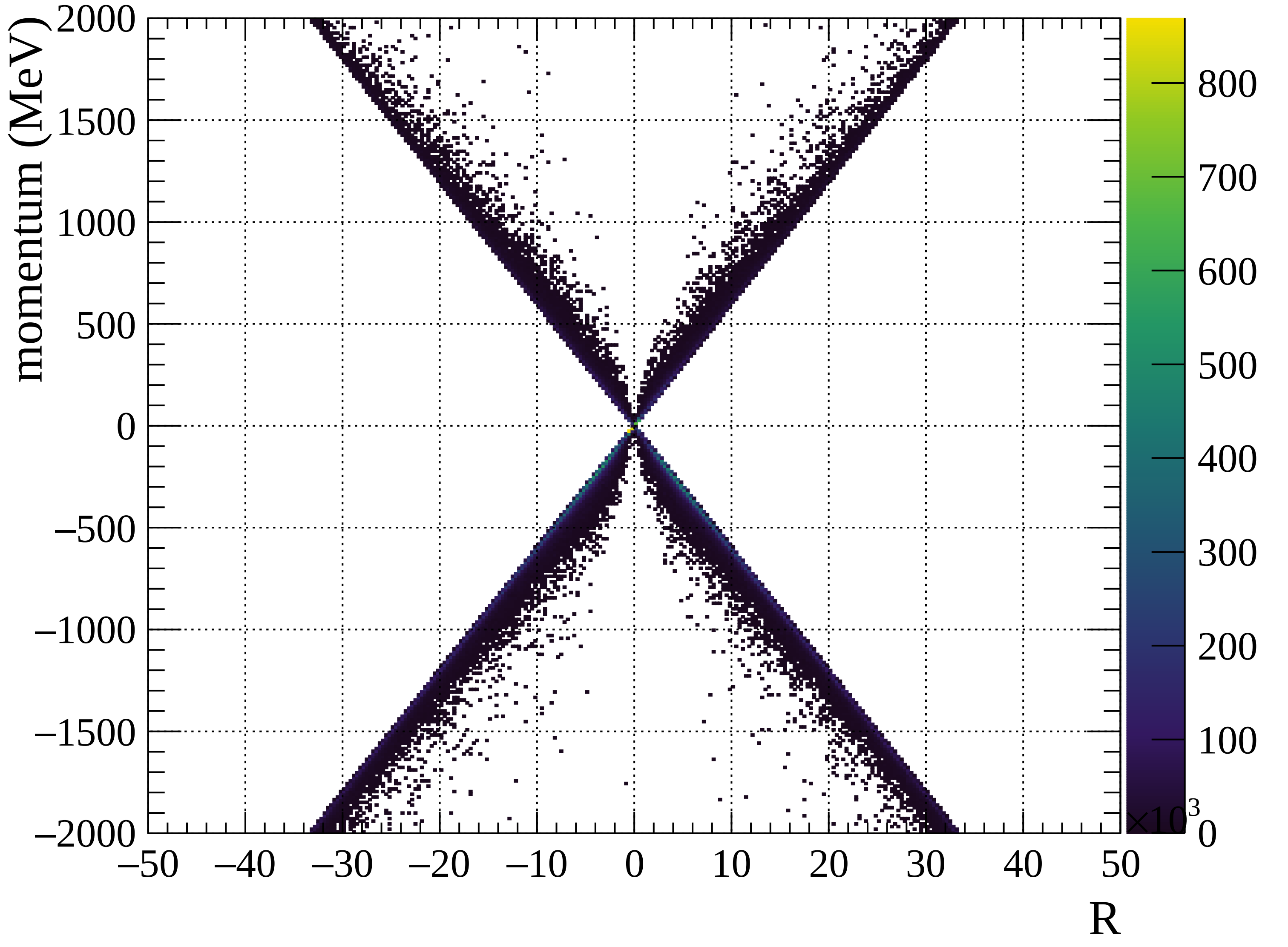
×10^3
Count
22
20
18
16
14
12
10
8
6
4
2
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 3.63 ± 0.01
μ_μ = 0.239 ± 0.003
σ_μ = 0.742 ± 0.003
μ_e = -3.037 ± 0.004
σ_e = 1.039 ± 0.003



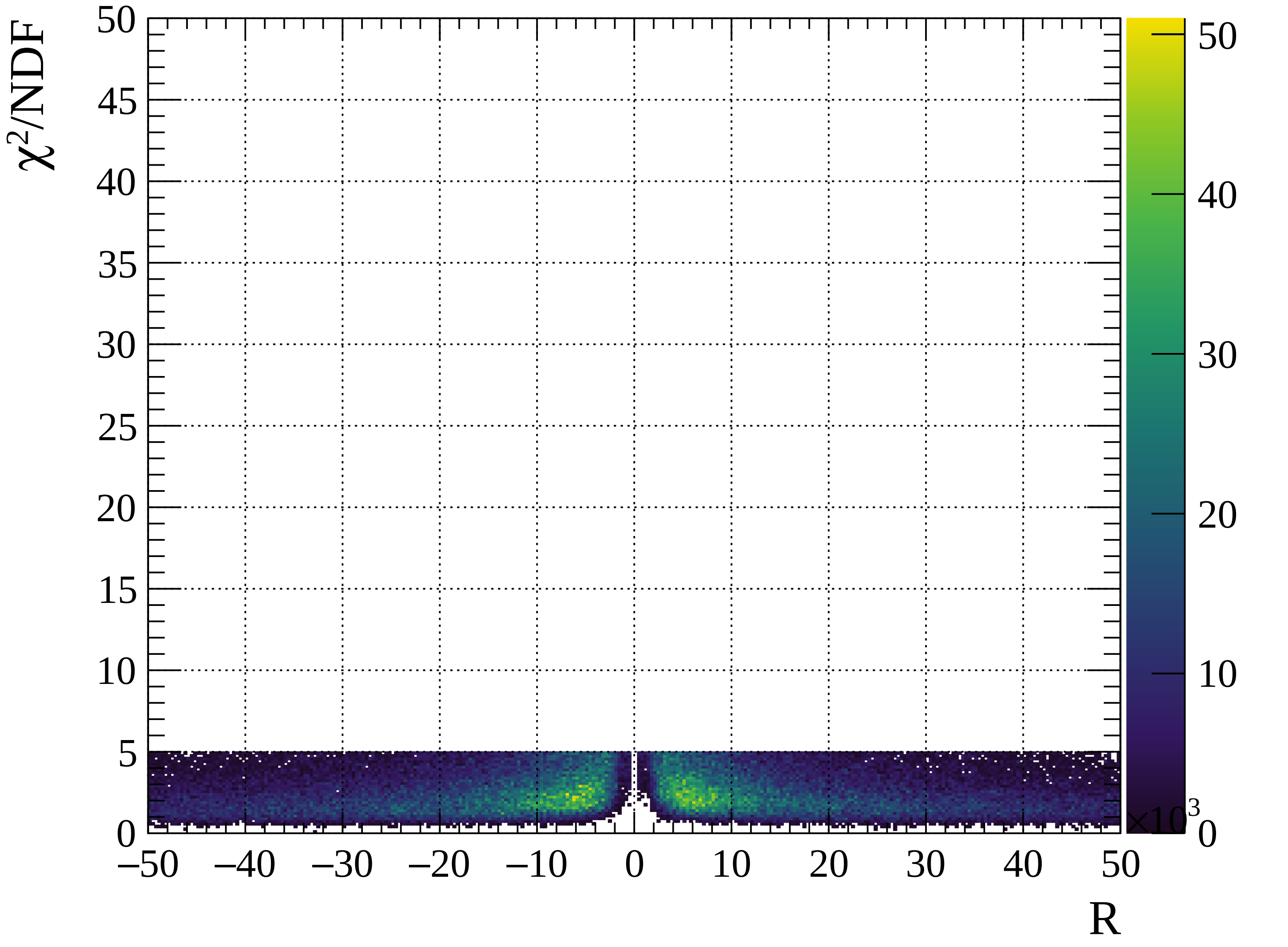
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

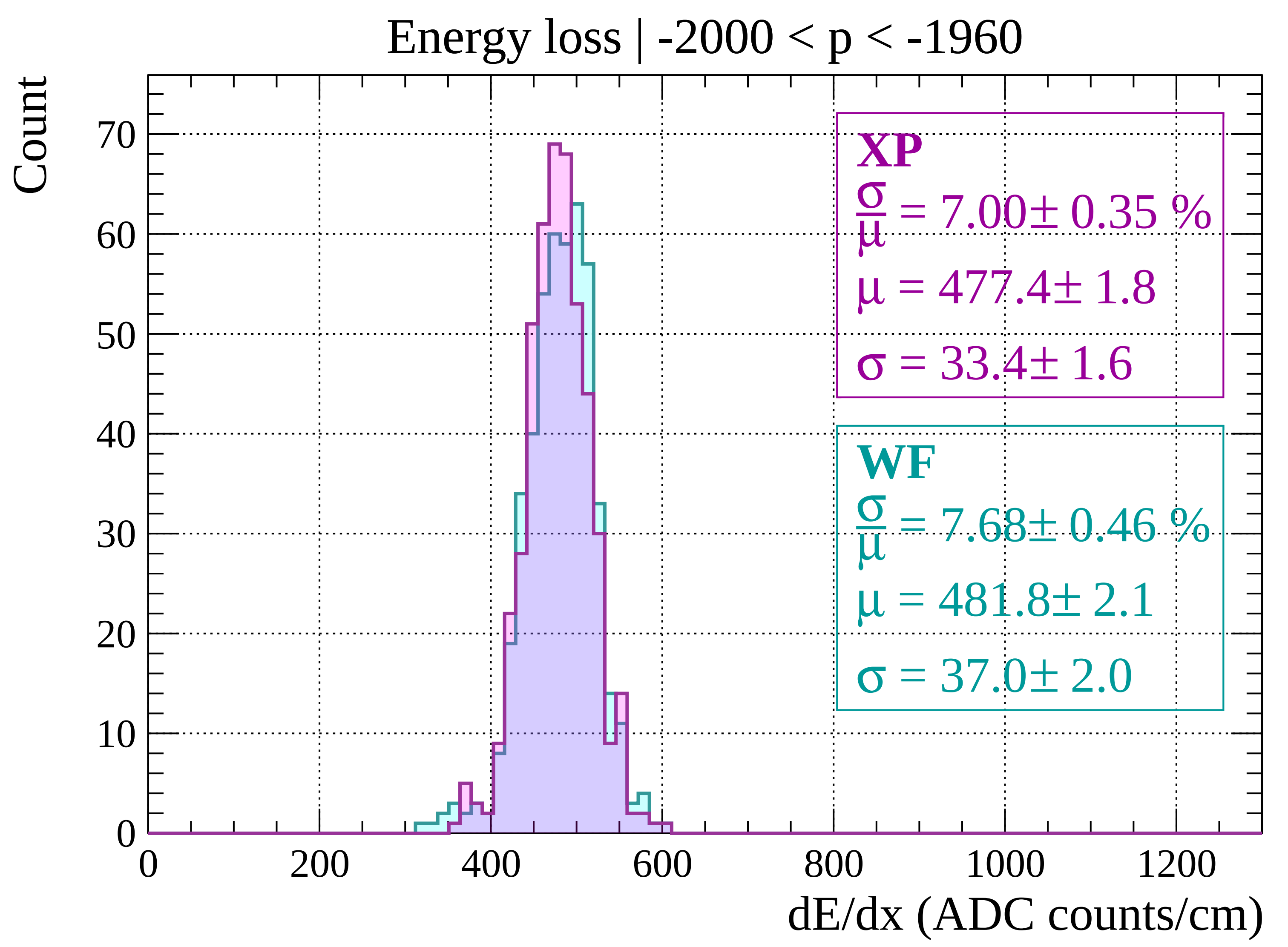

Count
3000
2500
2000
1500
1000
500
0
0
20
40
60
80
100
120
140
160
track length (cm)


Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

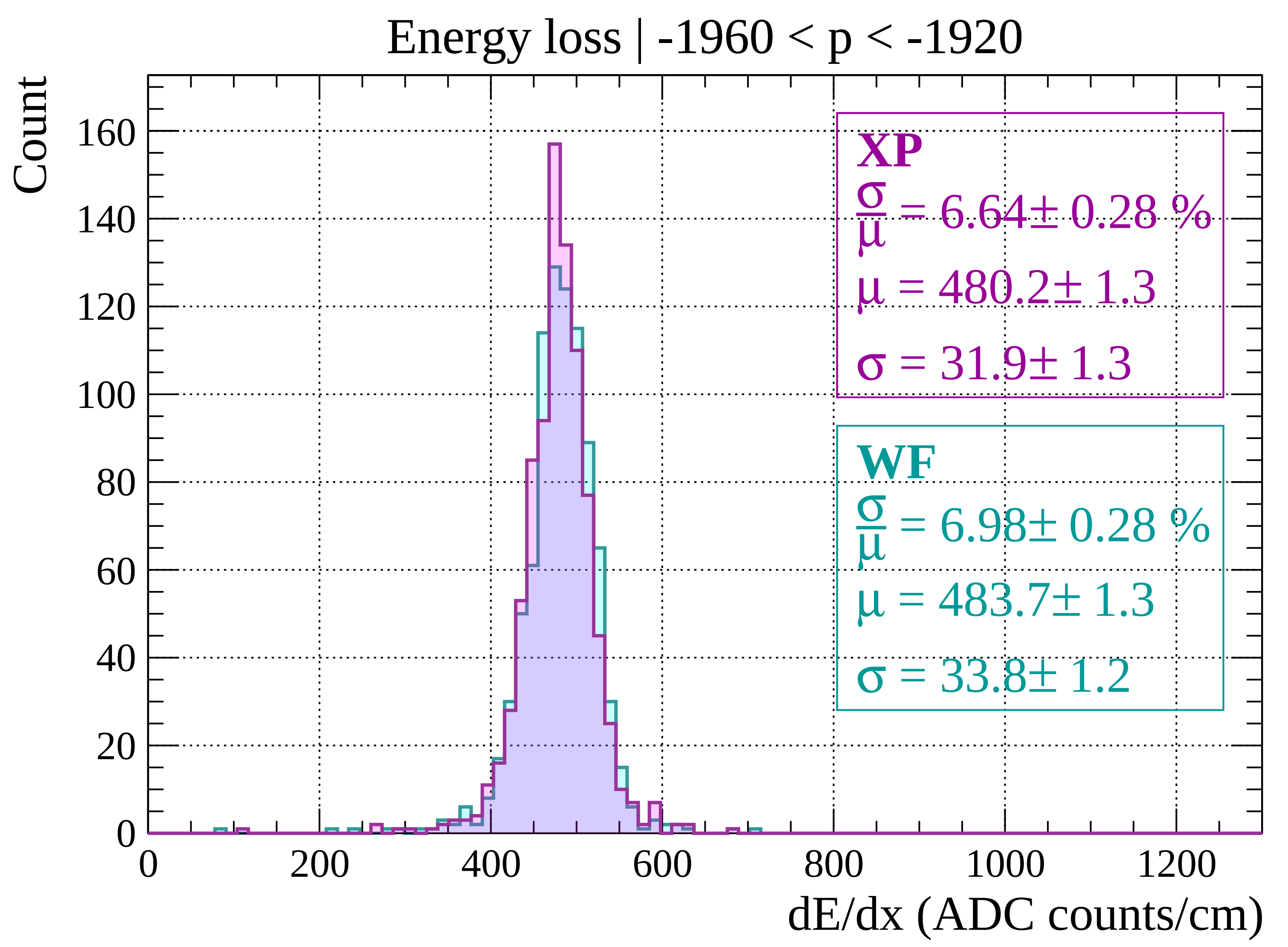


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
800
700
600
500
400
300
200
100
0
$\times10^3$

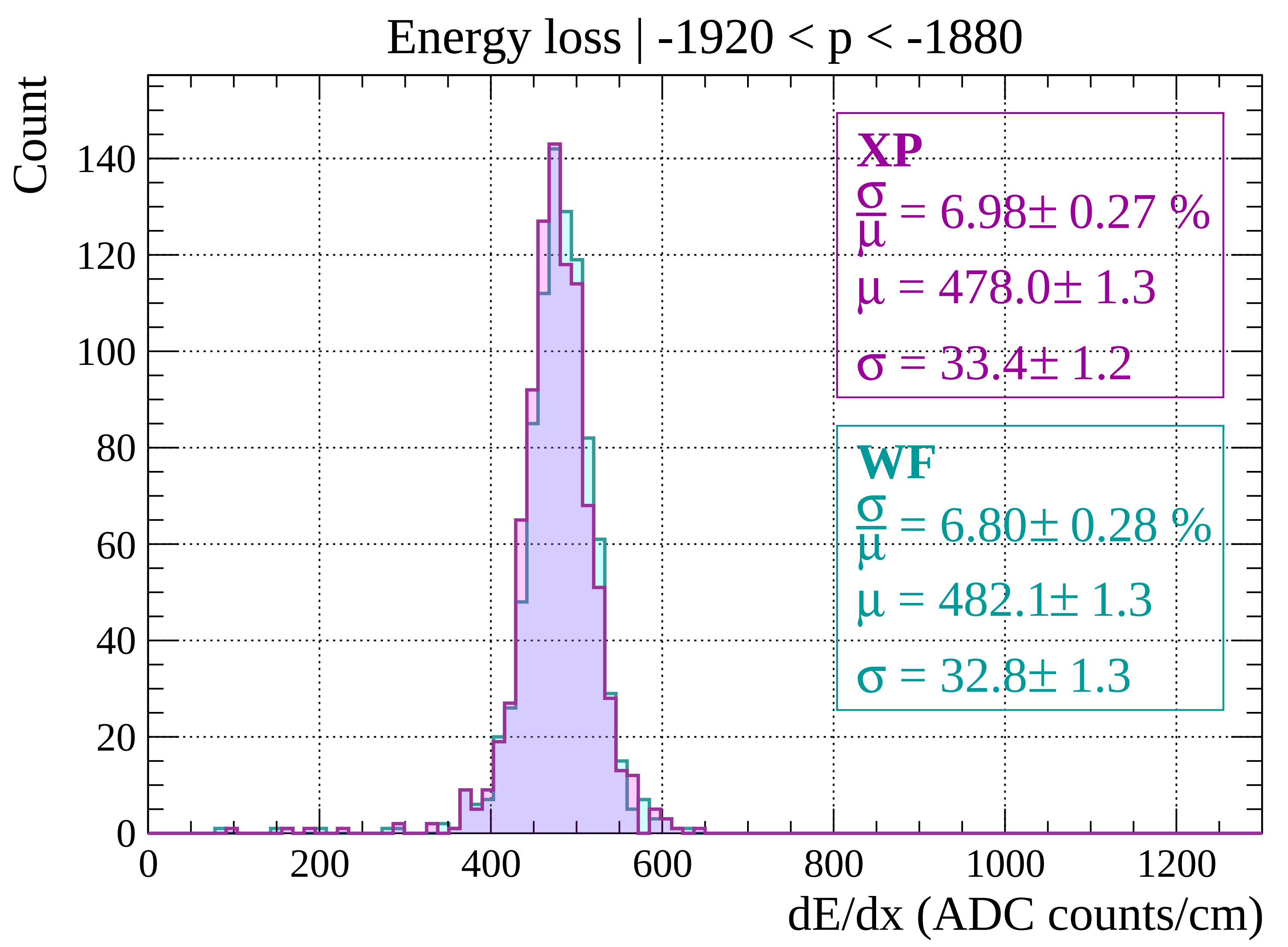

χ^2/NDF
R
$\times 10^3$
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50

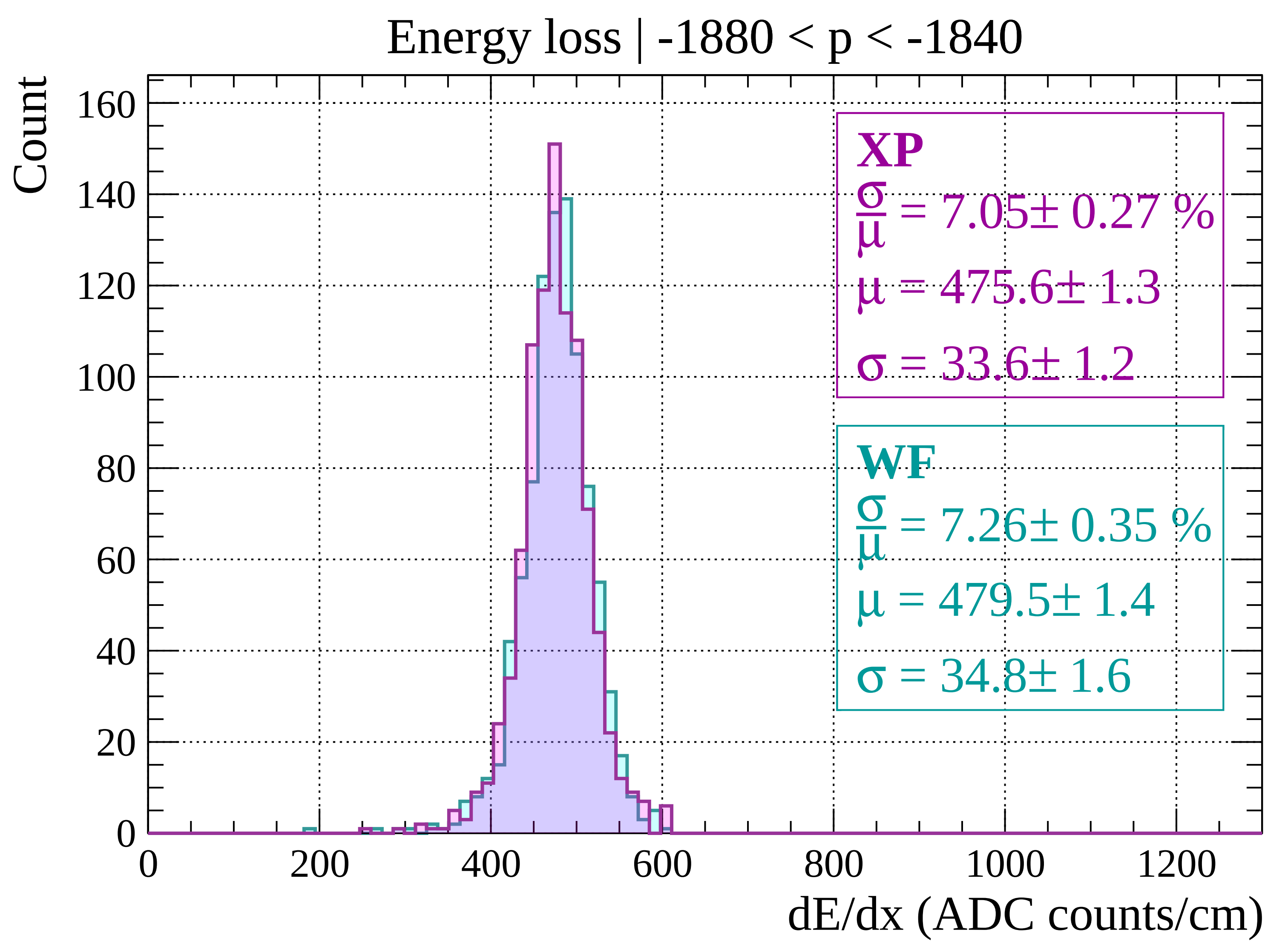


Energy loss | -2000 < p < -1960
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.00± 0.35 %
μ = 477.4± 1.8
σ = 33.4± 1.6
WF
σ/μ = 7.68± 0.46 %
μ = 481.8± 2.1
σ = 37.0± 2.0

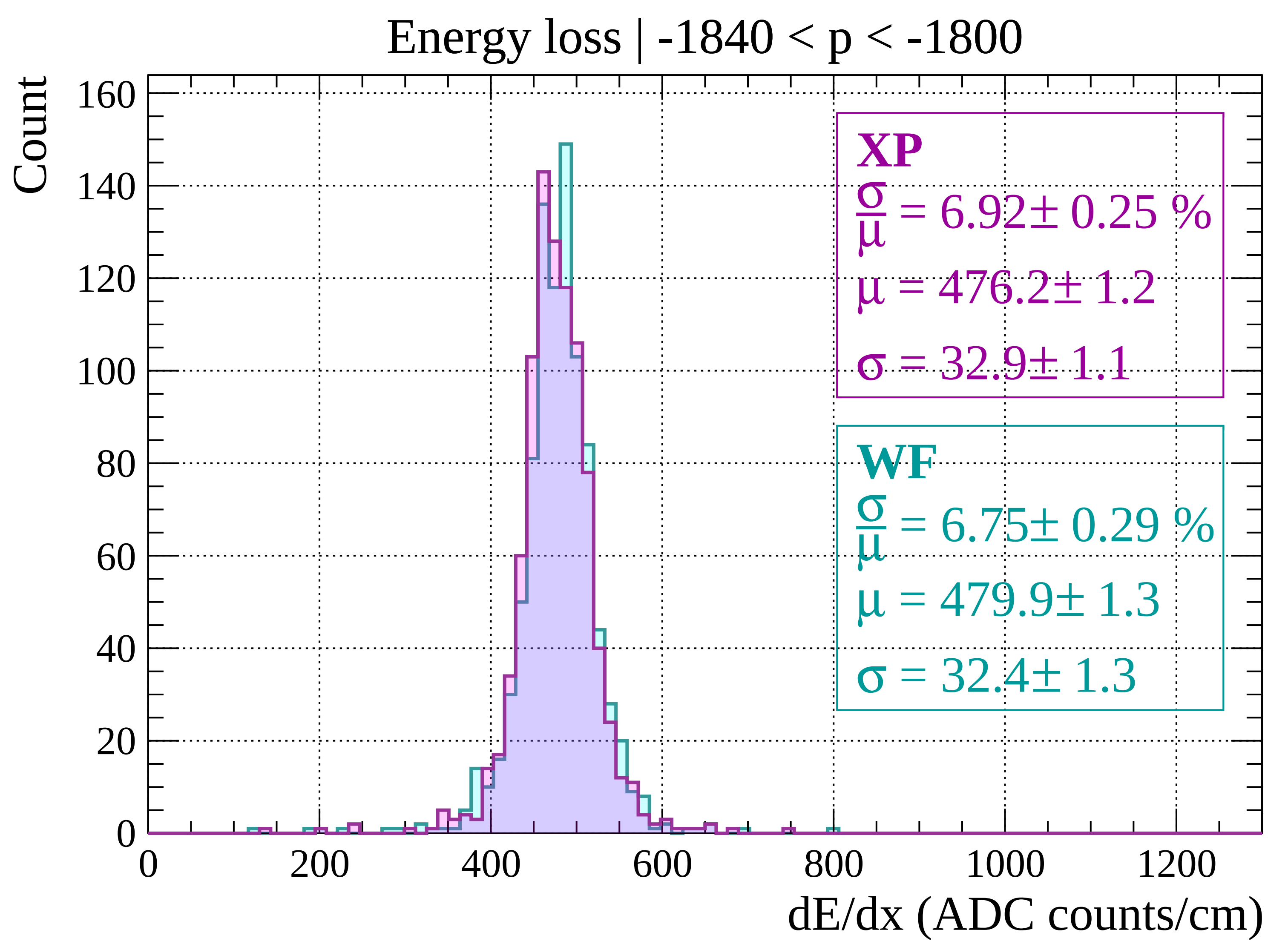


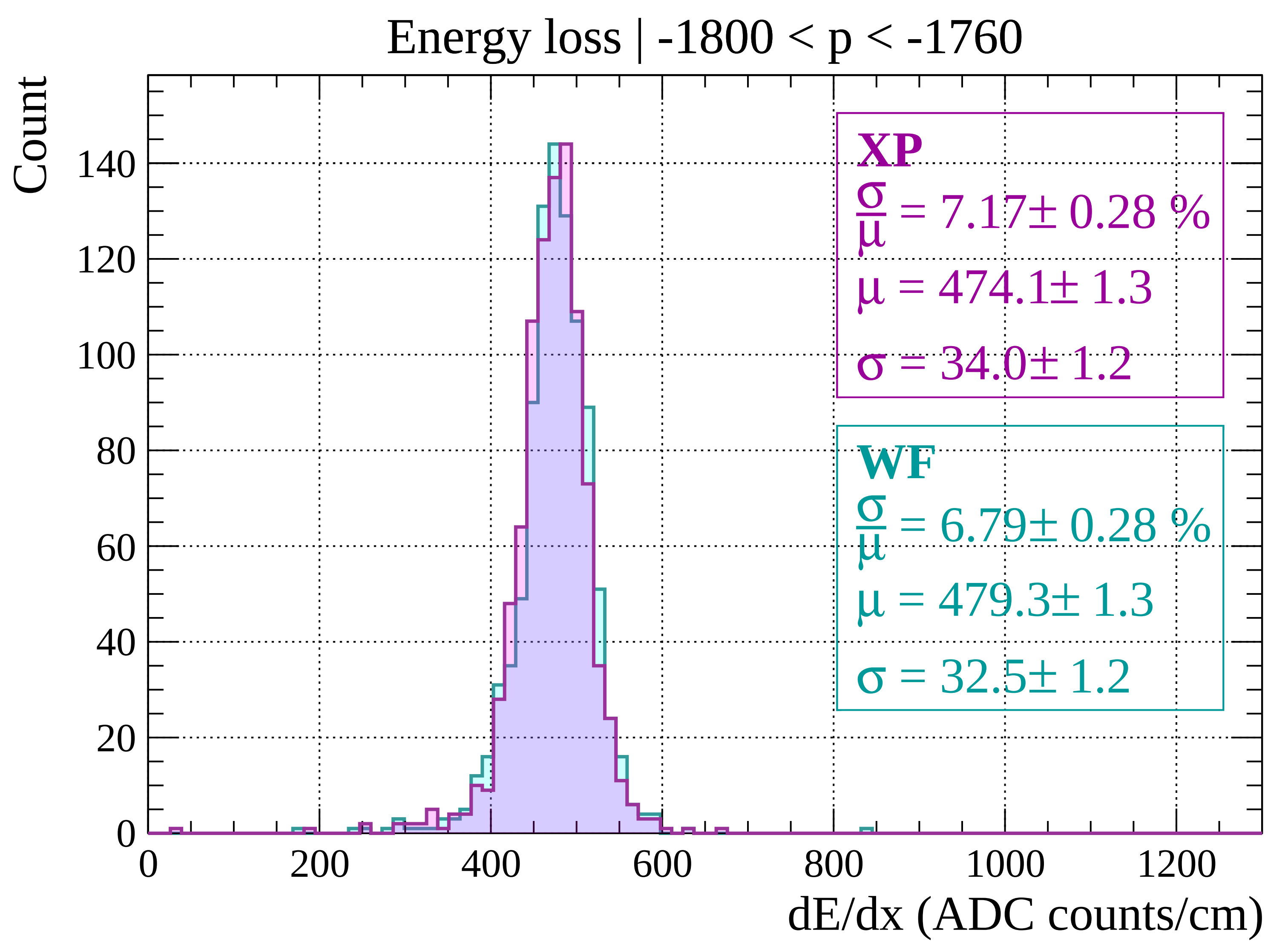
Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.64± 0.28 %
μ = 480.2± 1.3
σ = 31.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.98± 0.28 %
μ = 483.7± 1.3
σ = 33.8± 1.2

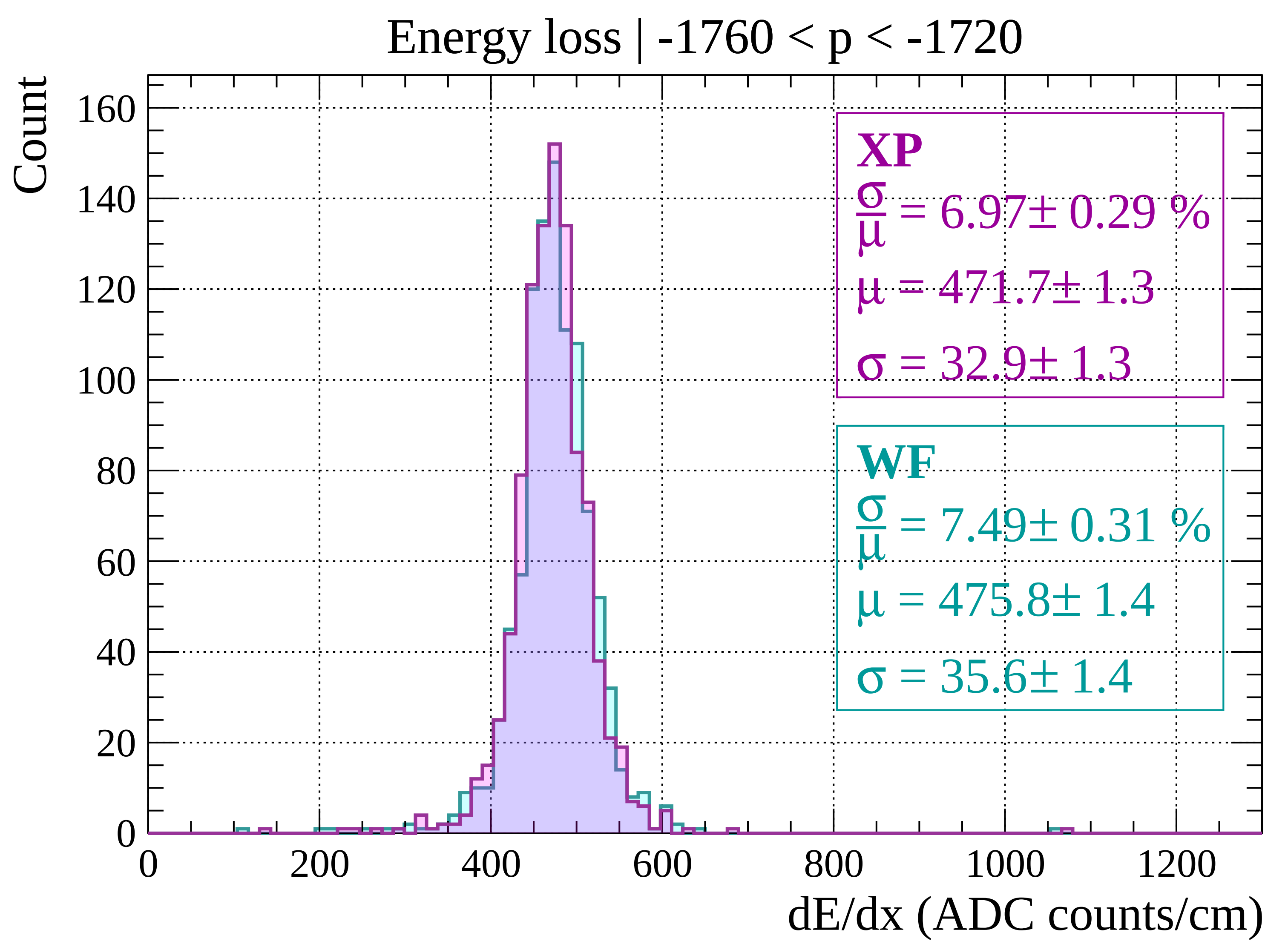

Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 6.98 \pm 0.27$ %
$\mu = 478.0 \pm 1.3$
$\sigma = 33.4 \pm 1.2$
WF
$\frac{\sigma}{\mu} = 6.80 \pm 0.28$ %
$\mu = 482.1 \pm 1.3$
$\sigma = 32.8 \pm 1.3$

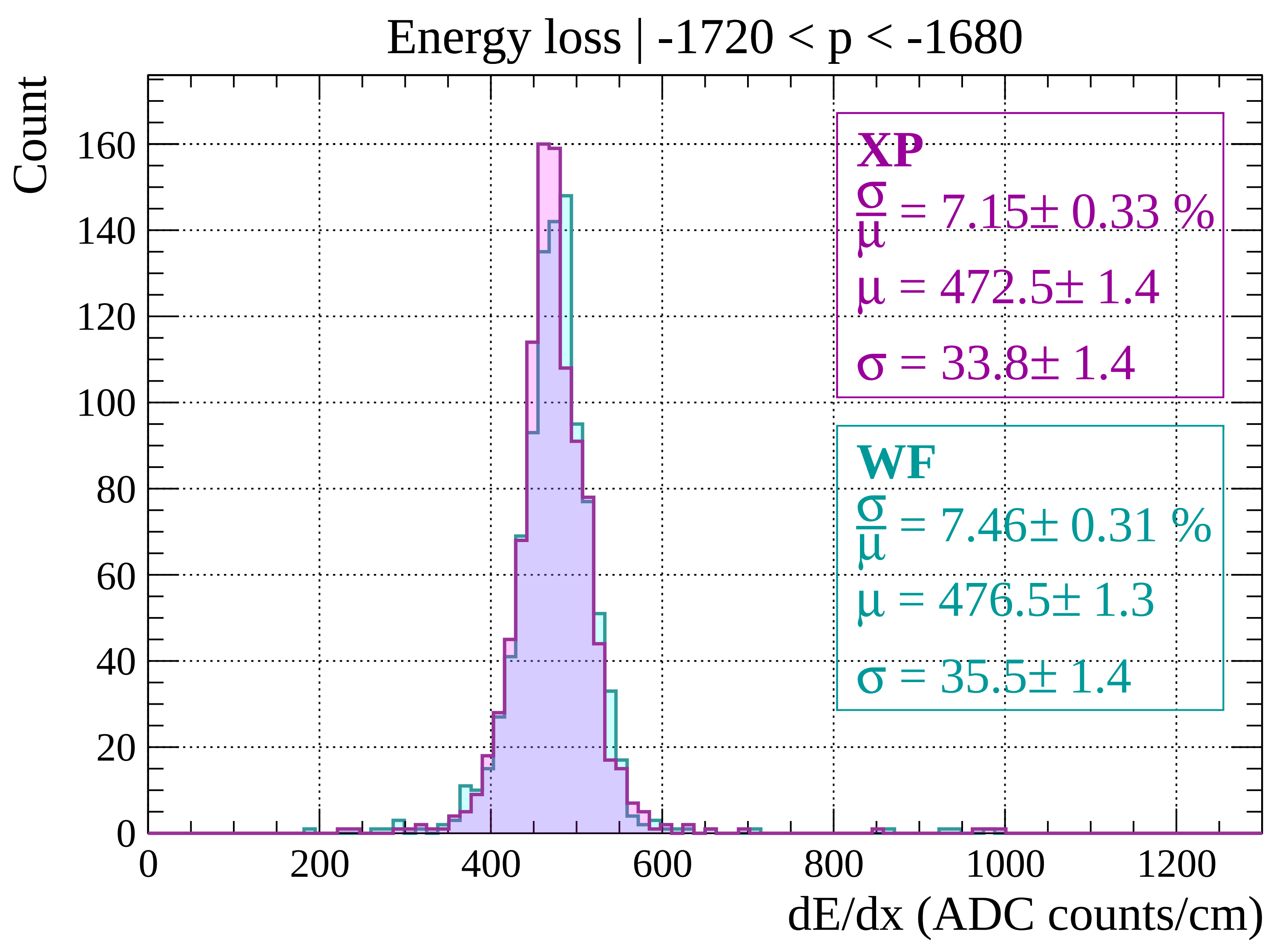


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.05± 0.27 %
μ = 475.6± 1.3
σ = 33.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 7.26± 0.35 %
μ = 479.5± 1.4
σ = 34.8± 1.6

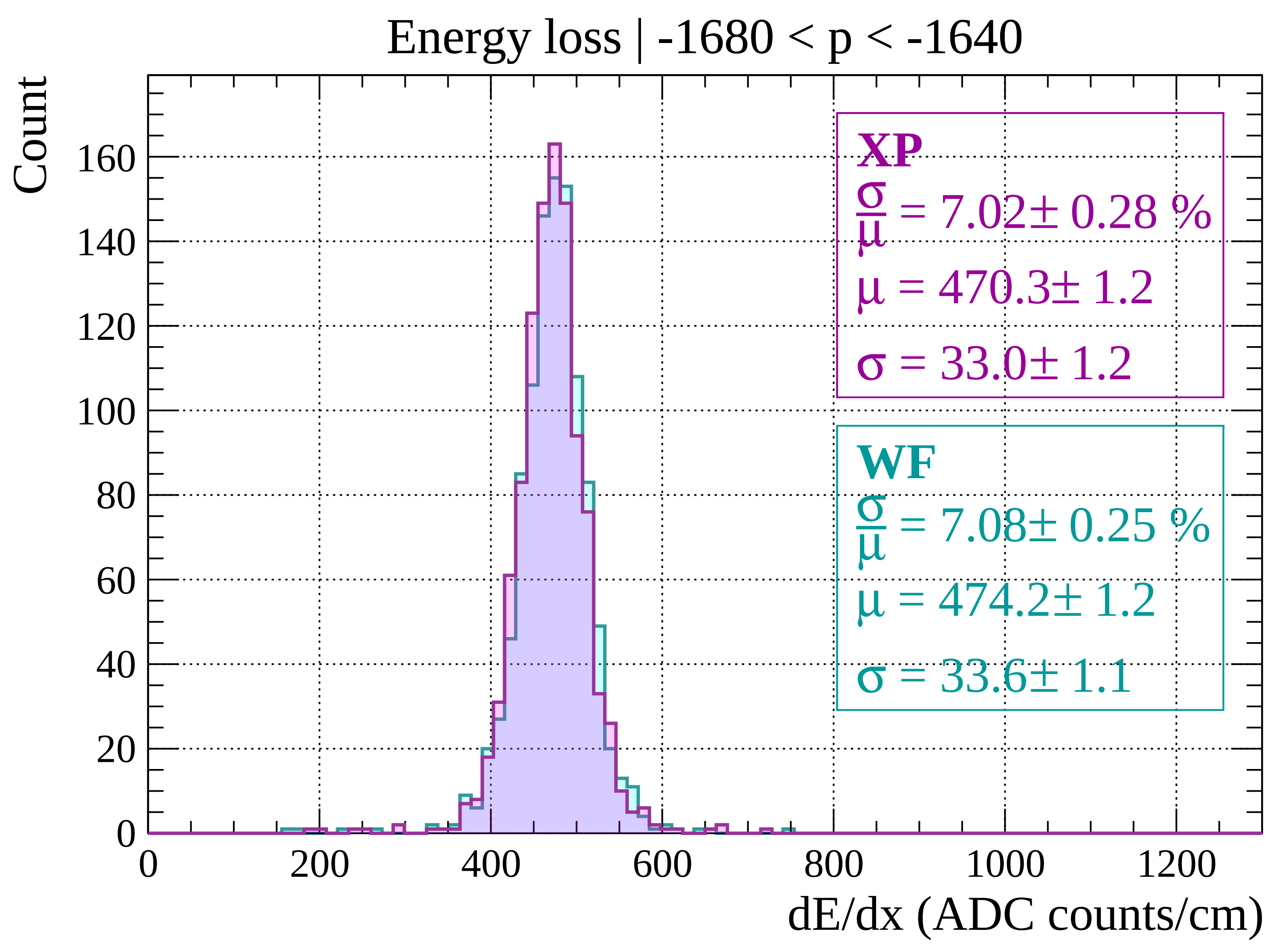

Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.92± 0.25 %
μ = 476.2± 1.2
σ = 32.9± 1.1
WF
σ/μ = 6.75± 0.29 %
μ = 479.9± 1.3
σ = 32.4± 1.3

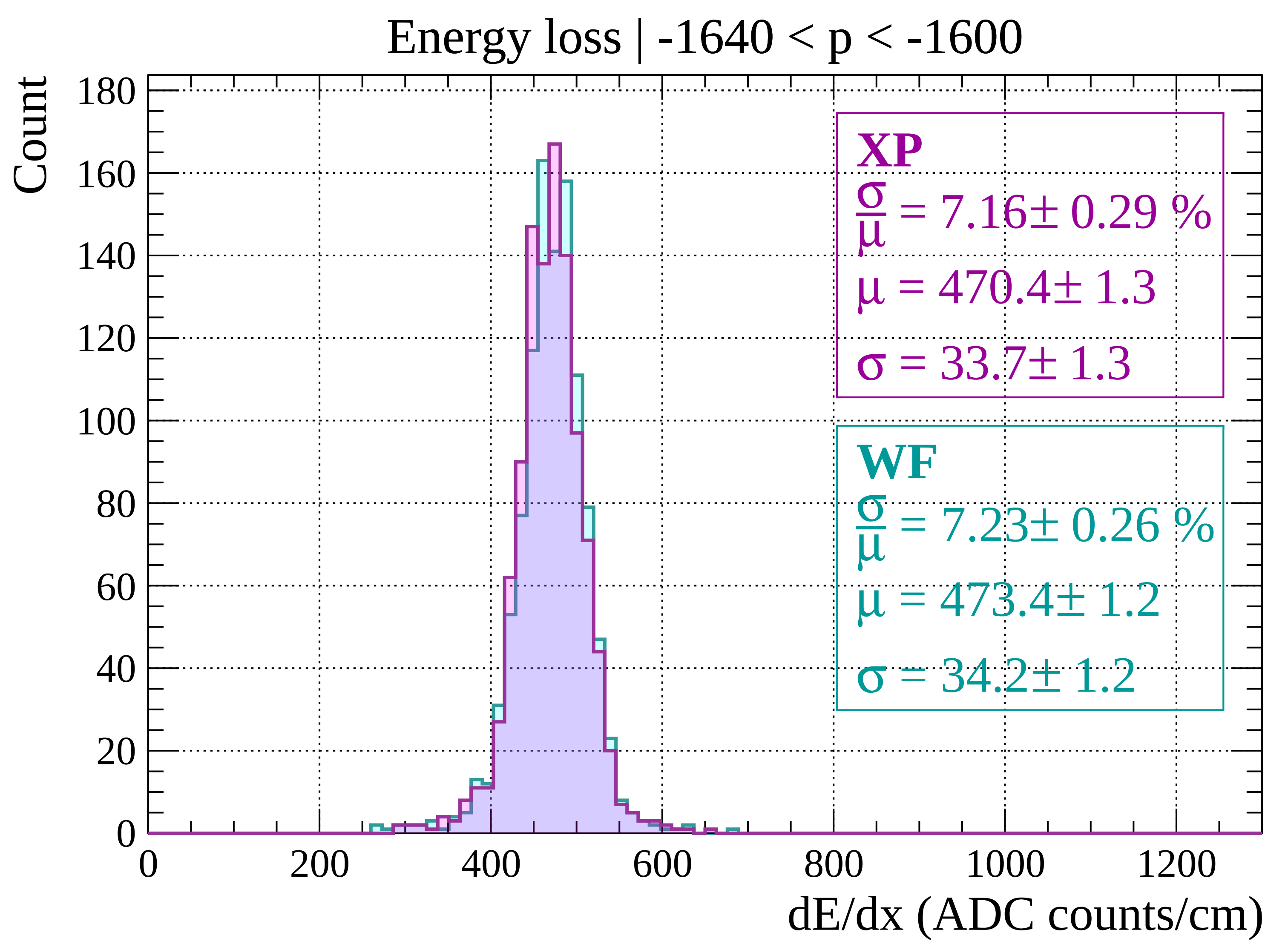

Energy loss | -1800 < p < -1760
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.17± 0.28 %
μ = 474.1± 1.3
σ = 34.0± 1.2
WF
σ/μ = 6.79± 0.28 %
μ = 479.3± 1.3
σ = 32.5± 1.2


Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 6.97± 0.29 %
μ = 471.7± 1.3
σ = 32.9± 1.3
WF
σ/μ = 7.49± 0.31 %
μ = 475.8± 1.4
σ = 35.6± 1.4

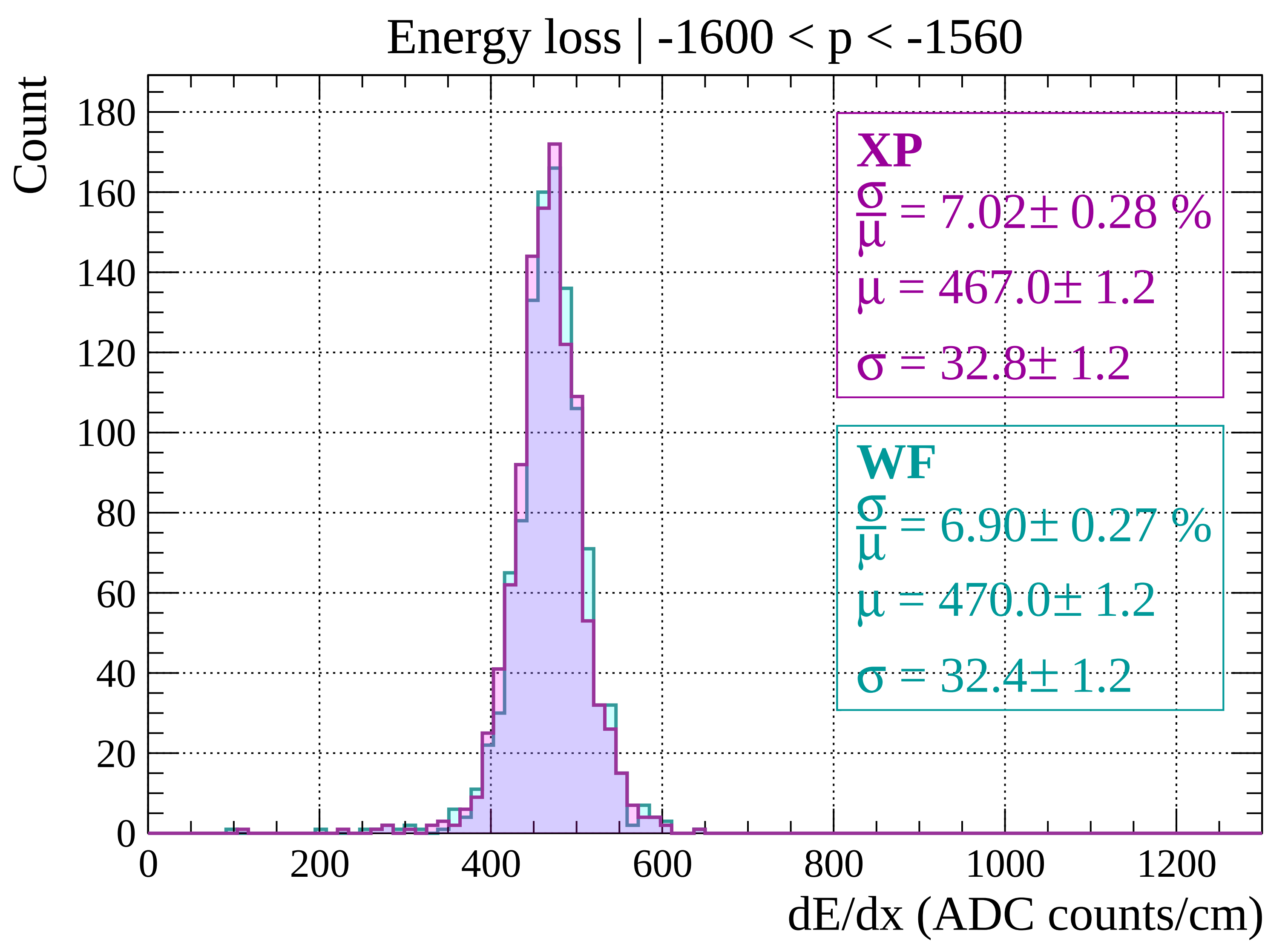


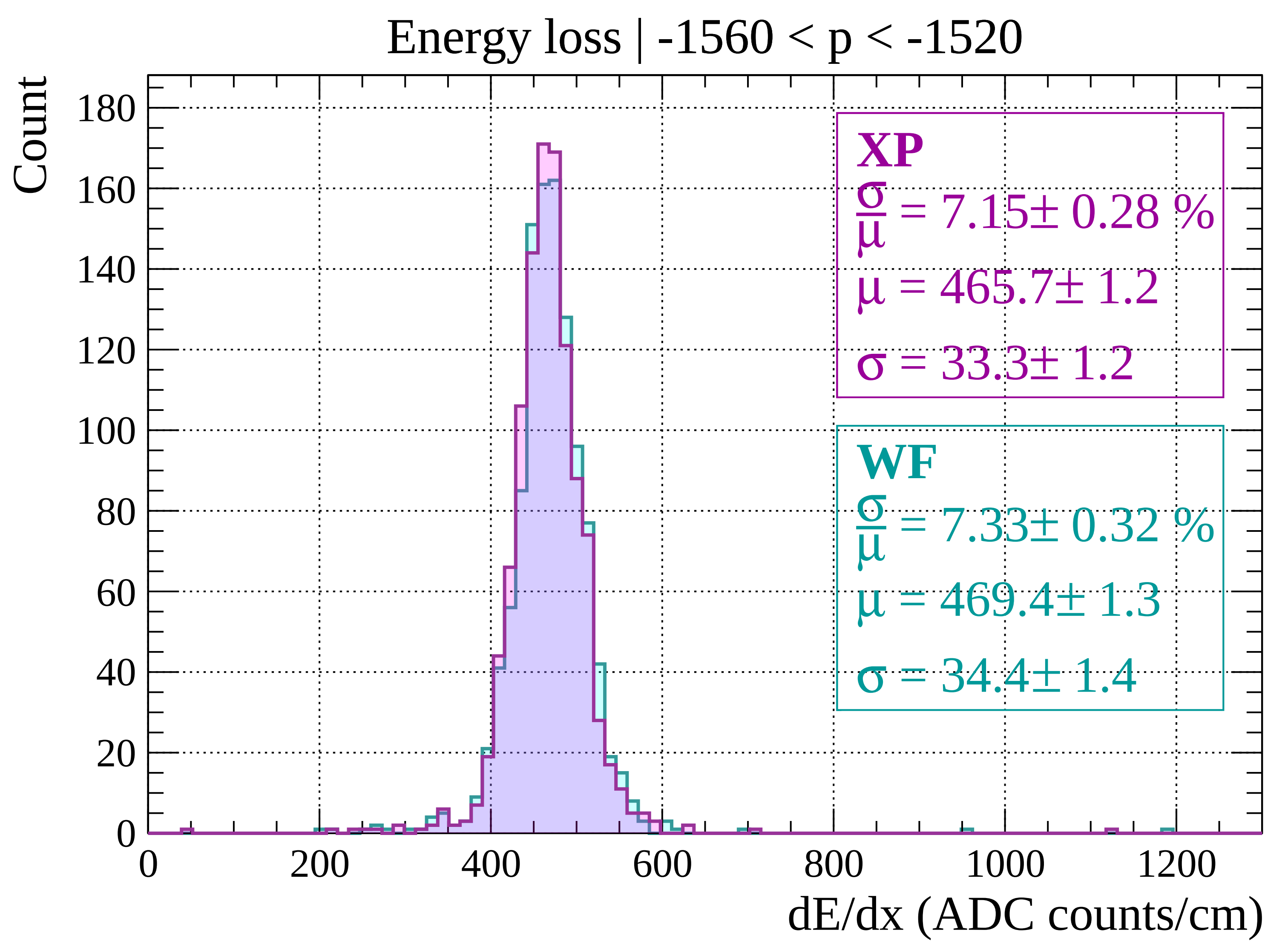
Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 7.15± 0.33 %
μ = 472.5± 1.4
σ = 33.8± 1.4
WF
σ/μ = 7.46± 0.31 %
μ = 476.5± 1.3
σ = 35.5± 1.4


Energy loss | -1680 < p < -1640
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.02± 0.28 %
μ = 470.3± 1.2
σ = 33.0± 1.2
WF
σ/μ = 7.08± 0.25 %
μ = 474.2± 1.2
σ = 33.6± 1.1

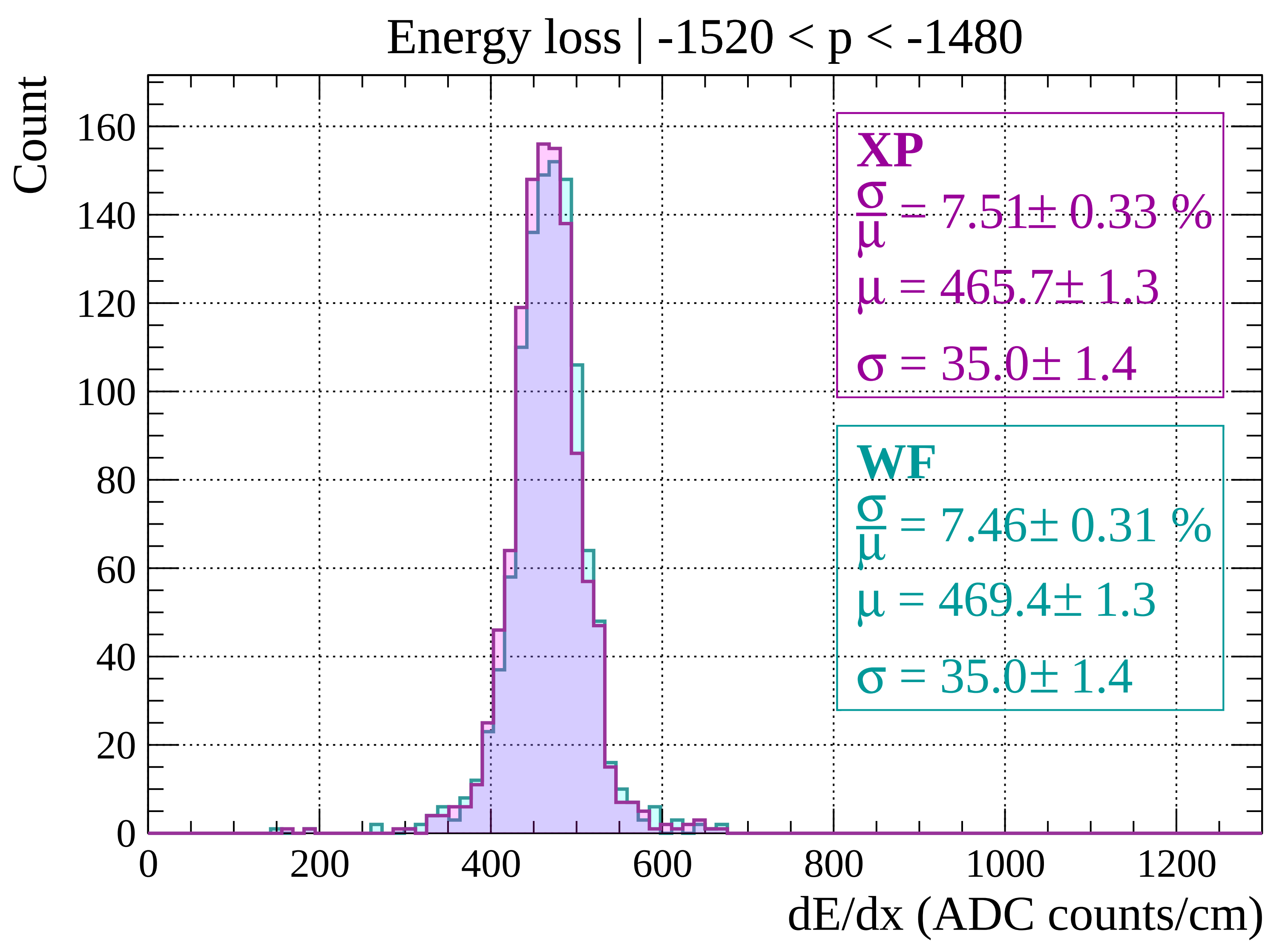


Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.16± 0.29 %
μ = 470.4± 1.3
σ = 33.7± 1.3
WF
$\frac{\sigma}{\mu}$ = 7.23± 0.26 %
μ = 473.4± 1.2
σ = 34.2± 1.2

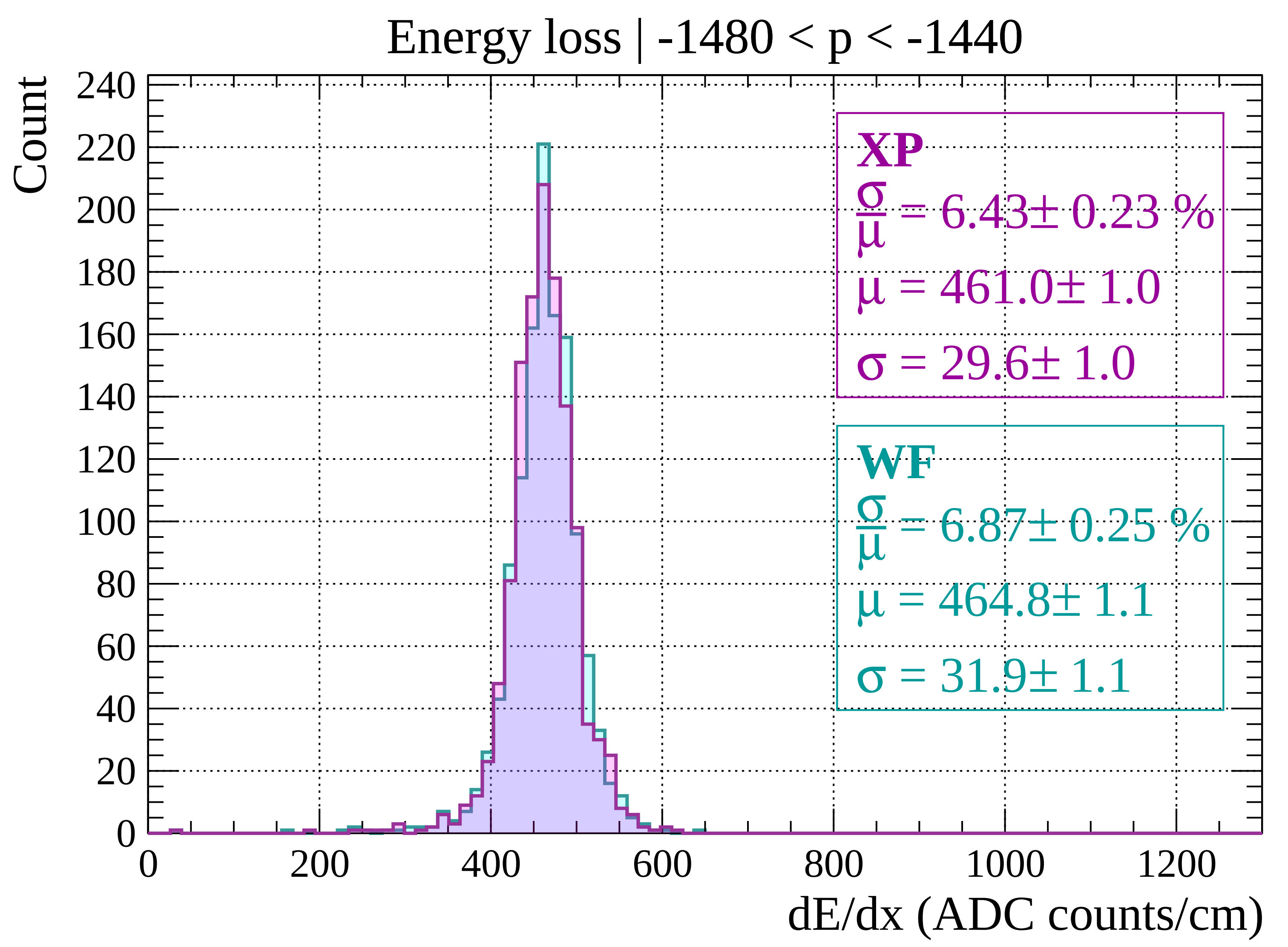

Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.02 \pm 0.28$ %
$\mu = 467.0 \pm 1.2$
$\sigma = 32.8 \pm 1.2$
WF
$\frac{\sigma}{\mu} = 6.90 \pm 0.27$ %
$\mu = 470.0 \pm 1.2$
$\sigma = 32.4 \pm 1.2$

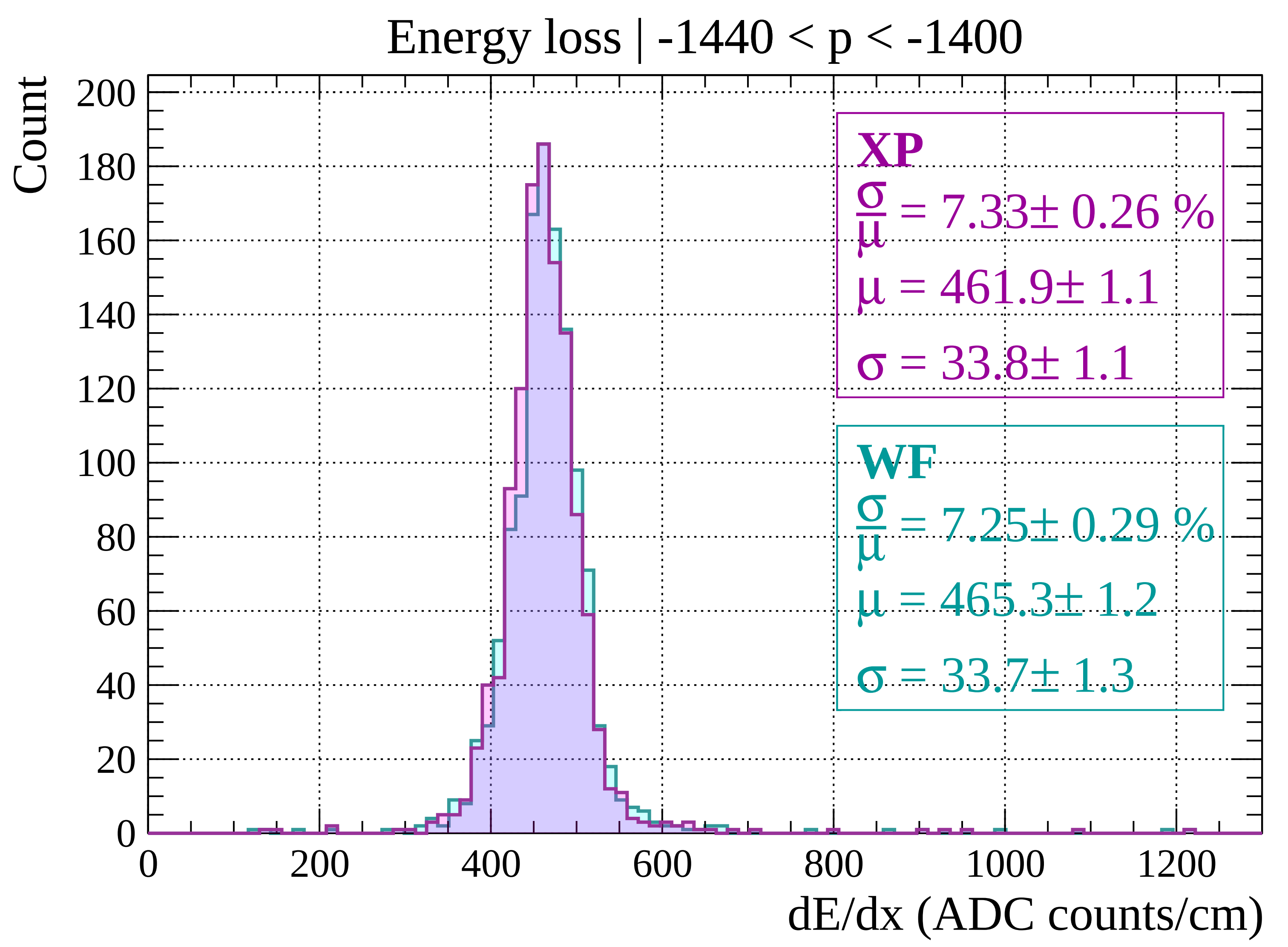


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 7.15± 0.28 %
μ = 465.7± 1.2
σ = 33.3± 1.2
WF
σ/μ = 7.33± 0.32 %
μ = 469.4± 1.3
σ = 34.4± 1.4

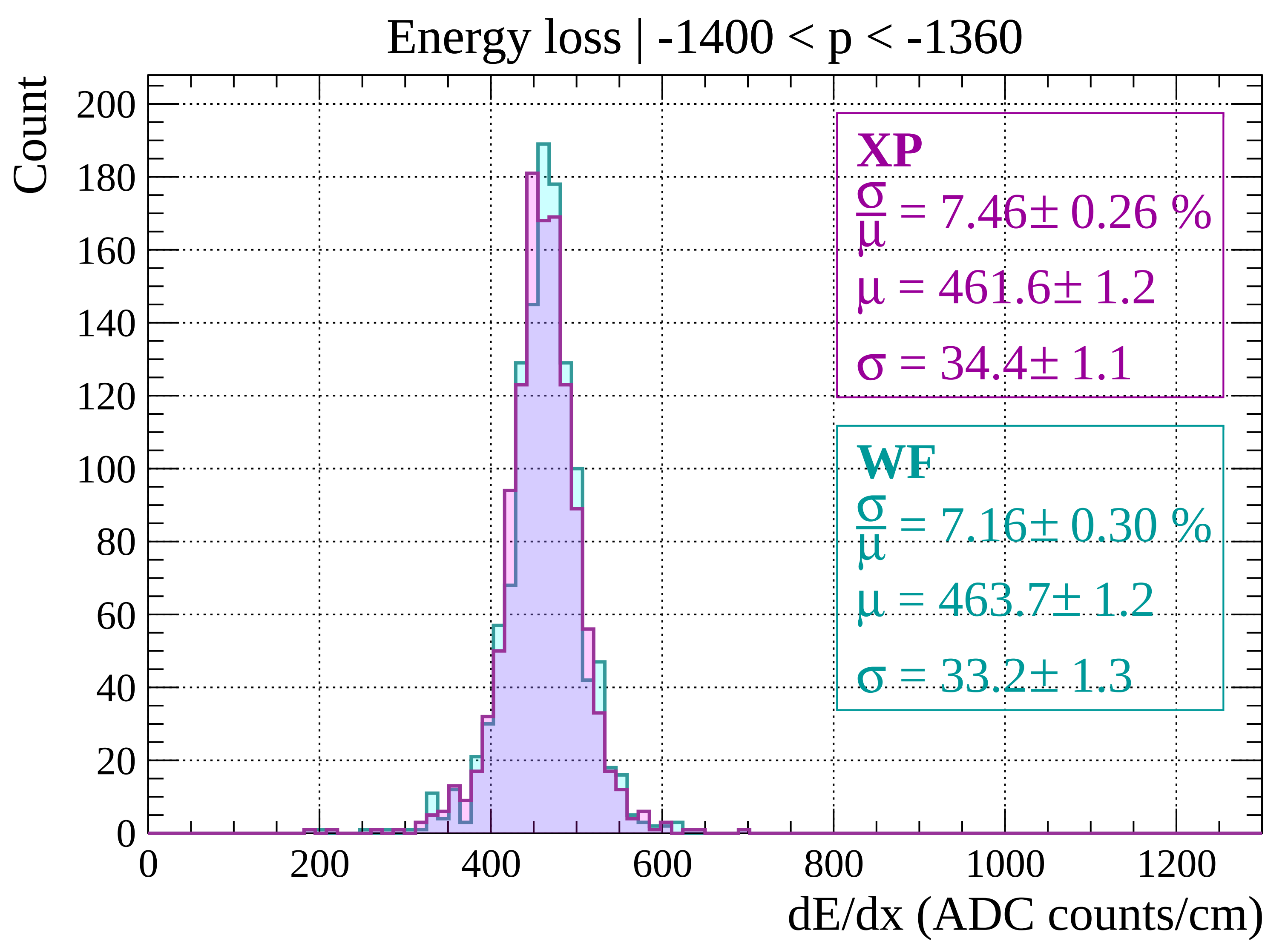


Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 7.51± 0.33 %
μ = 465.7± 1.3
σ = 35.0± 1.4
WF
σ/μ = 7.46± 0.31 %
μ = 469.4± 1.3
σ = 35.0± 1.4

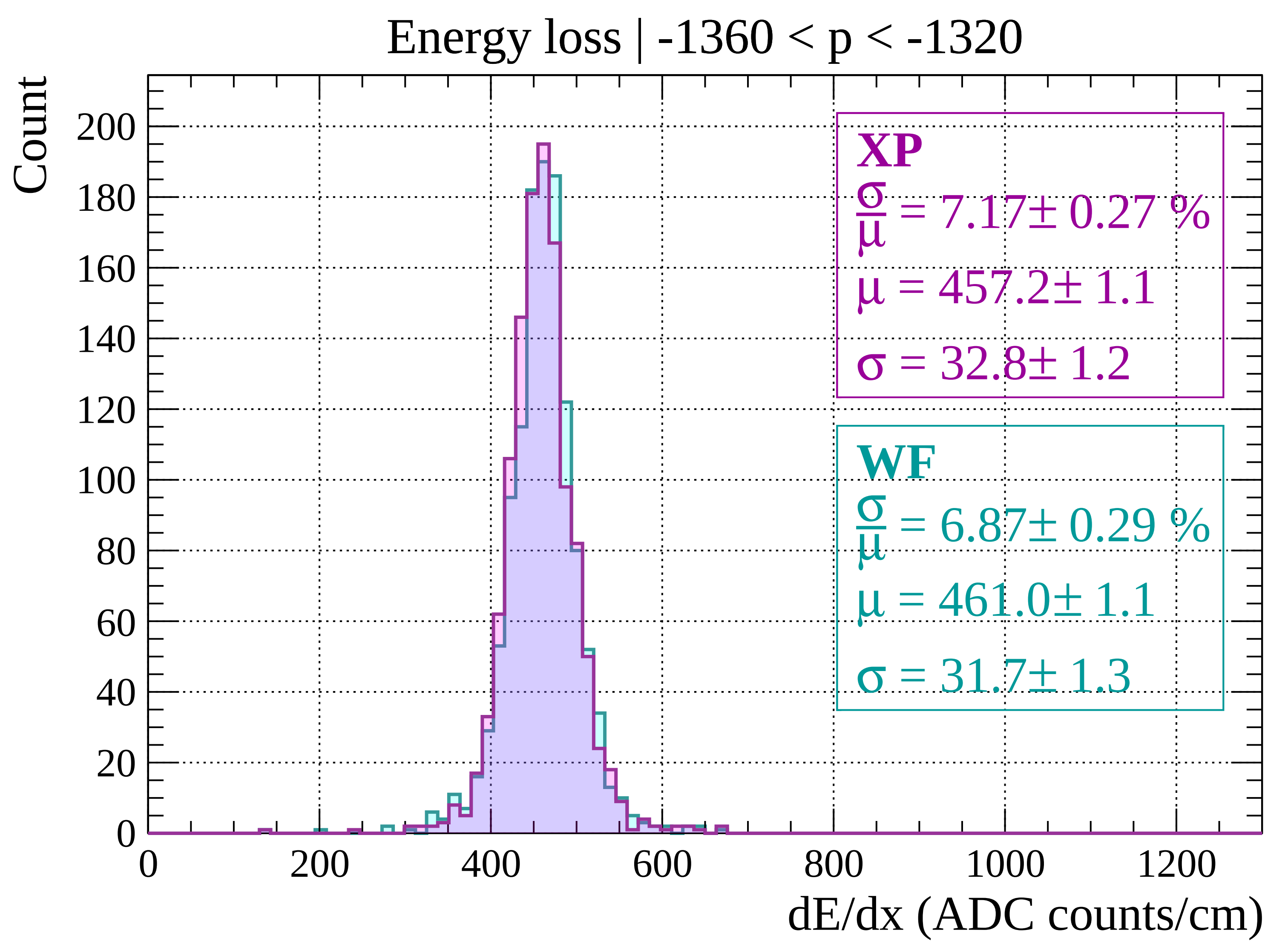

Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
180
200
220
240
XP
σ/μ = 6.43± 0.23 %
μ = 461.0± 1.0
σ = 29.6± 1.0
WF
σ/μ = 6.87± 0.25 %
μ = 464.8± 1.1
σ = 31.9± 1.1

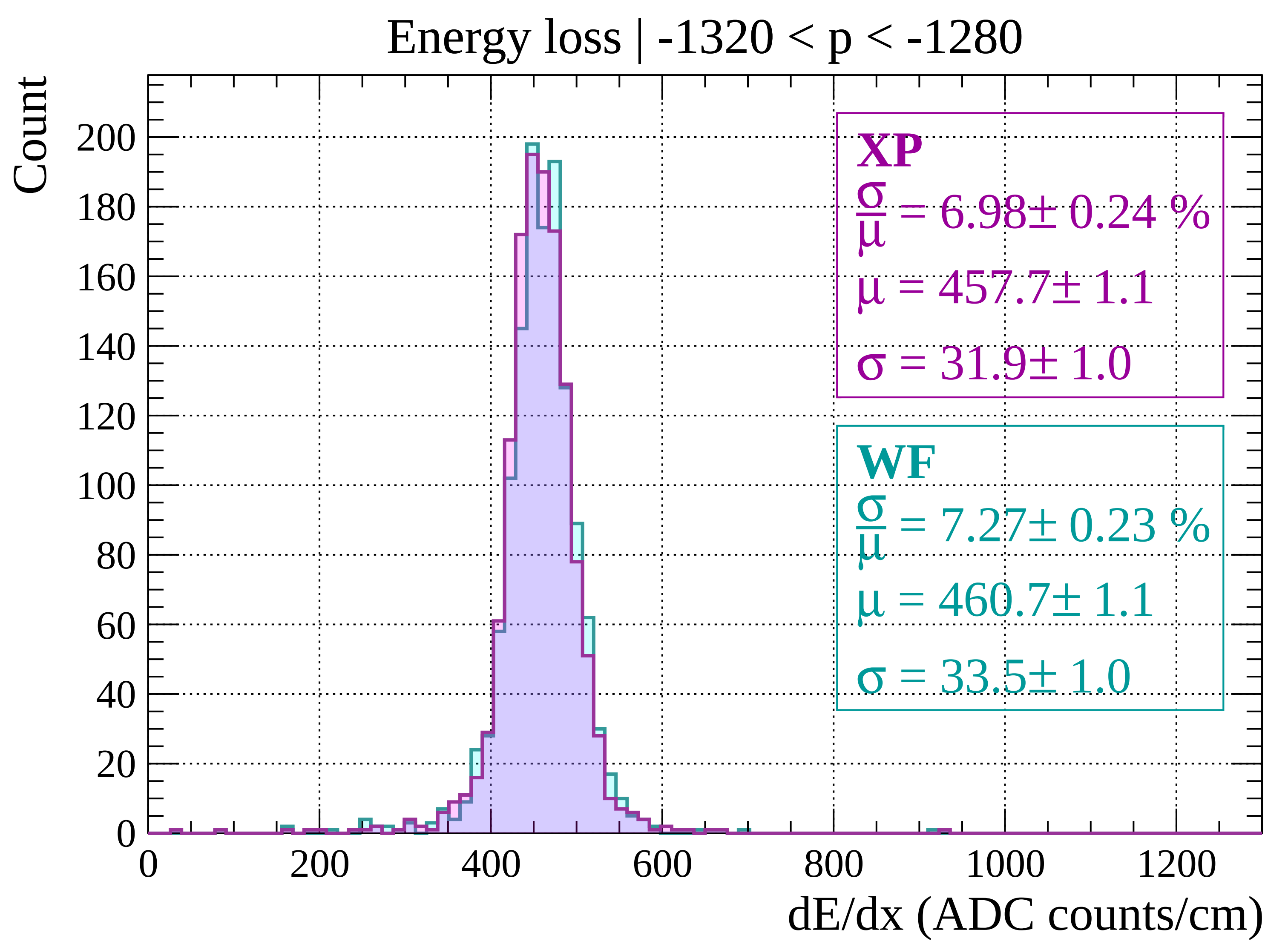


Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 7.33± 0.26 %
μ = 461.9± 1.1
σ = 33.8± 1.1
WF
σ/μ = 7.25± 0.29 %
μ = 465.3± 1.2
σ = 33.7± 1.3

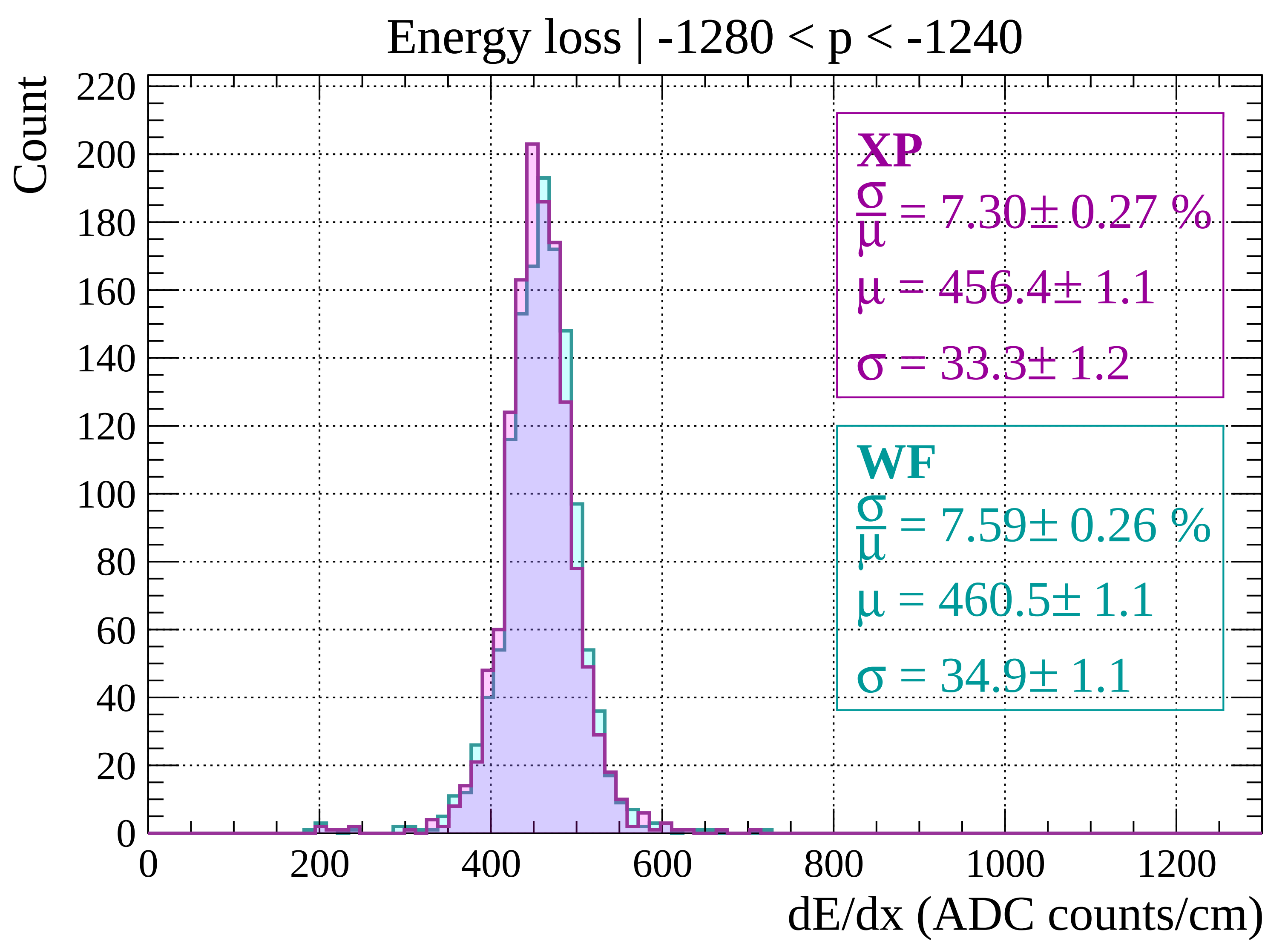


Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 7.46± 0.26 %
μ = 461.6± 1.2
σ = 34.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 7.16± 0.30 %
μ = 463.7± 1.2
σ = 33.2± 1.3

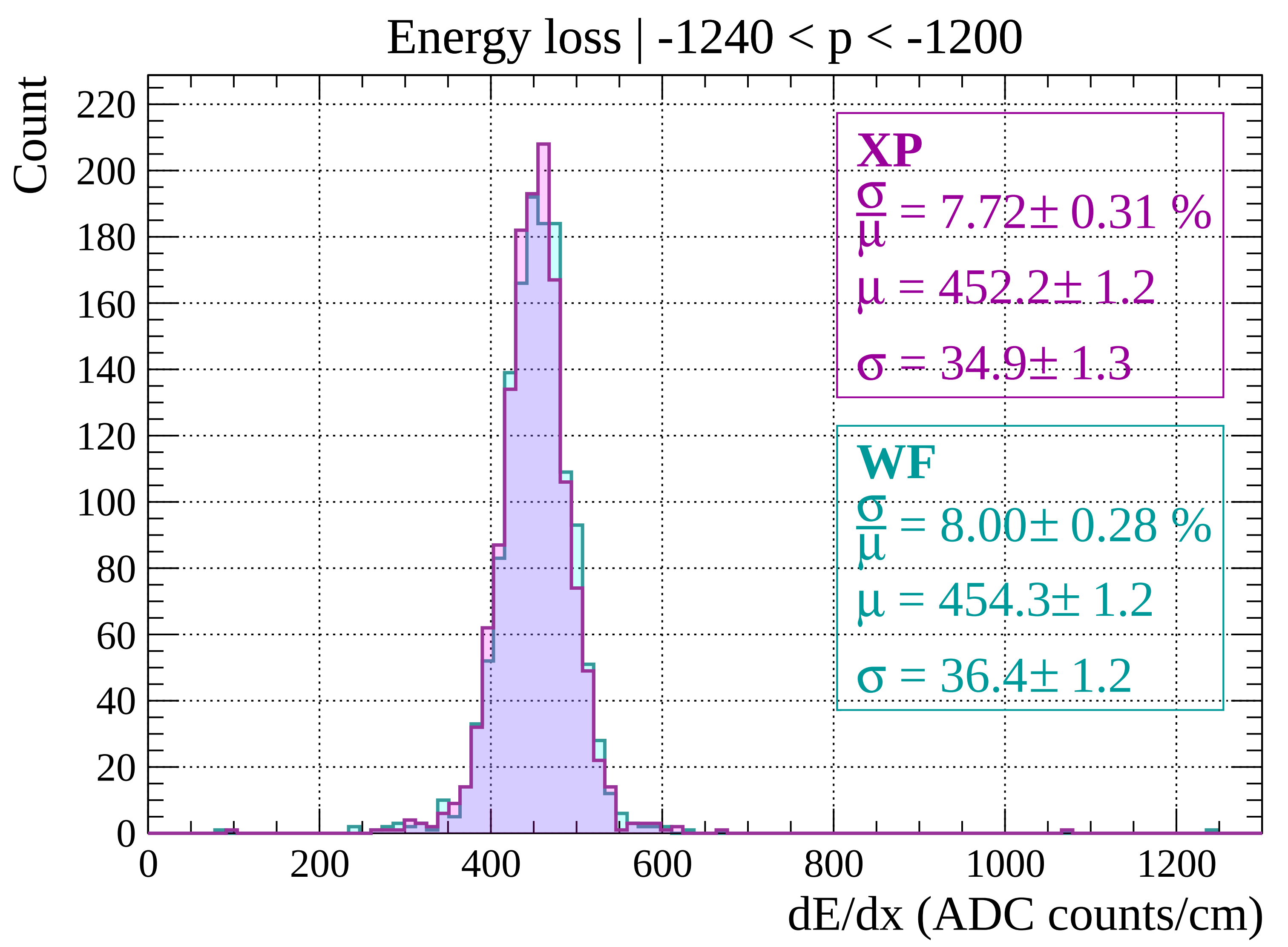

Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 7.17± 0.27 %
μ = 457.2± 1.1
σ = 32.8± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.87± 0.29 %
μ = 461.0± 1.1
σ = 31.7± 1.3



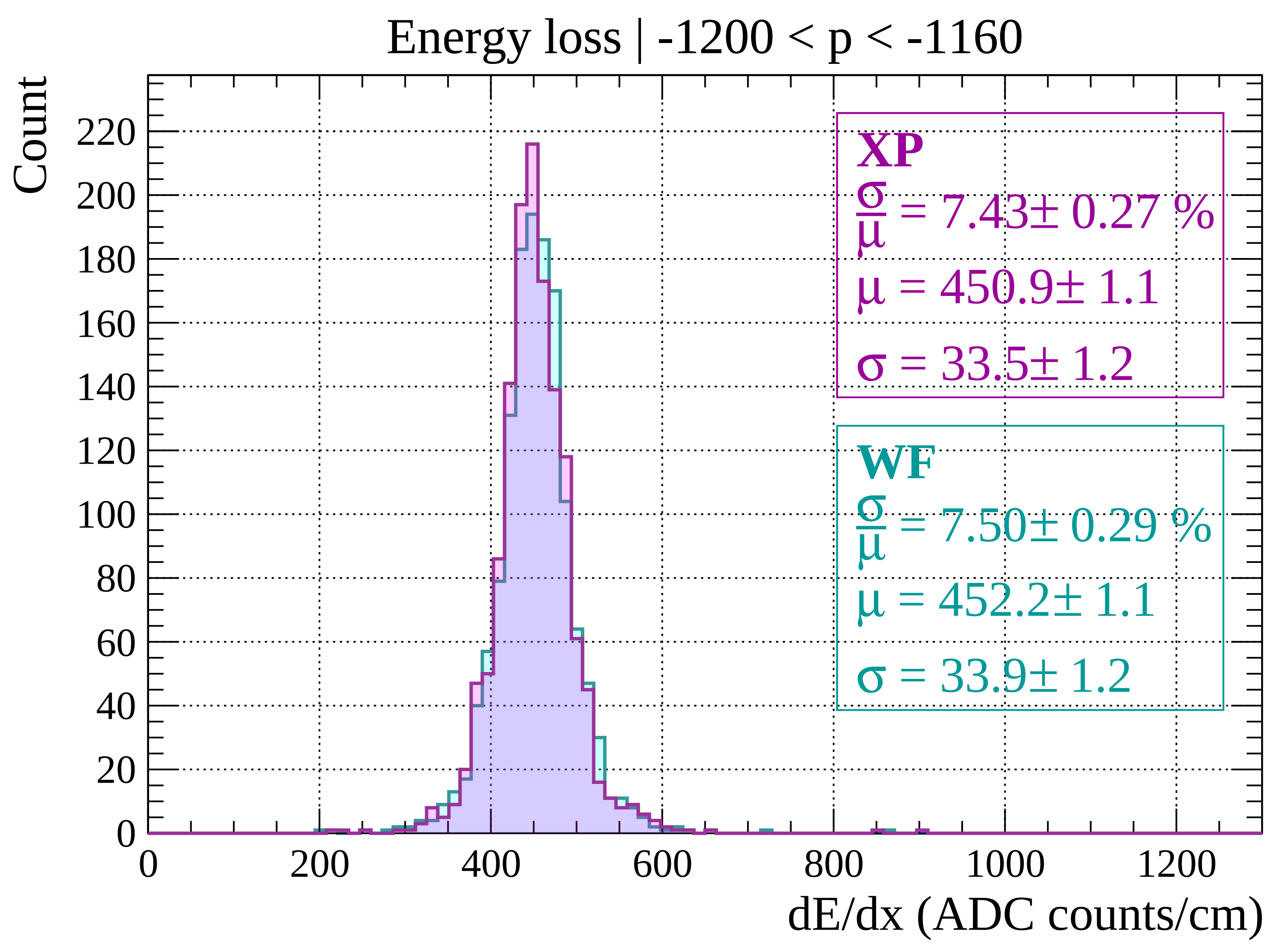
Energy loss | -1320 < p < -1280
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.98± 0.24 %
μ = 457.7± 1.1
σ = 31.9± 1.0
WF
σ/μ = 7.27± 0.23 %
μ = 460.7± 1.1
σ = 33.5± 1.0



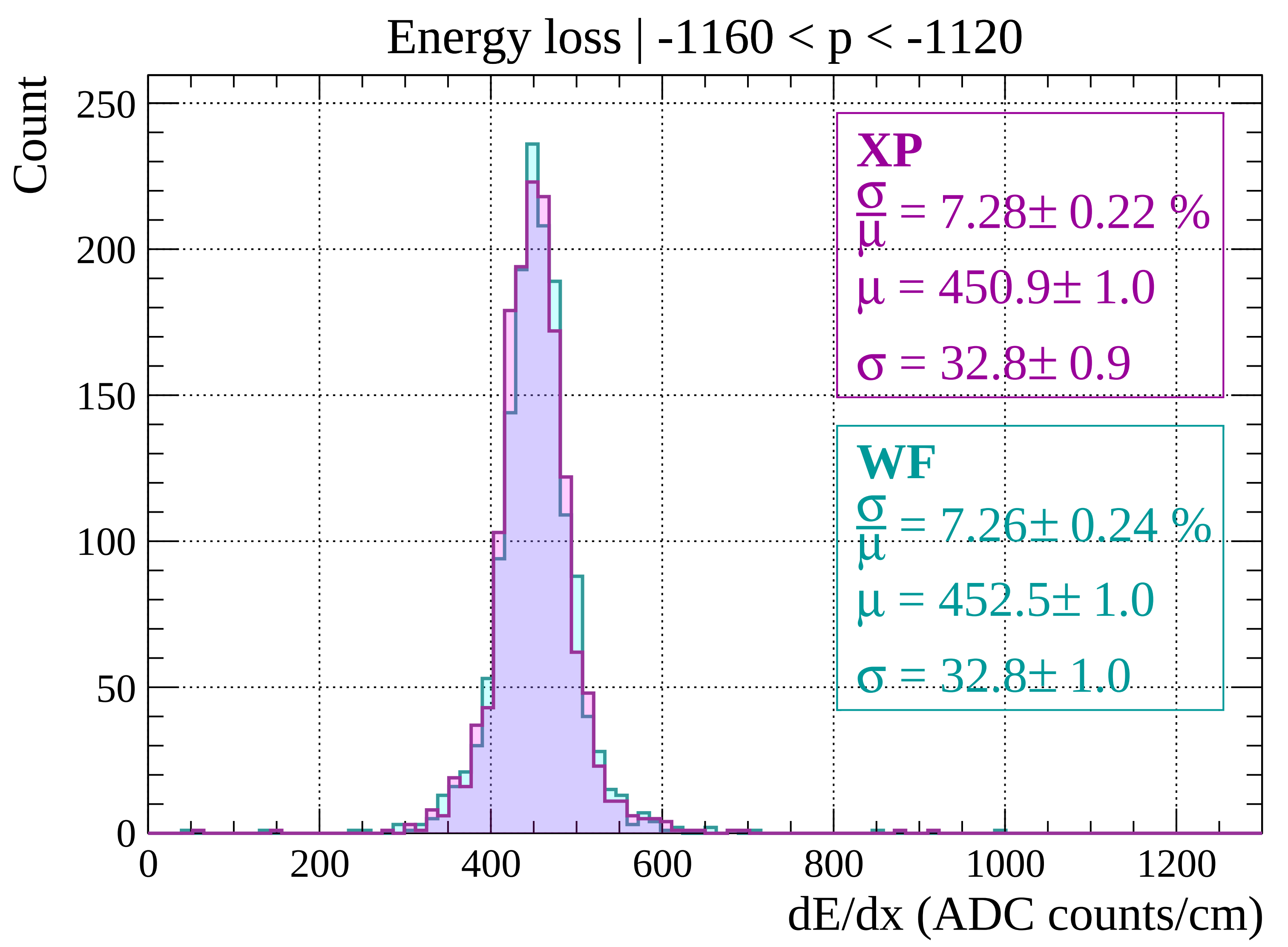
Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.30± 0.27 %
μ = 456.4± 1.1
σ = 33.3± 1.2
WF
$\frac{\sigma}{\mu}$ = 7.59± 0.26 %
μ = 460.5± 1.1
σ = 34.9± 1.1



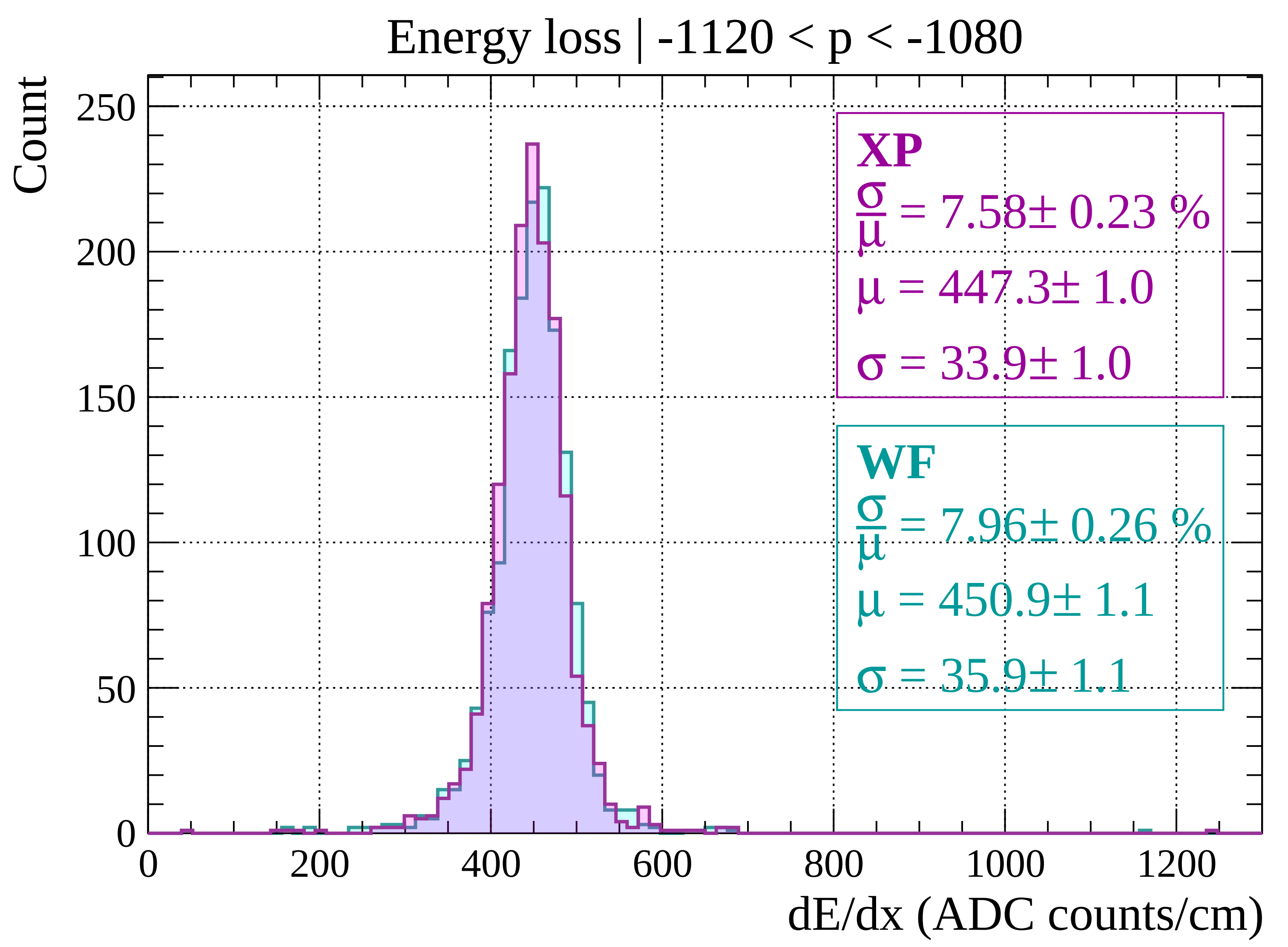
Energy loss | -1240 < p < -1200
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.72± 0.31 %
μ = 452.2± 1.2
σ = 34.9± 1.3
WF
σ/μ = 8.00± 0.28 %
μ = 454.3± 1.2
σ = 36.4± 1.2



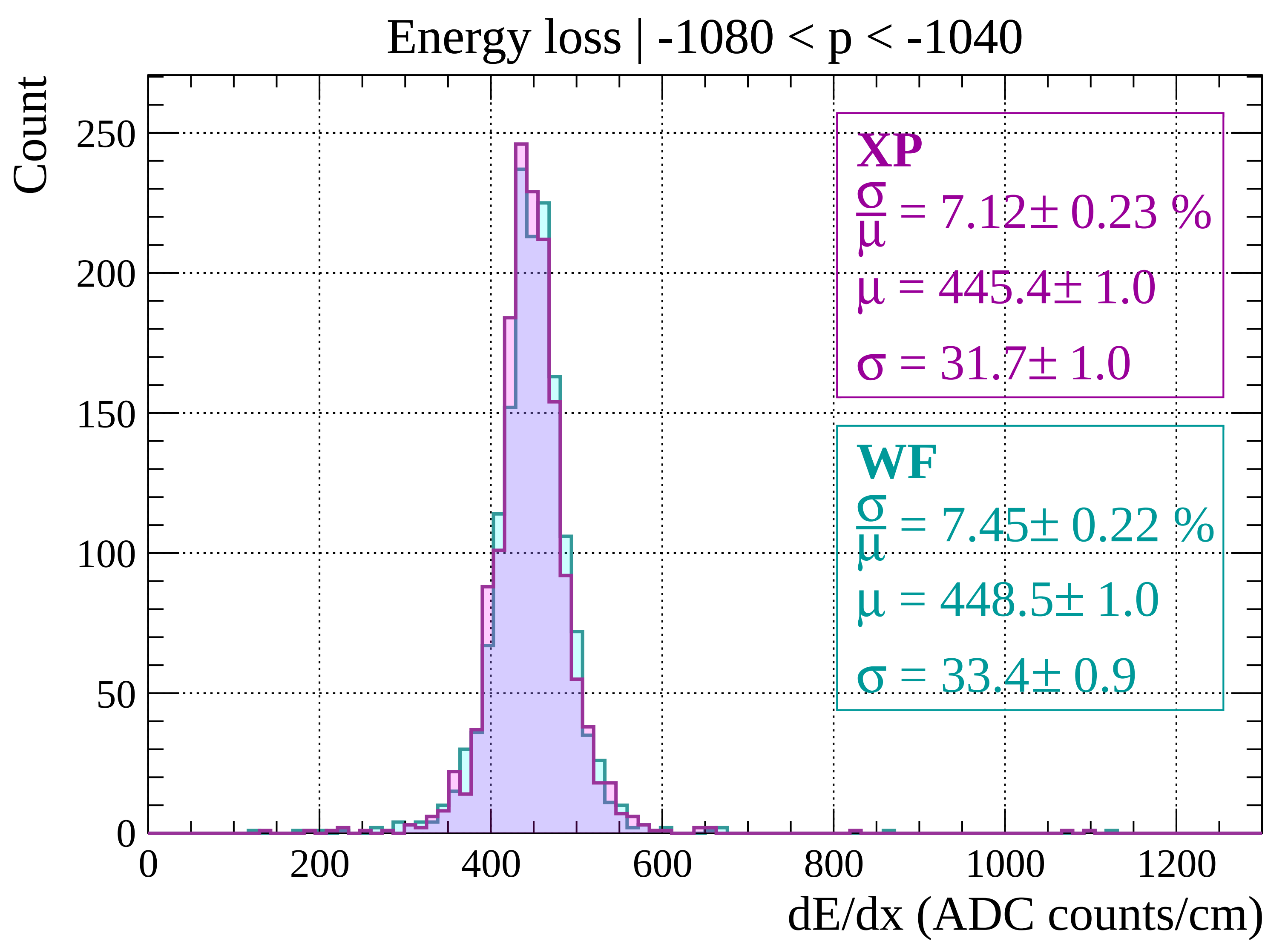
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 7.43± 0.27 %
μ = 450.9± 1.1
σ = 33.5± 1.2
WF
σ/μ = 7.50± 0.29 %
μ = 452.2± 1.1
σ = 33.9± 1.2



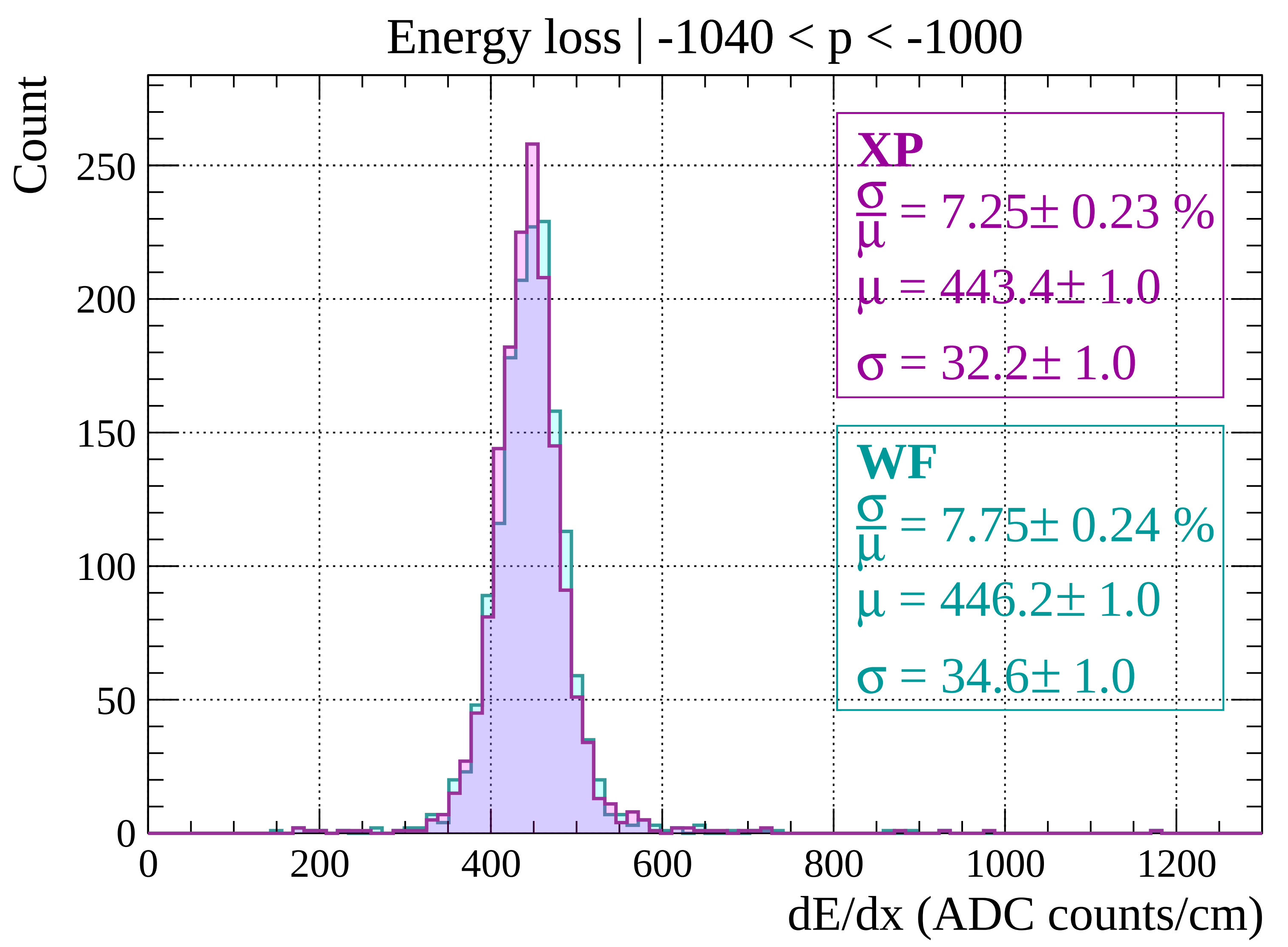
Energy loss | -1160 < p < -1120
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.28± 0.22 %
μ = 450.9± 1.0
σ = 32.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.26± 0.24 %
μ = 452.5± 1.0
σ = 32.8± 1.0



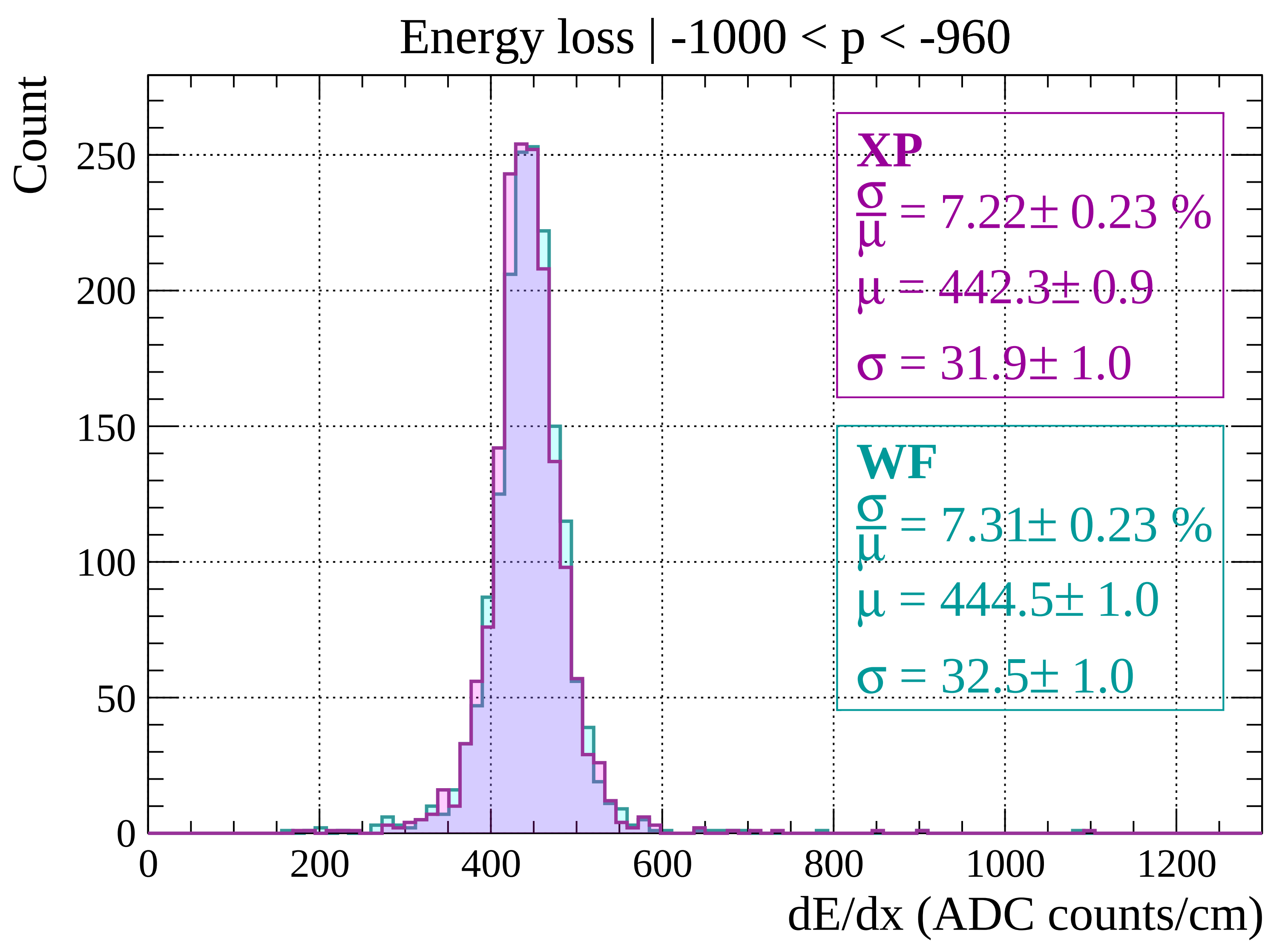
Energy loss | -1120 < p < -1080
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.58± 0.23 %
μ = 447.3± 1.0
σ = 33.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.96± 0.26 %
μ = 450.9± 1.1
σ = 35.9± 1.1

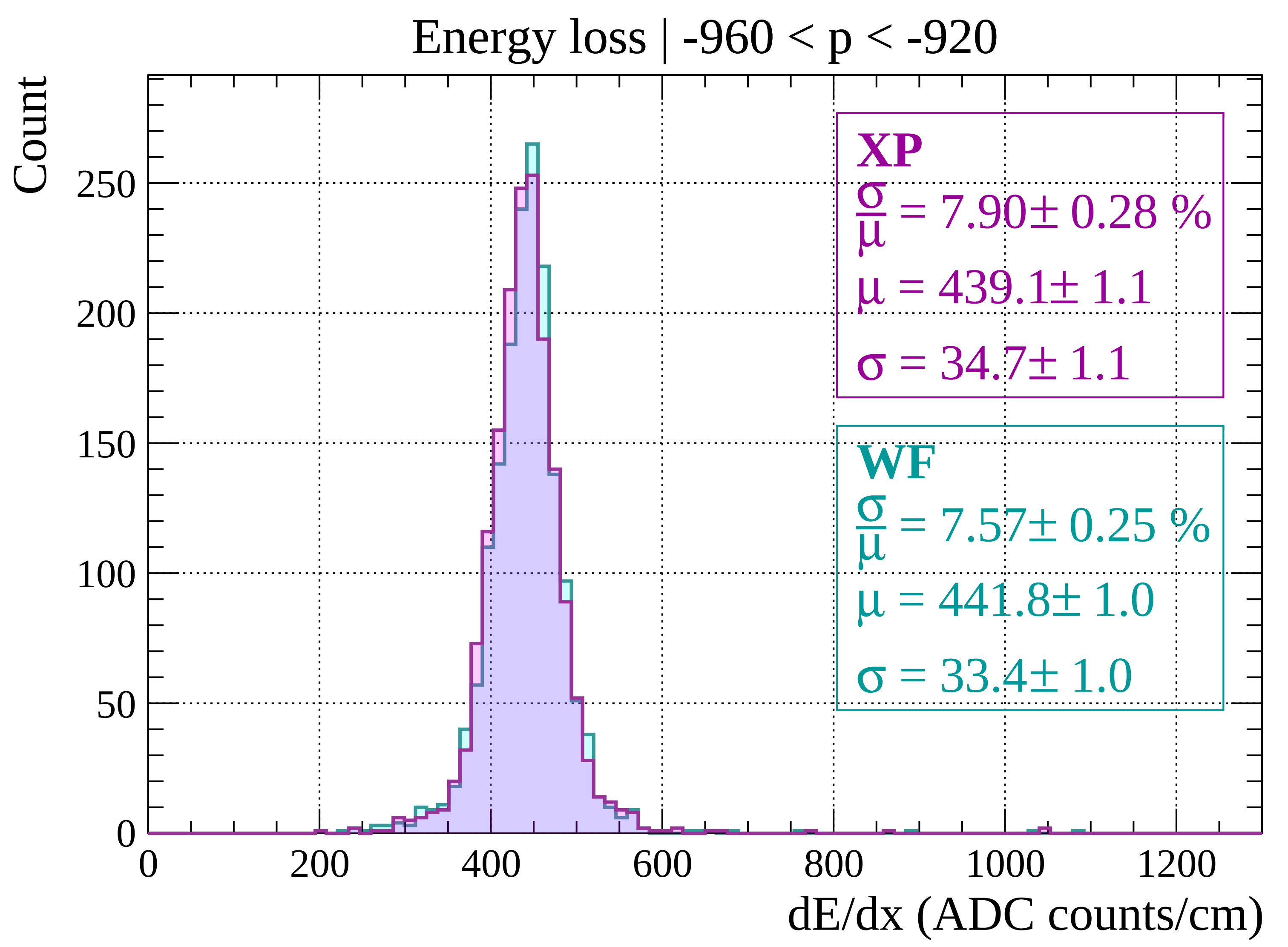


Energy loss | -1080 < p < -1040
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.12± 0.23 %
μ = 445.4± 1.0
σ = 31.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.45± 0.22 %
μ = 448.5± 1.0
σ = 33.4± 0.9

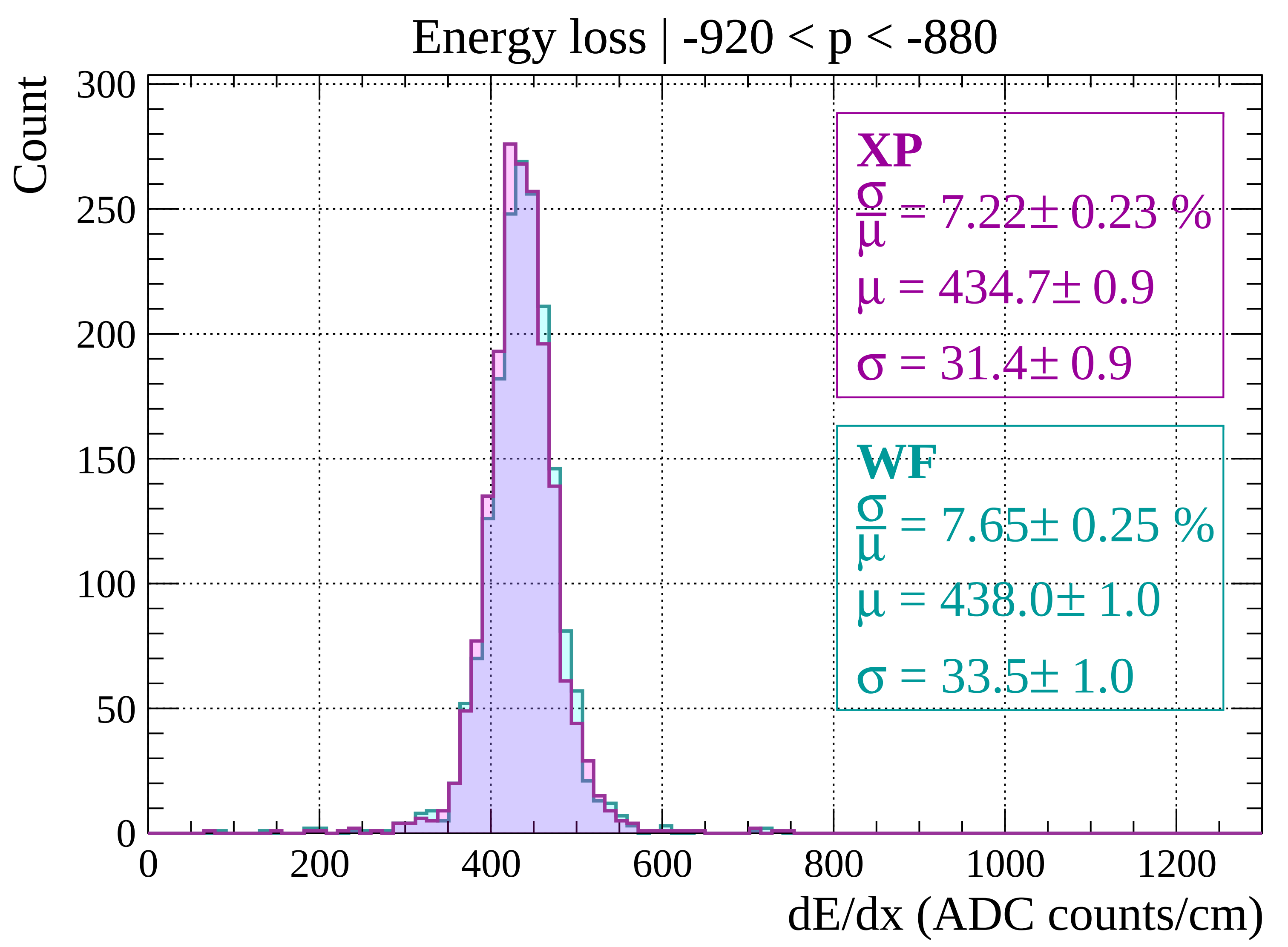


Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.25± 0.23 %
μ = 443.4± 1.0
σ = 32.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.75± 0.24 %
μ = 446.2± 1.0
σ = 34.6± 1.0
0
50
100
150
200
250
0
200
400
600
800
1000
1200

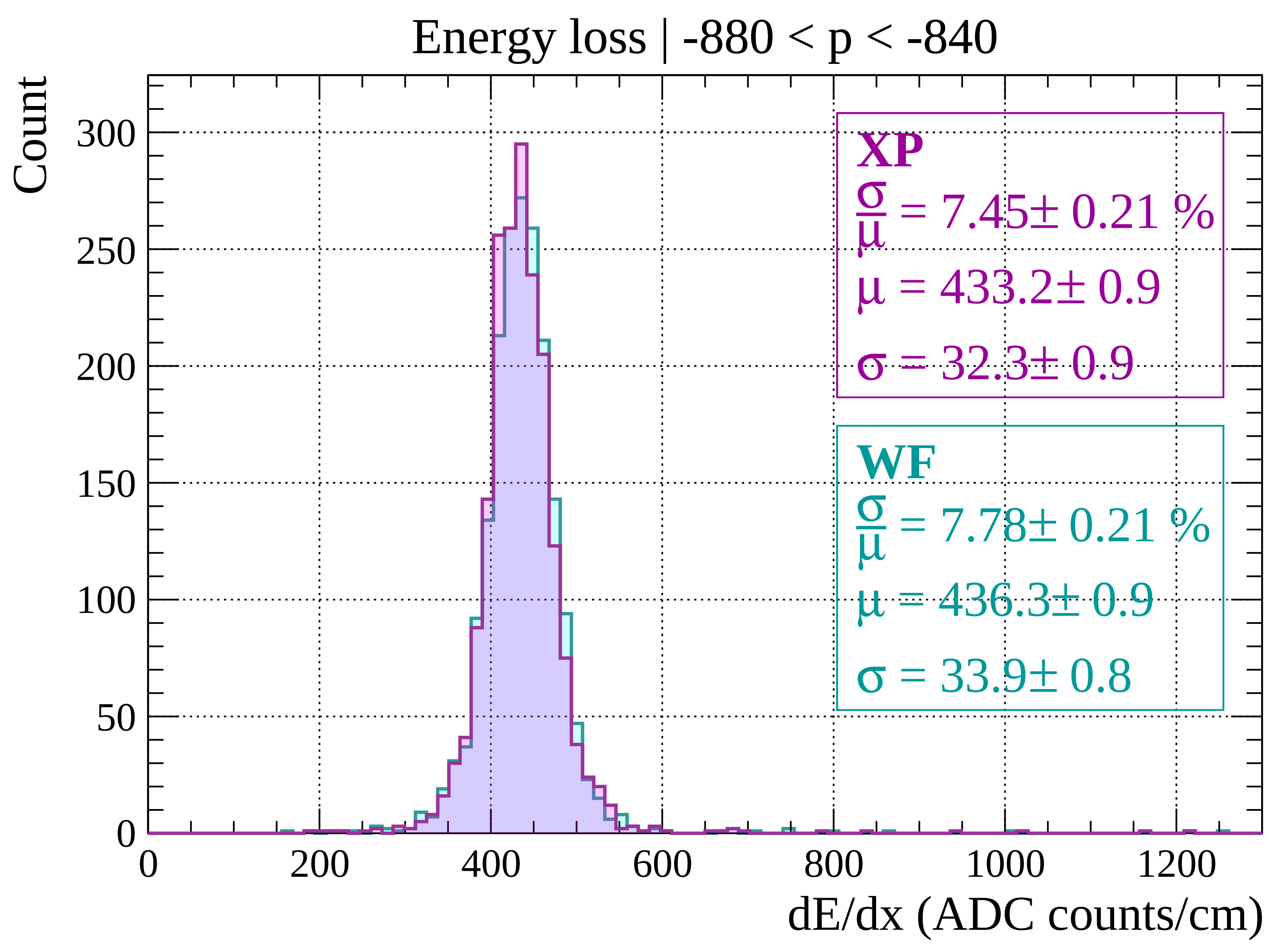

Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.22± 0.23 %
μ = 442.3± 0.9
σ = 31.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.31± 0.23 %
μ = 444.5± 1.0
σ = 32.5± 1.0
0
200
400
600
800
1000
1200
50
100
150
200
250



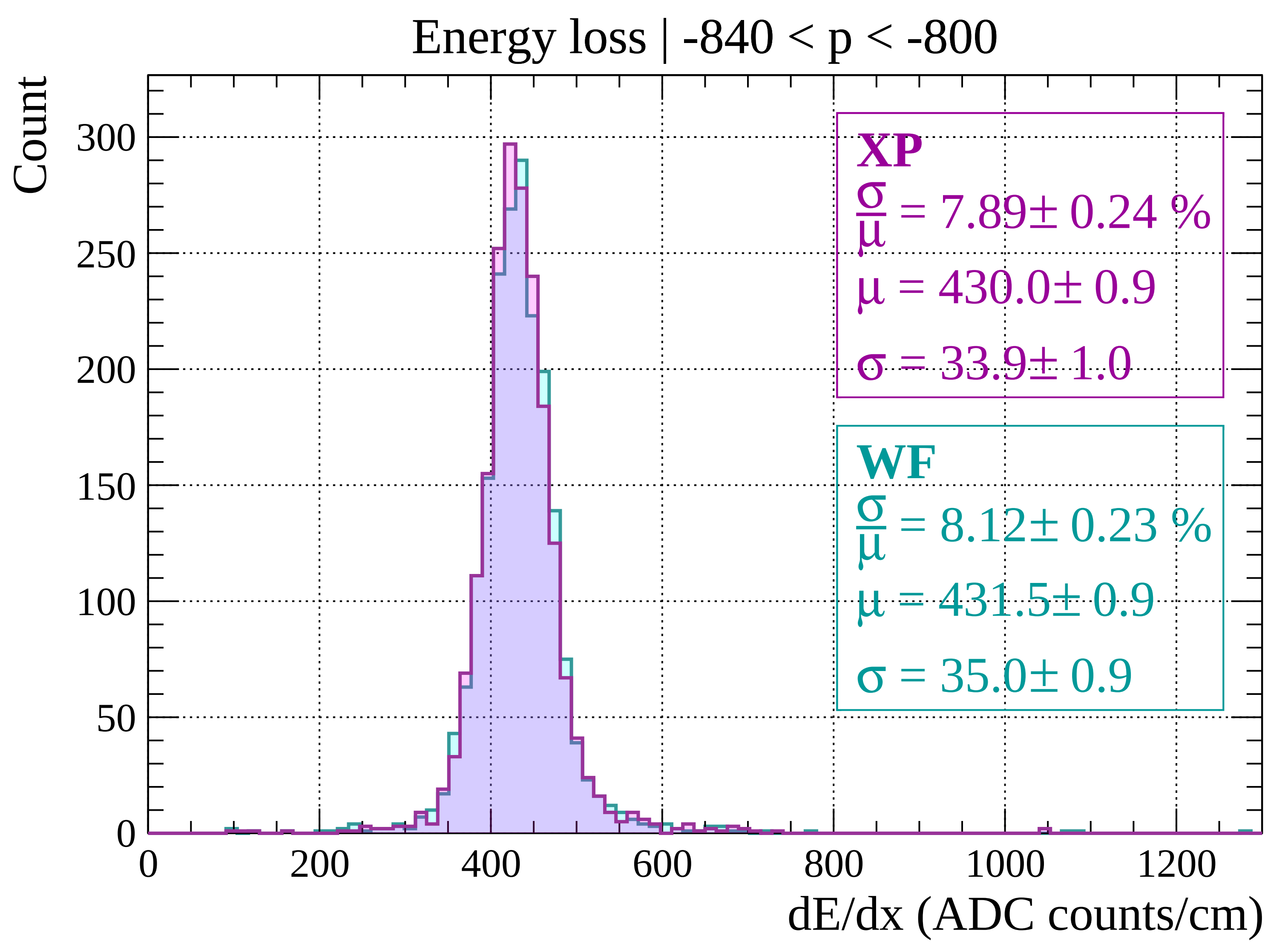
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.90± 0.28 %
μ = 439.1± 1.1
σ = 34.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 7.57± 0.25 %
μ = 441.8± 1.0
σ = 33.4± 1.0

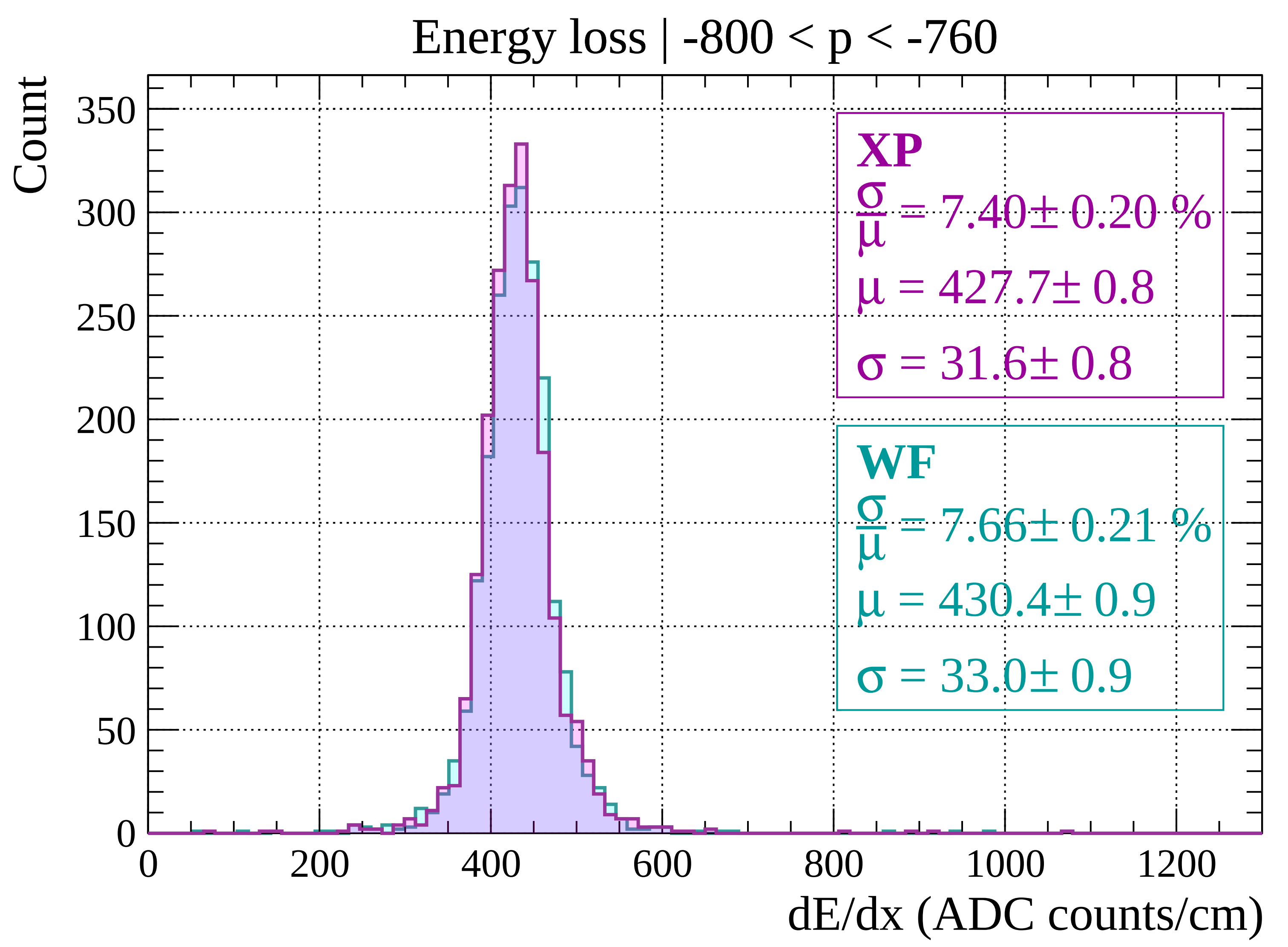


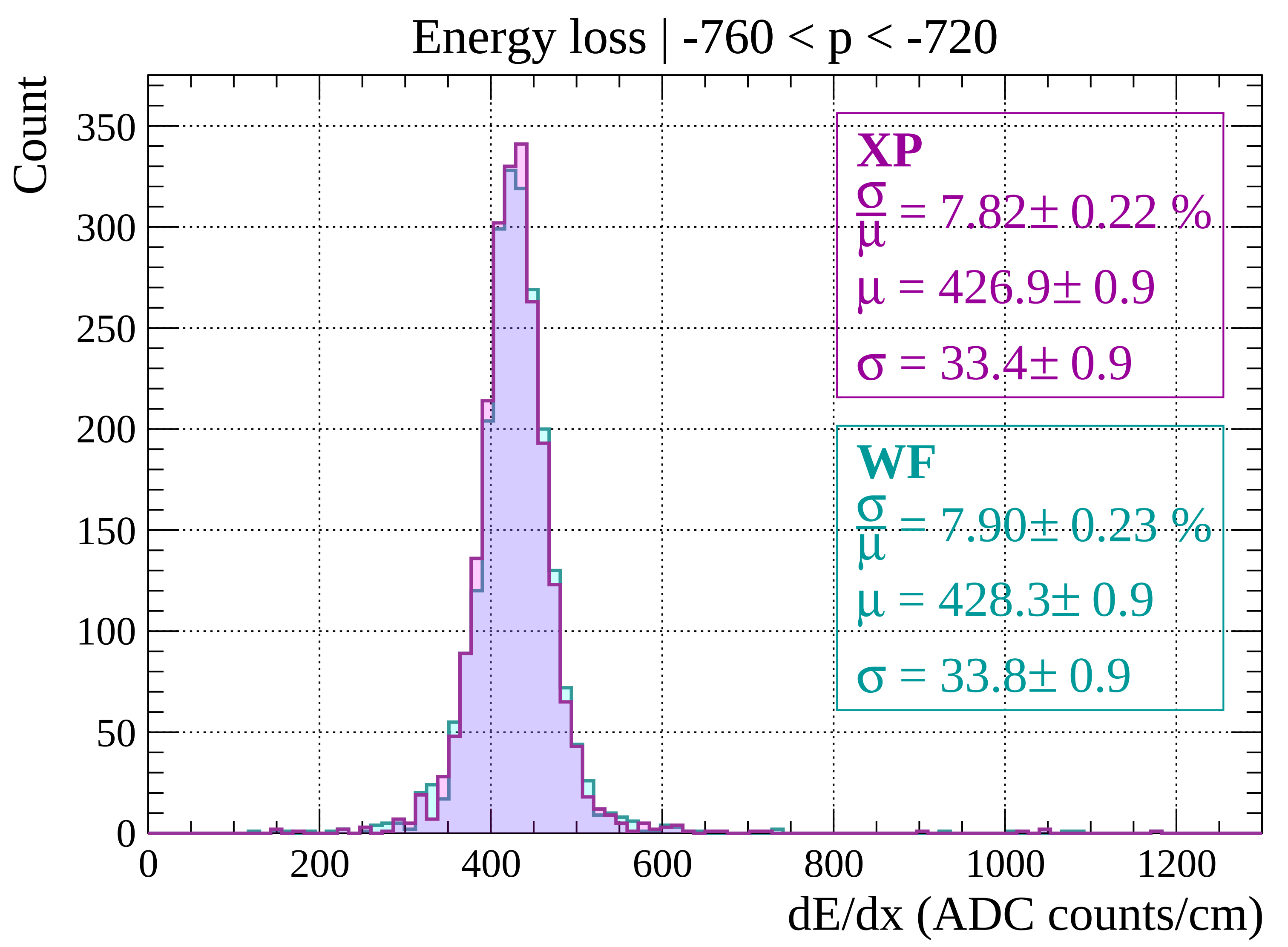
Energy loss | -920 < p < -880
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.22± 0.23 %
μ = 434.7± 0.9
σ = 31.4± 0.9
WF
σ/μ = 7.65± 0.25 %
μ = 438.0± 1.0
σ = 33.5± 1.0

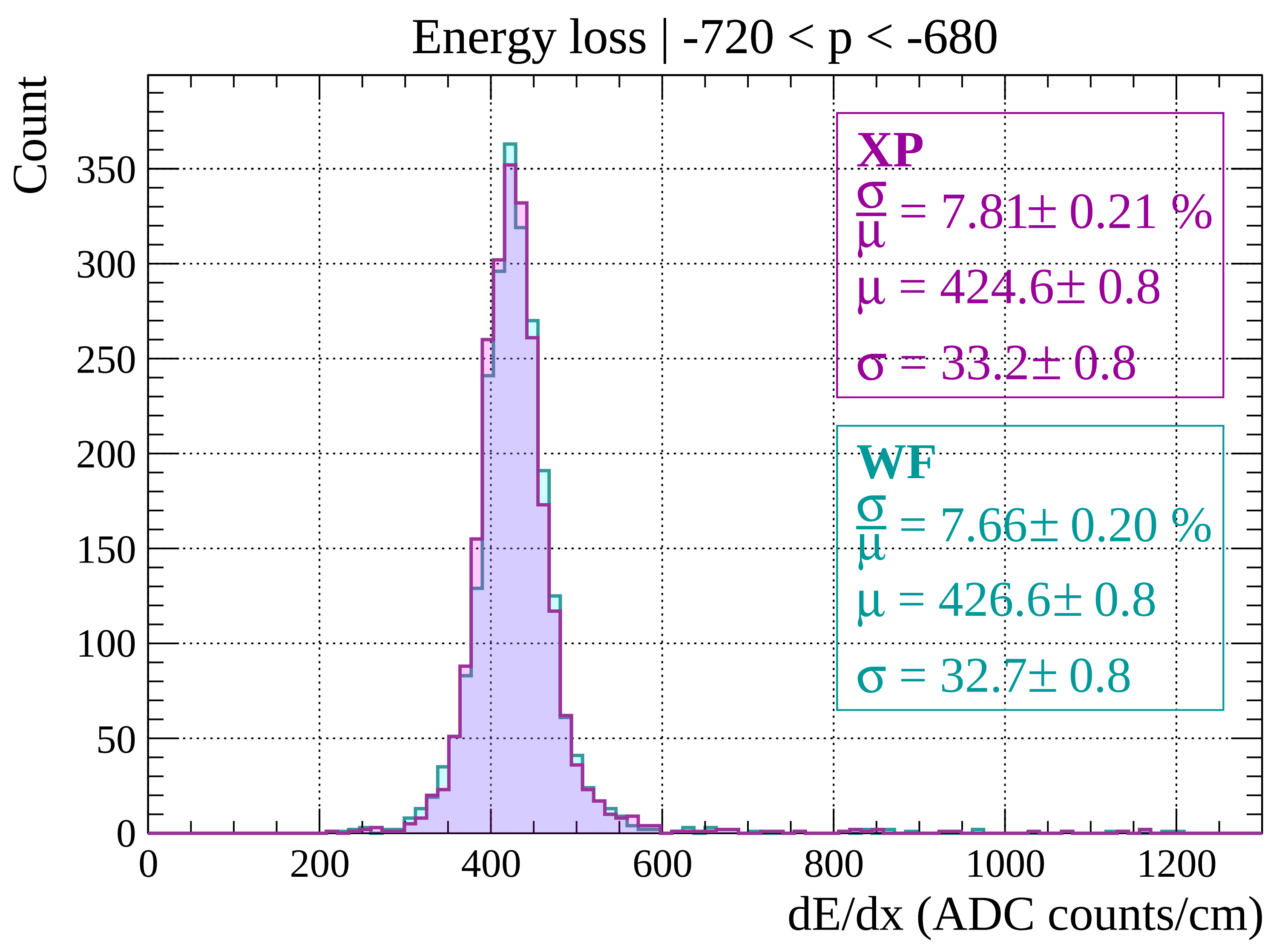


Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.45± 0.21 %
μ = 433.2± 0.9
σ = 32.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.78± 0.21 %
μ = 436.3± 0.9
σ = 33.9± 0.8

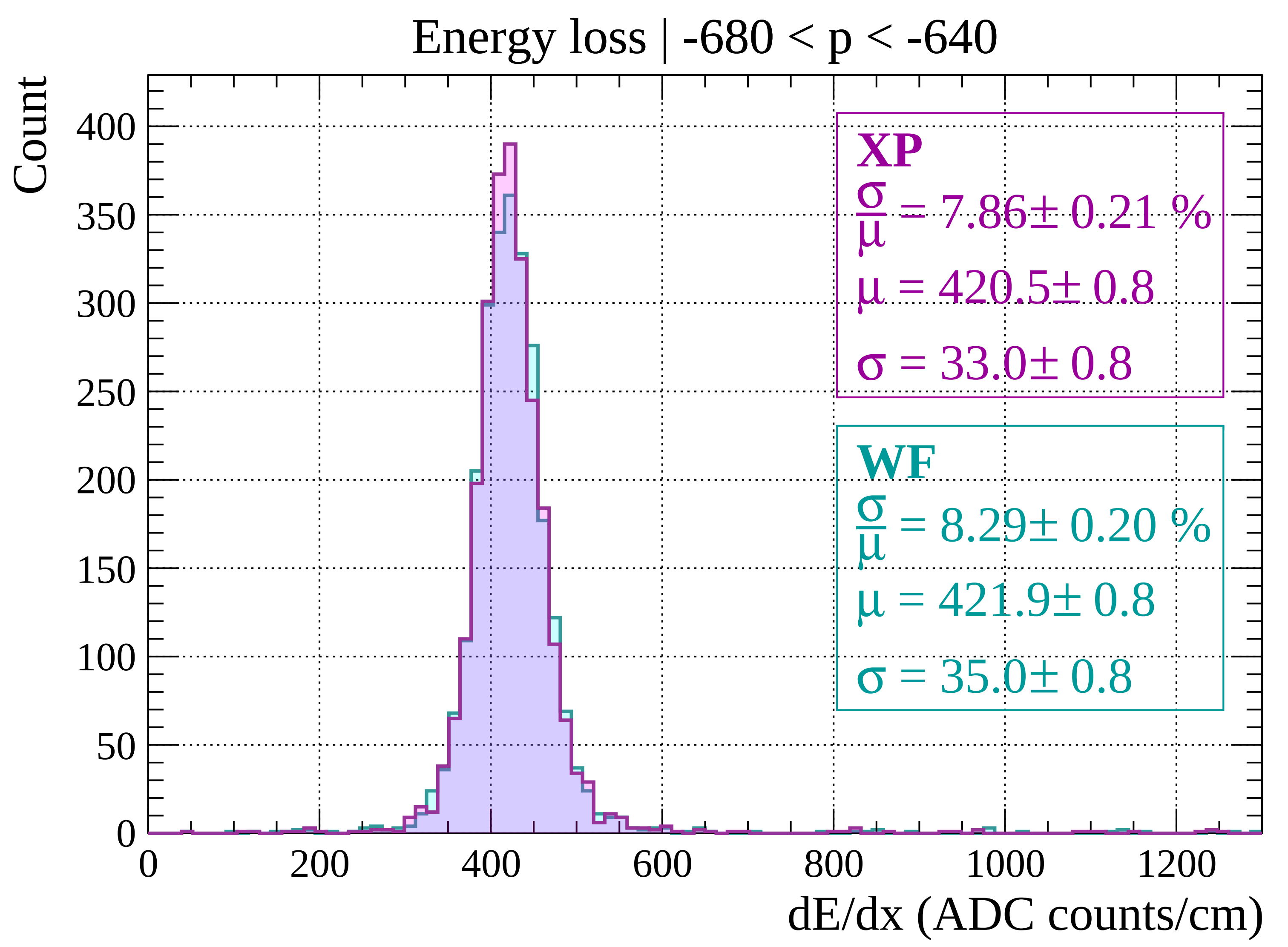

Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 7.89± 0.24 %
μ = 430.0± 0.9
σ = 33.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.12± 0.23 %
μ = 431.5± 0.9
σ = 35.0± 0.9


Energy loss | -800 < p < -760
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.40± 0.20 %
μ = 427.7± 0.8
σ = 31.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.66± 0.21 %
μ = 430.4± 0.9
σ = 33.0± 0.9

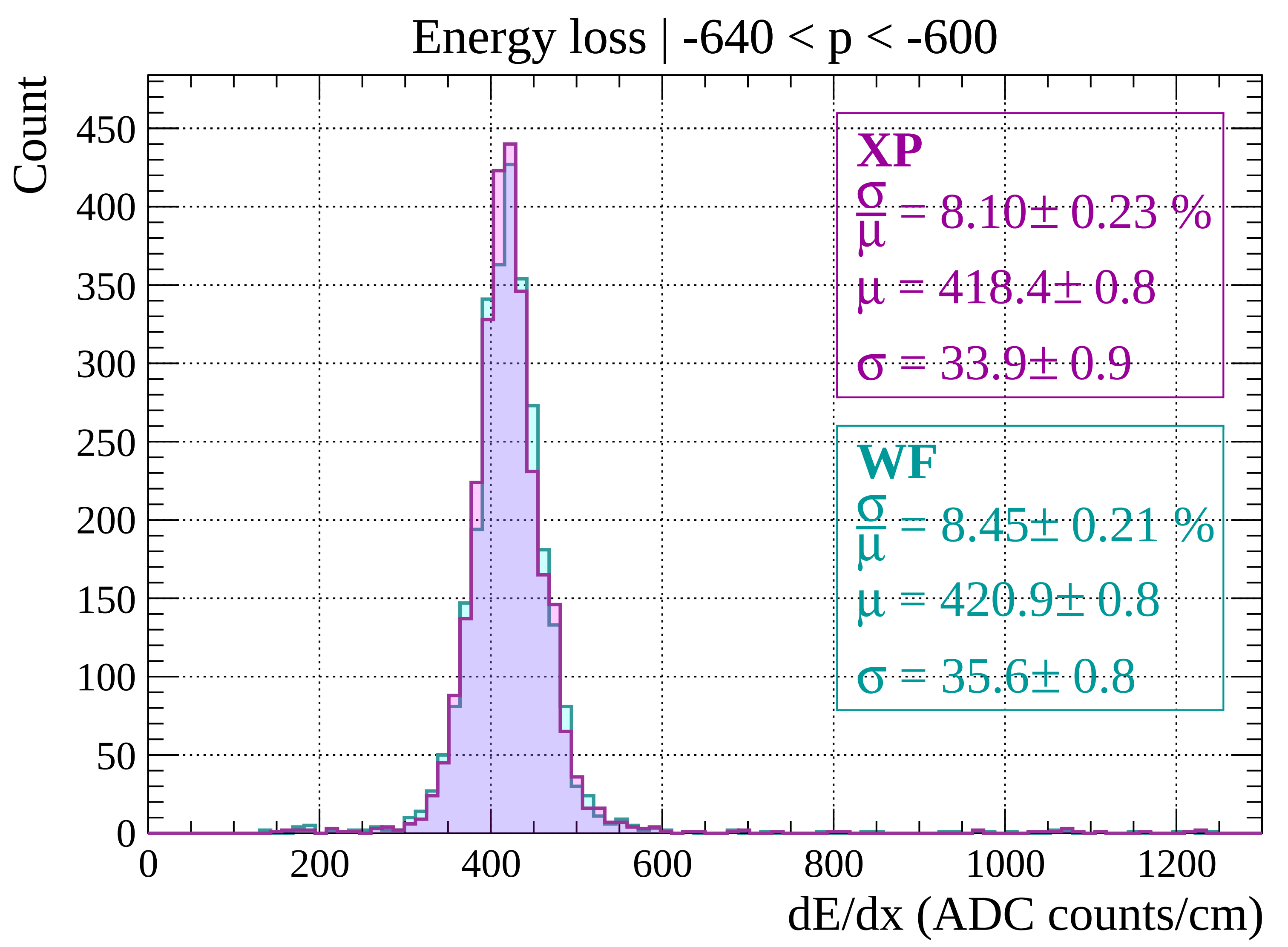

Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.82± 0.22 %
μ = 426.9± 0.9
σ = 33.4± 0.9
WF
σ/μ = 7.90± 0.23 %
μ = 428.3± 0.9
σ = 33.8± 0.9

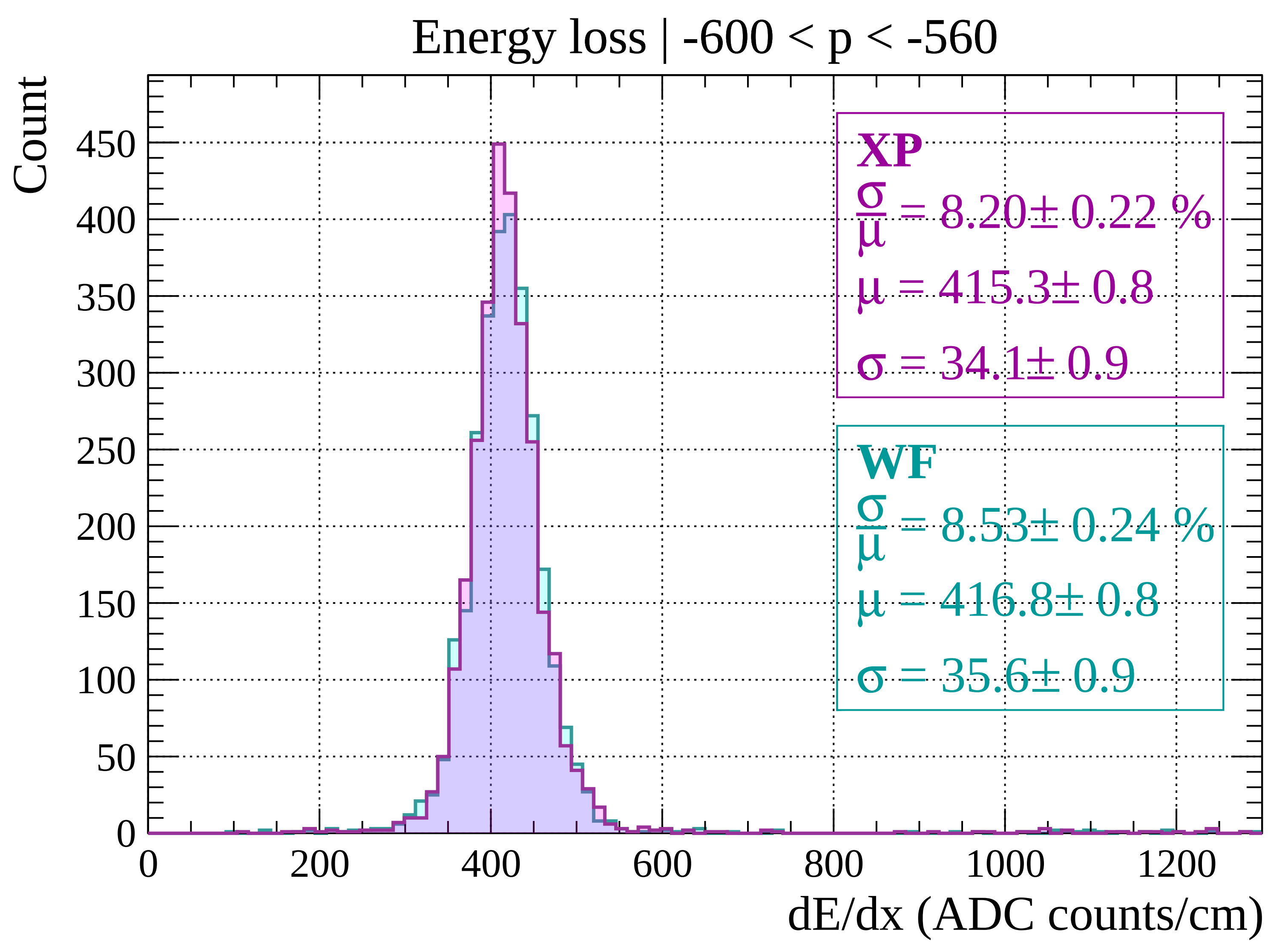


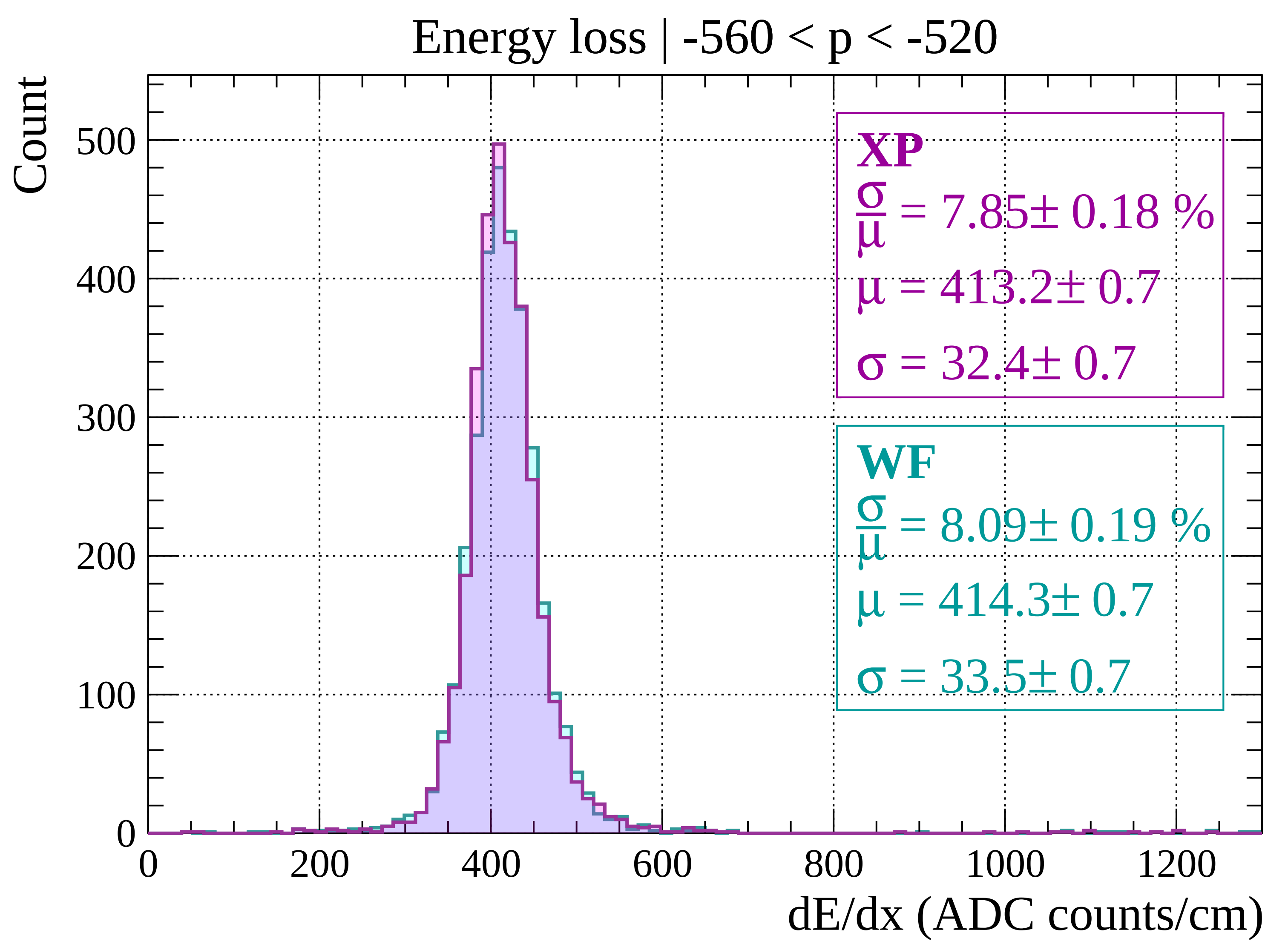
Energy loss | -720 < p < -680
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.81± 0.21 %
μ = 424.6± 0.8
σ = 33.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.66± 0.20 %
μ = 426.6± 0.8
σ = 32.7± 0.8

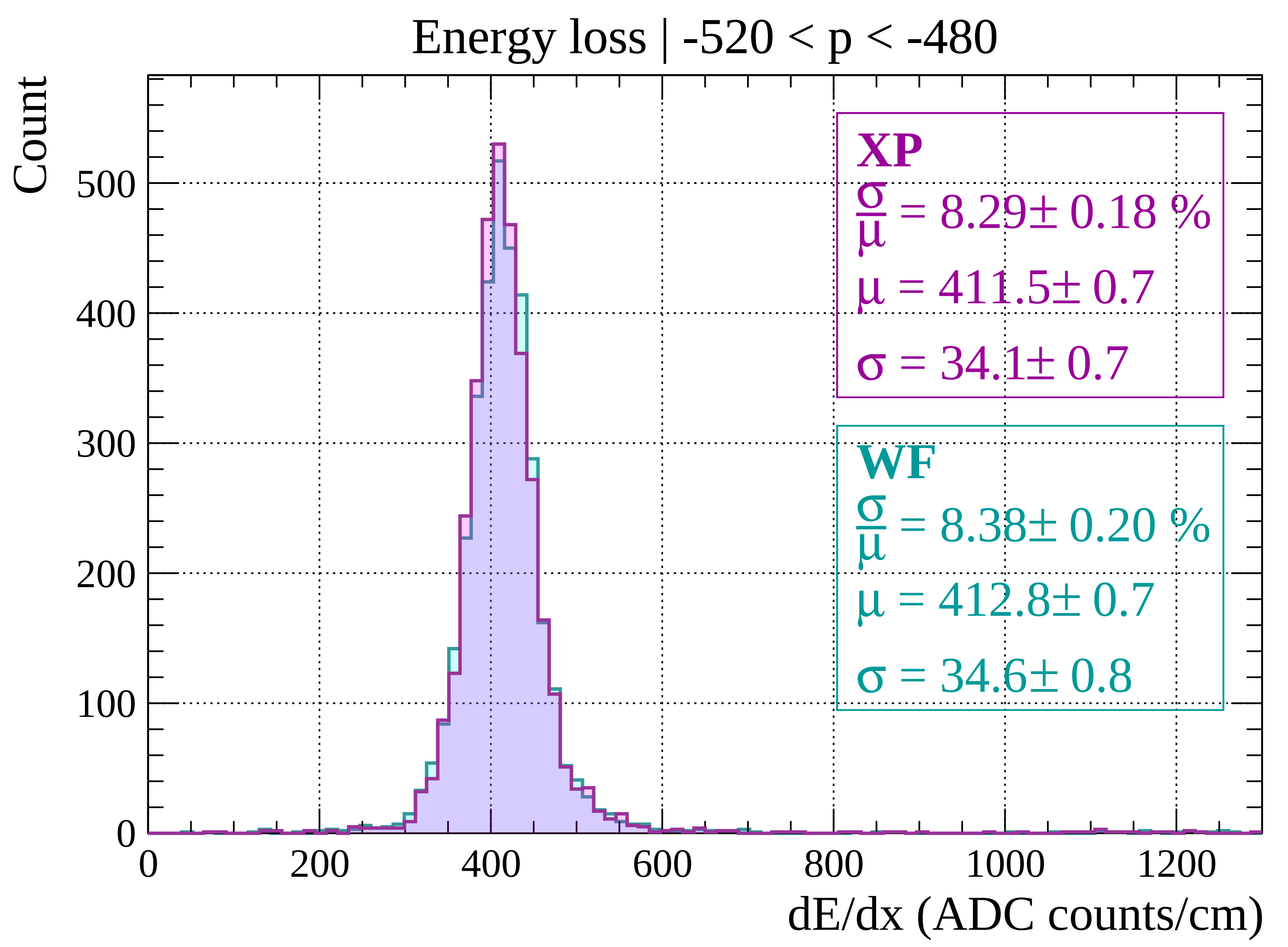


Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.86± 0.21 %
μ = 420.5± 0.8
σ = 33.0± 0.8
WF
σ/μ = 8.29± 0.20 %
μ = 421.9± 0.8
σ = 35.0± 0.8

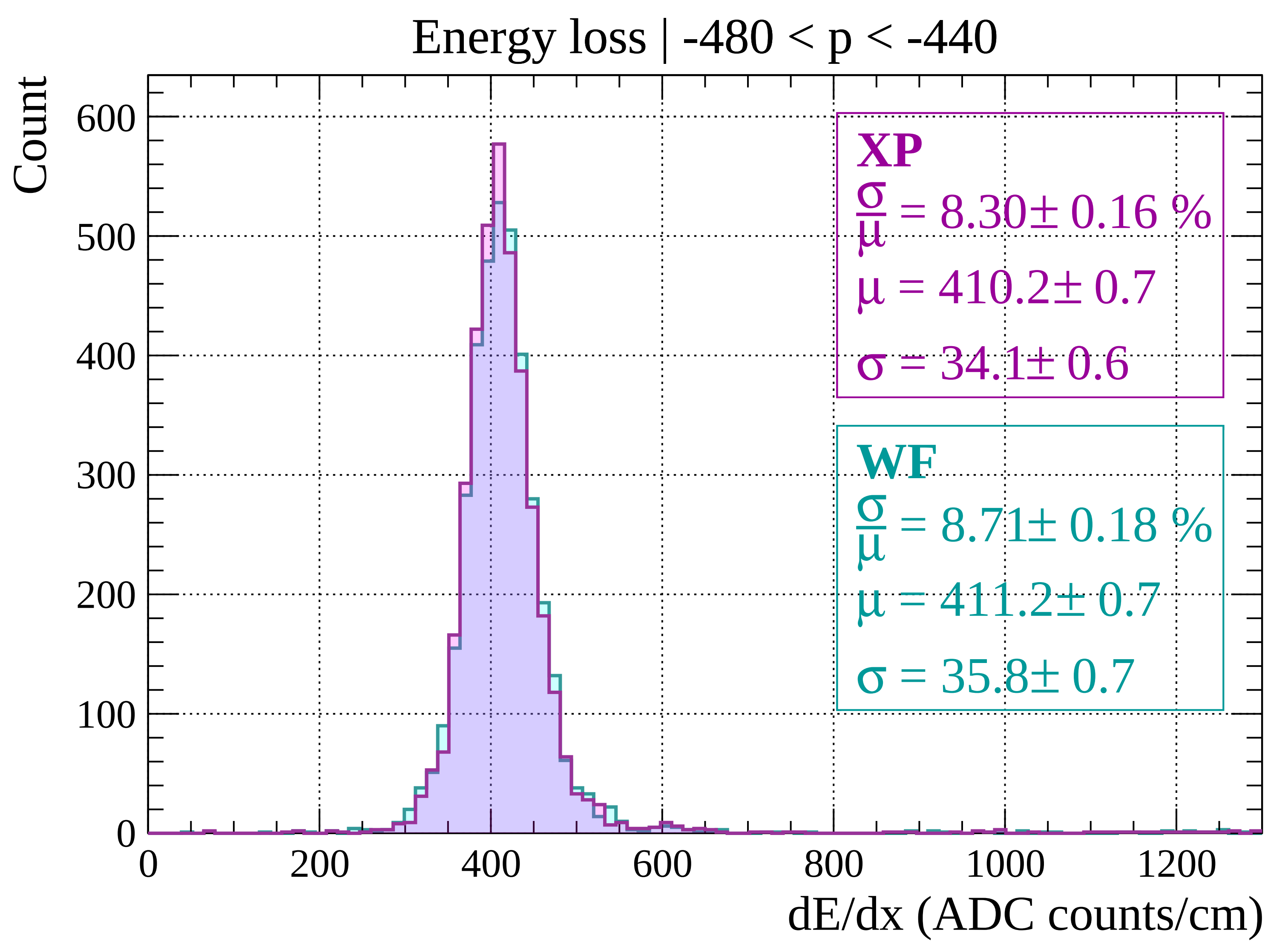

Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.10 \pm 0.23$ %
$\mu = 418.4 \pm 0.8$
$\sigma = 33.9 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 8.45 \pm 0.21$ %
$\mu = 420.9 \pm 0.8$
$\sigma = 35.6 \pm 0.8$

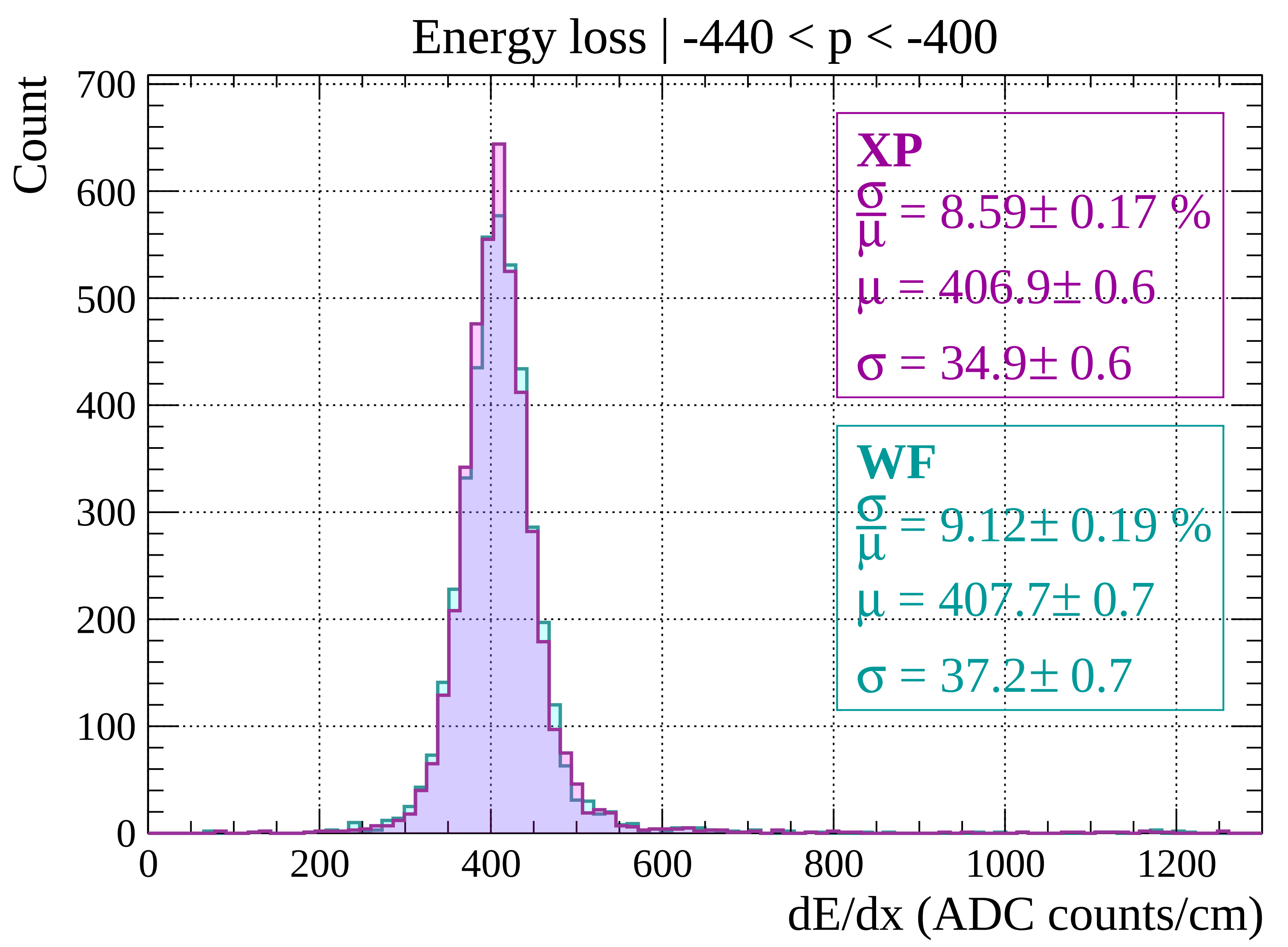

Energy loss | -600 < p < -560
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.20± 0.22 %
μ = 415.3± 0.8
σ = 34.1± 0.9
WF
σ/μ = 8.53± 0.24 %
μ = 416.8± 0.8
σ = 35.6± 0.9


Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 7.85± 0.18 %
μ = 413.2± 0.7
σ = 32.4± 0.7
WF
σ/μ = 8.09± 0.19 %
μ = 414.3± 0.7
σ = 33.5± 0.7

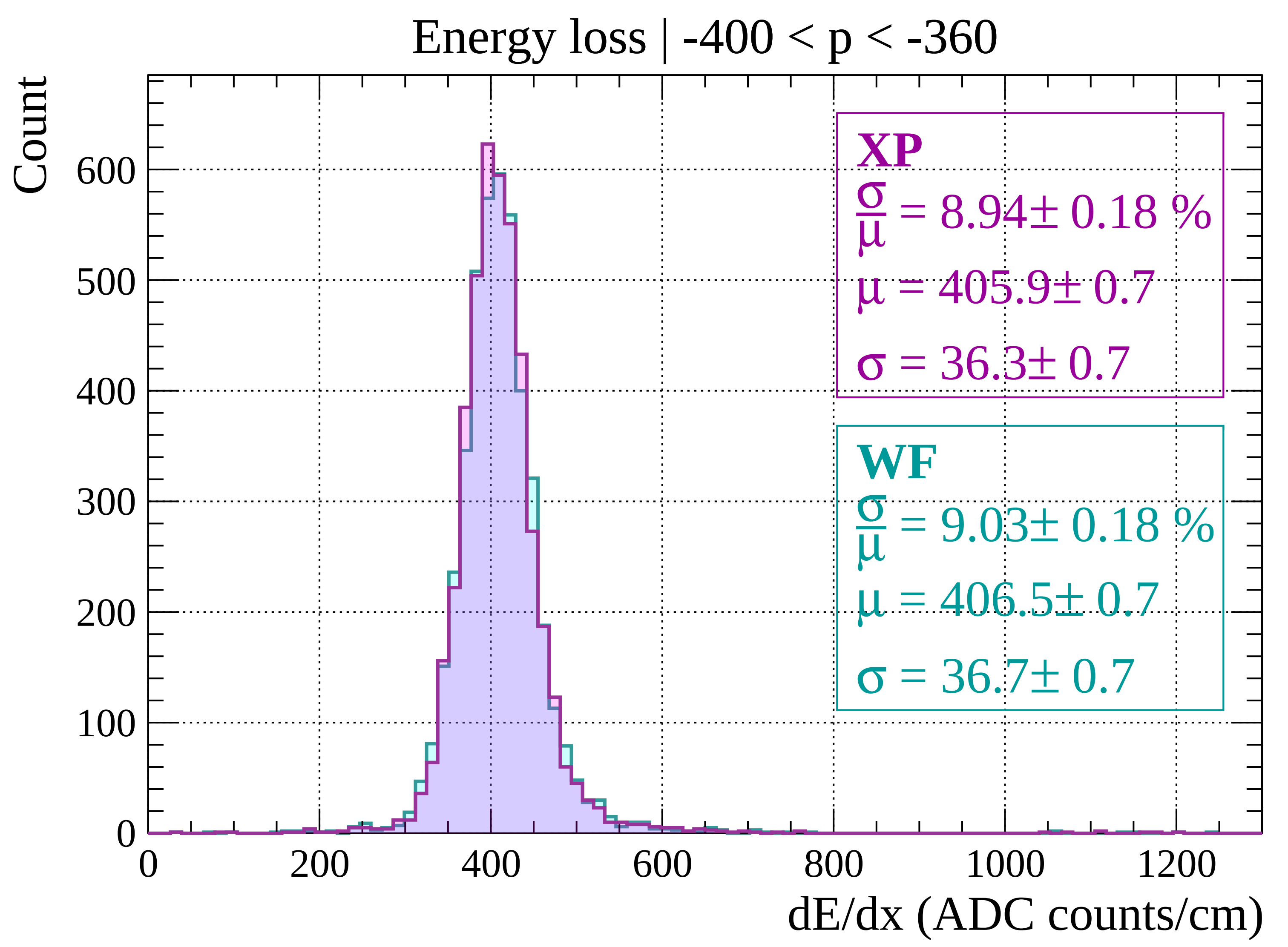


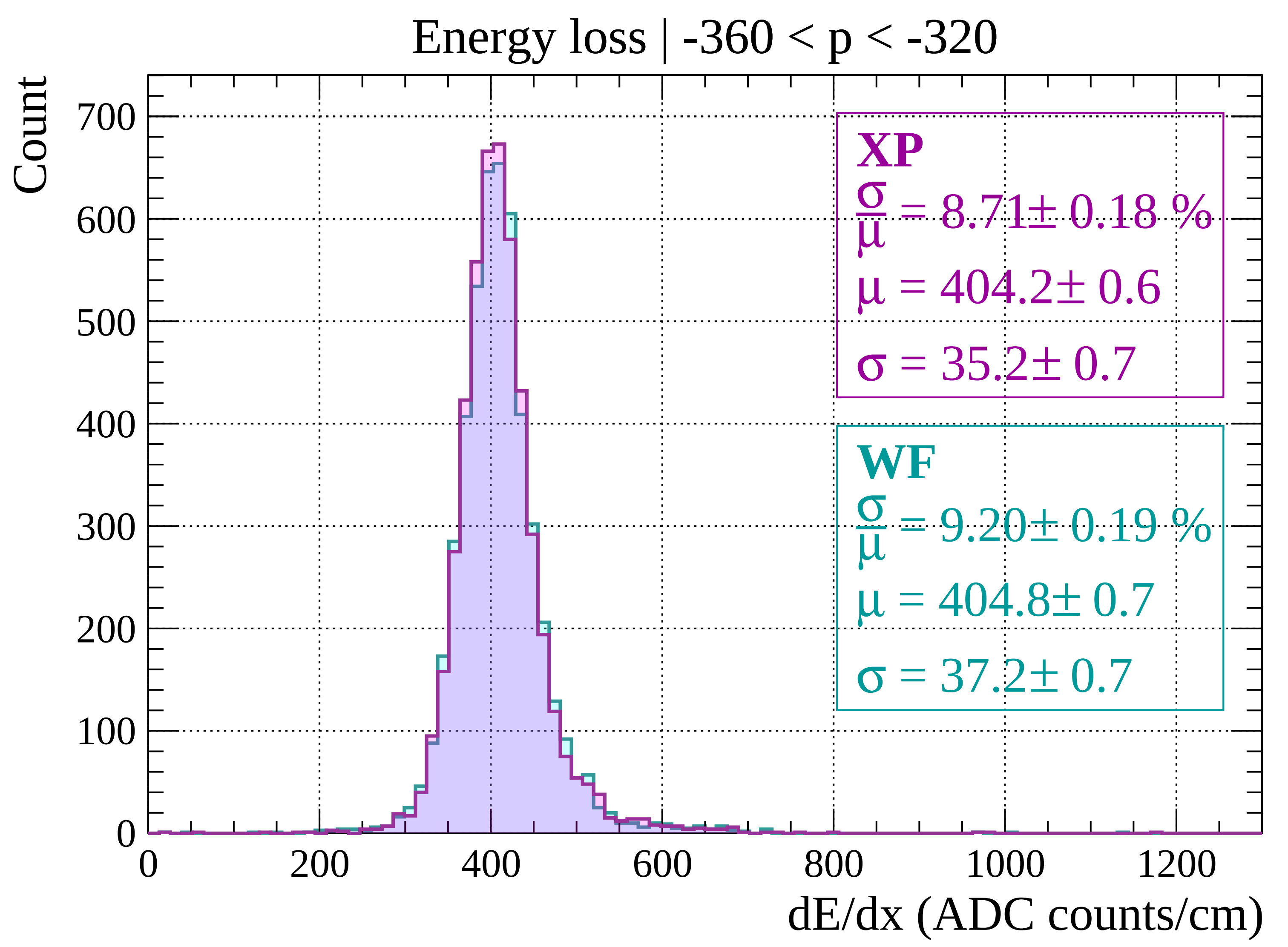
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.29± 0.18 %
μ = 411.5± 0.7
σ = 34.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.38± 0.20 %
μ = 412.8± 0.7
σ = 34.6± 0.8

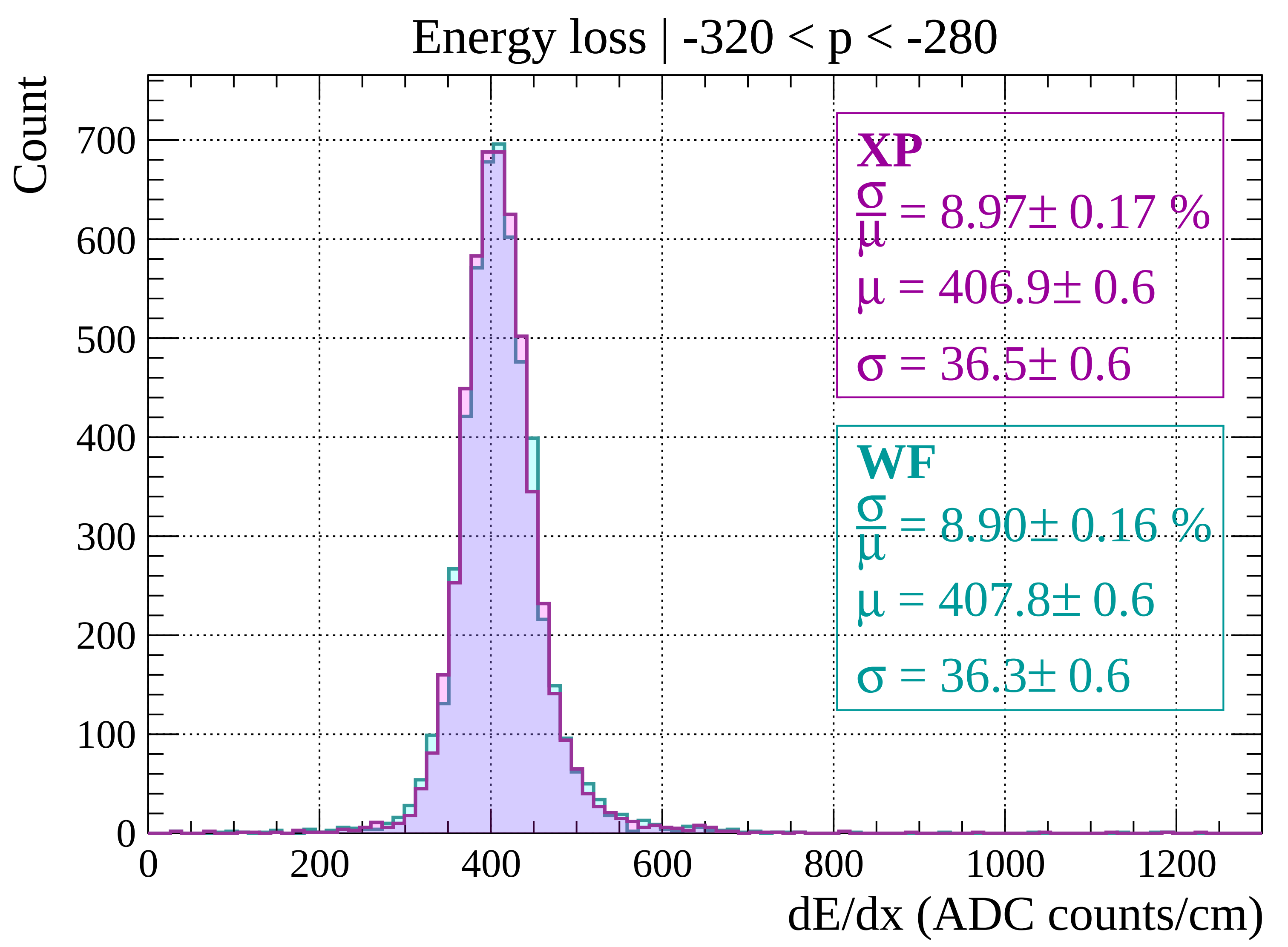

Energy loss | -480 < p < -440
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.30± 0.16 %
μ = 410.2± 0.7
σ = 34.1± 0.6
WF
σ/μ = 8.71± 0.18 %
μ = 411.2± 0.7
σ = 35.8± 0.7

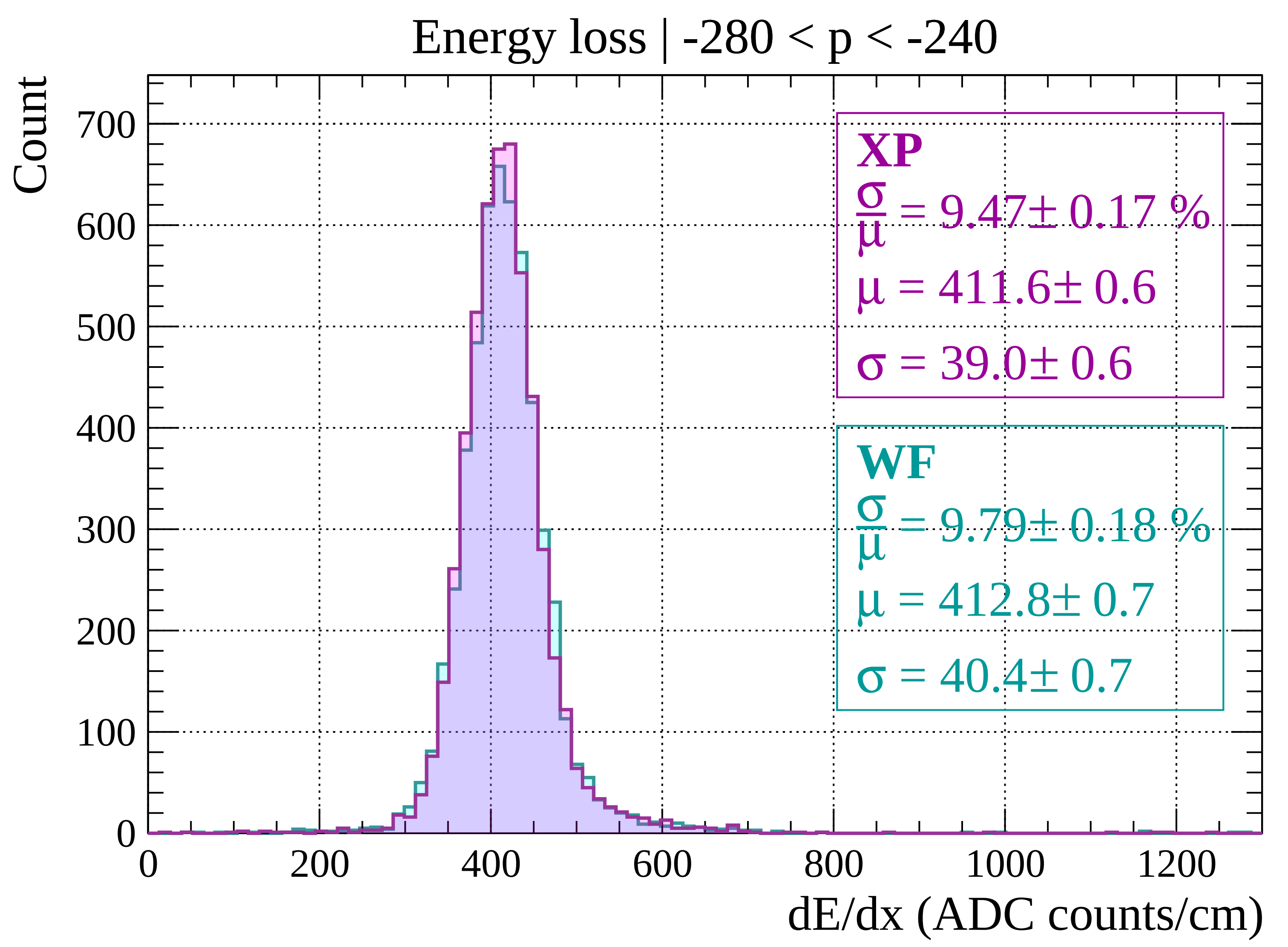


Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.59± 0.17 %
μ = 406.9± 0.6
σ = 34.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.12± 0.19 %
μ = 407.7± 0.7
σ = 37.2± 0.7

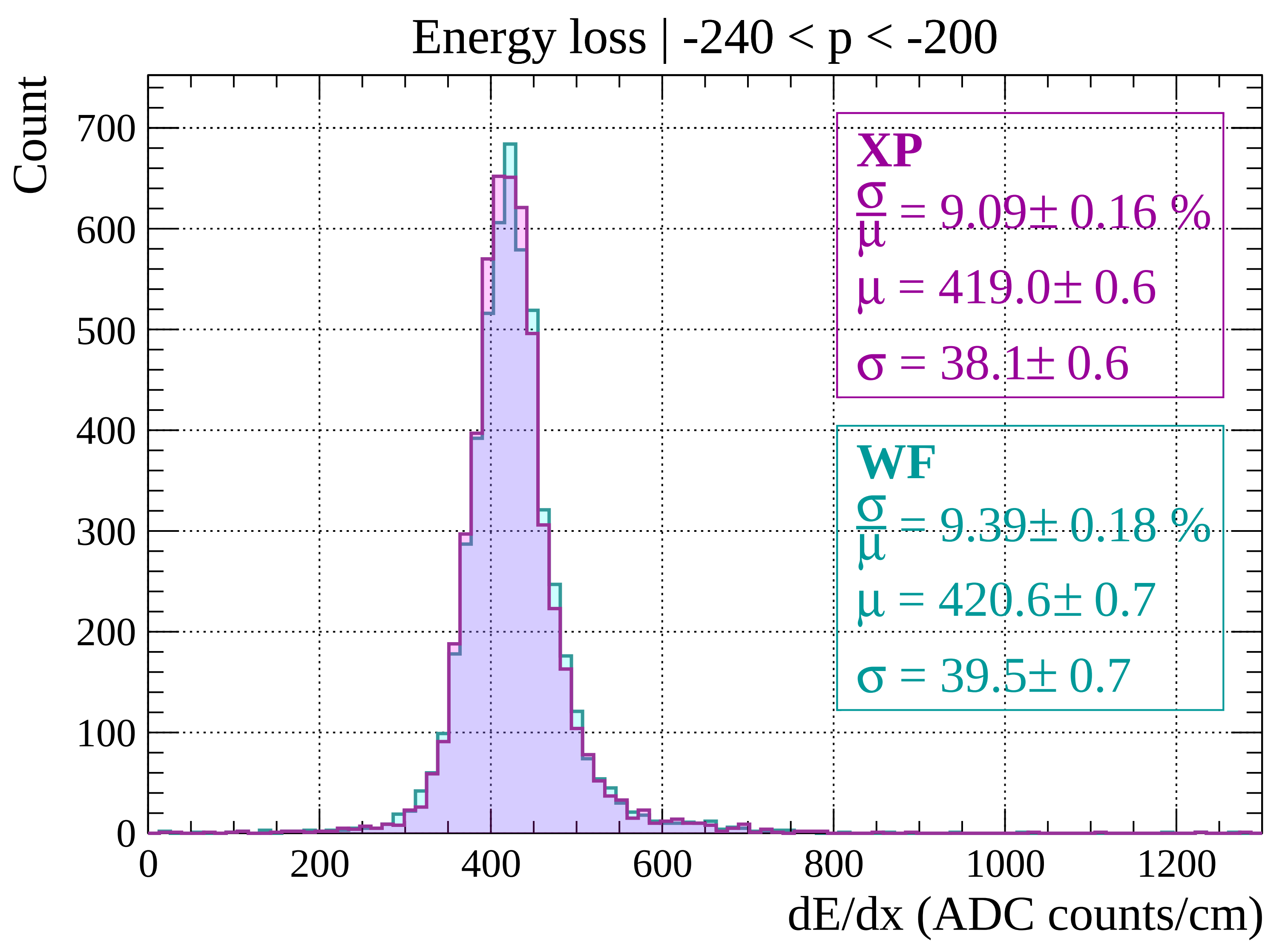

Energy loss | -400 < p < -360
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.94± 0.18 %
μ = 405.9± 0.7
σ = 36.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.03± 0.18 %
μ = 406.5± 0.7
σ = 36.7± 0.7


Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.71± 0.18 %
μ = 404.2± 0.6
σ = 35.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.20± 0.19 %
μ = 404.8± 0.7
σ = 37.2± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

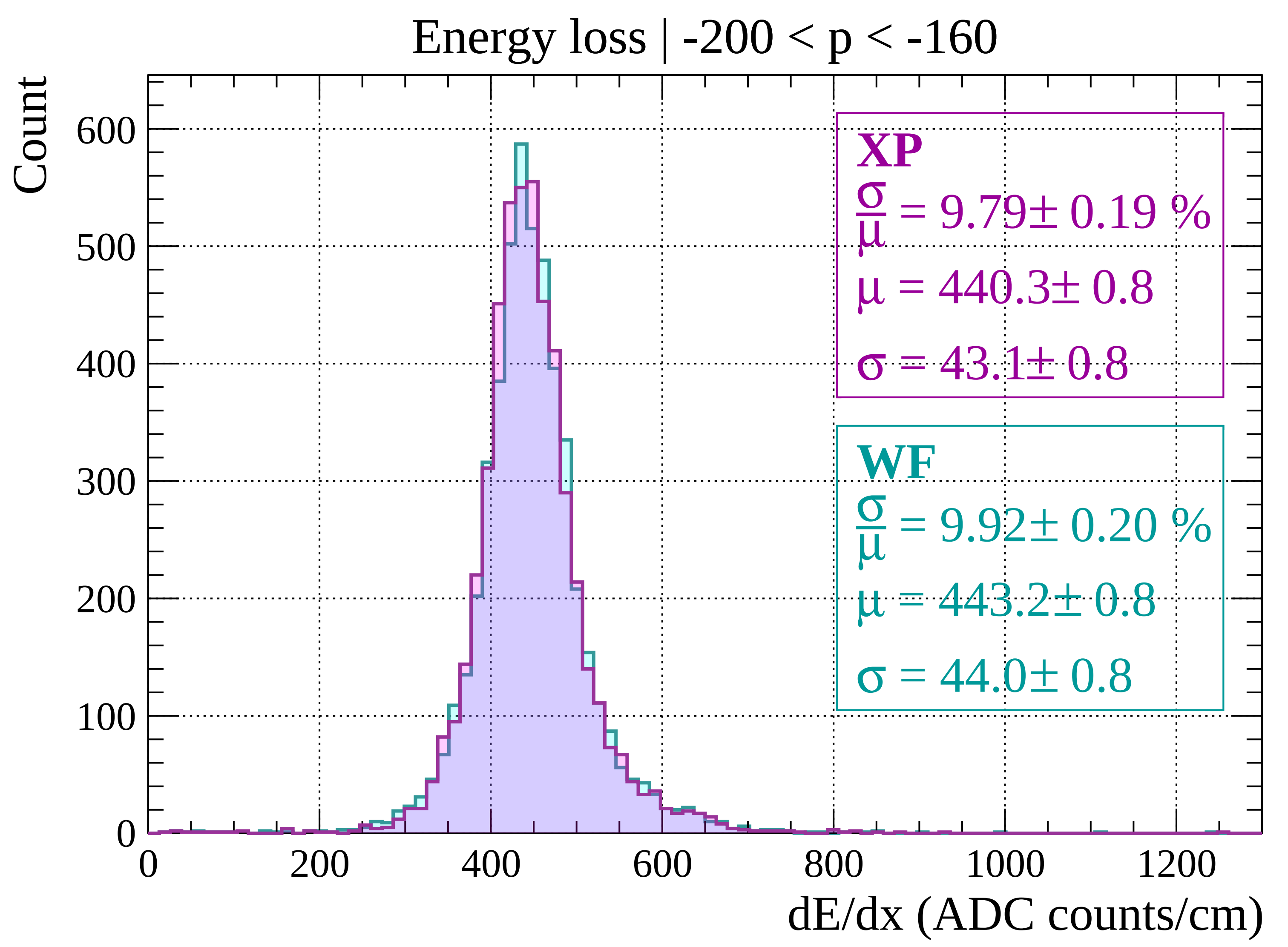

Energy loss | -320 < p < -280
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.97± 0.17 %
μ = 406.9± 0.6
σ = 36.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.90± 0.16 %
μ = 407.8± 0.6
σ = 36.3± 0.6



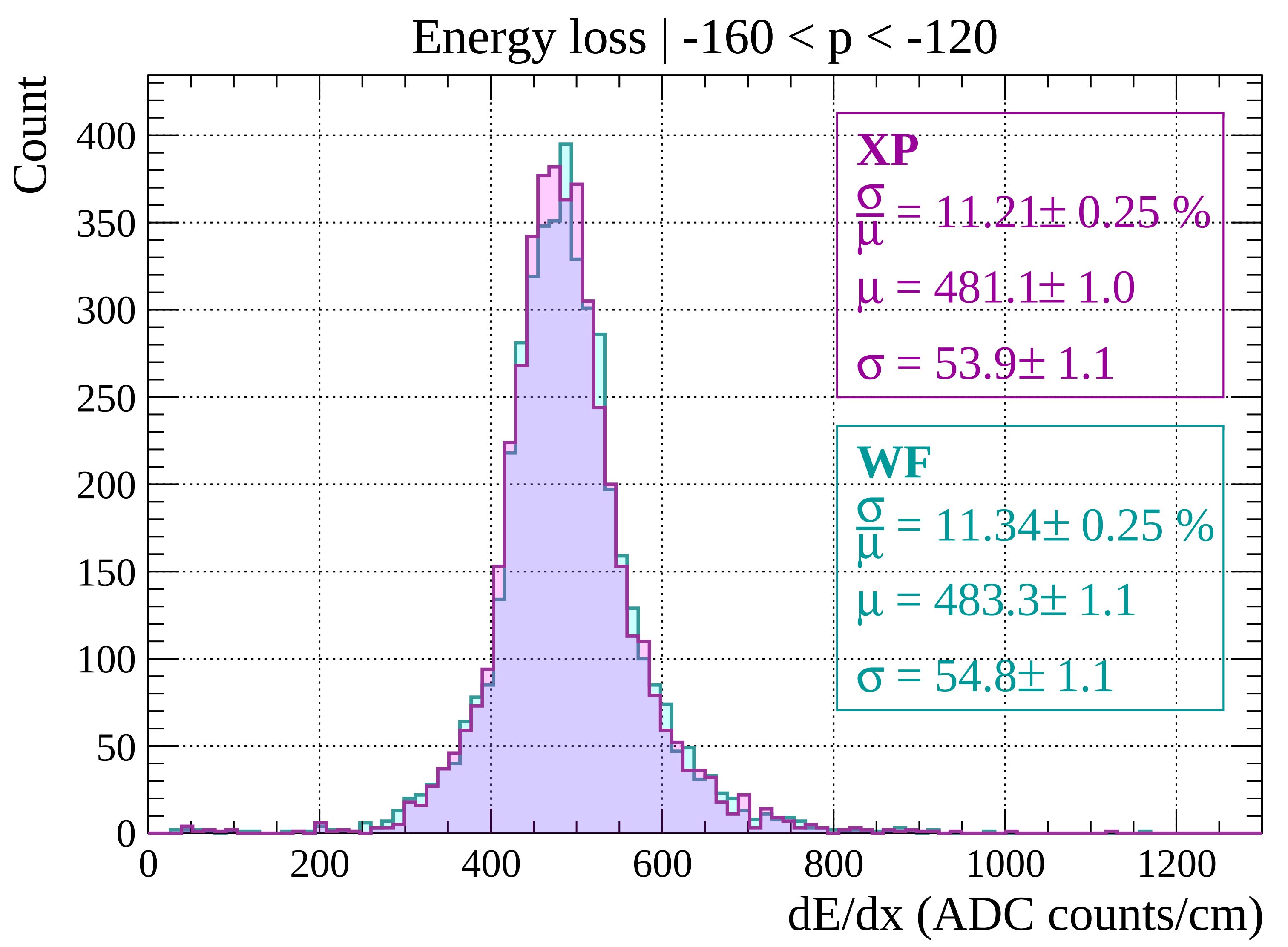
Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.47 \pm 0.17$ %
$\mu = 411.6 \pm 0.6$
$\sigma = 39.0 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.79 \pm 0.18$ %
$\mu = 412.8 \pm 0.7$
$\sigma = 40.4 \pm 0.7$

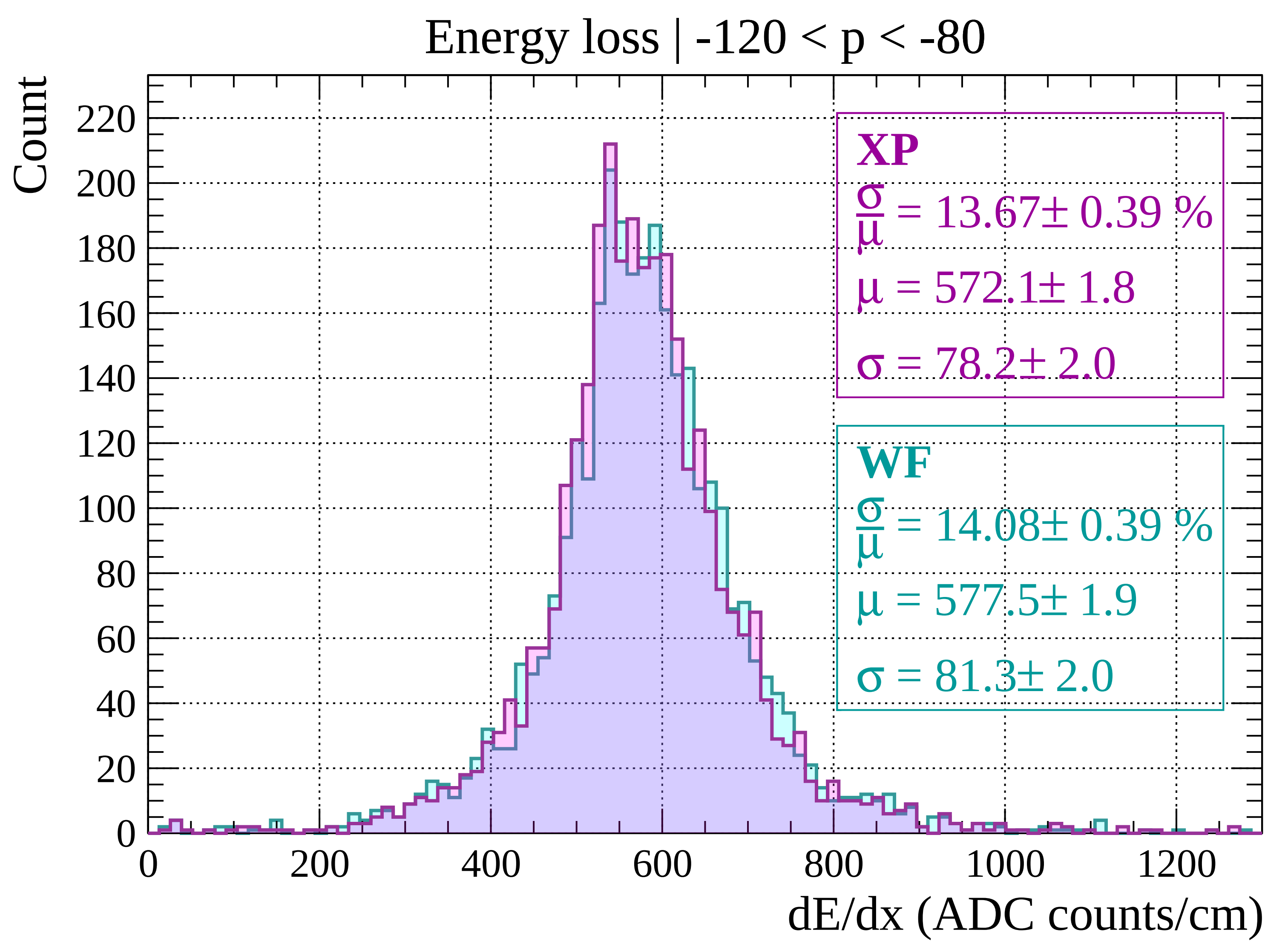


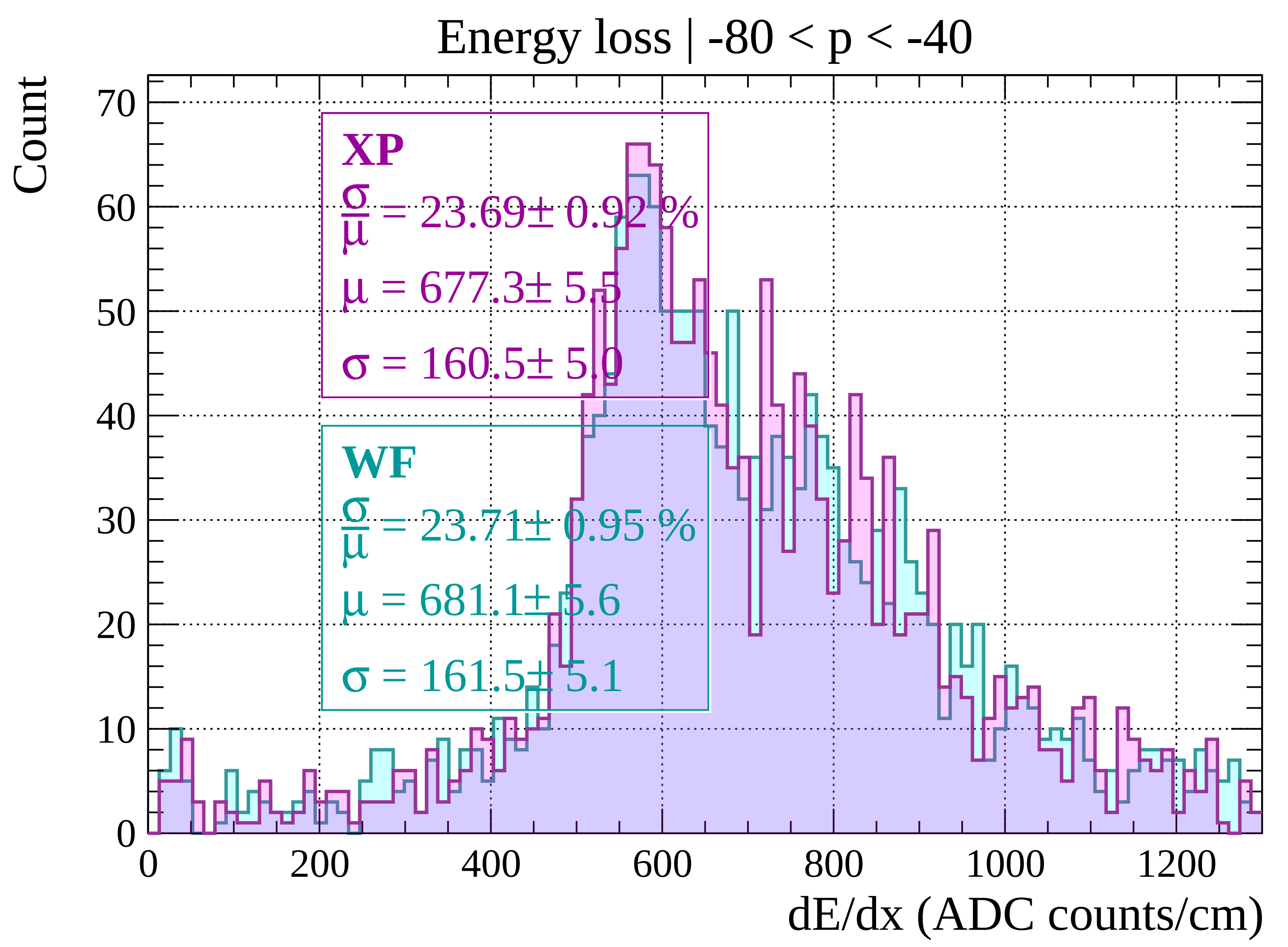
Energy loss | -240 < p < -200
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.09 ± 0.16 %
μ = 419.0 ± 0.6
σ = 38.1 ± 0.6
WF
σ/μ = 9.39 ± 0.18 %
μ = 420.6 ± 0.7
σ = 39.5 ± 0.7

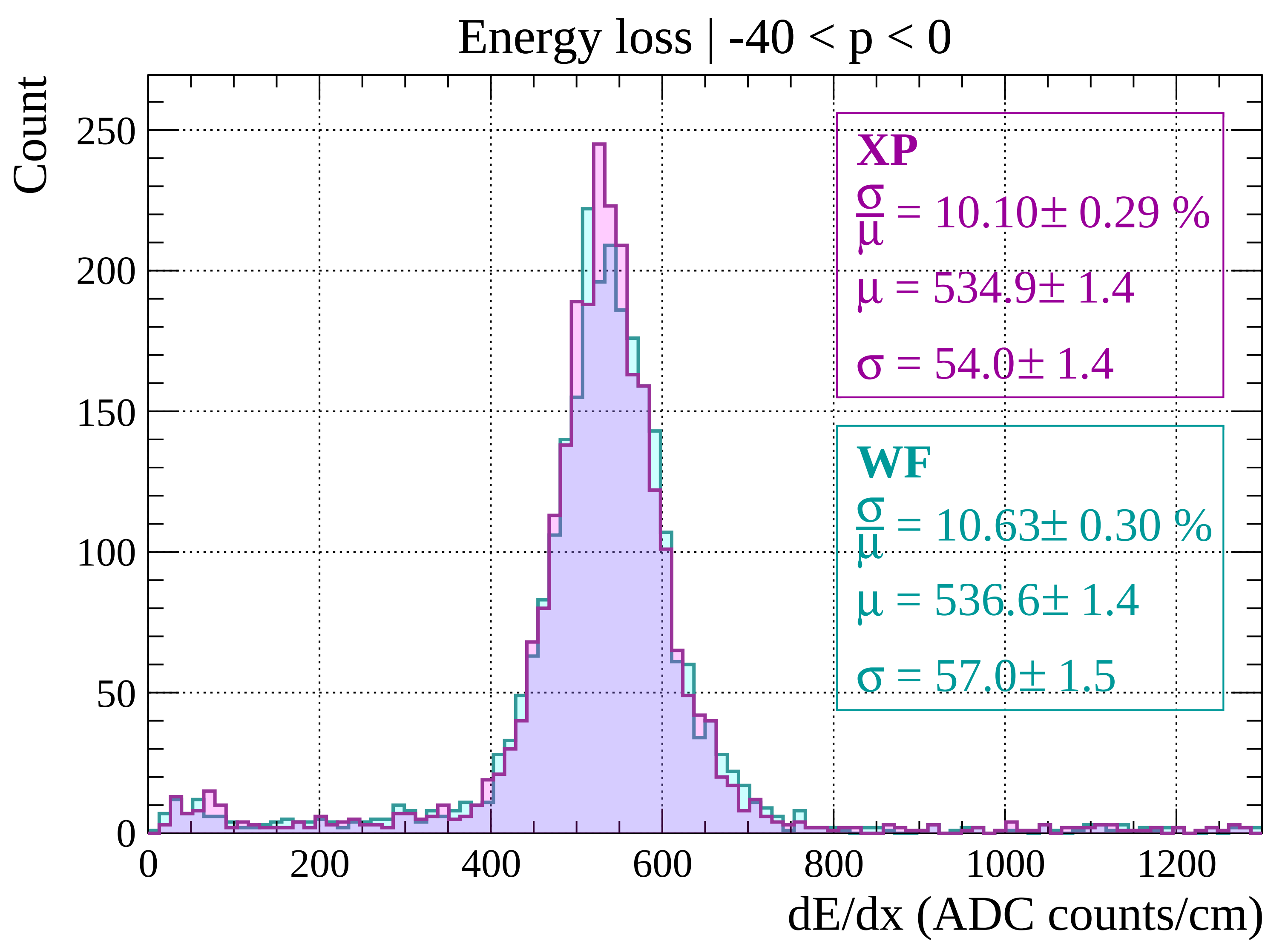


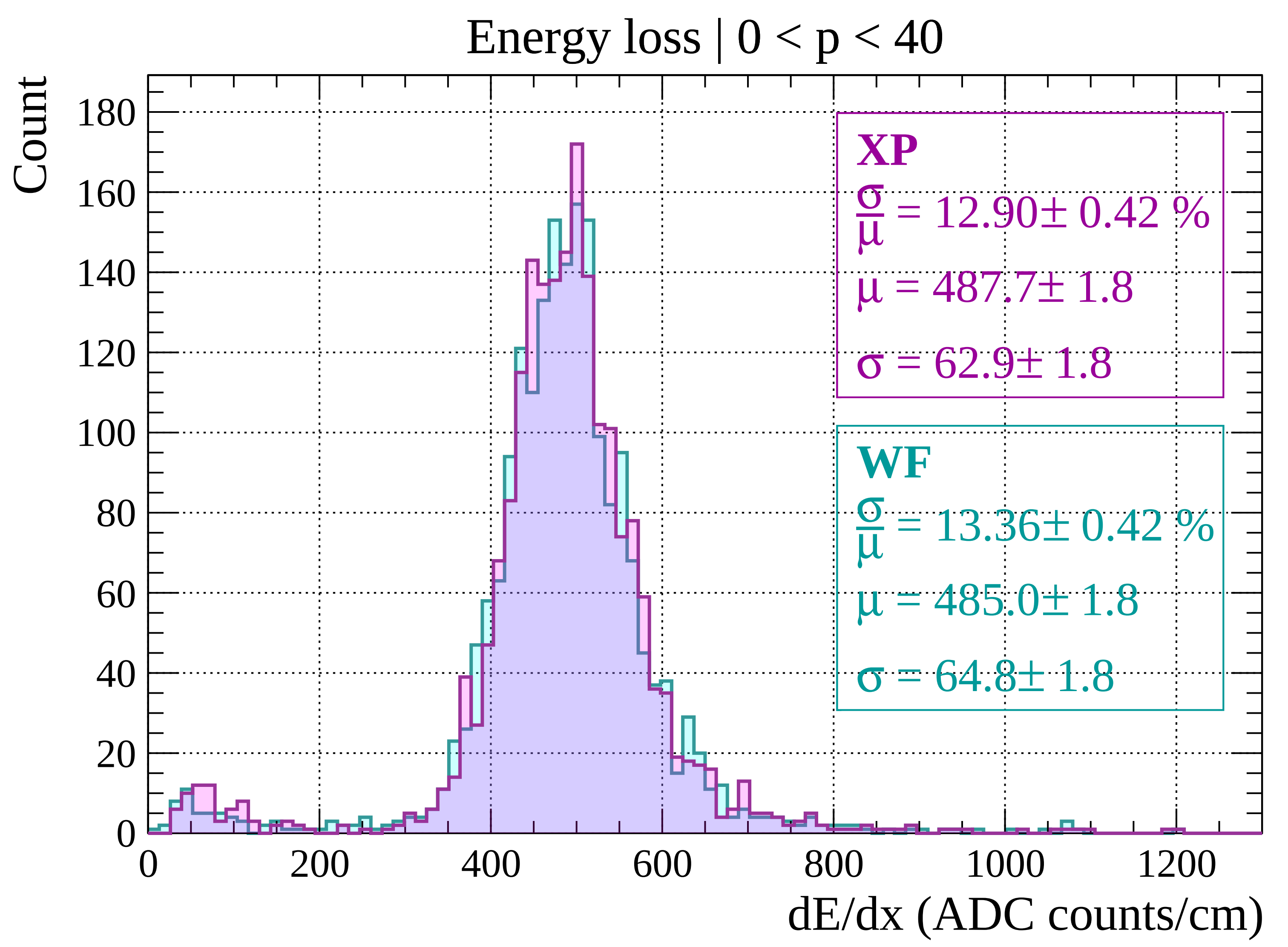
Energy loss | -200 < p < -160
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.79± 0.19 %
μ = 440.3± 0.8
σ = 43.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.92± 0.20 %
μ = 443.2± 0.8
σ = 44.0± 0.8


Energy loss | -160 < p < -120
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.21± 0.25 %
μ = 481.1± 1.0
σ = 53.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.34± 0.25 %
μ = 483.3± 1.1
σ = 54.8± 1.1

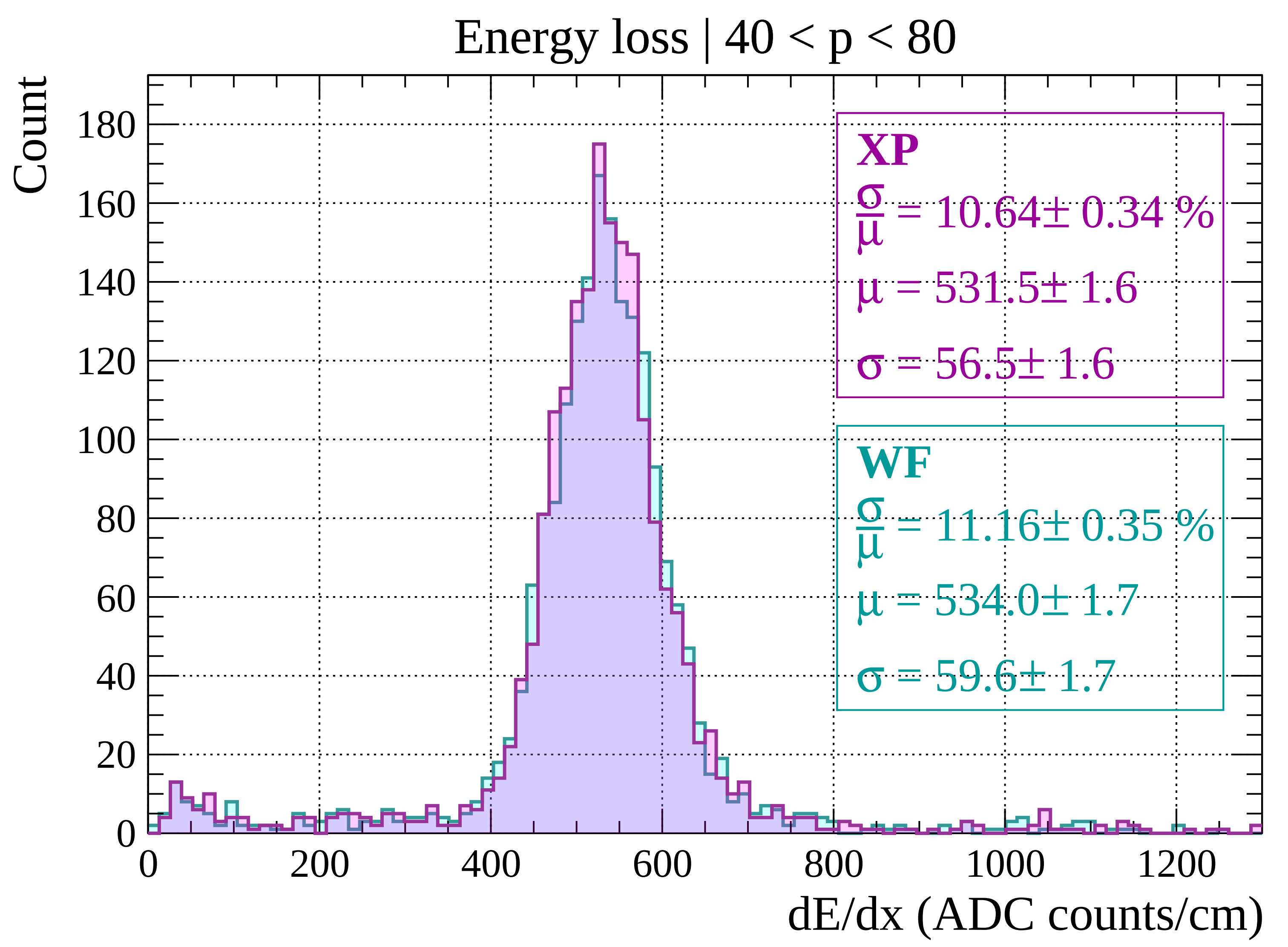

Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.67± 0.39 %
μ = 572.1± 1.8
σ = 78.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 14.08± 0.39 %
μ = 577.5± 1.9
σ = 81.3± 2.0

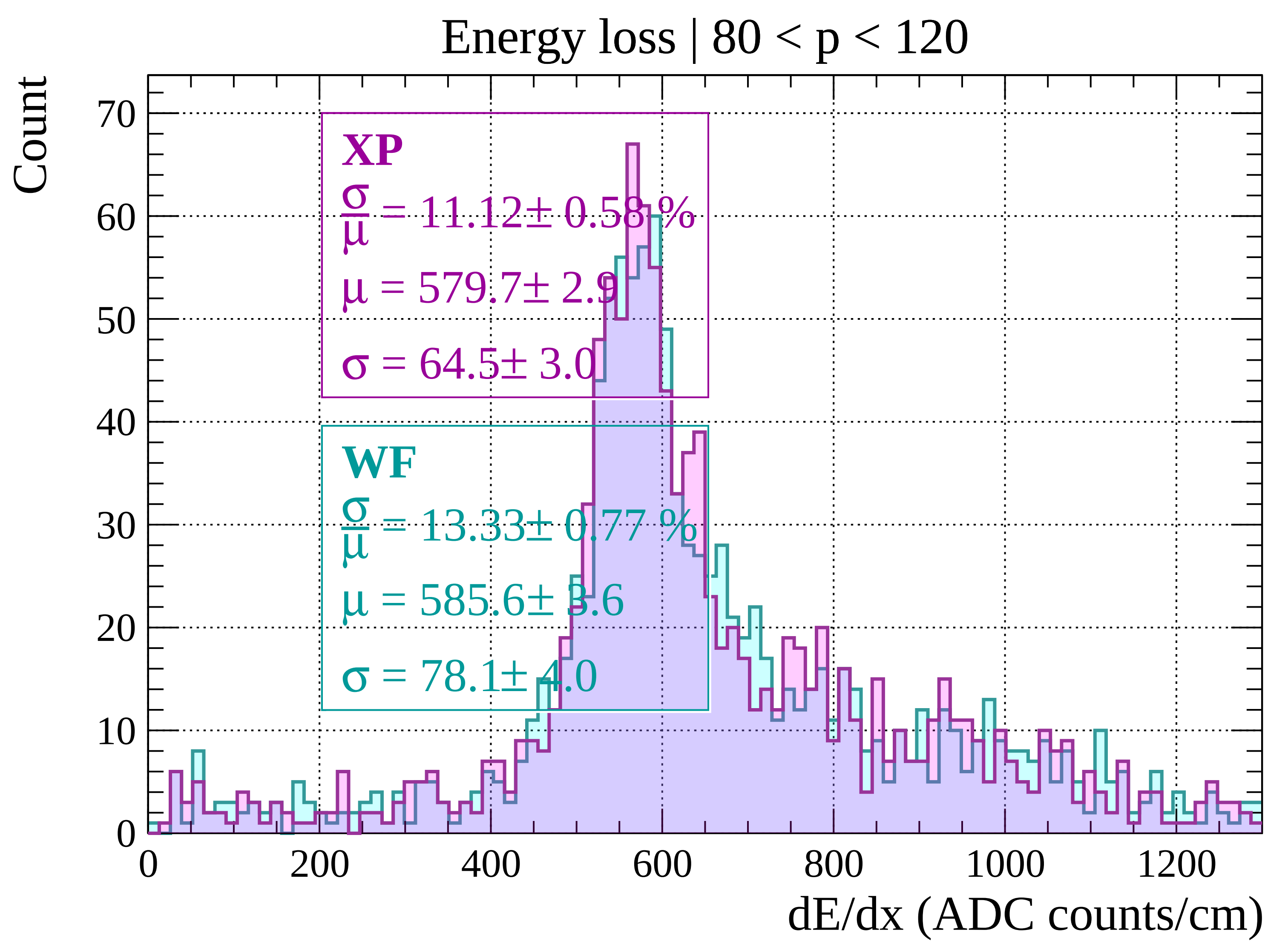

Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 23.69± 0.92 %
μ = 677.3± 5.5
σ = 160.5± 5.0
WF
$\frac{\sigma}{\mu}$ = 23.71± 0.95 %
μ = 681.1± 5.6
σ = 161.5± 5.1
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200


Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 10.10± 0.29 %
μ = 534.9± 1.4
σ = 54.0± 1.4
WF
σ/μ = 10.63± 0.30 %
μ = 536.6± 1.4
σ = 57.0± 1.5

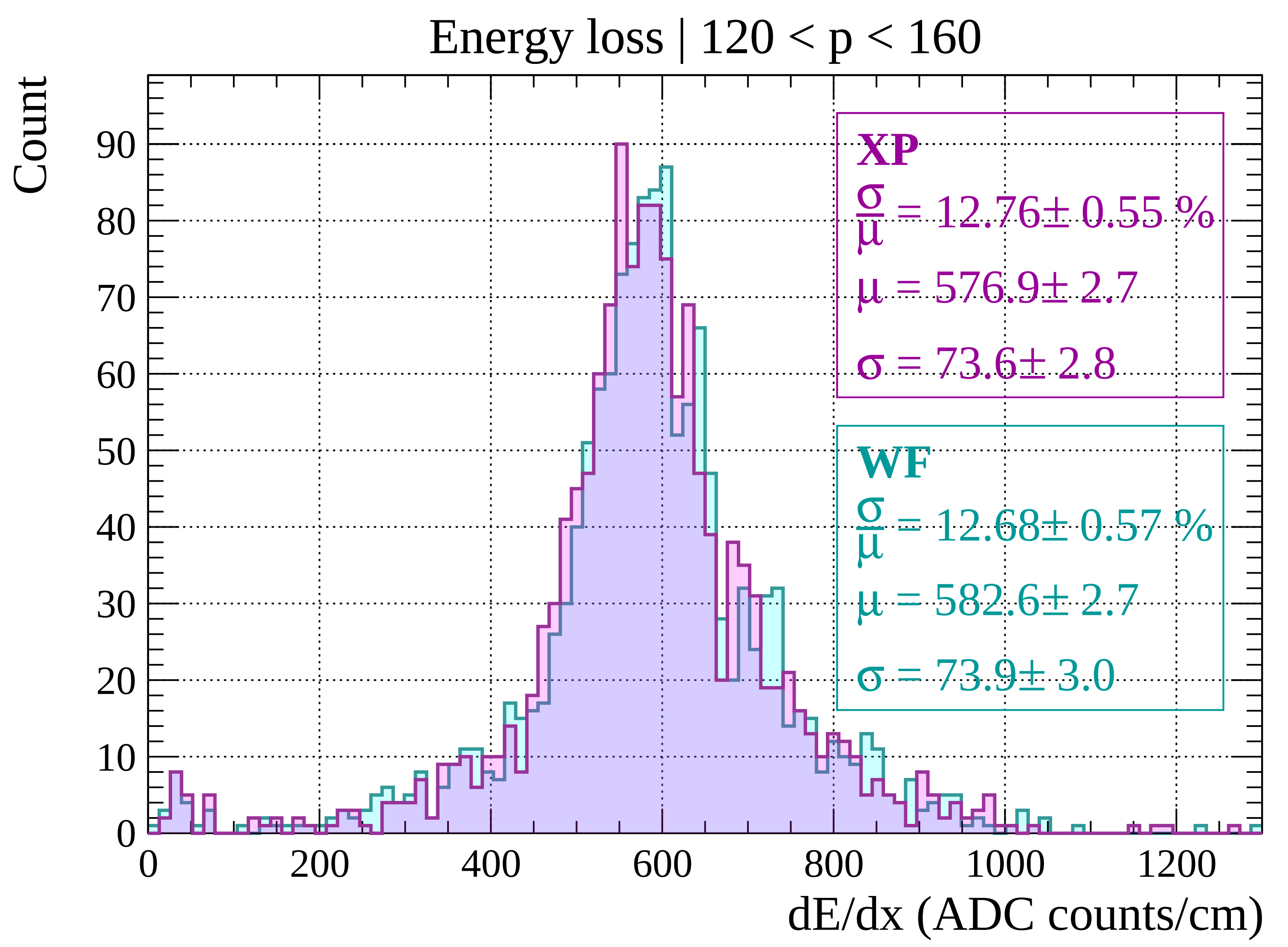


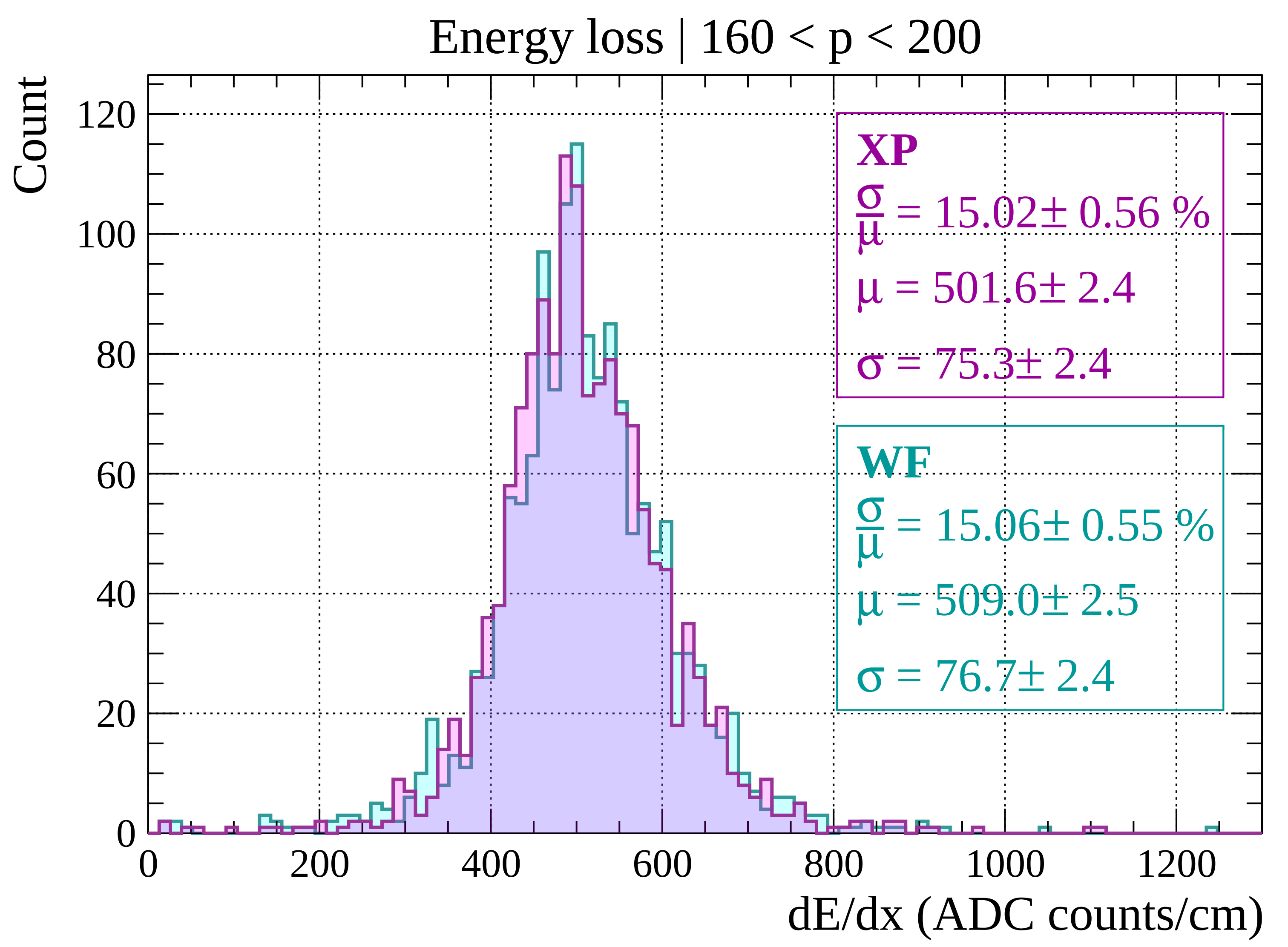
Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 12.90± 0.42 %
μ = 487.7± 1.8
σ = 62.9± 1.8
WF
σ/μ = 13.36± 0.42 %
μ = 485.0± 1.8
σ = 64.8± 1.8


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.64 ± 0.34 %
μ = 531.5 ± 1.6
σ = 56.5 ± 1.6
WF
$\frac{\sigma}{\mu}$ = 11.16 ± 0.35 %
μ = 534.0 ± 1.7
σ = 59.6 ± 1.7

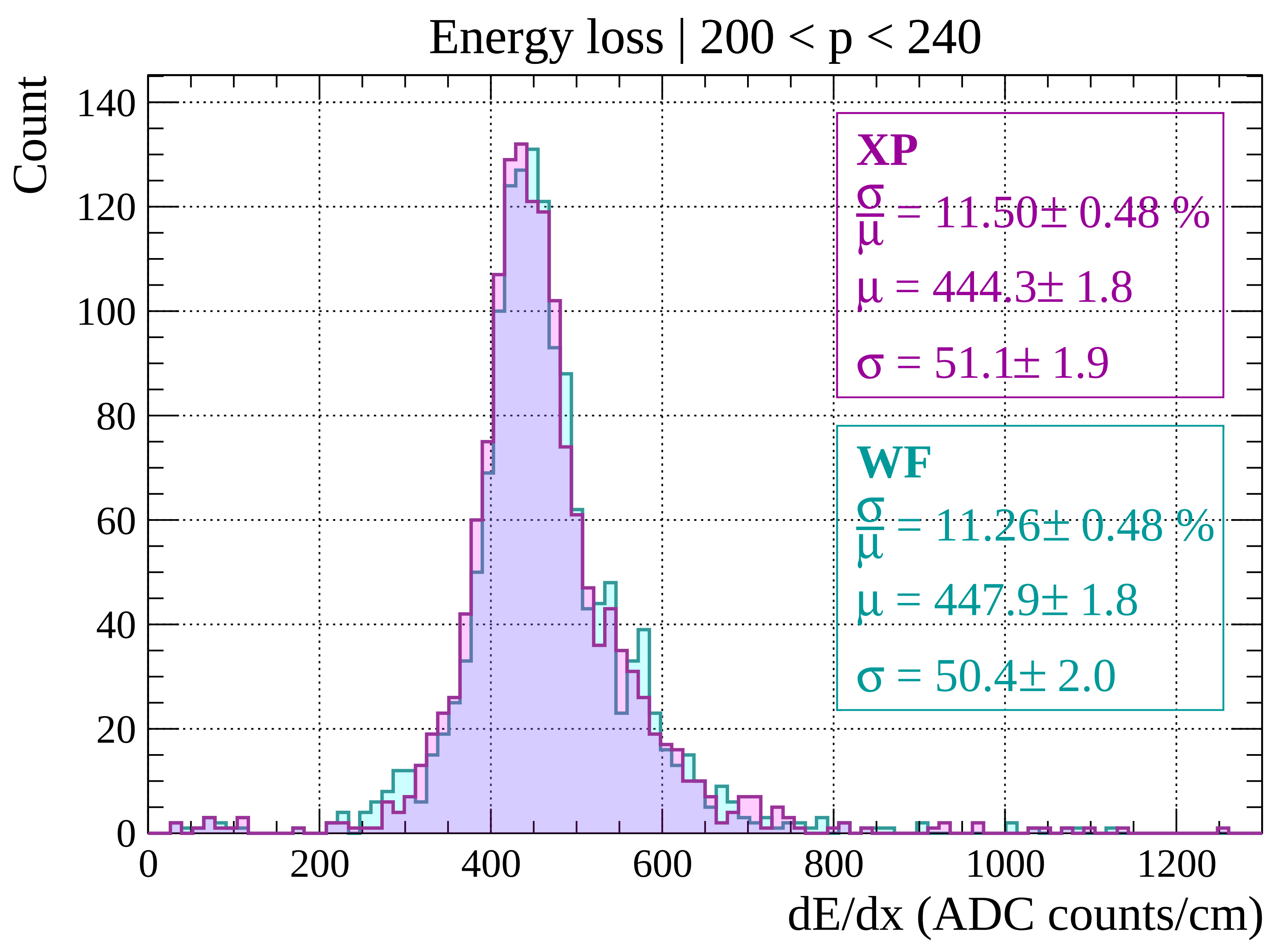


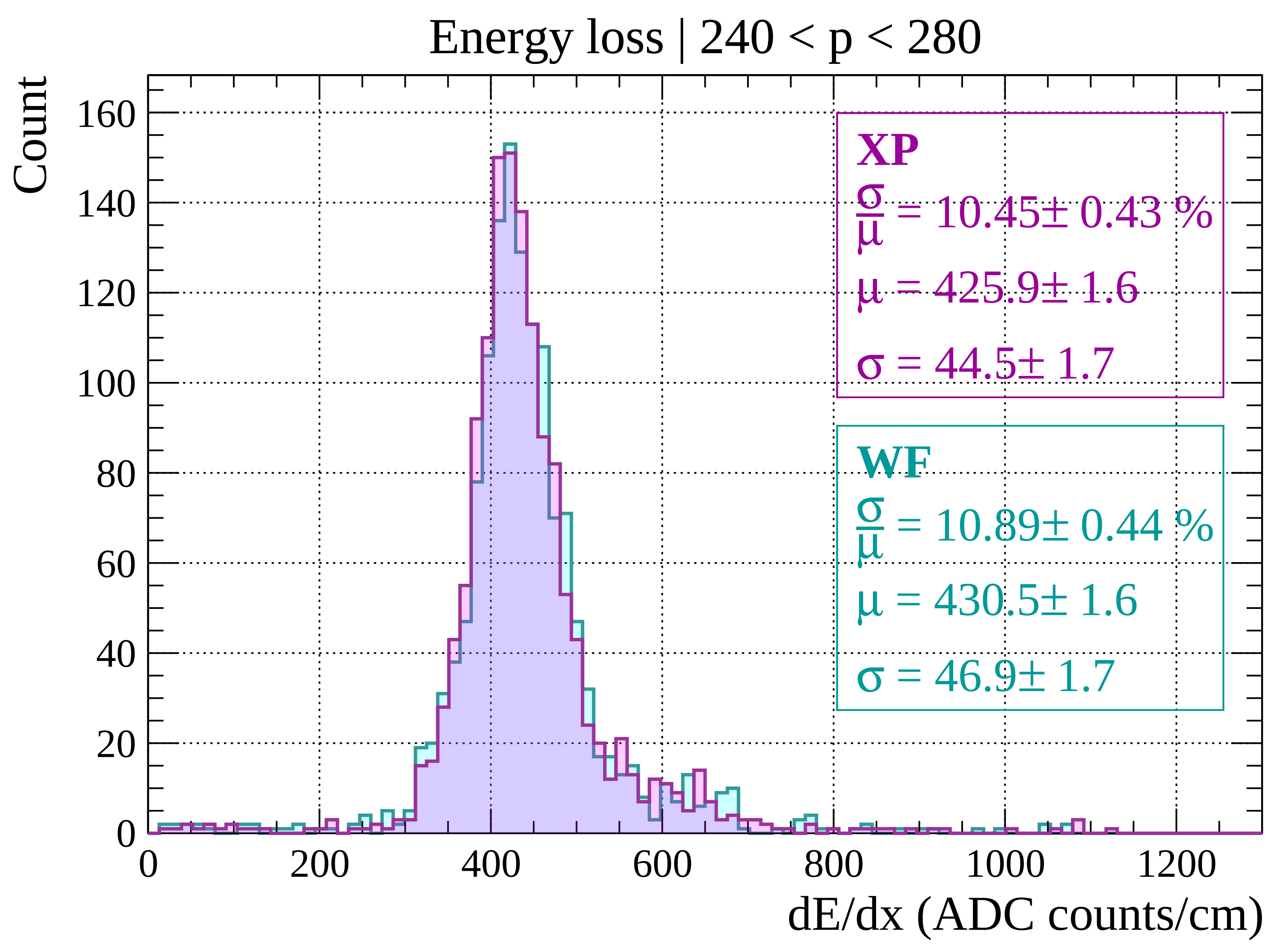
Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.12± 0.58 %
μ = 579.7± 2.9
σ = 64.5± 3.0
WF
σ/μ = 13.33± 0.77 %
μ = 585.6± 3.6
σ = 78.1± 4.0
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

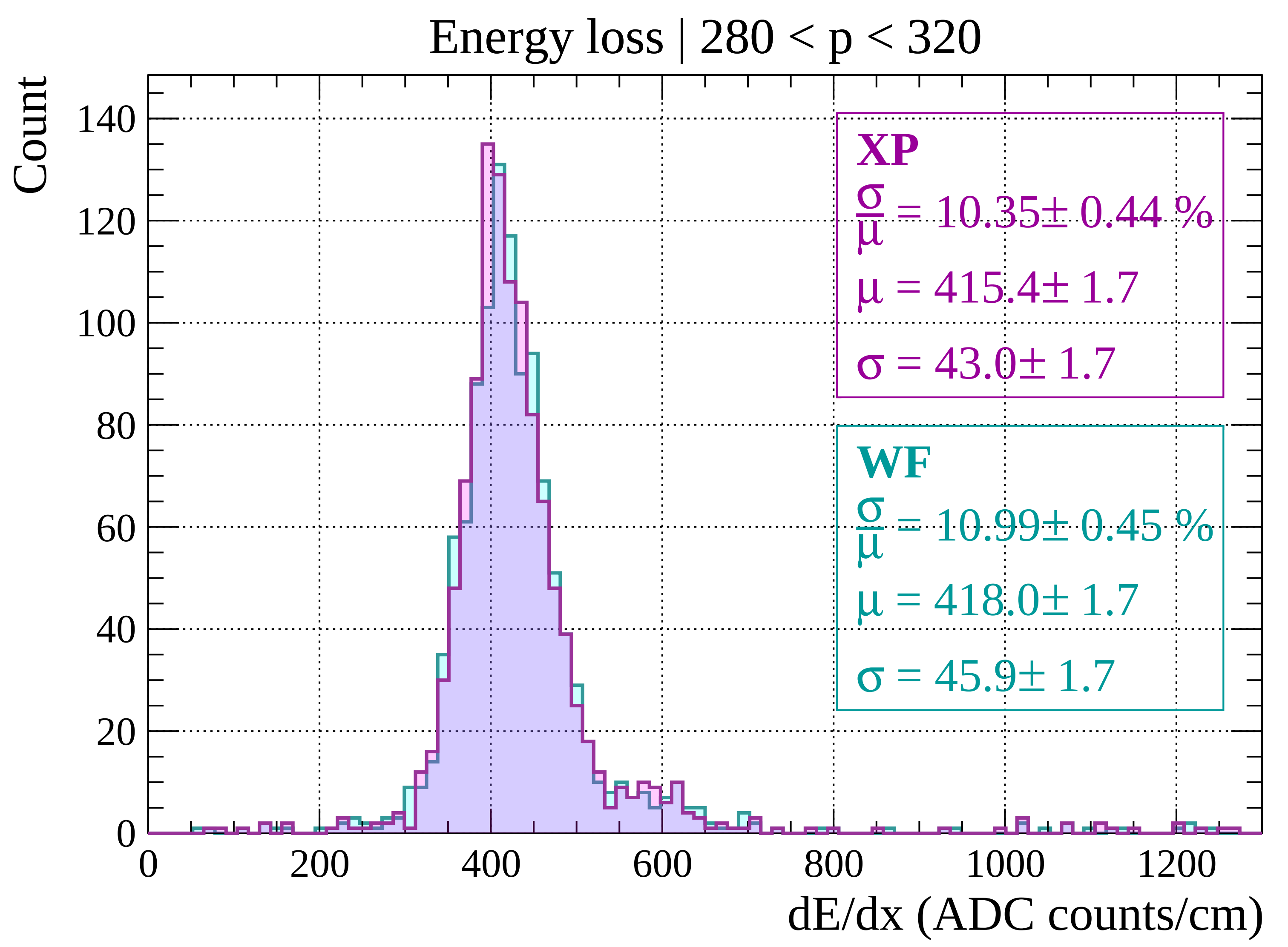

Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 12.76± 0.55 %
μ = 576.9± 2.7
σ = 73.6± 2.8
WF
σ/μ = 12.68± 0.57 %
μ = 582.6± 2.7
σ = 73.9± 3.0

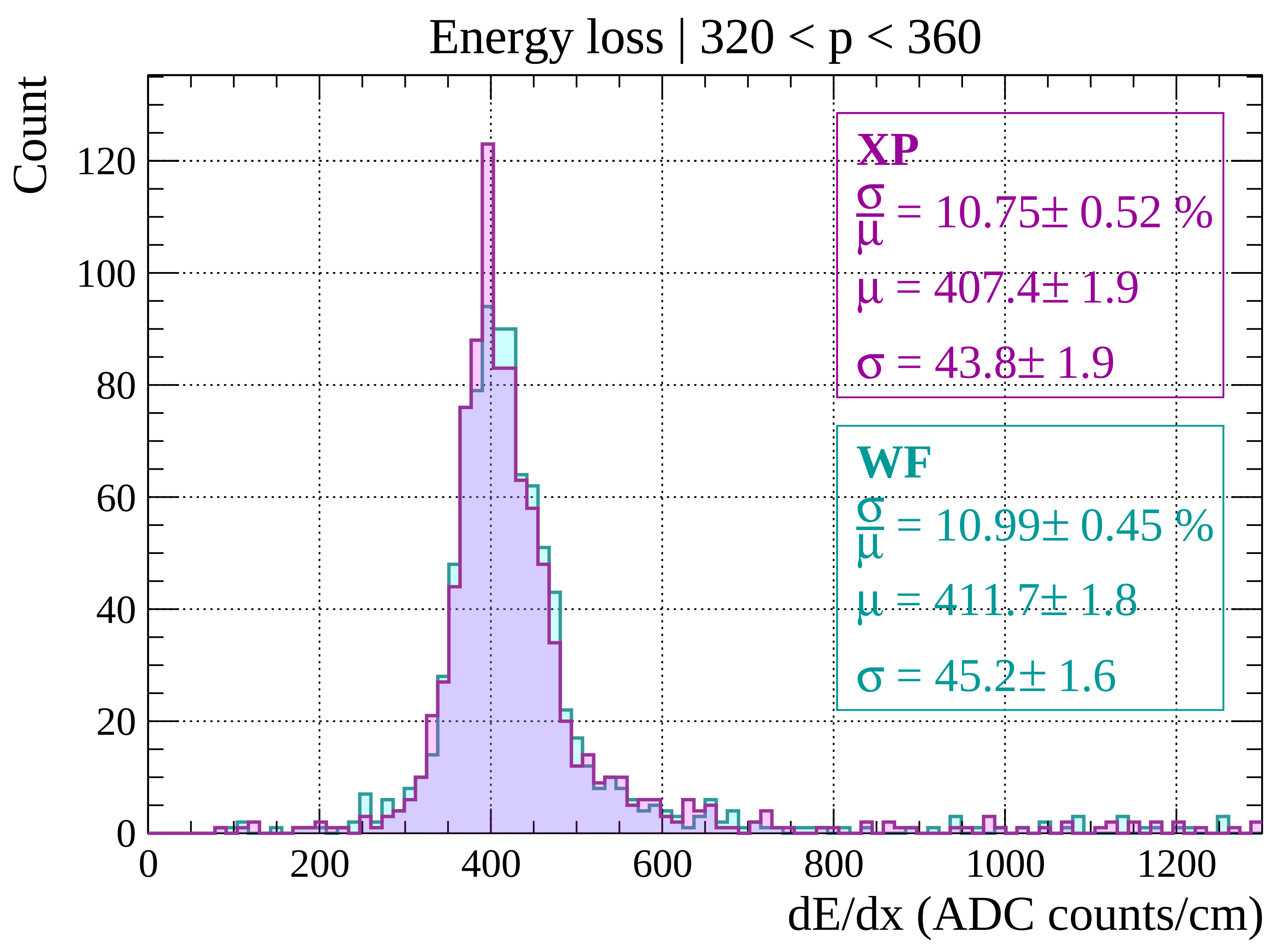


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.02± 0.56 %
μ = 501.6± 2.4
σ = 75.3± 2.4
WF
$\frac{\sigma}{\mu}$ = 15.06± 0.55 %
μ = 509.0± 2.5
σ = 76.7± 2.4

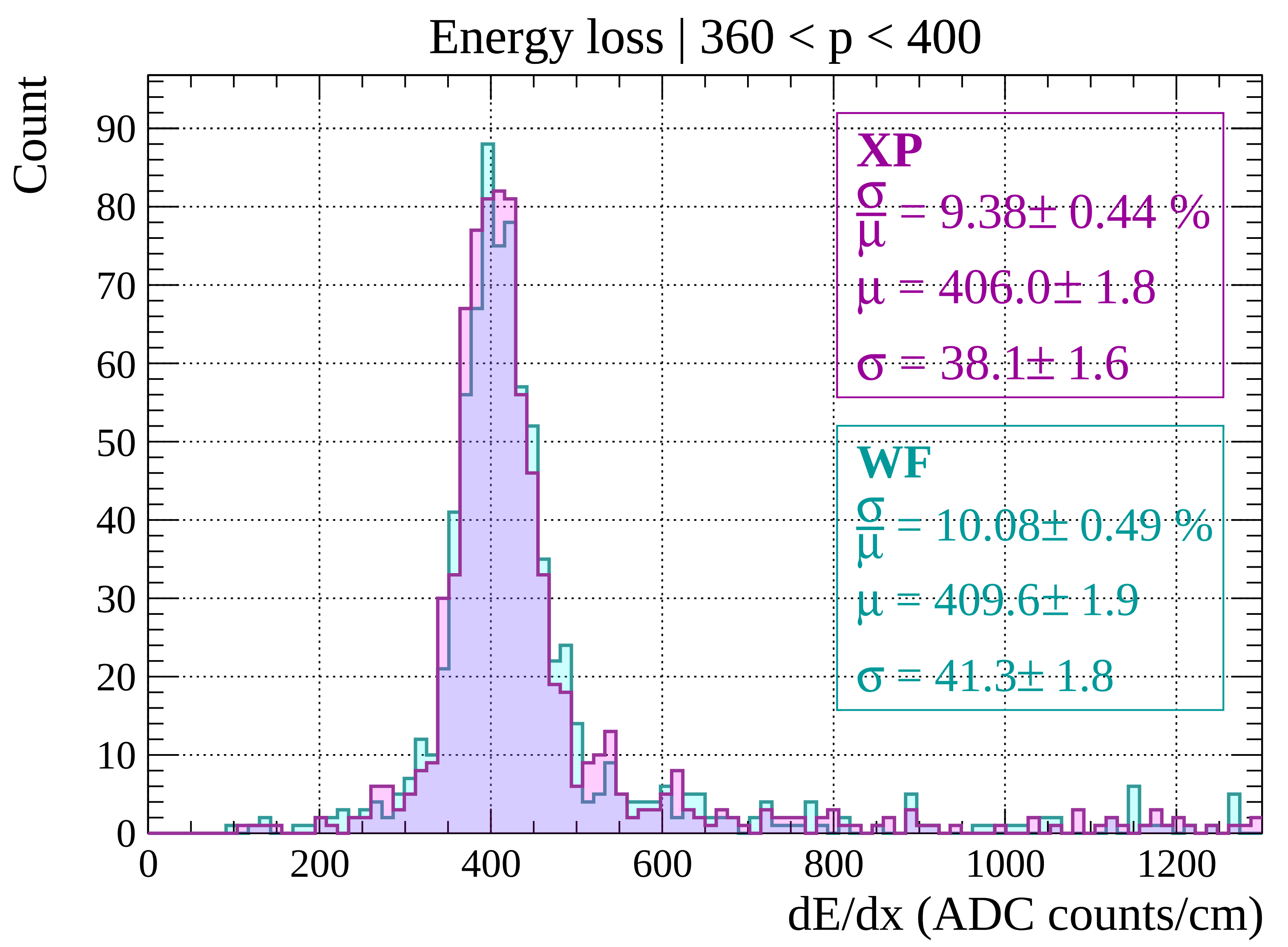

Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 11.50± 0.48 %
μ = 444.3± 1.8
σ = 51.1± 1.9
WF
σ/μ = 11.26± 0.48 %
μ = 447.9± 1.8
σ = 50.4± 2.0


Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 10.45± 0.43 %
μ = 425.9± 1.6
σ = 44.5± 1.7
WF
σ/μ = 10.89± 0.44 %
μ = 430.5± 1.6
σ = 46.9± 1.7

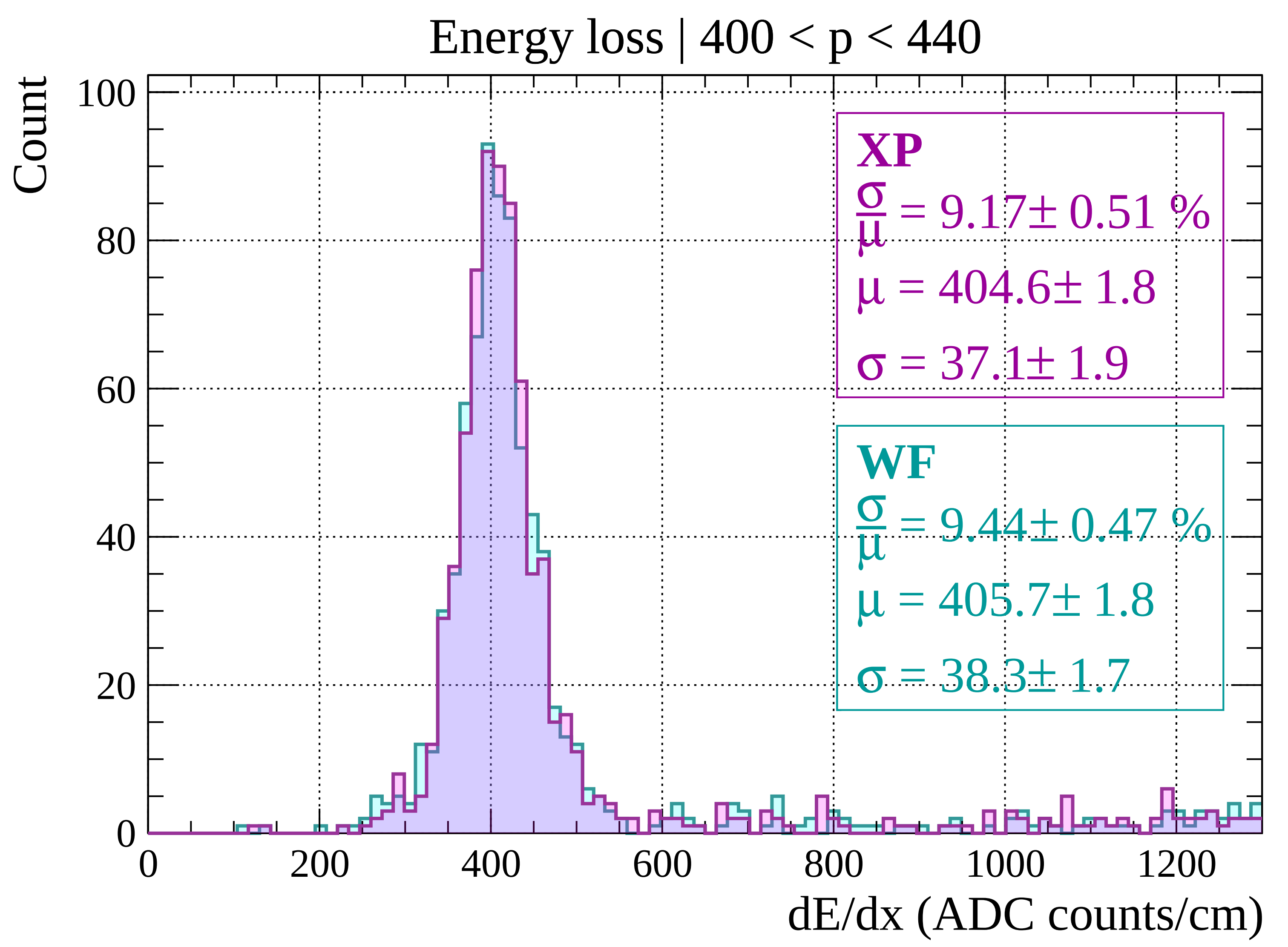

Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 10.35± 0.44 %
μ = 415.4± 1.7
σ = 43.0± 1.7
WF
σ/μ = 10.99± 0.45 %
μ = 418.0± 1.7
σ = 45.9± 1.7

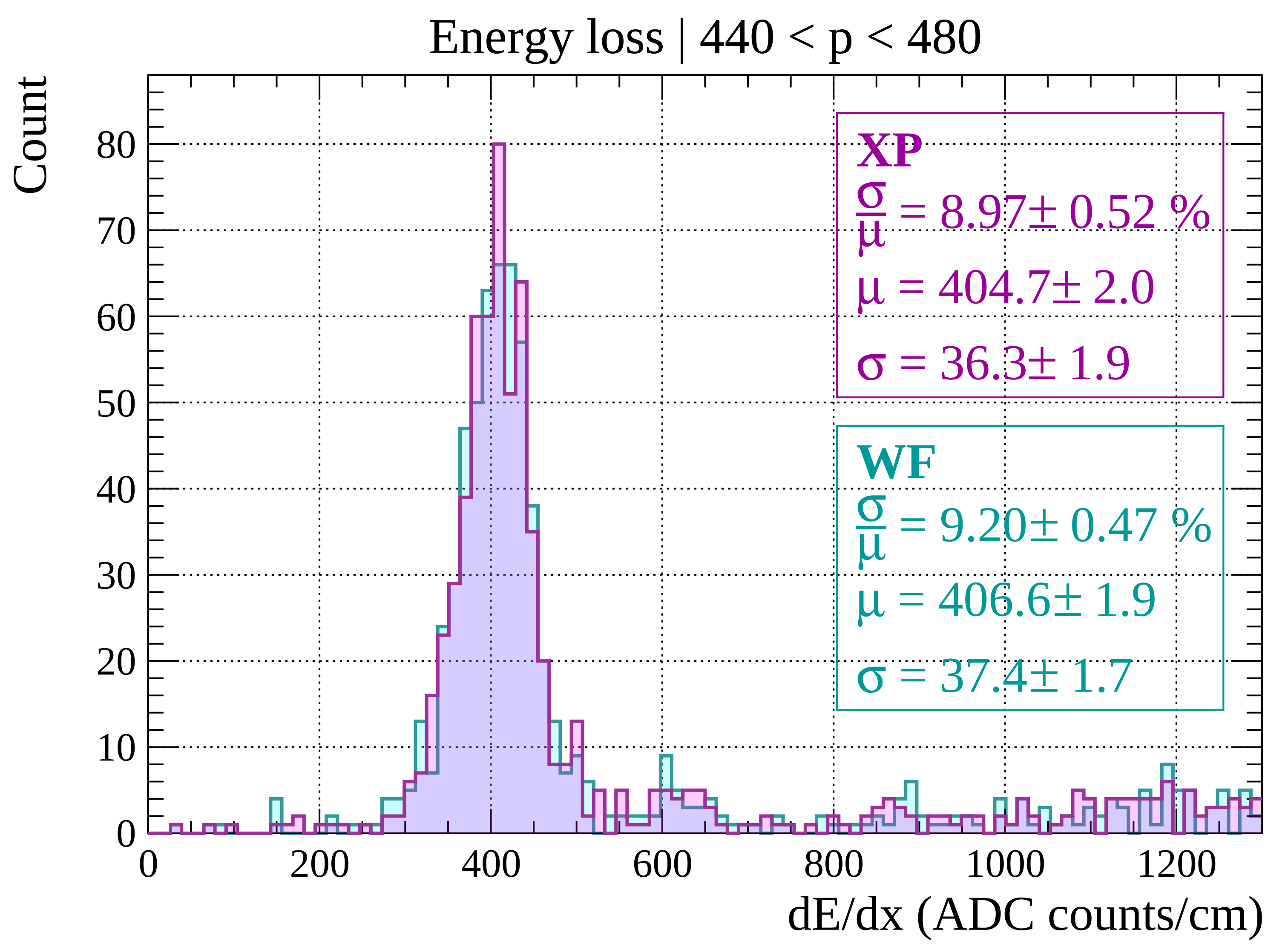


Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.75± 0.52 %
μ = 407.4± 1.9
σ = 43.8± 1.9
WF
$\frac{\sigma}{\mu}$ = 10.99± 0.45 %
μ = 411.7± 1.8
σ = 45.2± 1.6

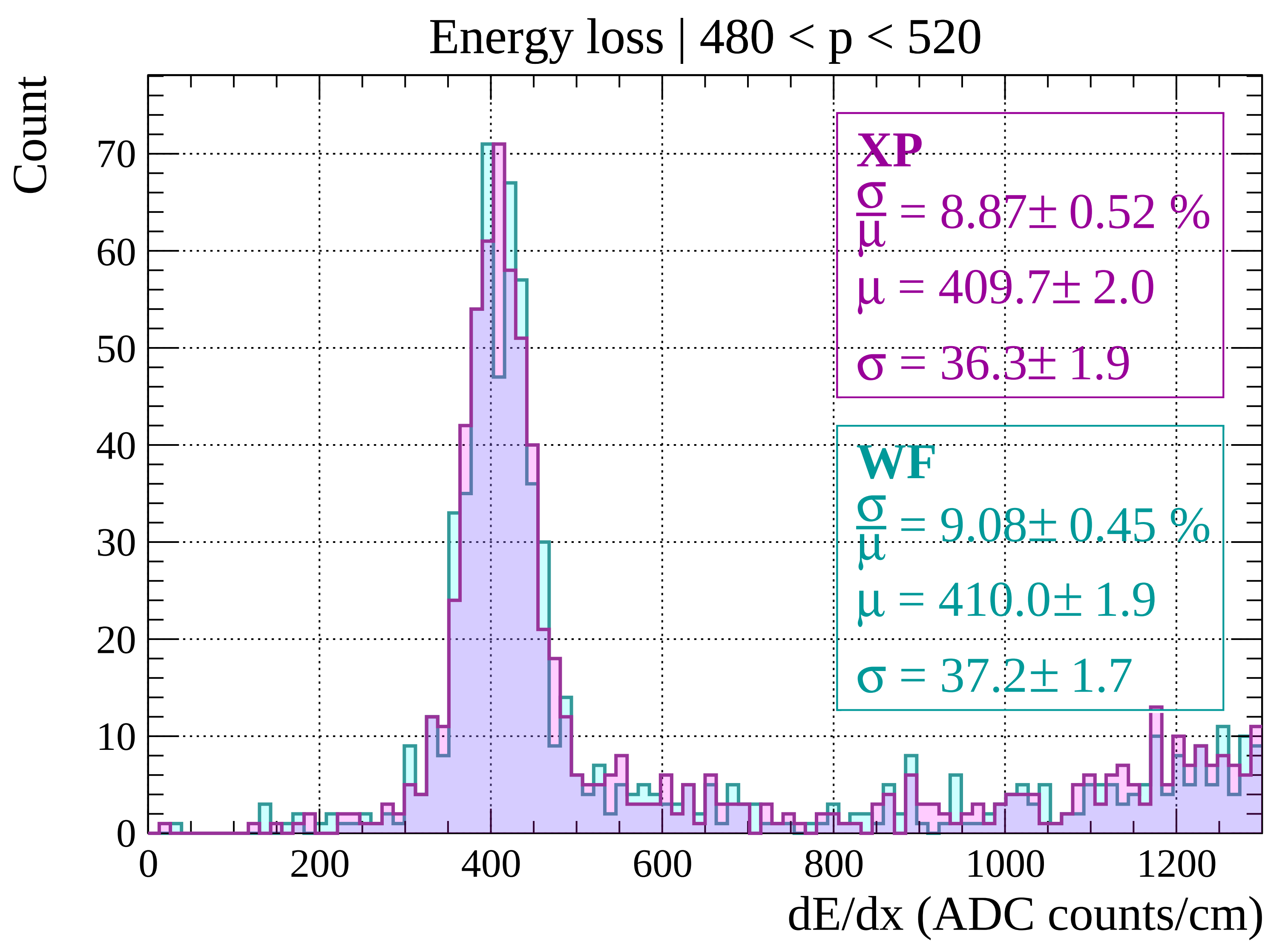


Energy loss | 360 < p < 400
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.38± 0.44 %
μ = 406.0± 1.8
σ = 38.1± 1.6
WF
$\frac{\sigma}{\mu}$ = 10.08± 0.49 %
μ = 409.6± 1.9
σ = 41.3± 1.8

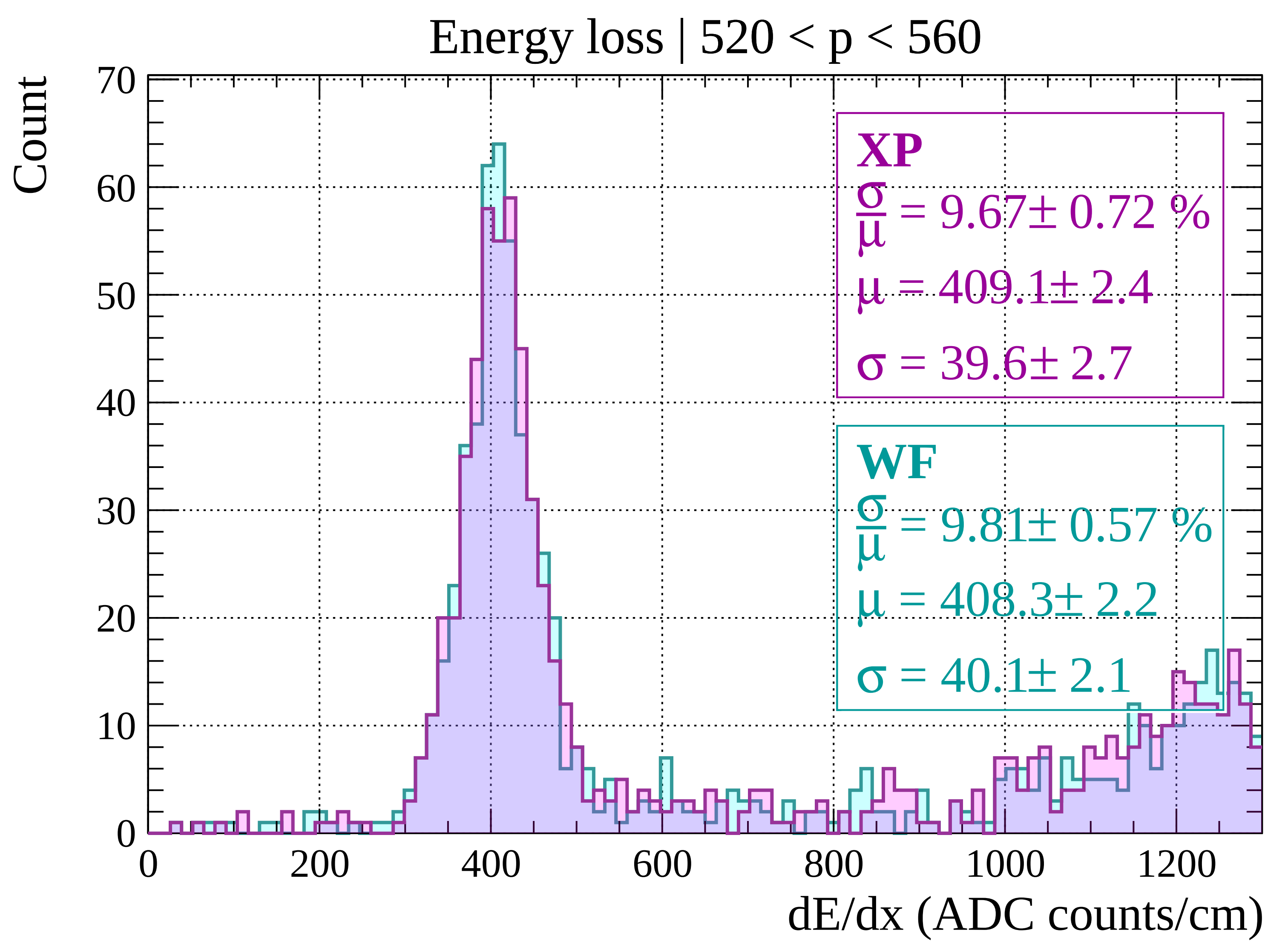

Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.17± 0.51 %
μ = 404.6± 1.8
σ = 37.1± 1.9
WF
$\frac{\sigma}{\mu}$ = 9.44± 0.47 %
μ = 405.7± 1.8
σ = 38.3± 1.7



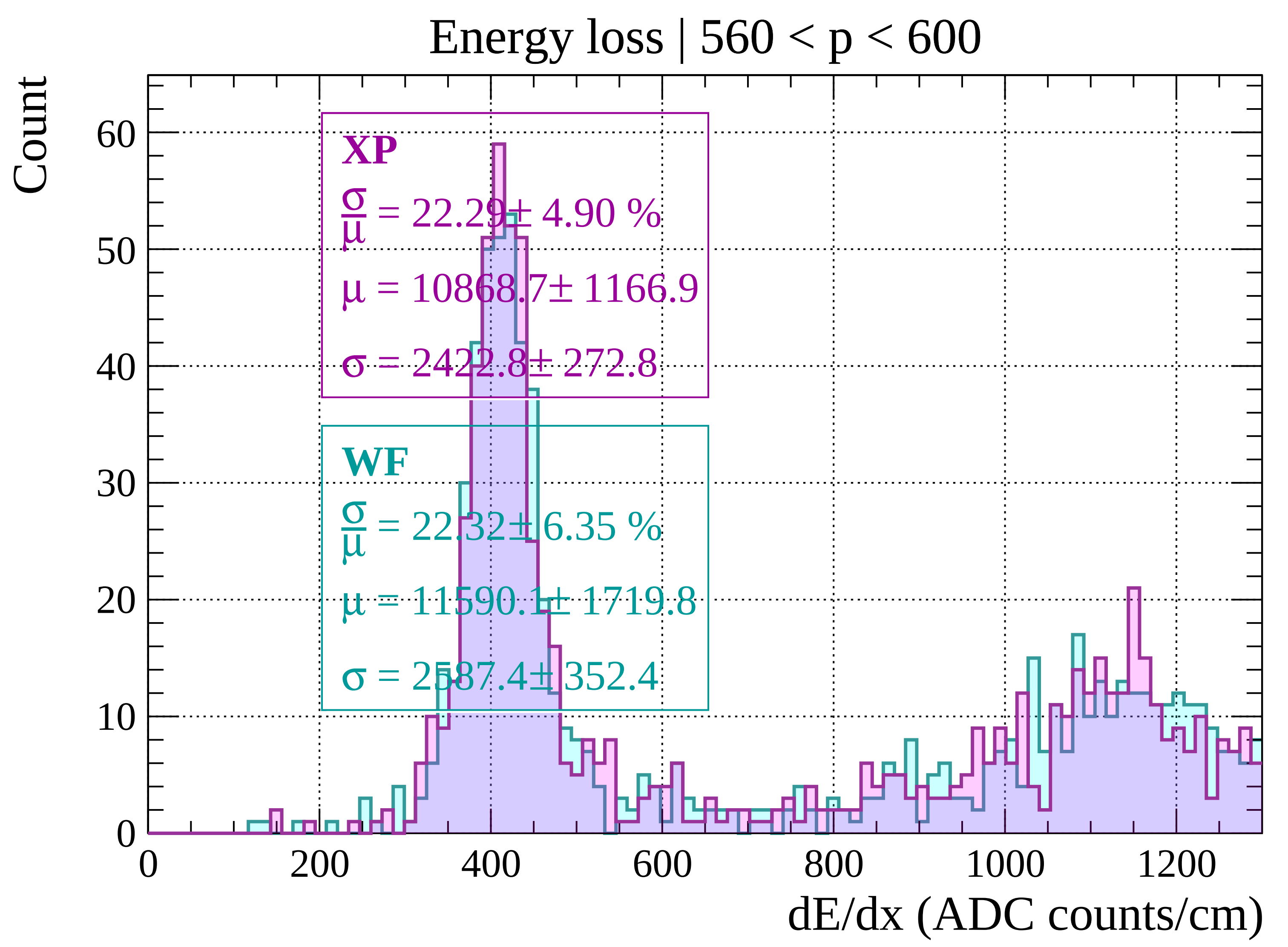
Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
$\frac{\sigma}{\mu}$ = 8.97± 0.52 %
μ = 404.7± 2.0
σ = 36.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 9.20± 0.47 %
μ = 406.6± 1.9
σ = 37.4± 1.7



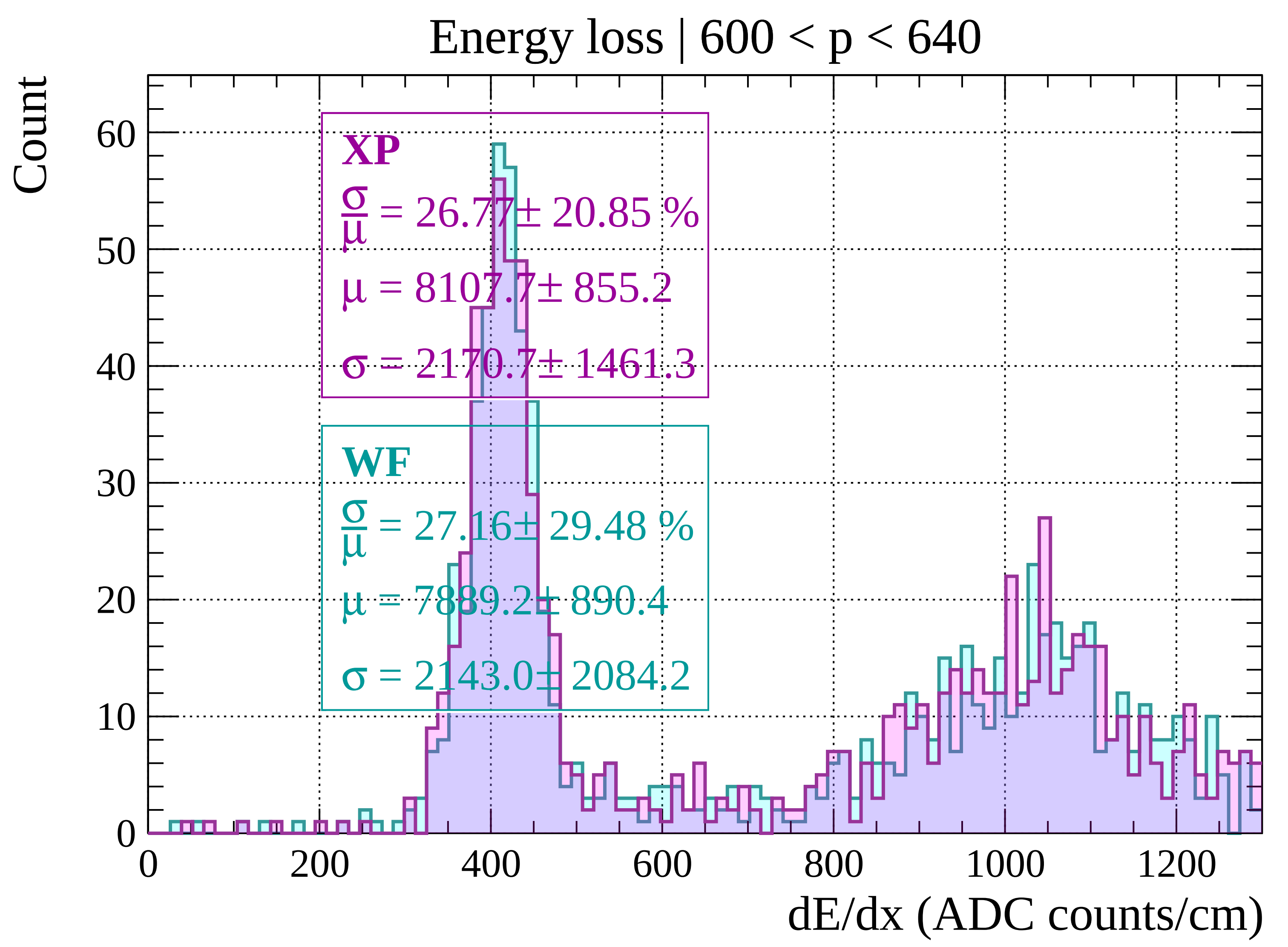
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.87± 0.52 %
μ = 409.7± 2.0
σ = 36.3± 1.9
WF
σ/μ = 9.08± 0.45 %
μ = 410.0± 1.9
σ = 37.2± 1.7



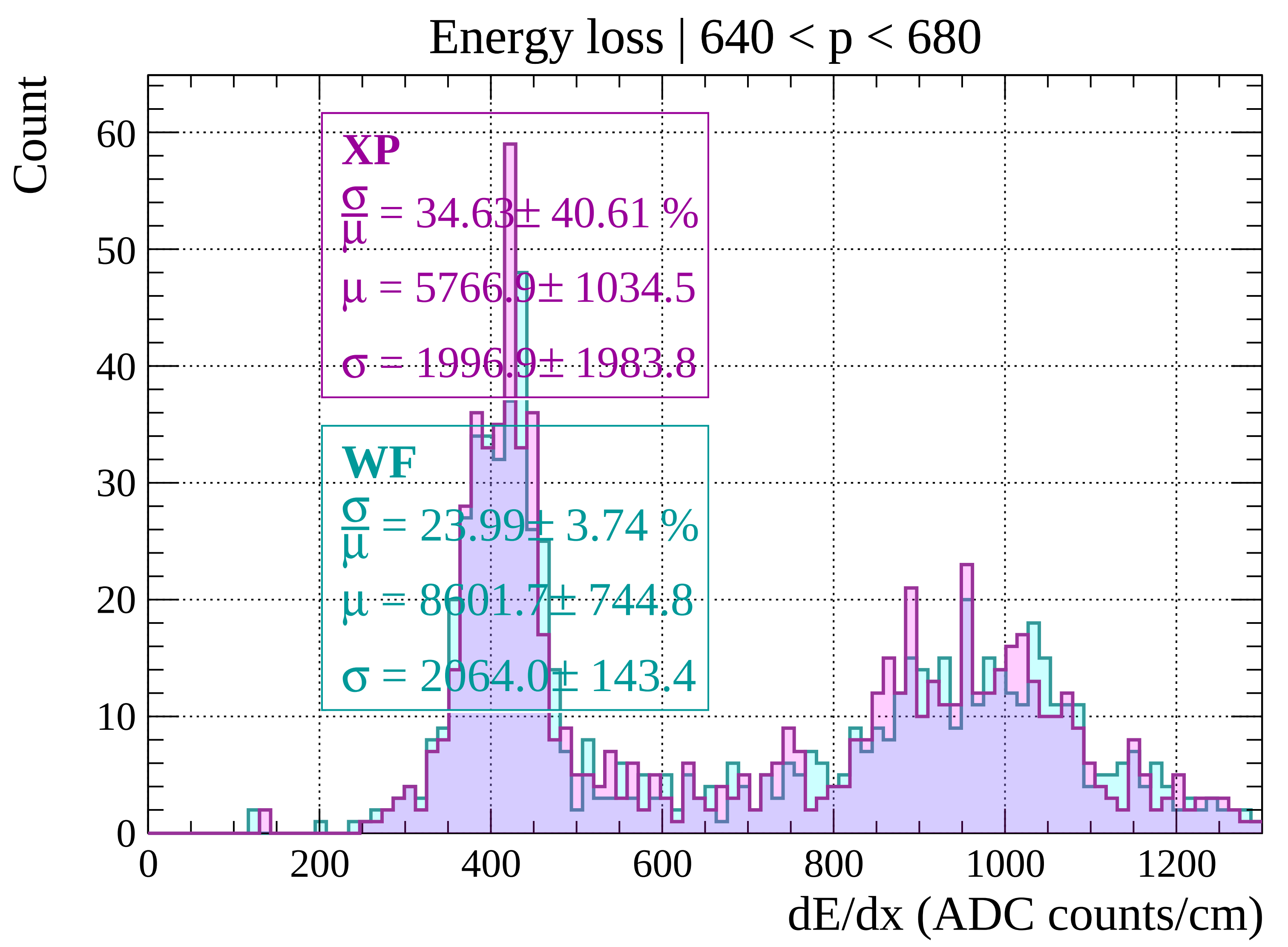
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.67± 0.72 %
μ = 409.1± 2.4
σ = 39.6± 2.7
WF
$\frac{\sigma}{\mu}$ = 9.81± 0.57 %
μ = 408.3± 2.2
σ = 40.1± 2.1



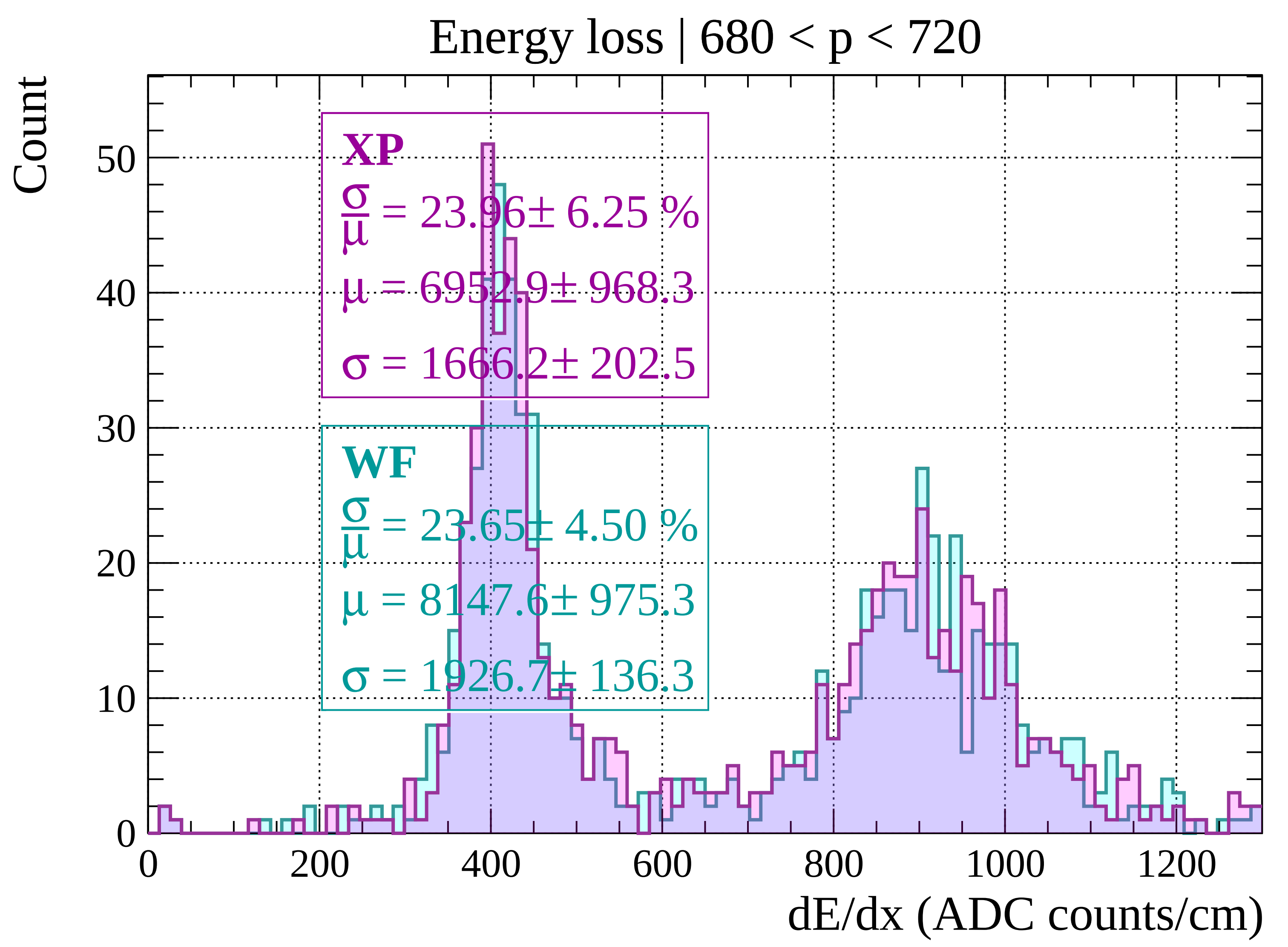
Energy loss | 560 < p < 600
Count
XP
σ/μ = 22.29± 4.90 %
μ = 10868.7± 1166.9
σ = 2422.8± 272.8
WF
σ/μ = 22.32± 6.35 %
μ = 11590.1± 1719.8
σ = 2587.4± 352.4
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



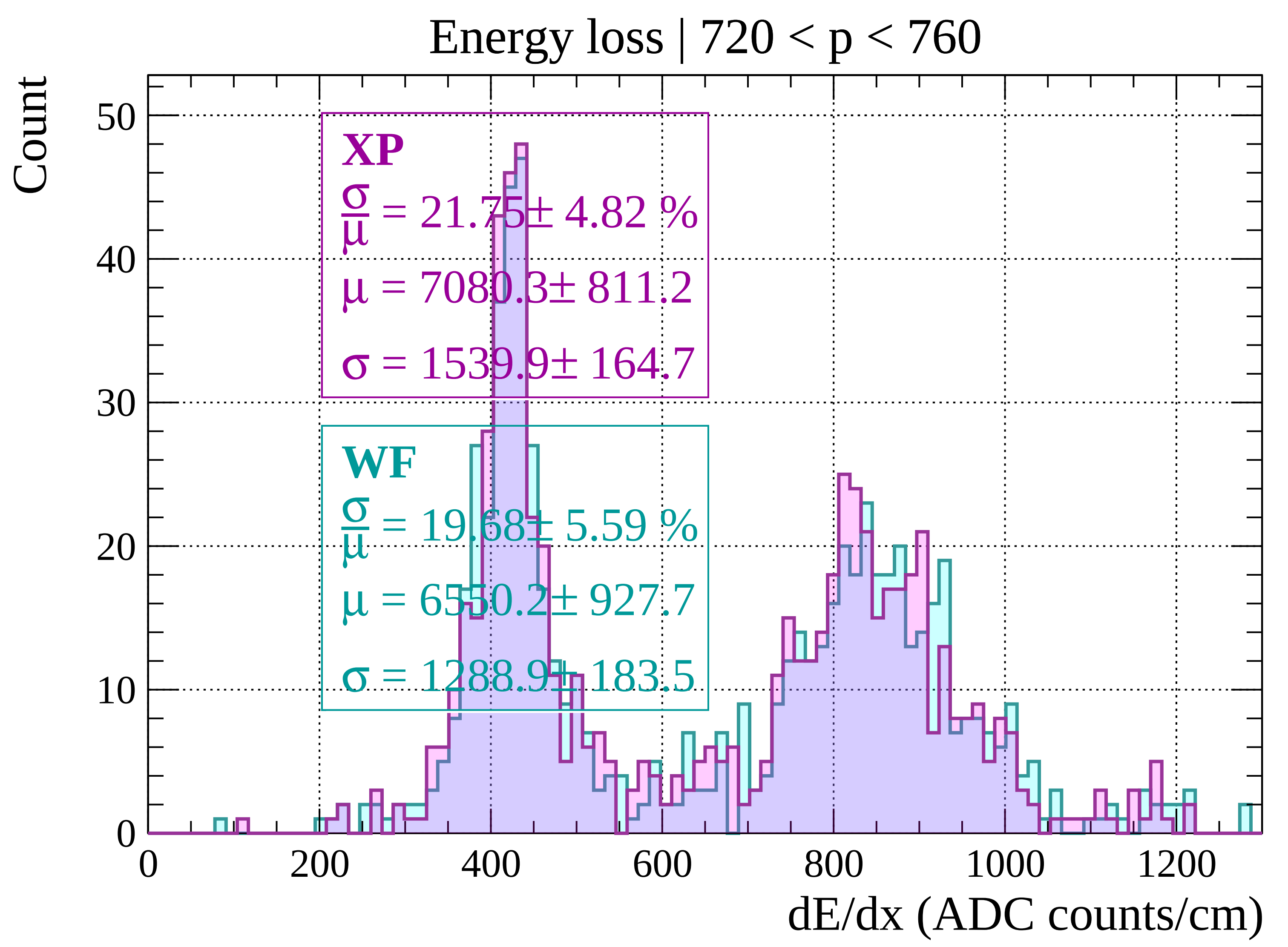
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 26.77± 20.85 %
μ = 8107.7± 855.2
σ = 2170.7± 1461.3
WF
σ/μ = 27.16± 29.48 %
μ = 7889.2± 890.4
σ = 2143.0± 2084.2



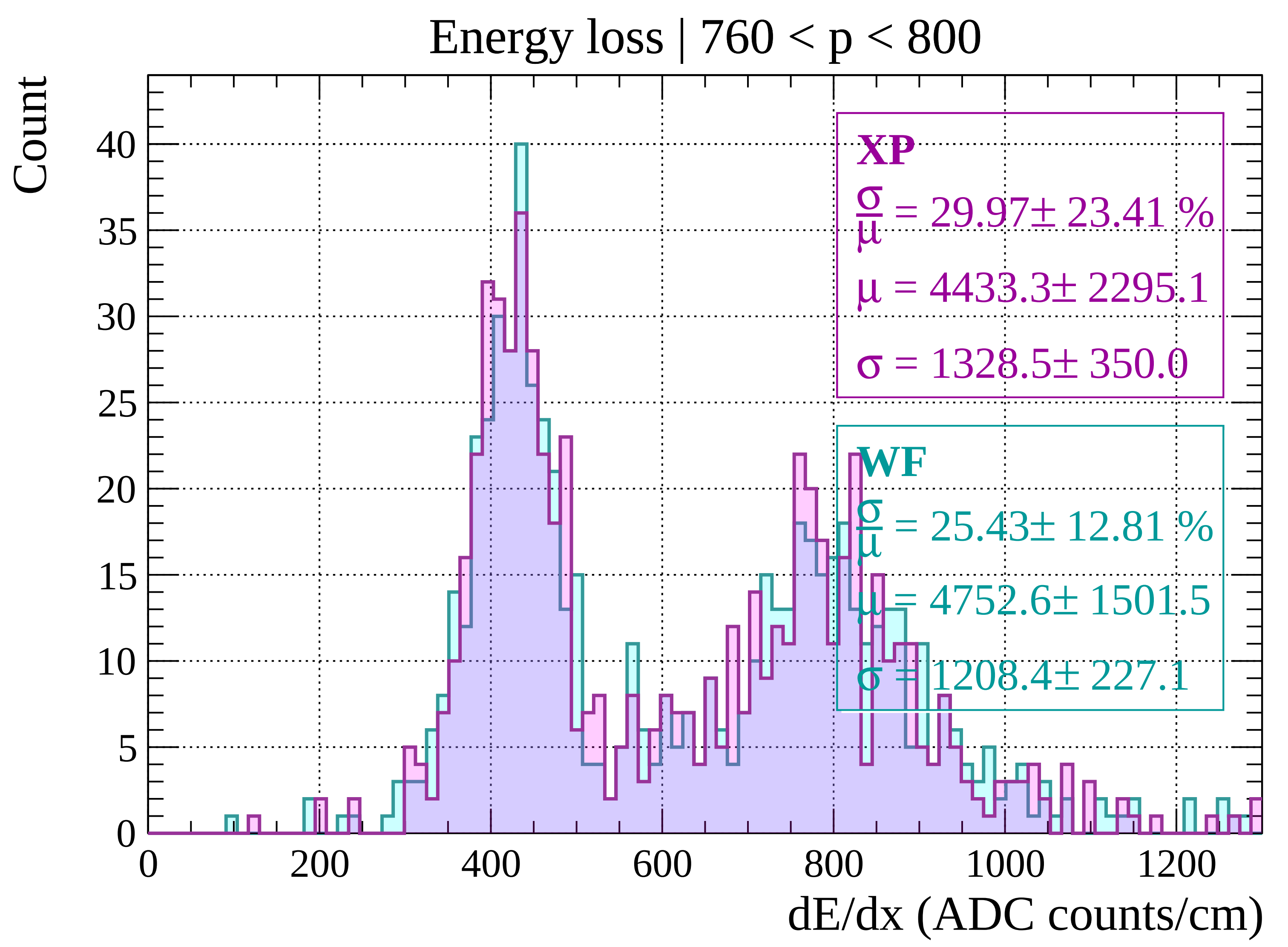
Energy loss | 640 < p < 680
Count
XP
σ/μ = 34.63± 40.61 %
μ = 5766.9± 1034.5
σ = 1996.9± 1983.8
WF
σ/μ = 23.99± 3.74 %
μ = 8601.7± 744.8
σ = 2064.0± 143.4
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



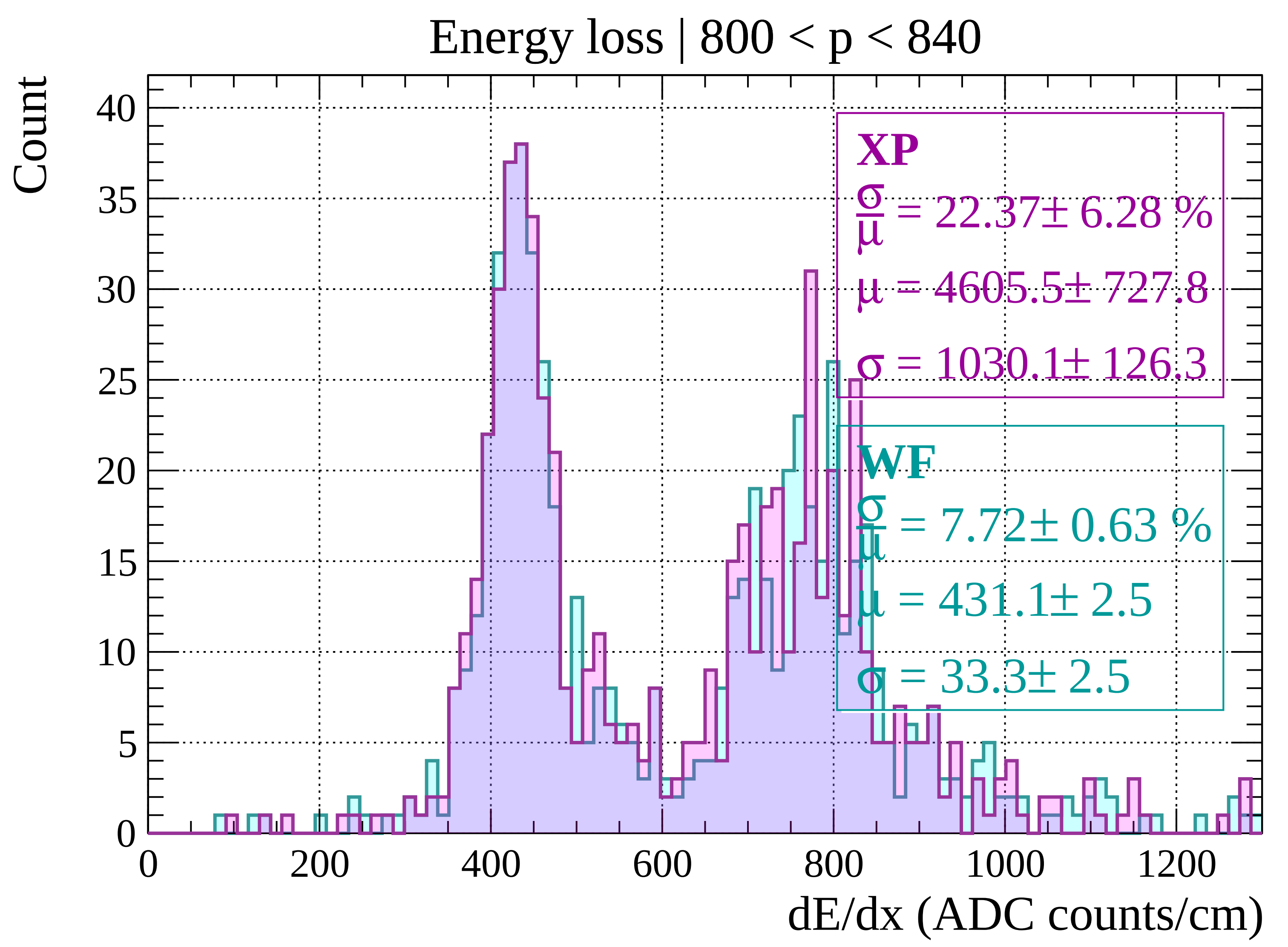
Energy loss | 680 < p < 720
Count
XP
$\frac{\sigma}{\mu}$ = 23.96± 6.25 %
μ = 6952.9± 968.3
σ = 1666.2± 202.5
WF
$\frac{\sigma}{\mu}$ = 23.65± 4.50 %
μ = 8147.6± 975.3
σ = 1926.7± 136.3
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



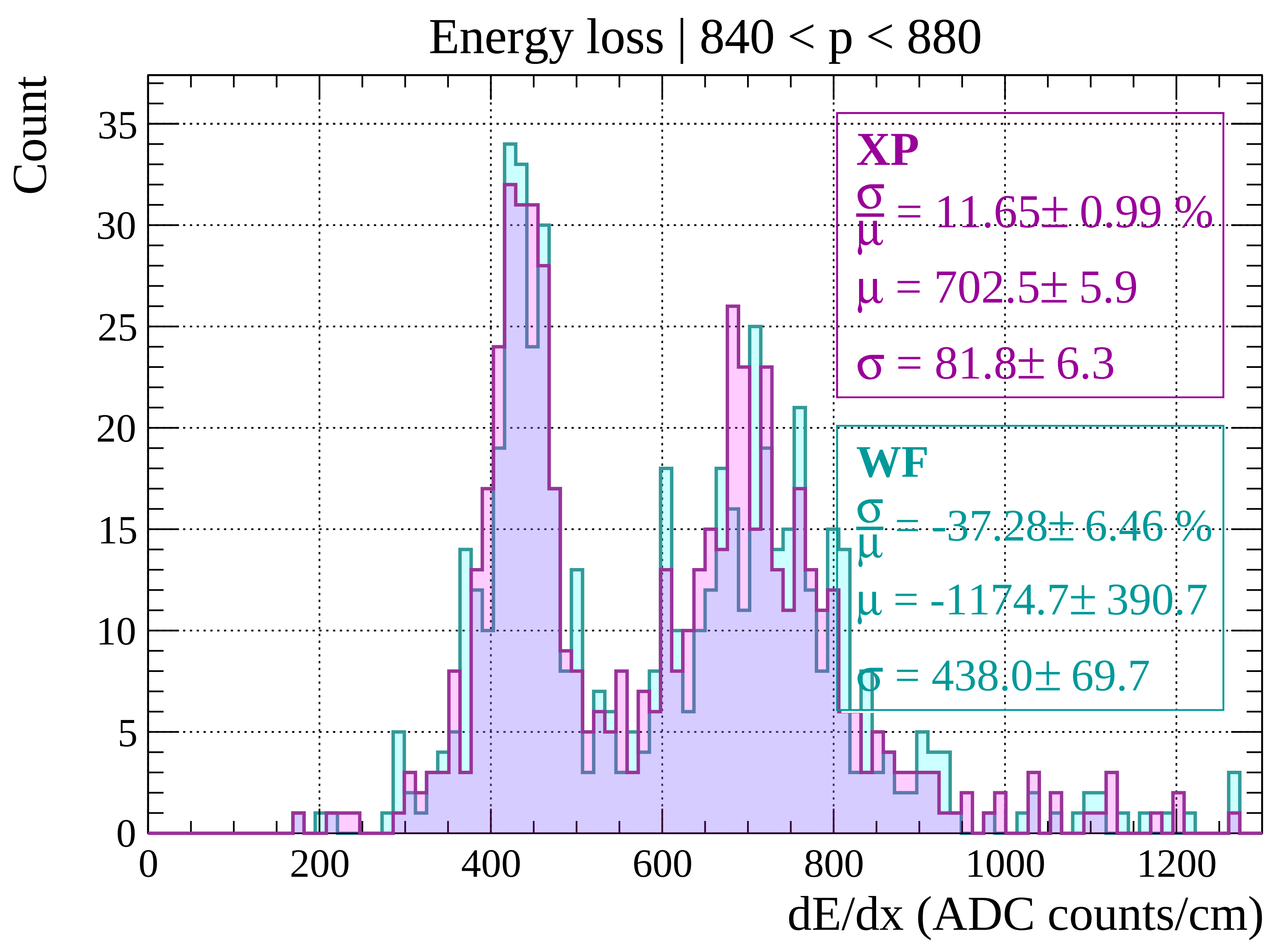
Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 21.75 ± 4.82 %
μ = 7080.3 ± 811.2
σ = 1539.9 ± 164.7
WF
$\frac{\sigma}{\mu}$ = 19.68 ± 5.59 %
μ = 6550.2 ± 927.7
σ = 1288.9 ± 183.5

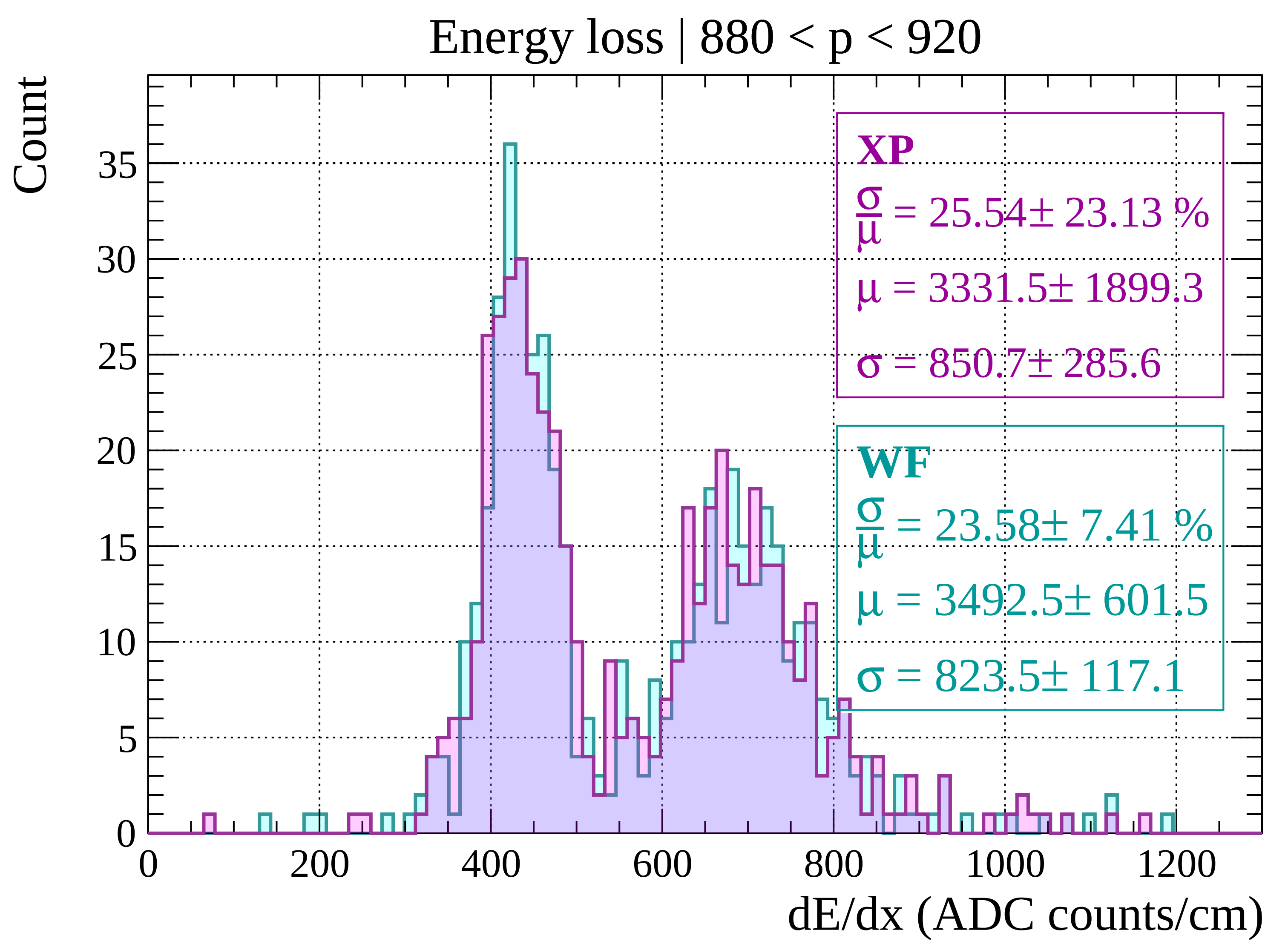


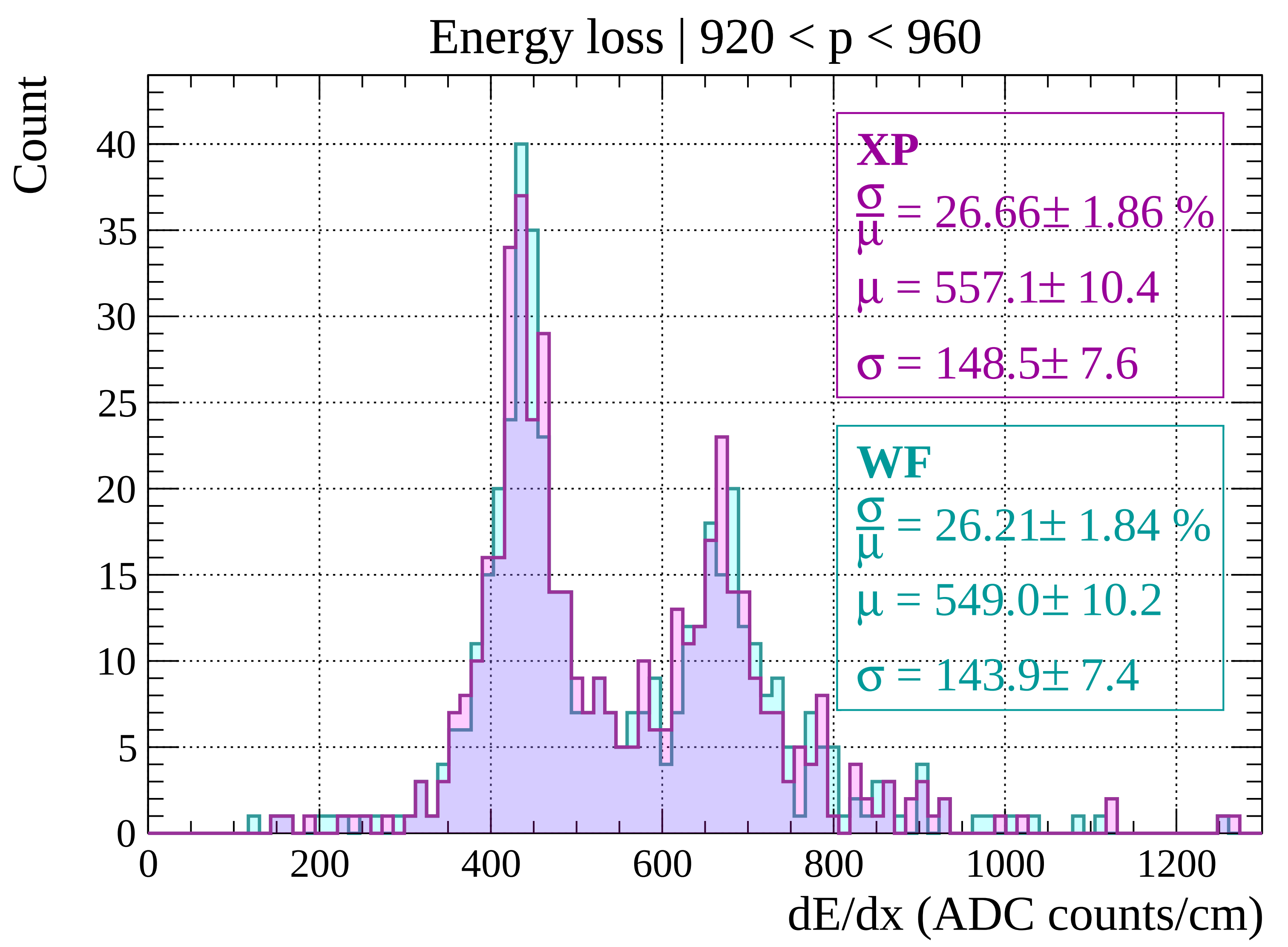
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 29.97± 23.41 %
μ = 4433.3± 2295.1
σ = 1328.5± 350.0
WF
σ/μ = 25.43± 12.81 %
μ = 4752.6± 1501.5
σ = 1208.4± 227.1

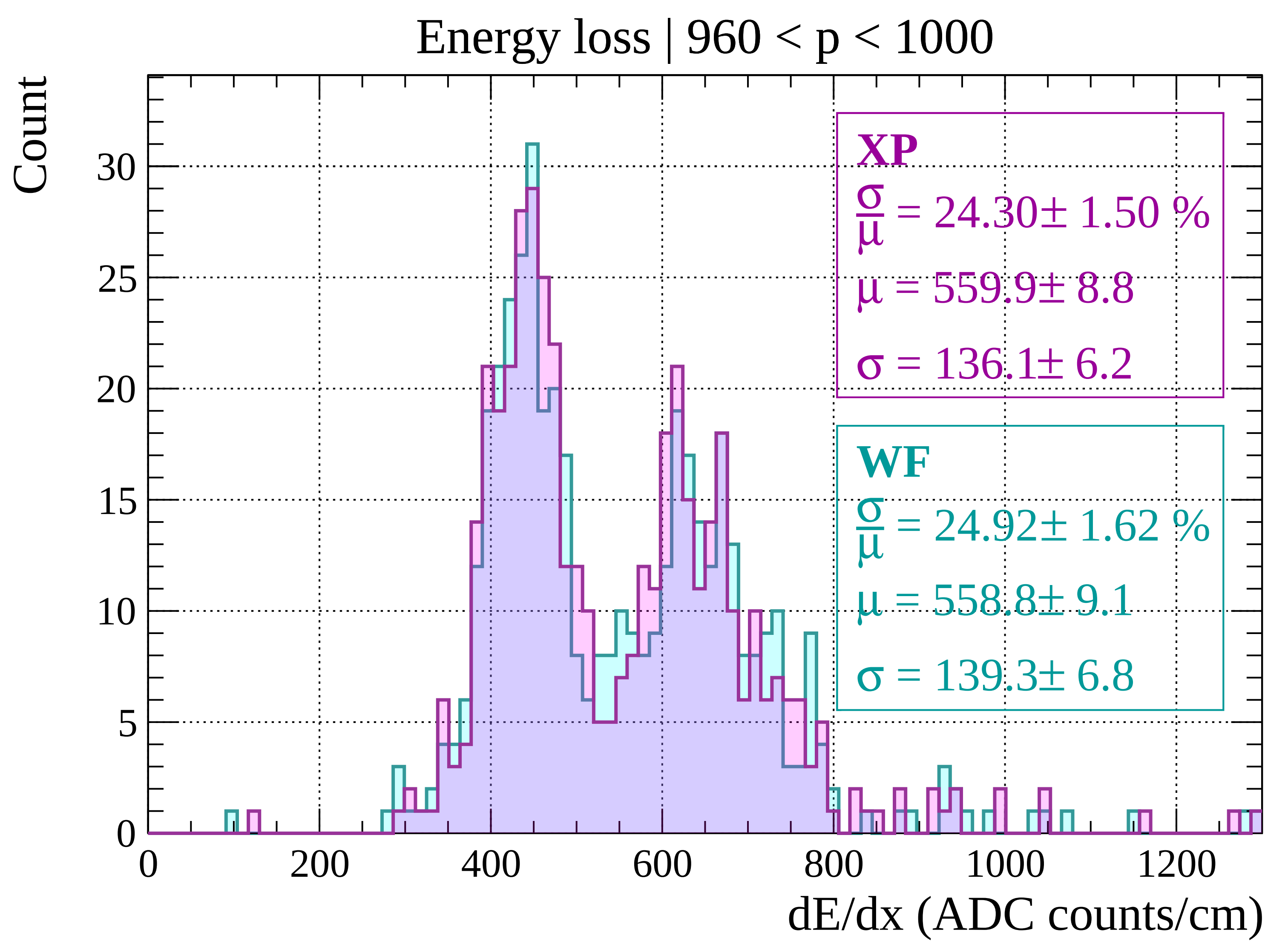


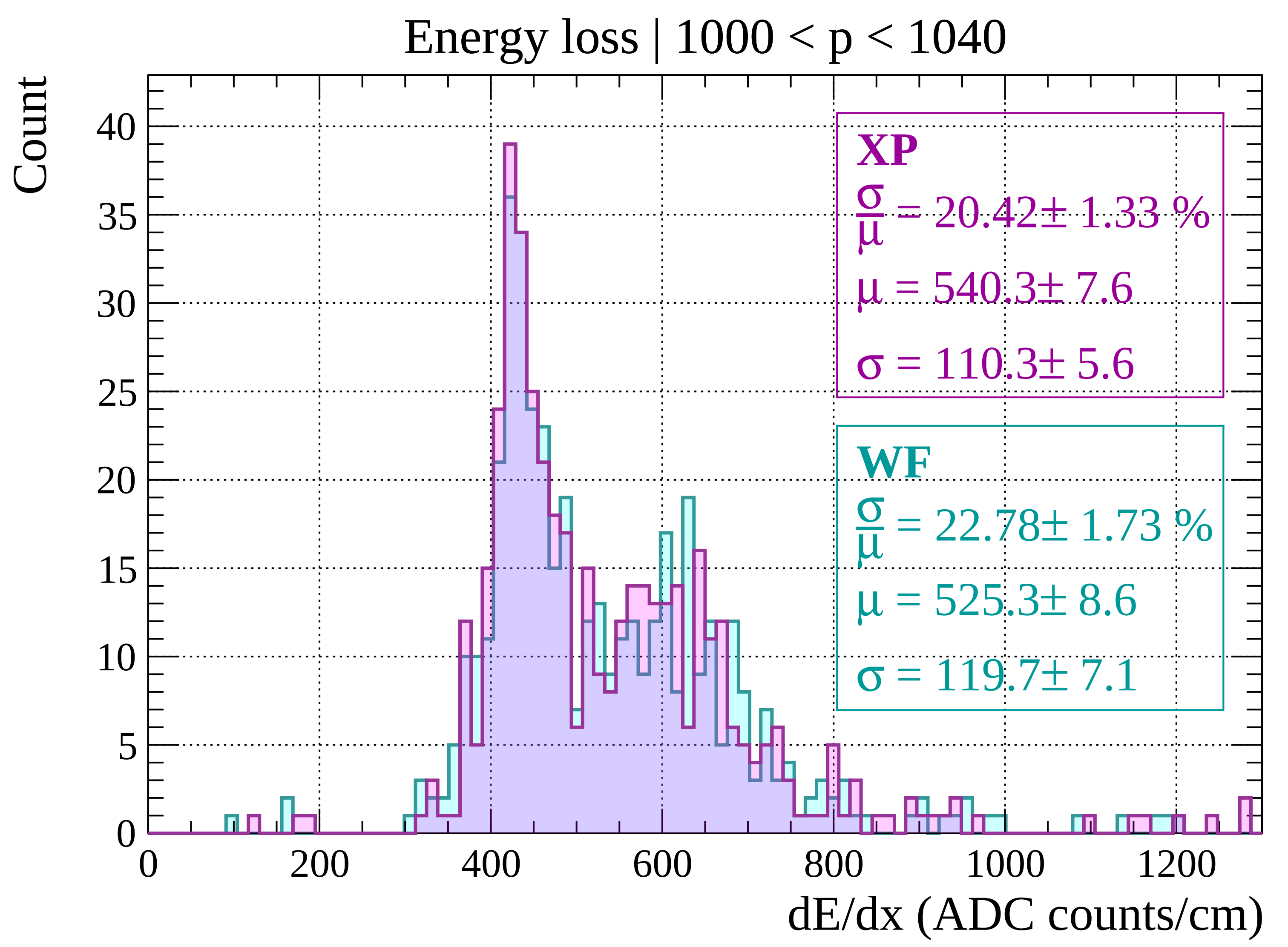
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 22.37± 6.28 %
μ = 4605.5± 727.8
σ = 1030.1± 126.3
WF
σ/μ = 7.72± 0.63 %
μ = 431.1± 2.5
σ = 33.3± 2.5


Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 11.65± 0.99 %
μ = 702.5± 5.9
σ = 81.8± 6.3
WF
σ/μ = -37.28± 6.46 %
μ = -1174.7± 390.7
σ = 438.0± 69.7

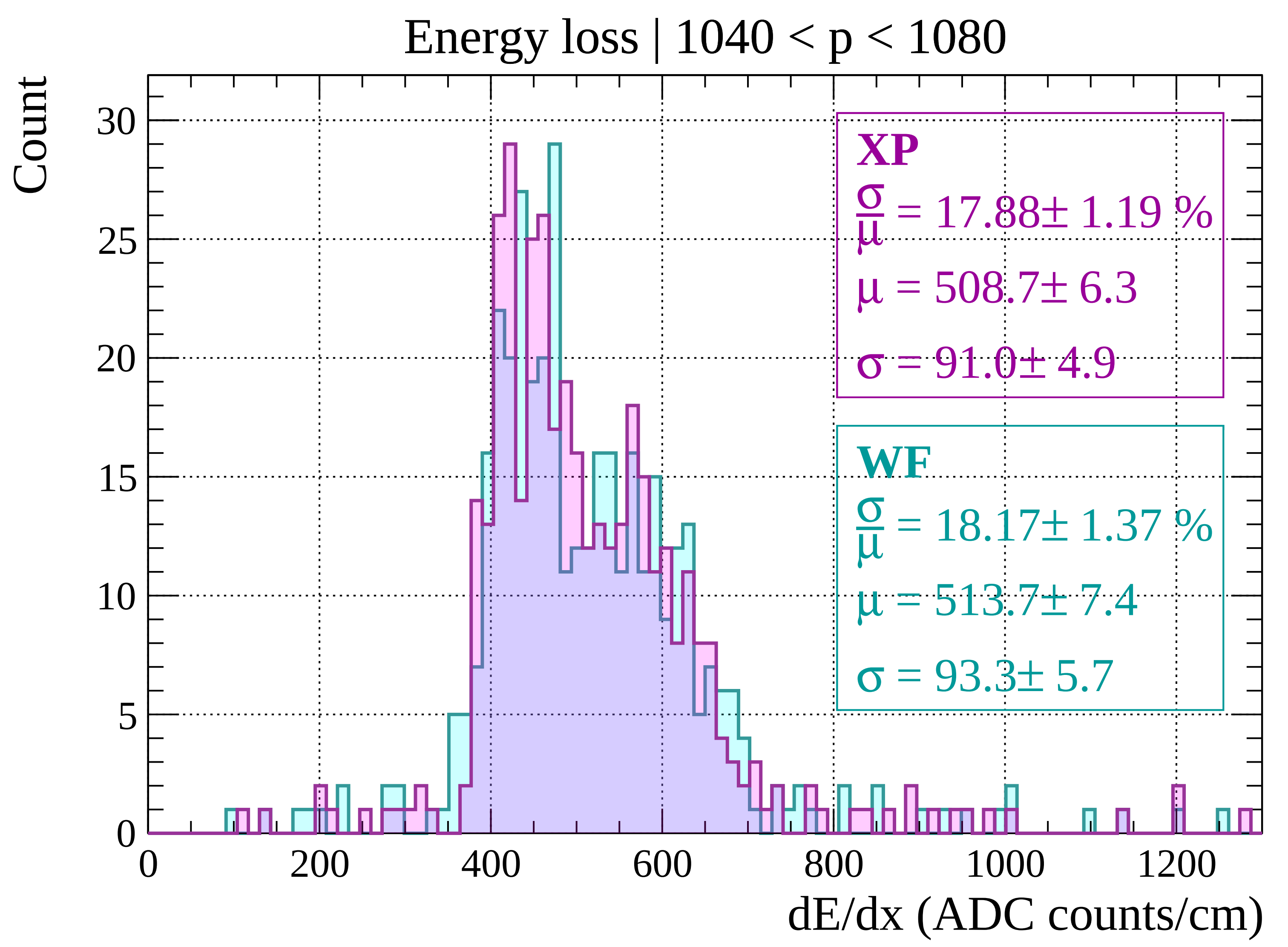

Energy loss | 880 < p < 920
Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 25.54± 23.13 %
μ = 3331.5± 1899.3
σ = 850.7± 285.6
WF
σ/μ = 23.58± 7.41 %
μ = 3492.5± 601.5
σ = 823.5± 117.1


Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
XP
σ/μ = 26.66± 1.86 %
μ = 557.1± 10.4
σ = 148.5± 7.6
WF
σ/μ = 26.21± 1.84 %
μ = 549.0± 10.2
σ = 143.9± 7.4

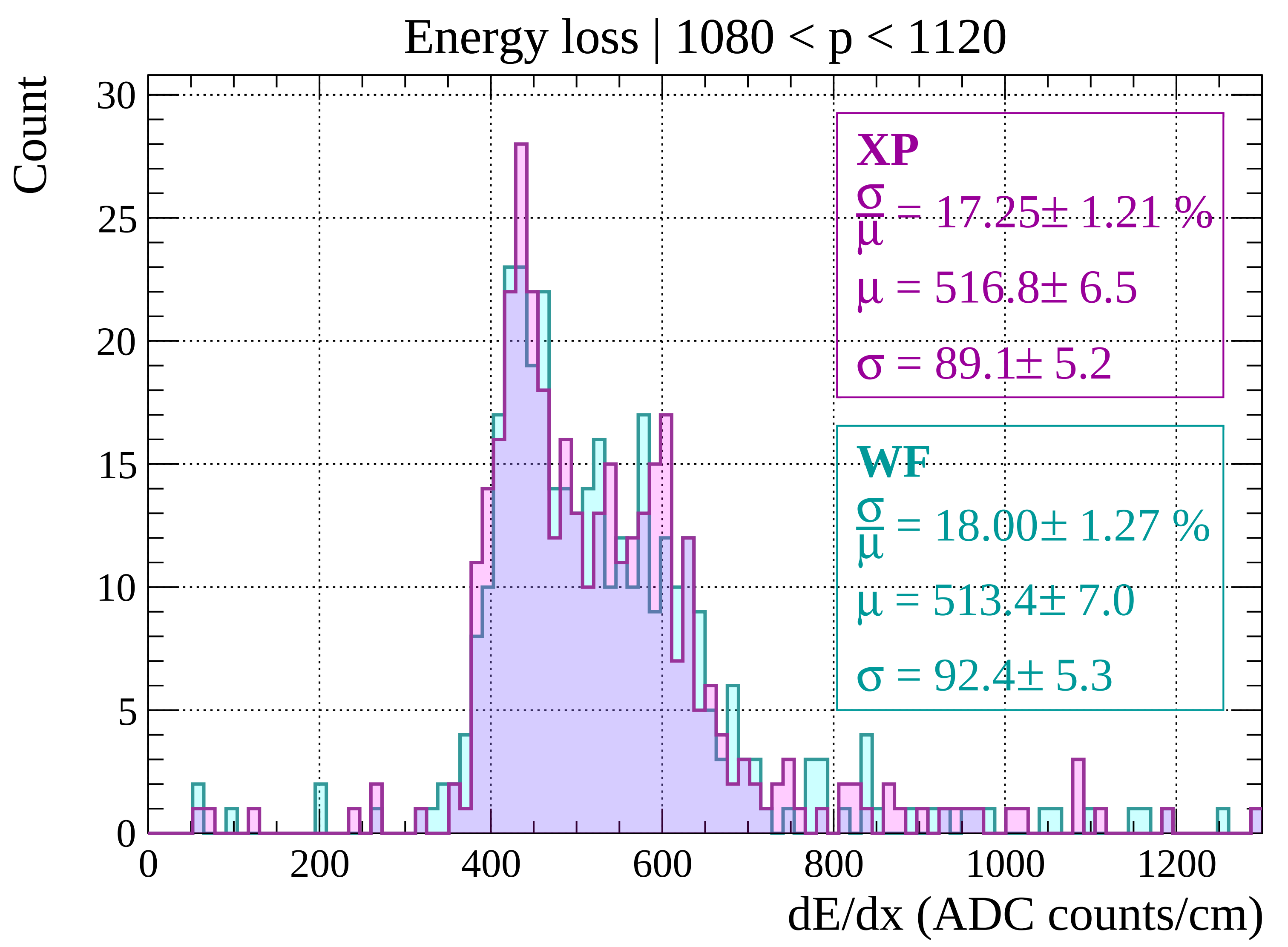

Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 24.30± 1.50 %
μ = 559.9± 8.8
σ = 136.1± 6.2
WF
σ/μ = 24.92± 1.62 %
μ = 558.8± 9.1
σ = 139.3± 6.8

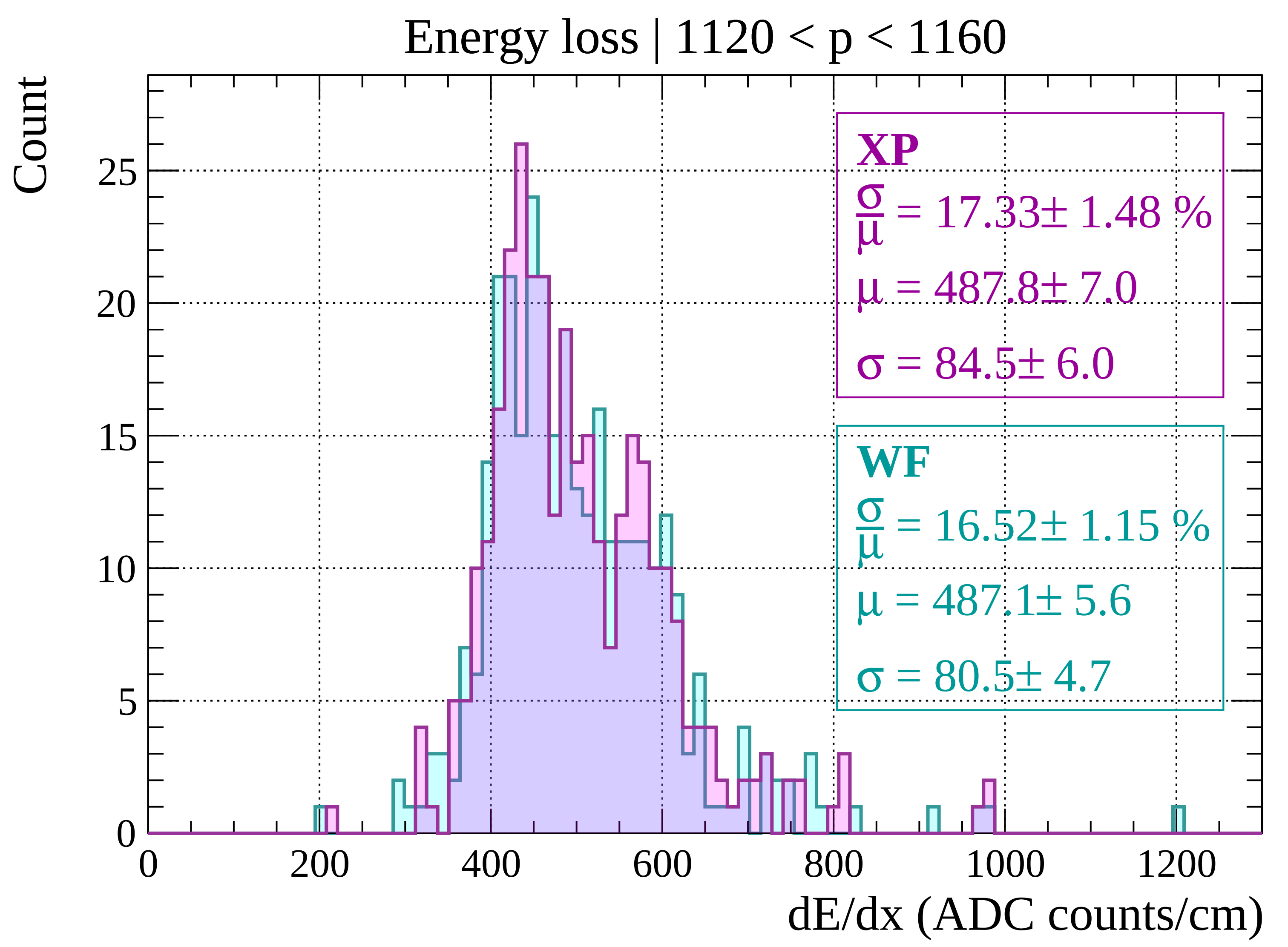


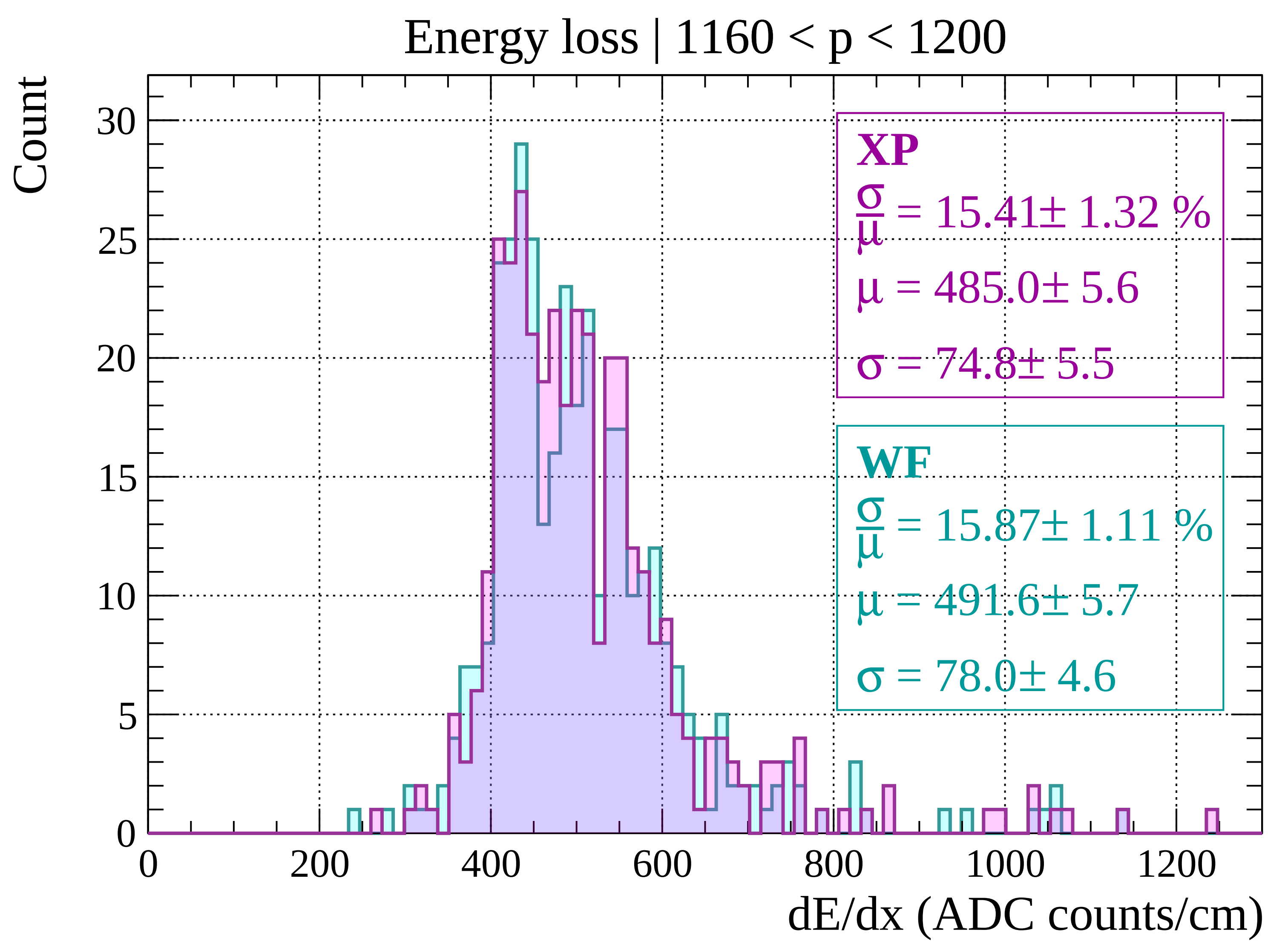
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 20.42± 1.33 %
μ = 540.3± 7.6
σ = 110.3± 5.6
WF
$\frac{\sigma}{\mu}$ = 22.78± 1.73 %
μ = 525.3± 8.6
σ = 119.7± 7.1
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200



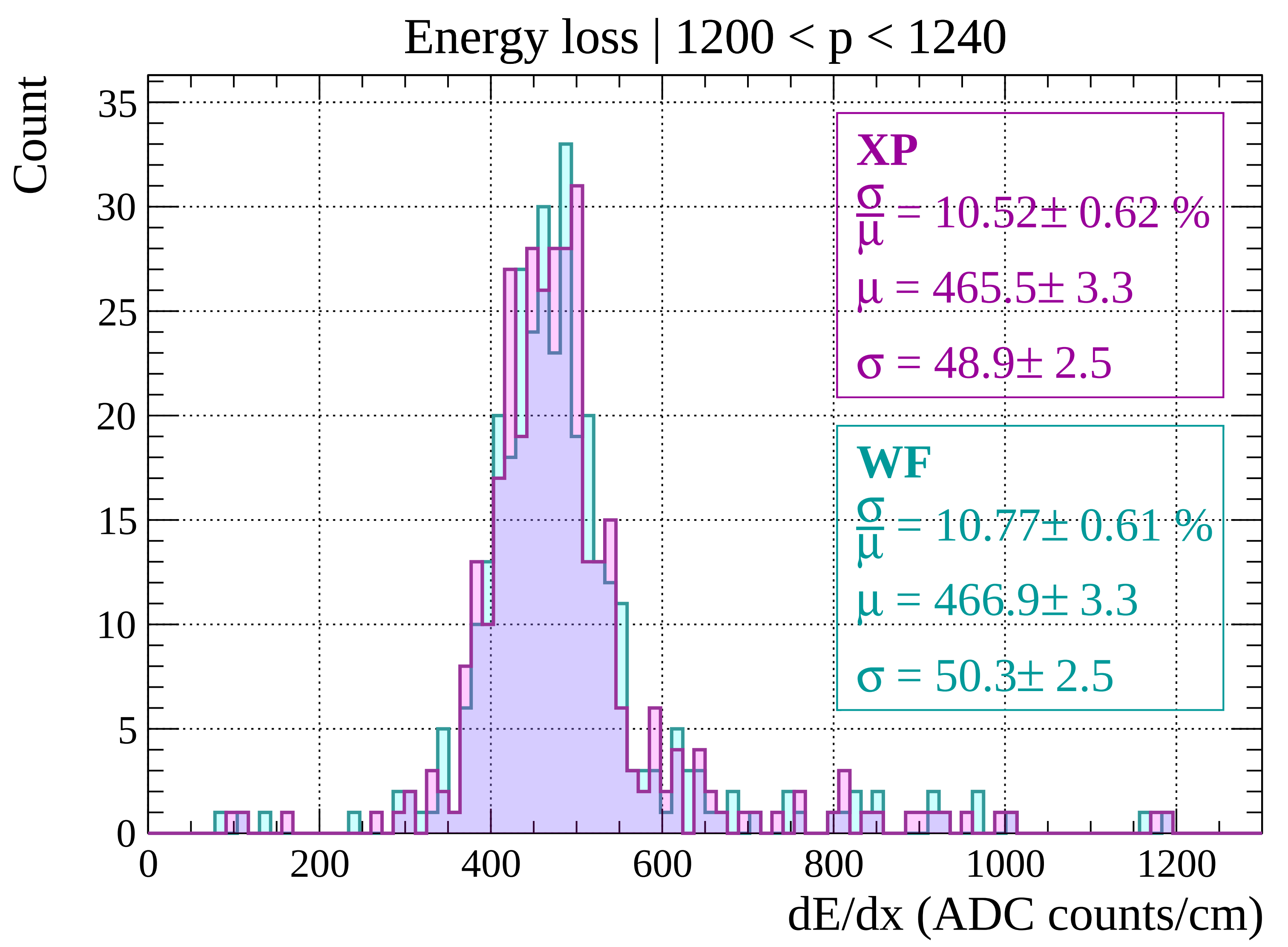
Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
$\frac{\sigma}{\mu}$ = 17.88± 1.19 %
μ = 508.7± 6.3
σ = 91.0± 4.9
WF
$\frac{\sigma}{\mu}$ = 18.17± 1.37 %
μ = 513.7± 7.4
σ = 93.3± 5.7

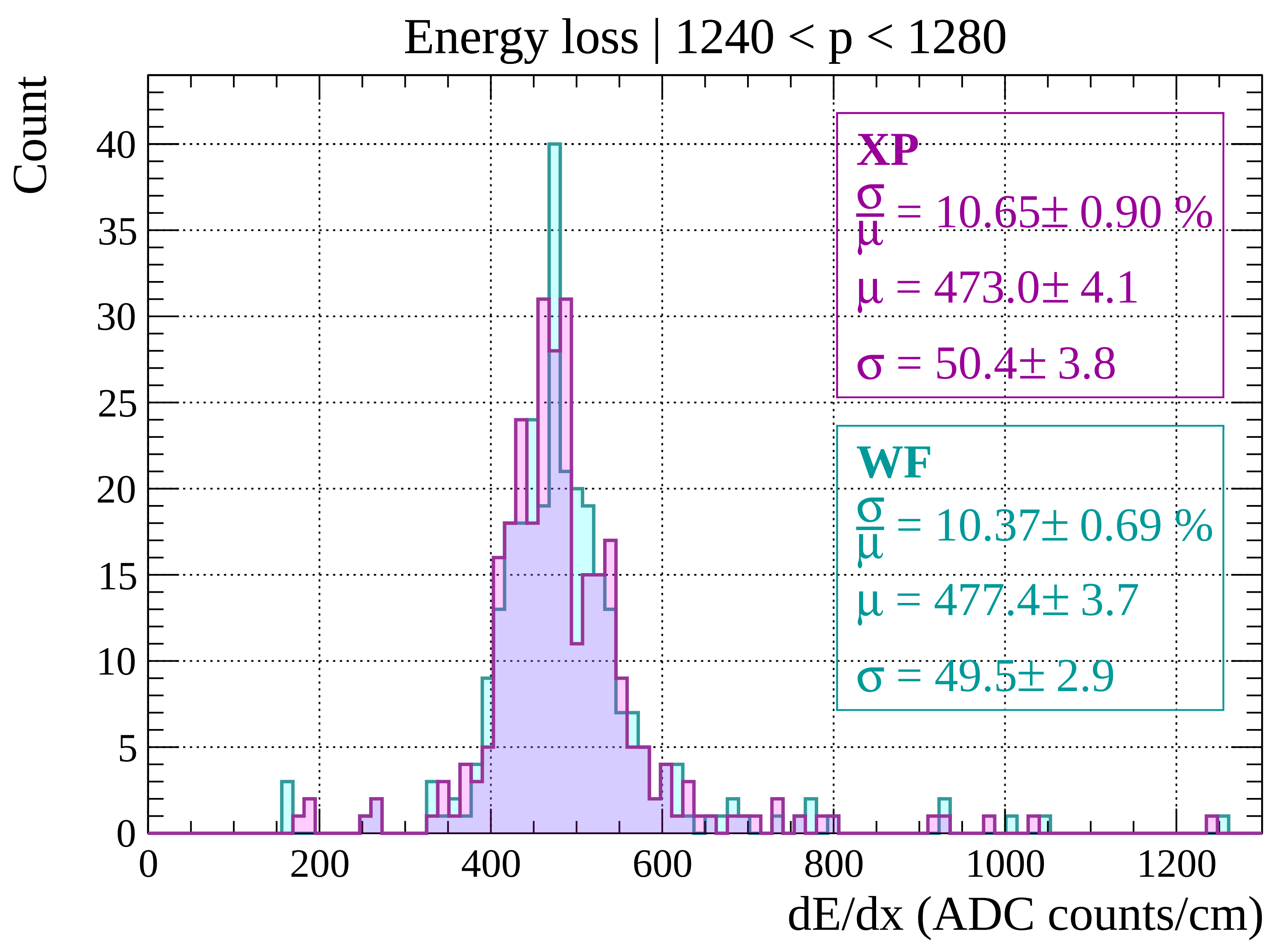

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
$\frac{\sigma}{\mu}$ = 17.25± 1.21 %
μ = 516.8± 6.5
σ = 89.1± 5.2
WF
$\frac{\sigma}{\mu}$ = 18.00± 1.27 %
μ = 513.4± 7.0
σ = 92.4± 5.3


Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 17.33± 1.48 %
μ = 487.8± 7.0
σ = 84.5± 6.0
WF
σ/μ = 16.52± 1.15 %
μ = 487.1± 5.6
σ = 80.5± 4.7

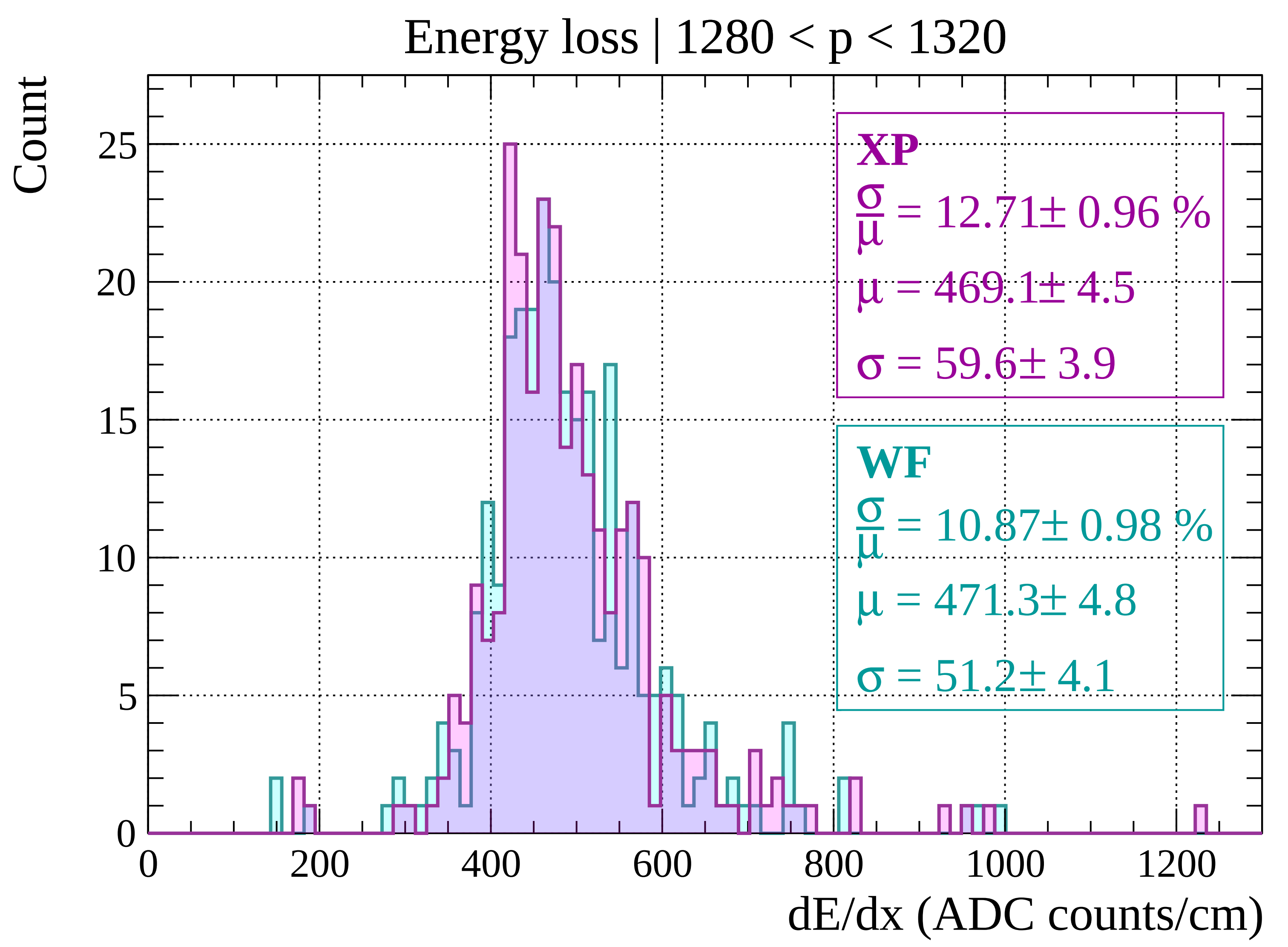


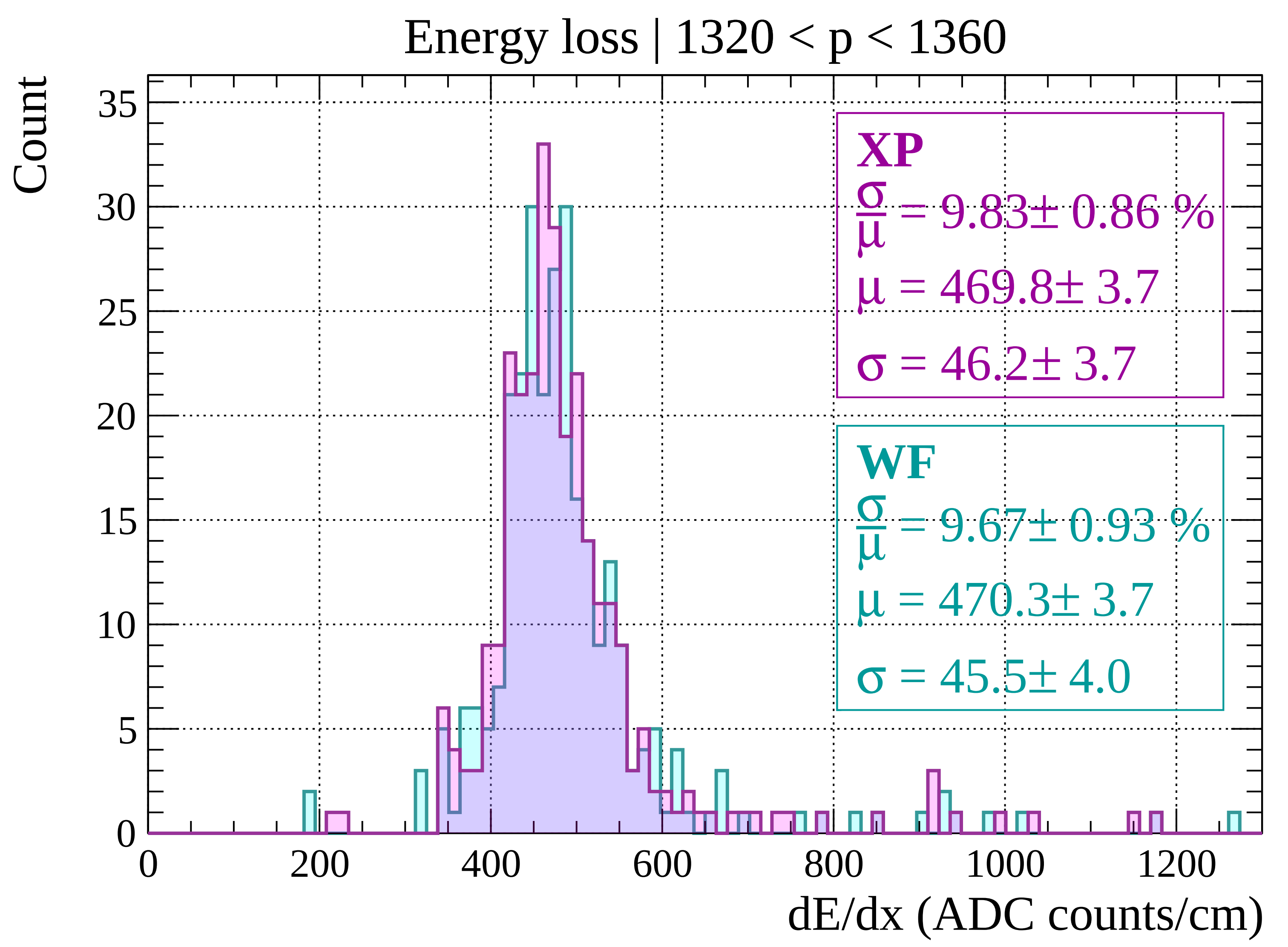
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
XP
σ/μ = 15.41± 1.32 %
μ = 485.0± 5.6
σ = 74.8± 5.5
WF
σ/μ = 15.87± 1.11 %
μ = 491.6± 5.7
σ = 78.0± 4.6


Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.52 ± 0.62 %
μ = 465.5 ± 3.3
σ = 48.9 ± 2.5
WF
σ/μ = 10.77 ± 0.61 %
μ = 466.9 ± 3.3
σ = 50.3 ± 2.5

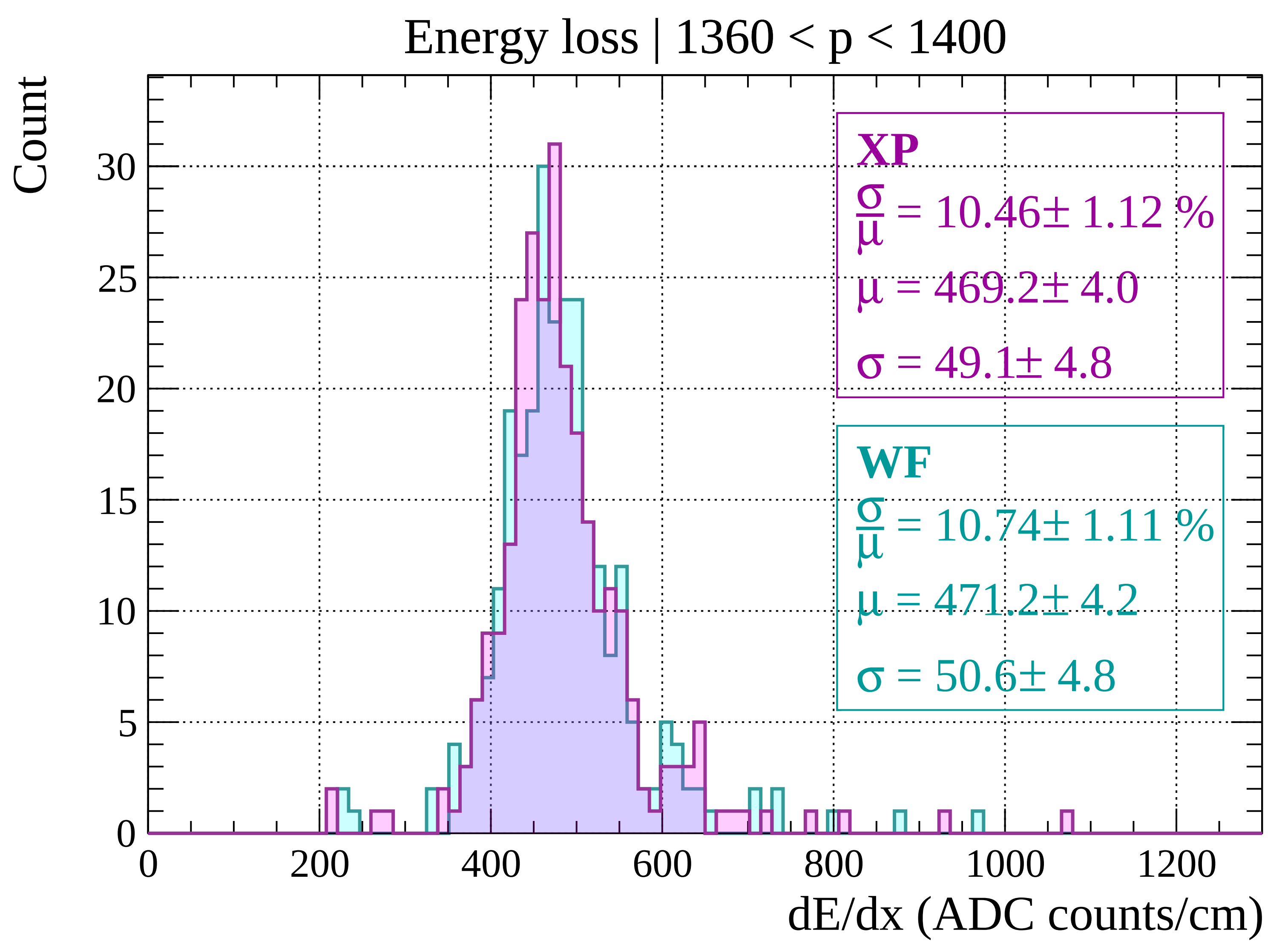


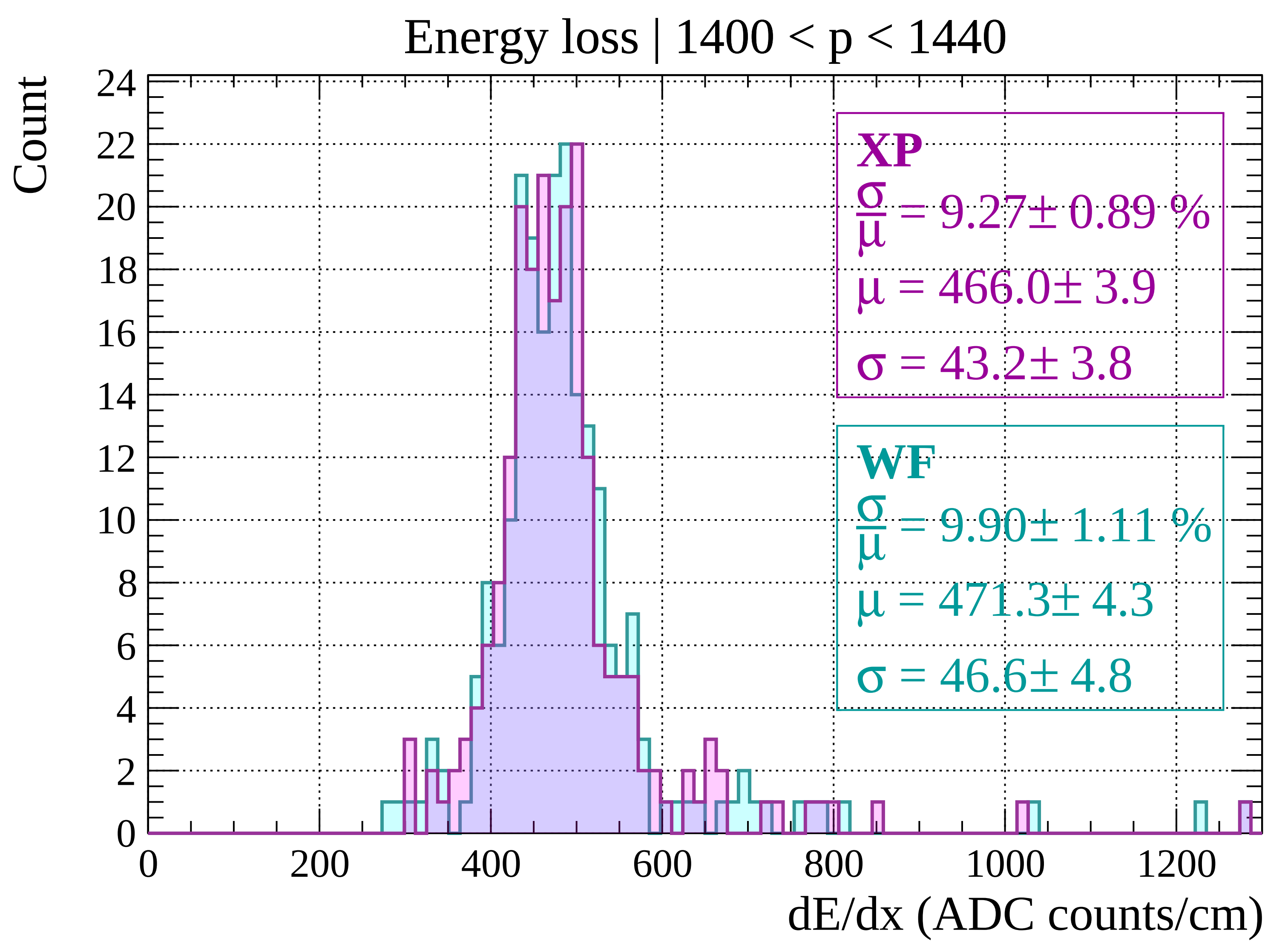
Energy loss | 1240 < p < 1280
Count
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.65± 0.90 %
μ = 473.0± 4.1
σ = 50.4± 3.8
WF
σ/μ = 10.37± 0.69 %
μ = 477.4± 3.7
σ = 49.5± 2.9

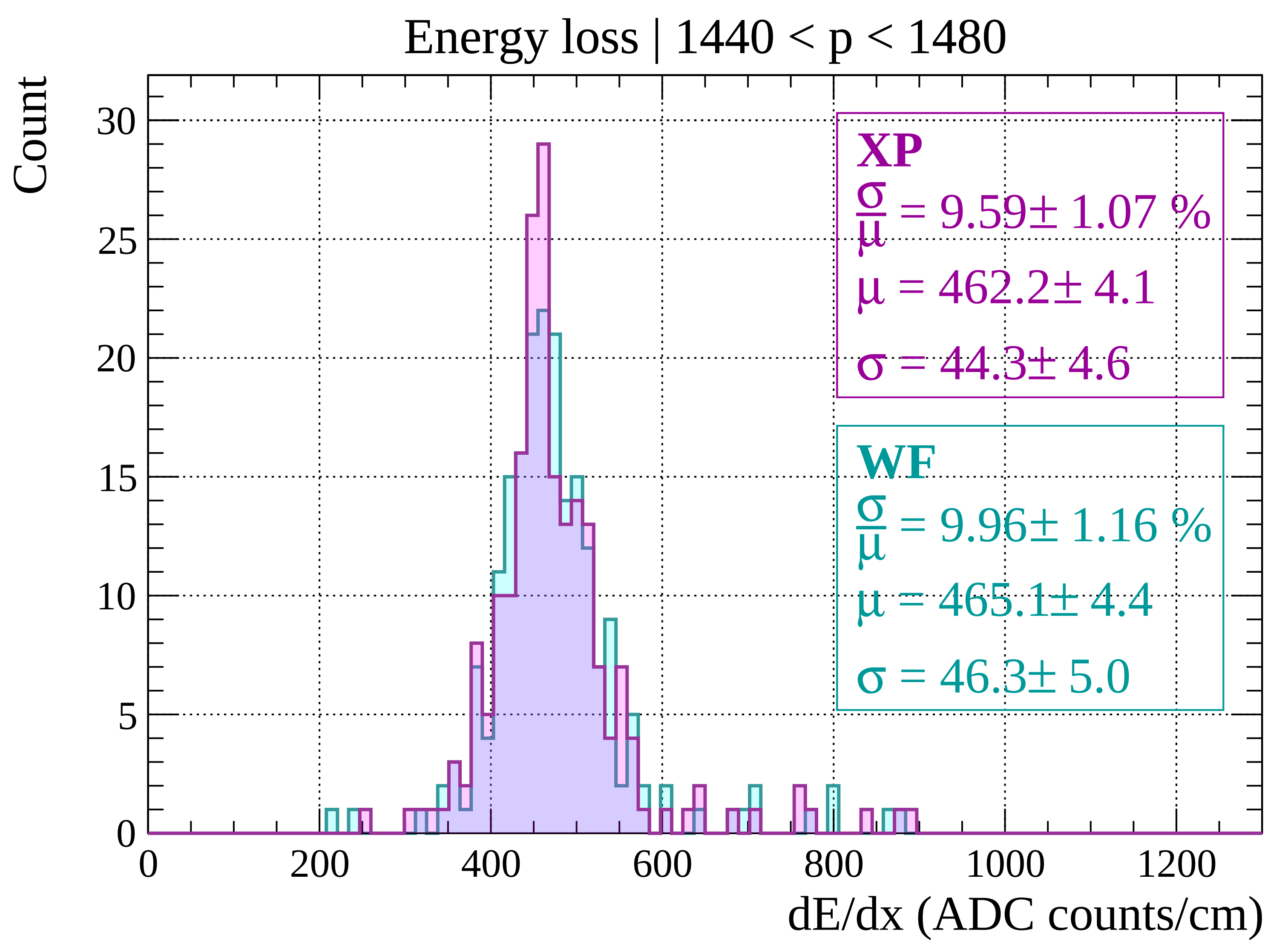

Energy loss | 1280 < p < 1320
Count
0
5
10
15
20
25
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.71± 0.96 %
μ = 469.1± 4.5
σ = 59.6± 3.9
WF
σ/μ = 10.87± 0.98 %
μ = 471.3± 4.8
σ = 51.2± 4.1



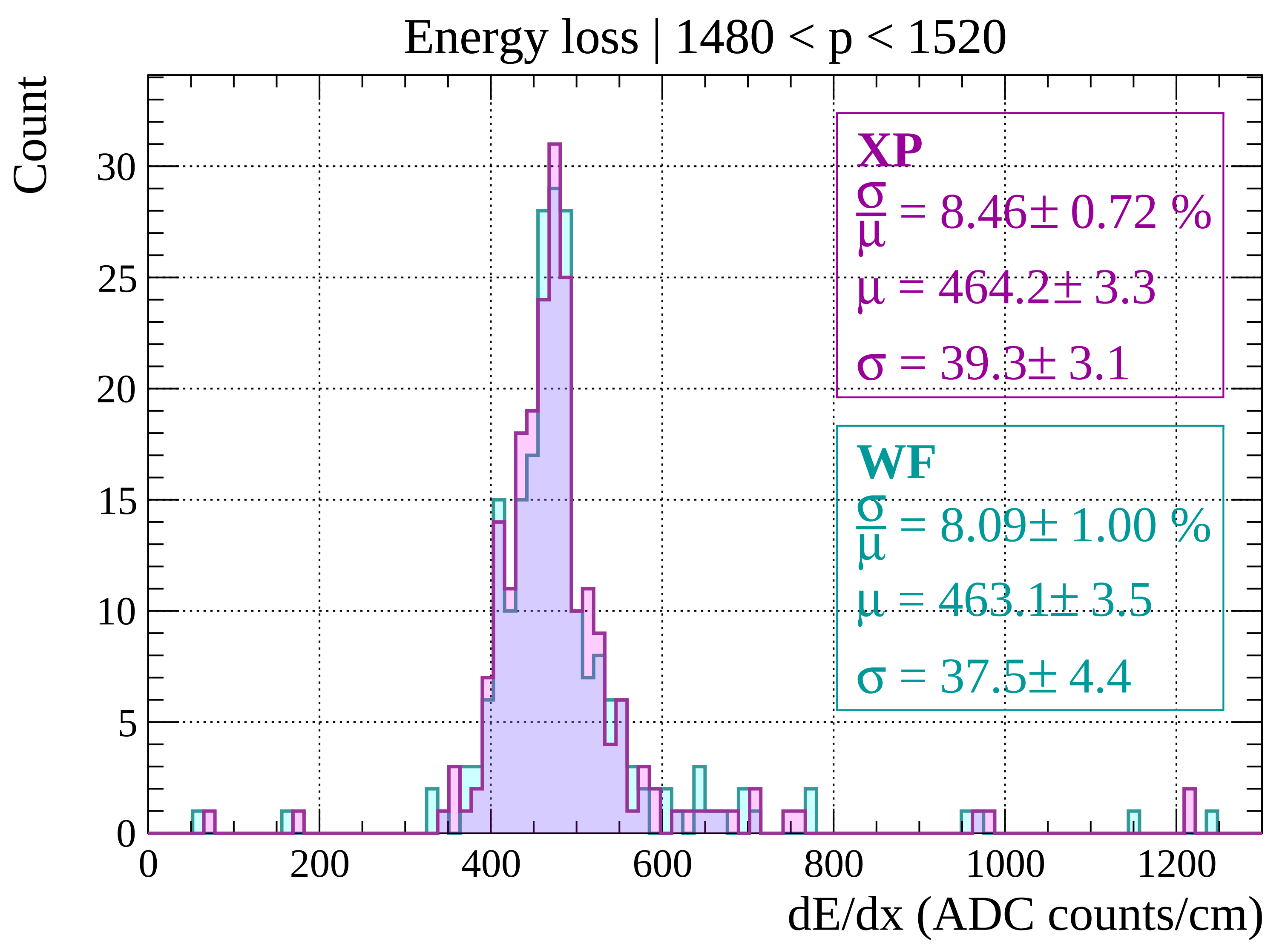
Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.83± 0.86 %
μ = 469.8± 3.7
σ = 46.2± 3.7
WF
$\frac{\sigma}{\mu}$ = 9.67± 0.93 %
μ = 470.3± 3.7
σ = 45.5± 4.0


Energy loss | 1360 < p < 1400
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.46± 1.12 %
μ = 469.2± 4.0
σ = 49.1± 4.8
WF
σ/μ = 10.74± 1.11 %
μ = 471.2± 4.2
σ = 50.6± 4.8

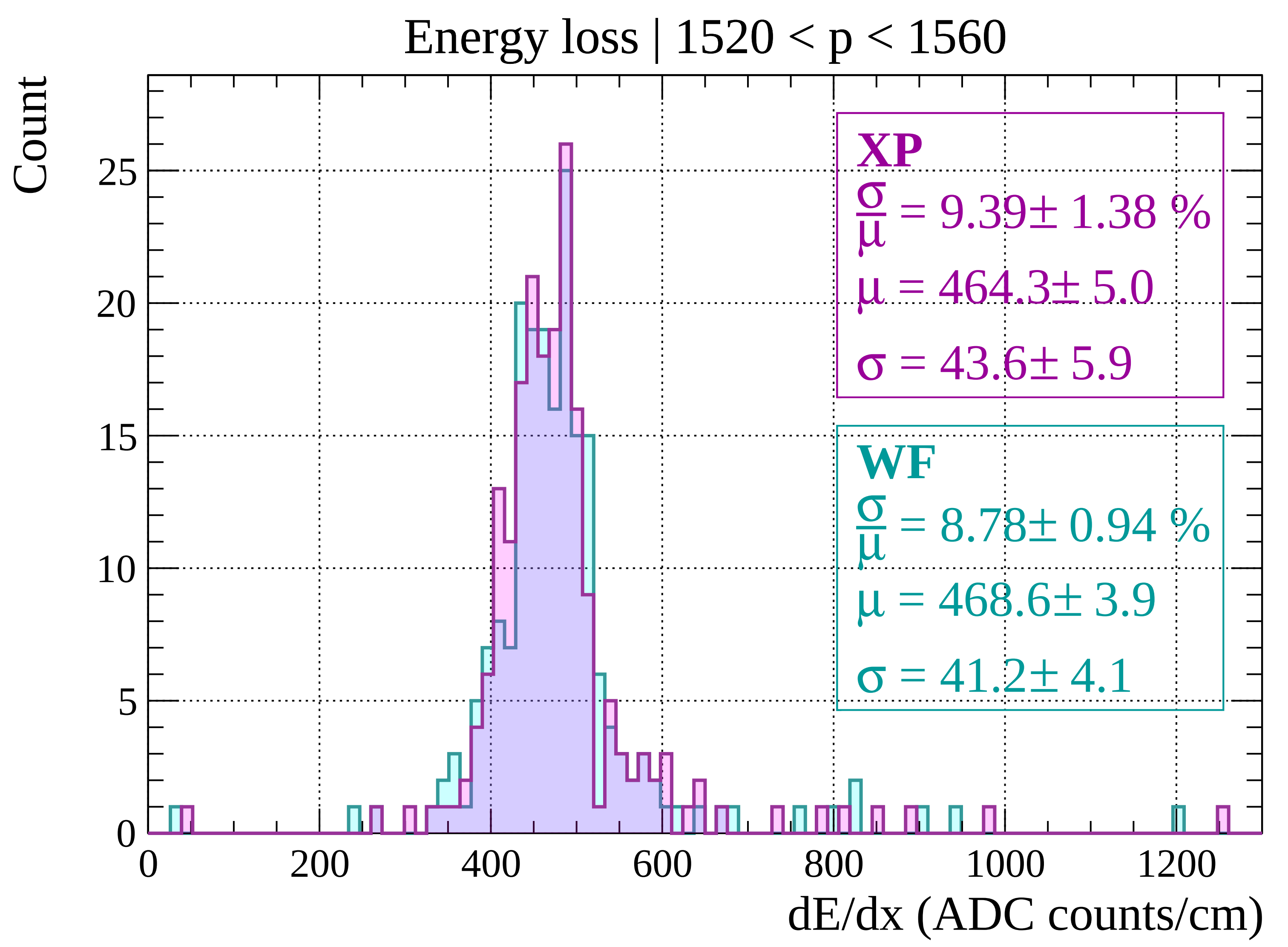

Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
24
XP
$\frac{\sigma}{\mu}$ = 9.27± 0.89 %
μ = 466.0± 3.9
σ = 43.2± 3.8
WF
$\frac{\sigma}{\mu}$ = 9.90± 1.11 %
μ = 471.3± 4.3
σ = 46.6± 4.8



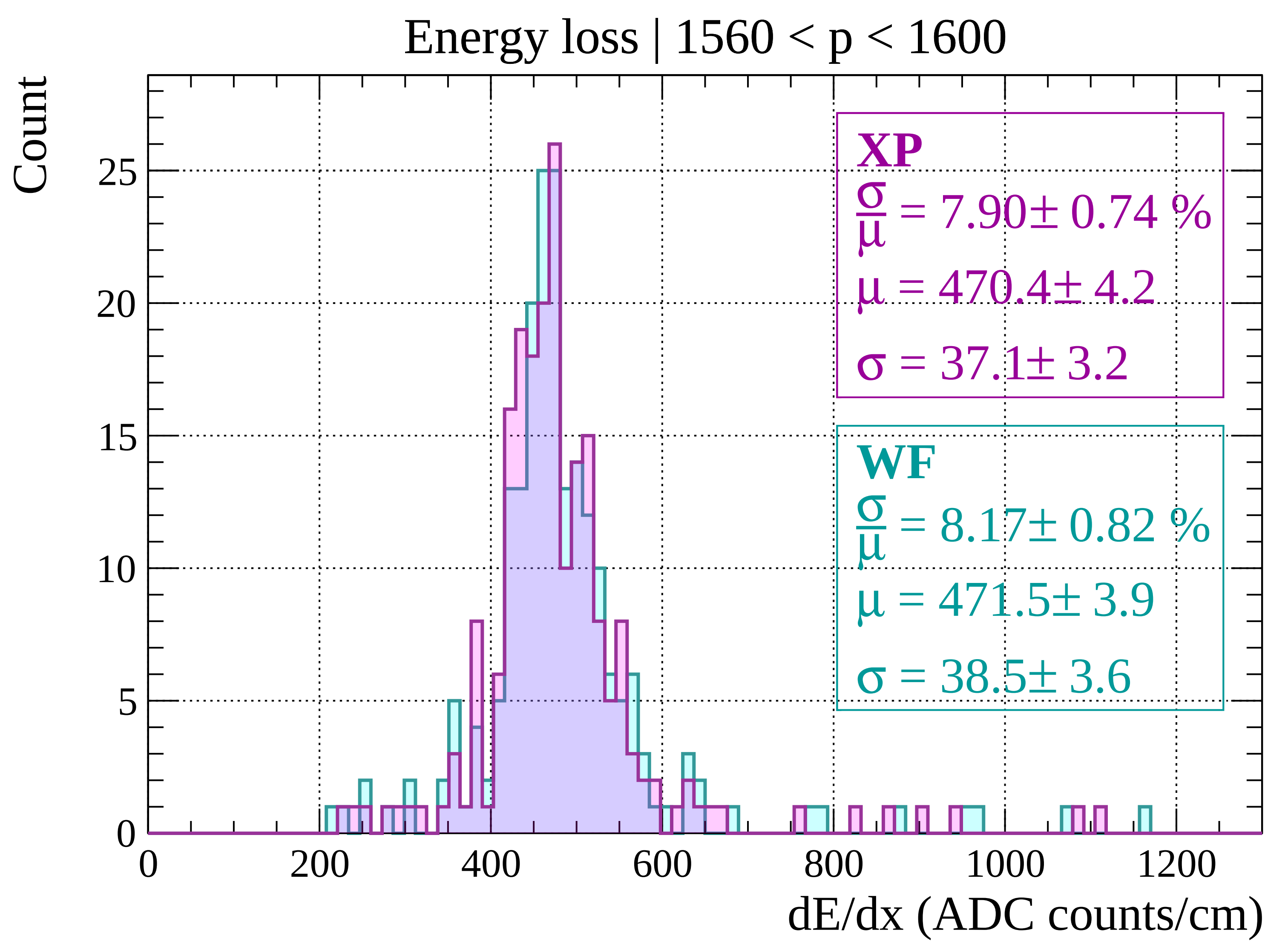
Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 9.59± 1.07 %
μ = 462.2± 4.1
σ = 44.3± 4.6
WF
σ/μ = 9.96± 1.16 %
μ = 465.1± 4.4
σ = 46.3± 5.0

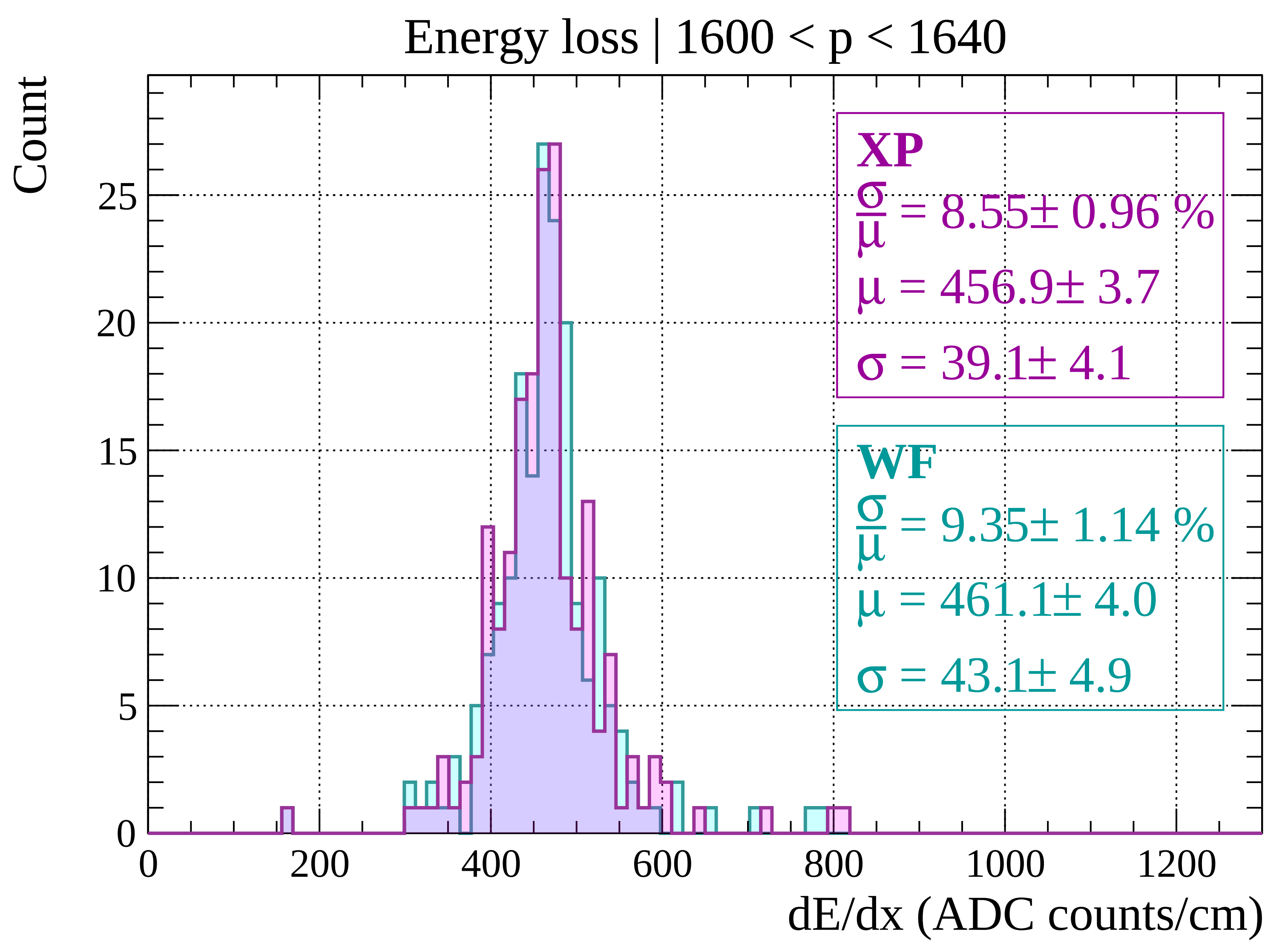


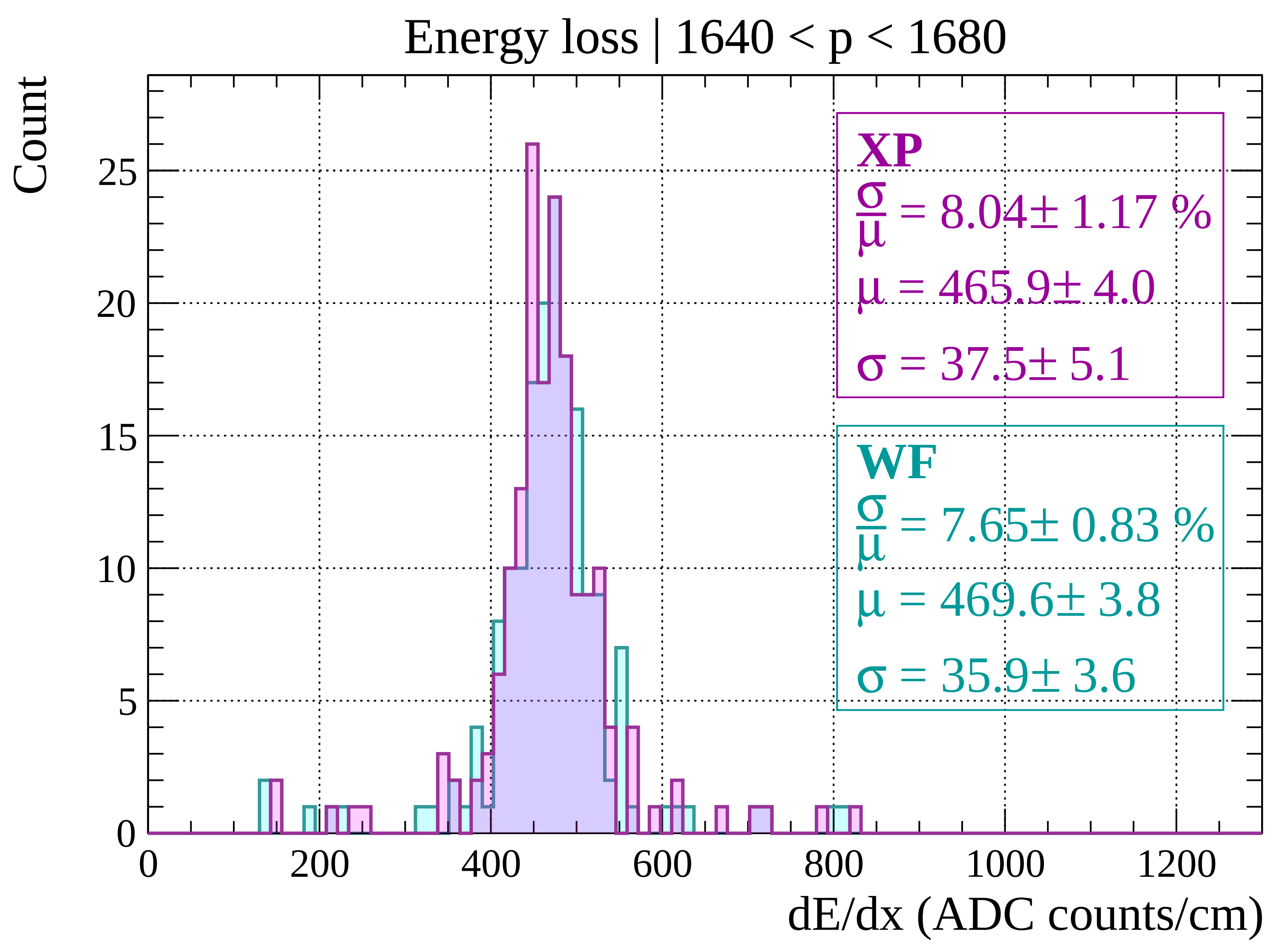
Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.46± 0.72 %
μ = 464.2± 3.3
σ = 39.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 8.09± 1.00 %
μ = 463.1± 3.5
σ = 37.5± 4.4
0
200
400
600
800
1000
1200
5
10
15
20
25
30



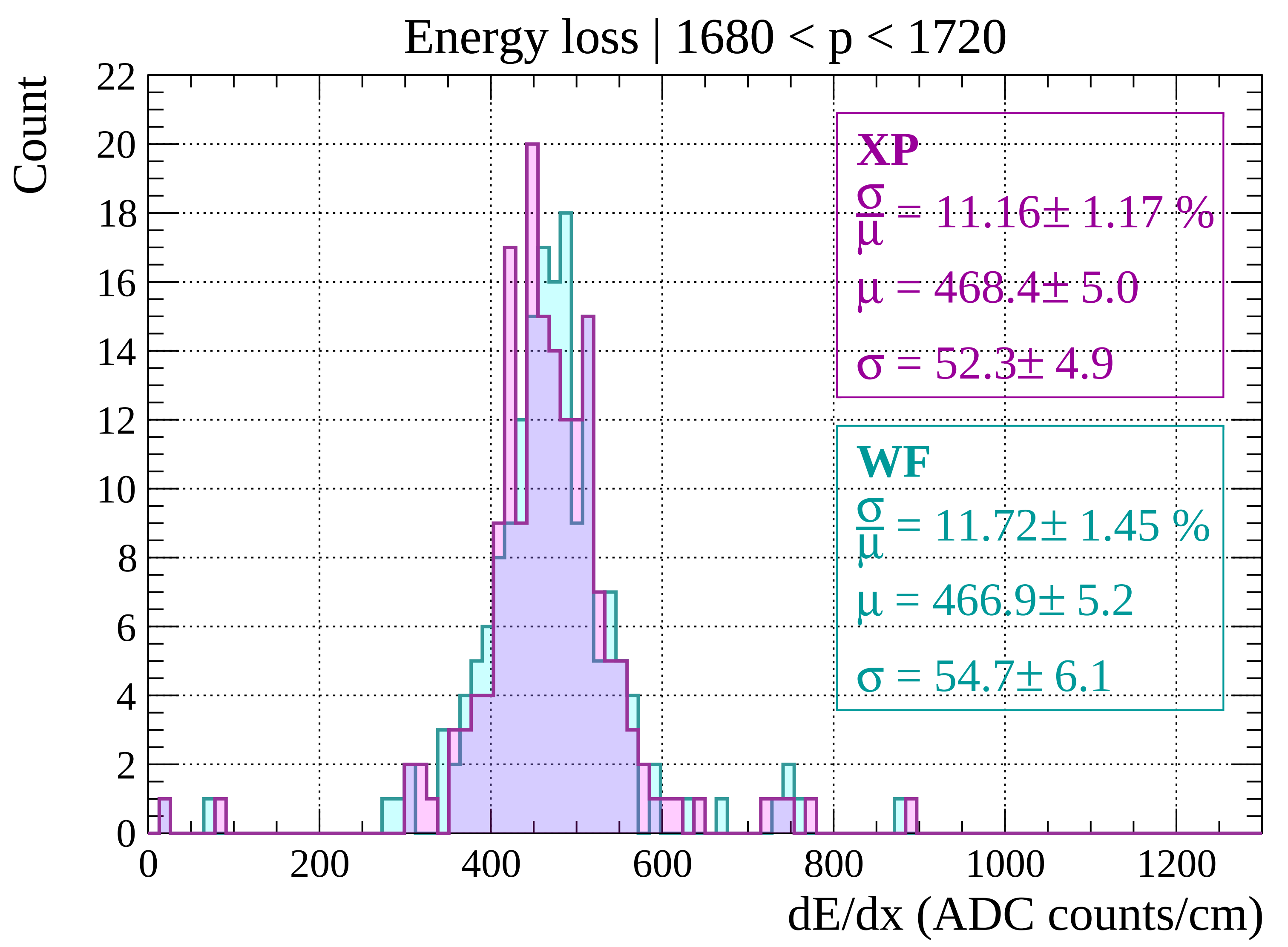
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 9.39± 1.38 %
μ = 464.3± 5.0
σ = 43.6± 5.9
WF
σ/μ = 8.78± 0.94 %
μ = 468.6± 3.9
σ = 41.2± 4.1


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 7.90± 0.74 %
μ = 470.4± 4.2
σ = 37.1± 3.2
WF
σ/μ = 8.17± 0.82 %
μ = 471.5± 3.9
σ = 38.5± 3.6

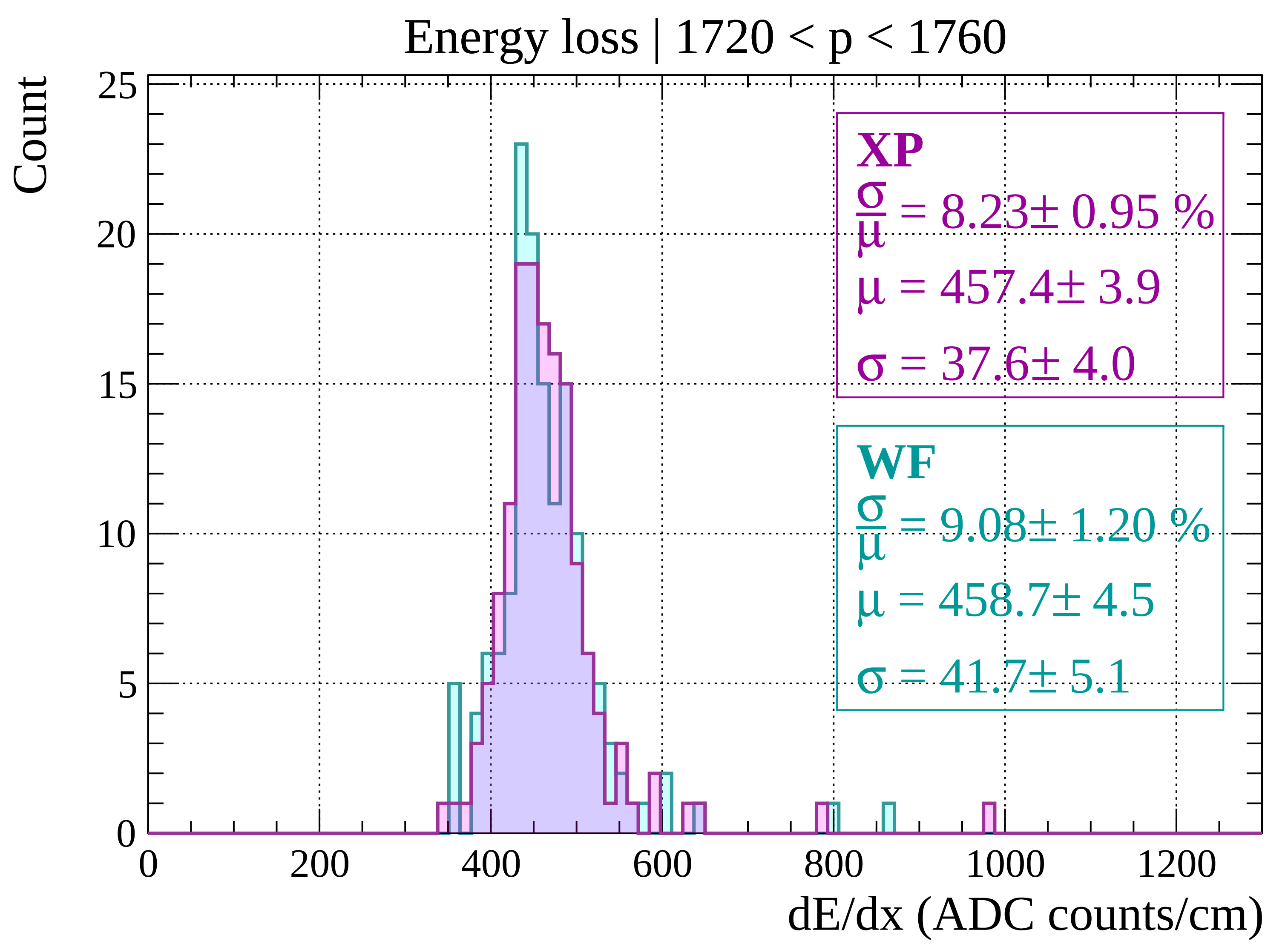

Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 8.55± 0.96 %
μ = 456.9± 3.7
σ = 39.1± 4.1
WF
$\frac{\sigma}{\mu}$ = 9.35± 1.14 %
μ = 461.1± 4.0
σ = 43.1± 4.9



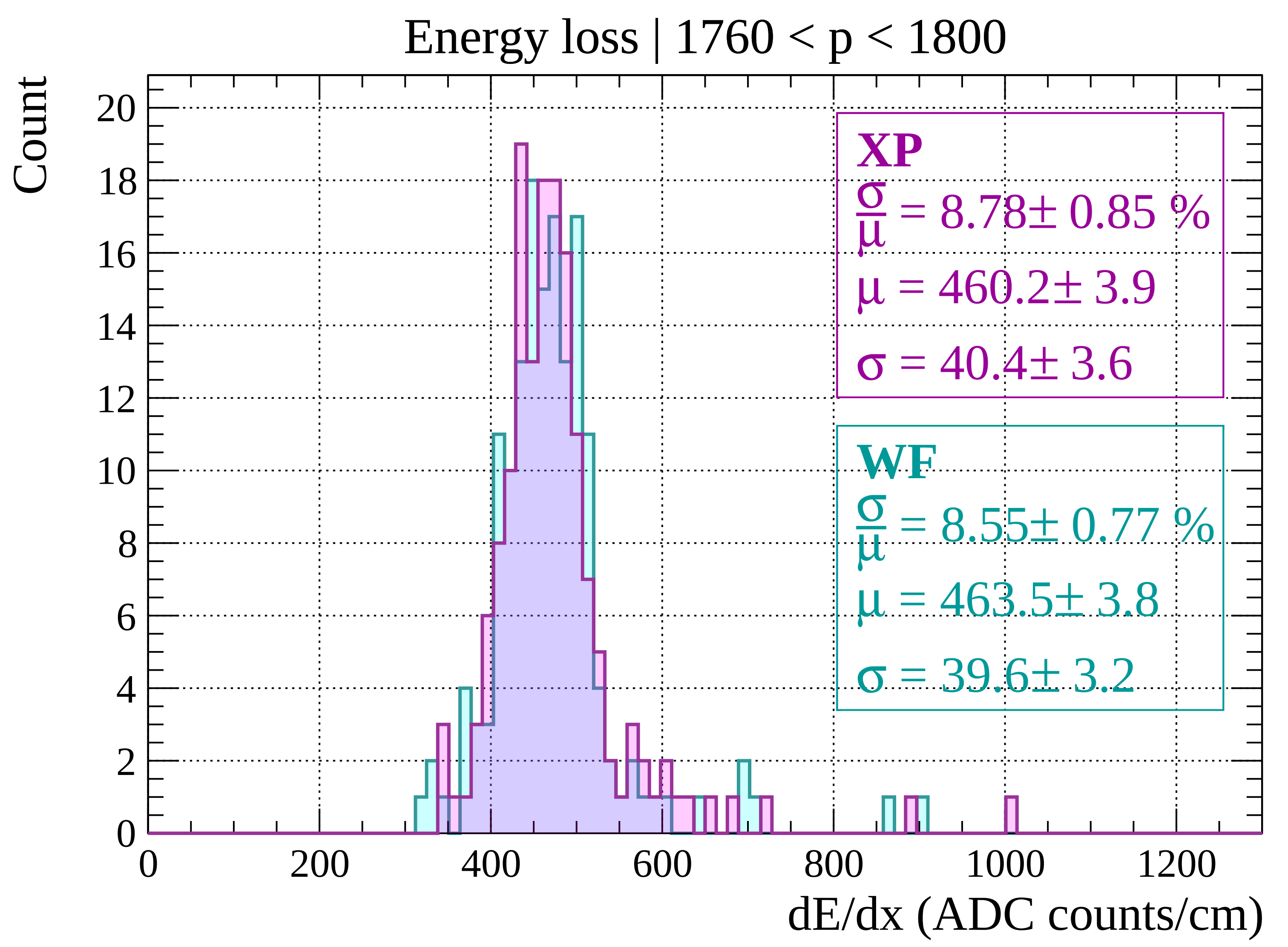
Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 8.04± 1.17 %
μ = 465.9± 4.0
σ = 37.5± 5.1
WF
$\frac{\sigma}{\mu}$ = 7.65± 0.83 %
μ = 469.6± 3.8
σ = 35.9± 3.6

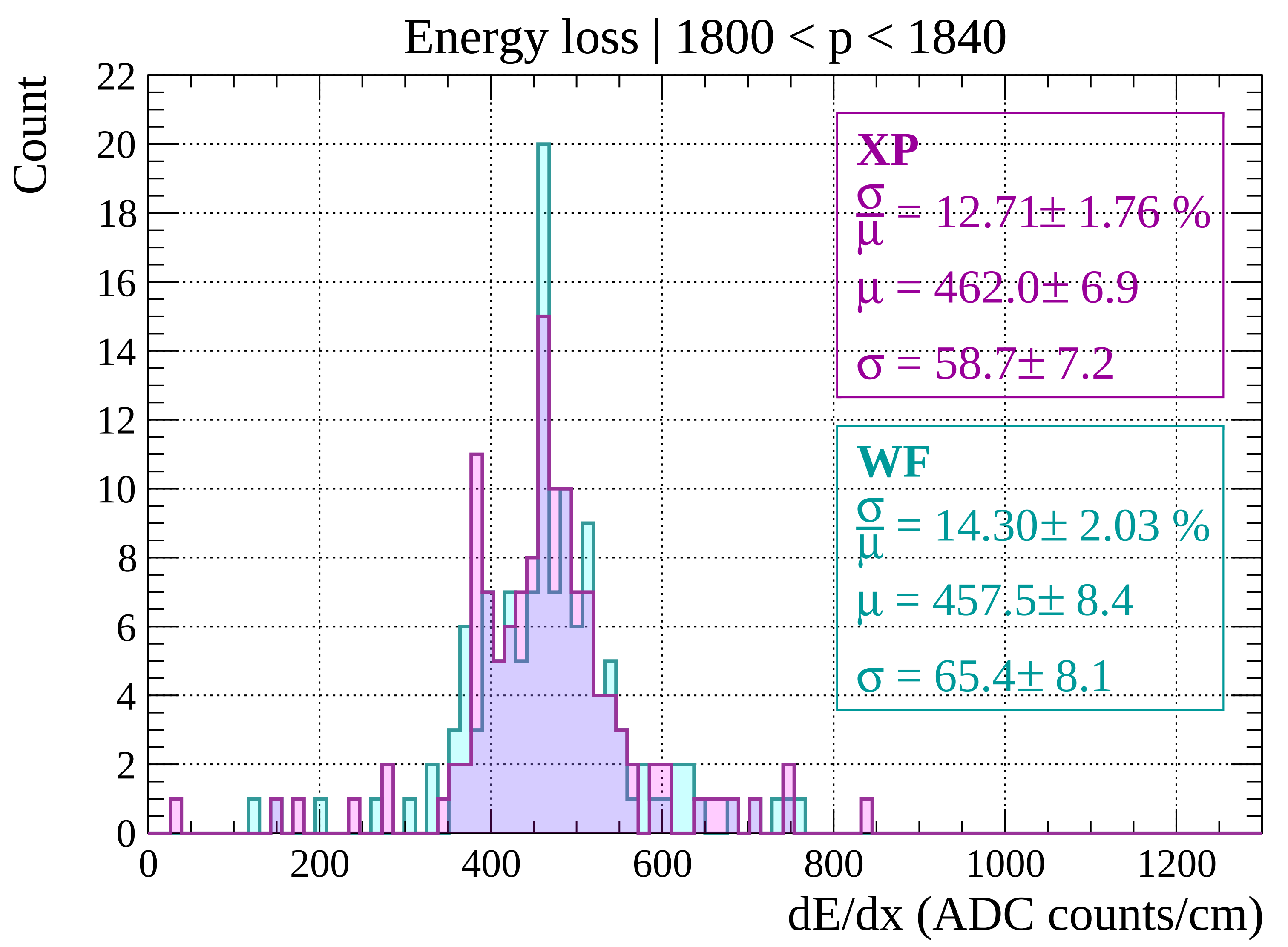


Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
XP
$\frac{\sigma}{\mu}$ = 11.16± 1.17 %
μ = 468.4± 5.0
σ = 52.3± 4.9
WF
$\frac{\sigma}{\mu}$ = 11.72± 1.45 %
μ = 466.9± 5.2
σ = 54.7± 6.1

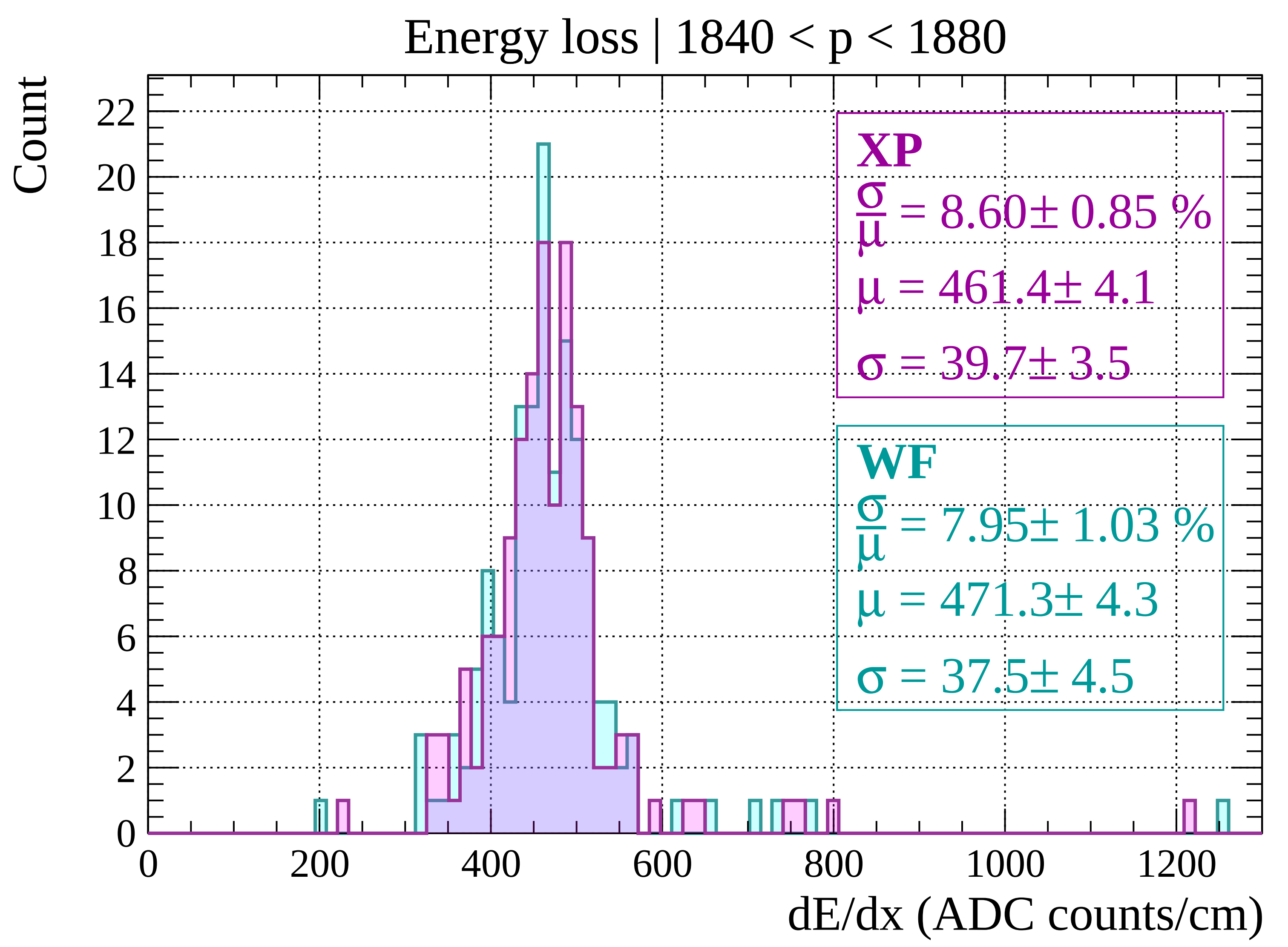


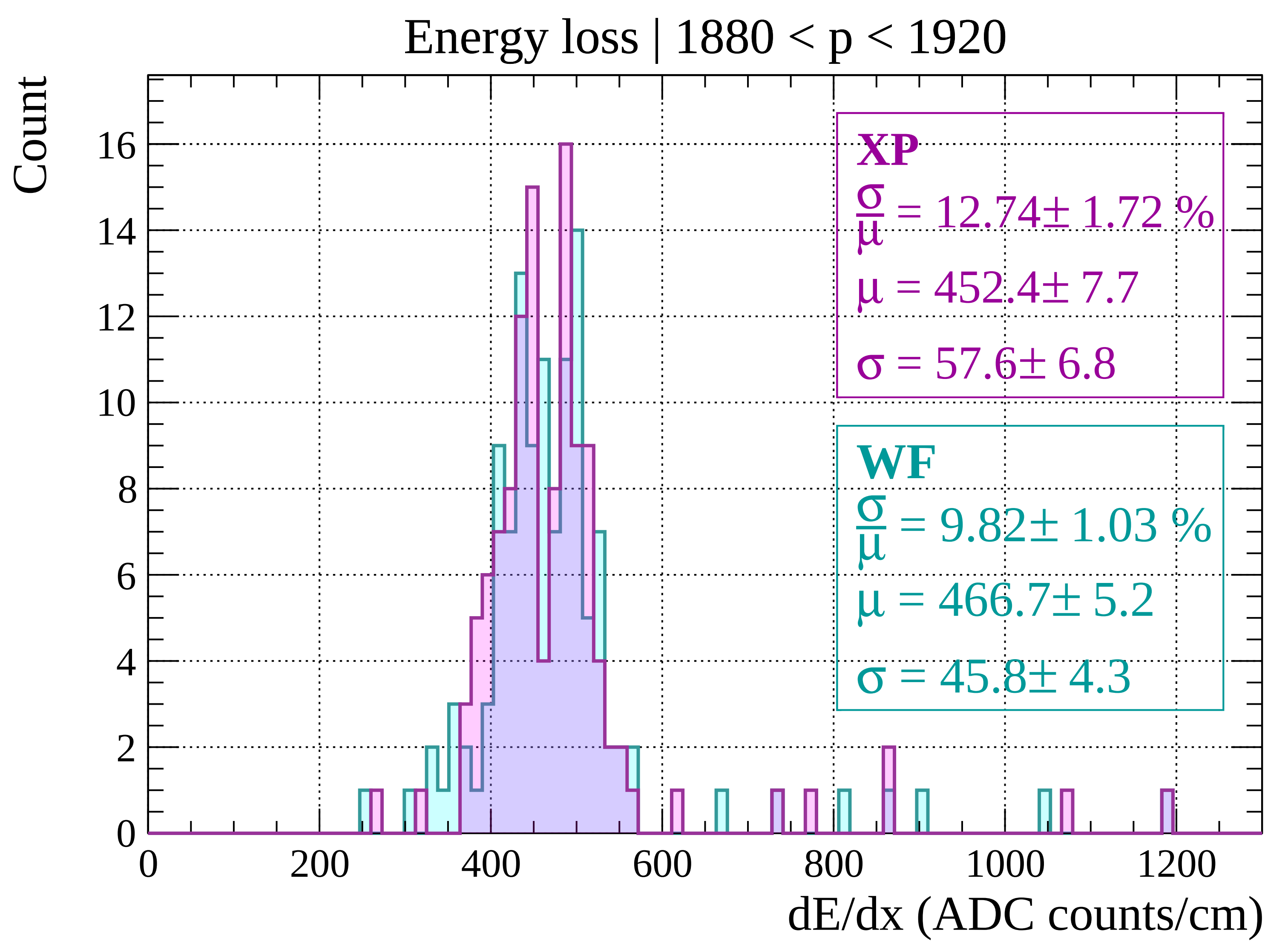
Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 8.23± 0.95 %
μ = 457.4± 3.9
σ = 37.6± 4.0
WF
$\frac{\sigma}{\mu}$ = 9.08± 1.20 %
μ = 458.7± 4.5
σ = 41.7± 5.1


Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP
σ/μ = 8.78± 0.85 %
μ = 460.2± 3.9
σ = 40.4± 3.6
WF
σ/μ = 8.55± 0.77 %
μ = 463.5± 3.8
σ = 39.6± 3.2

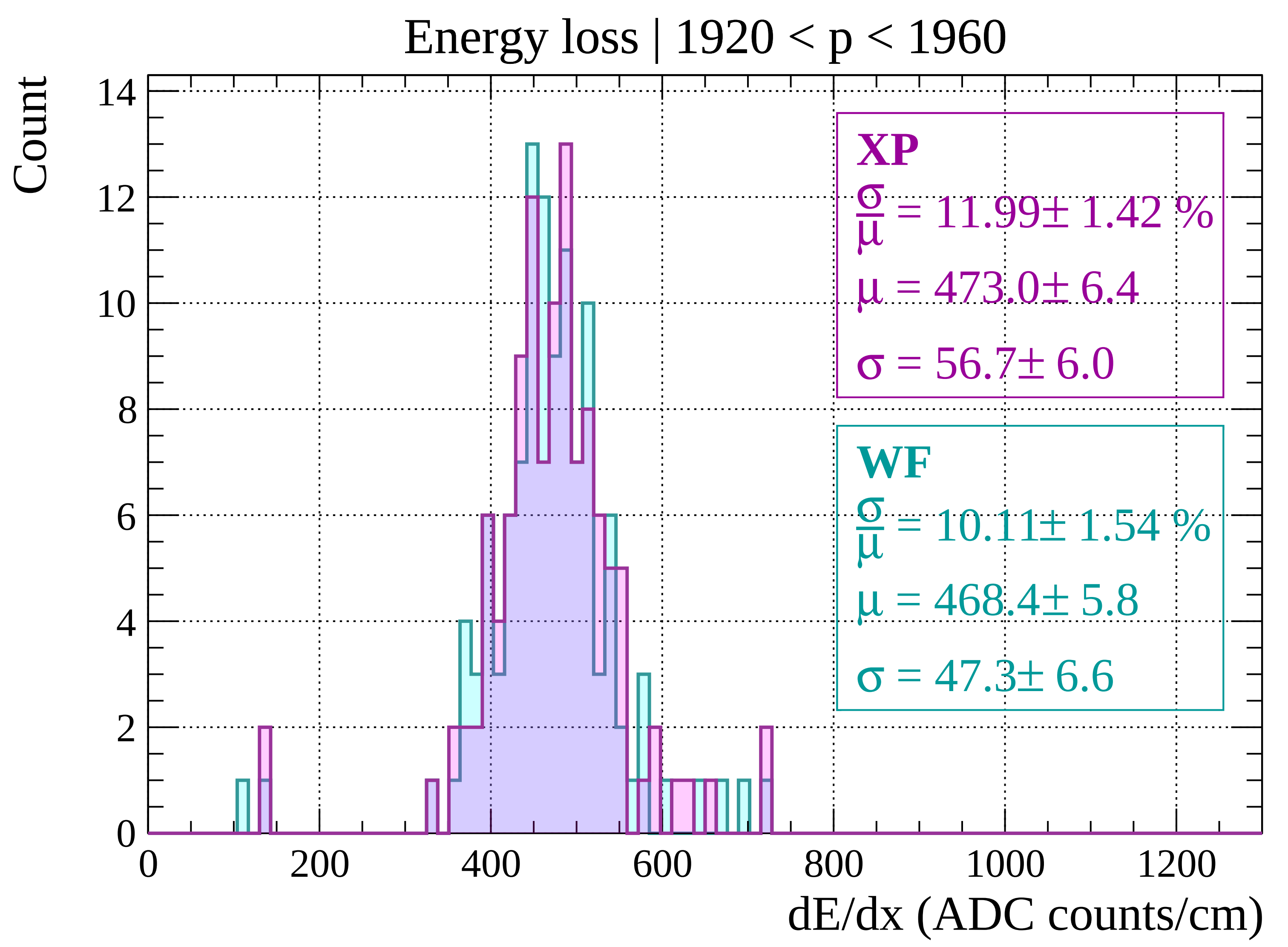


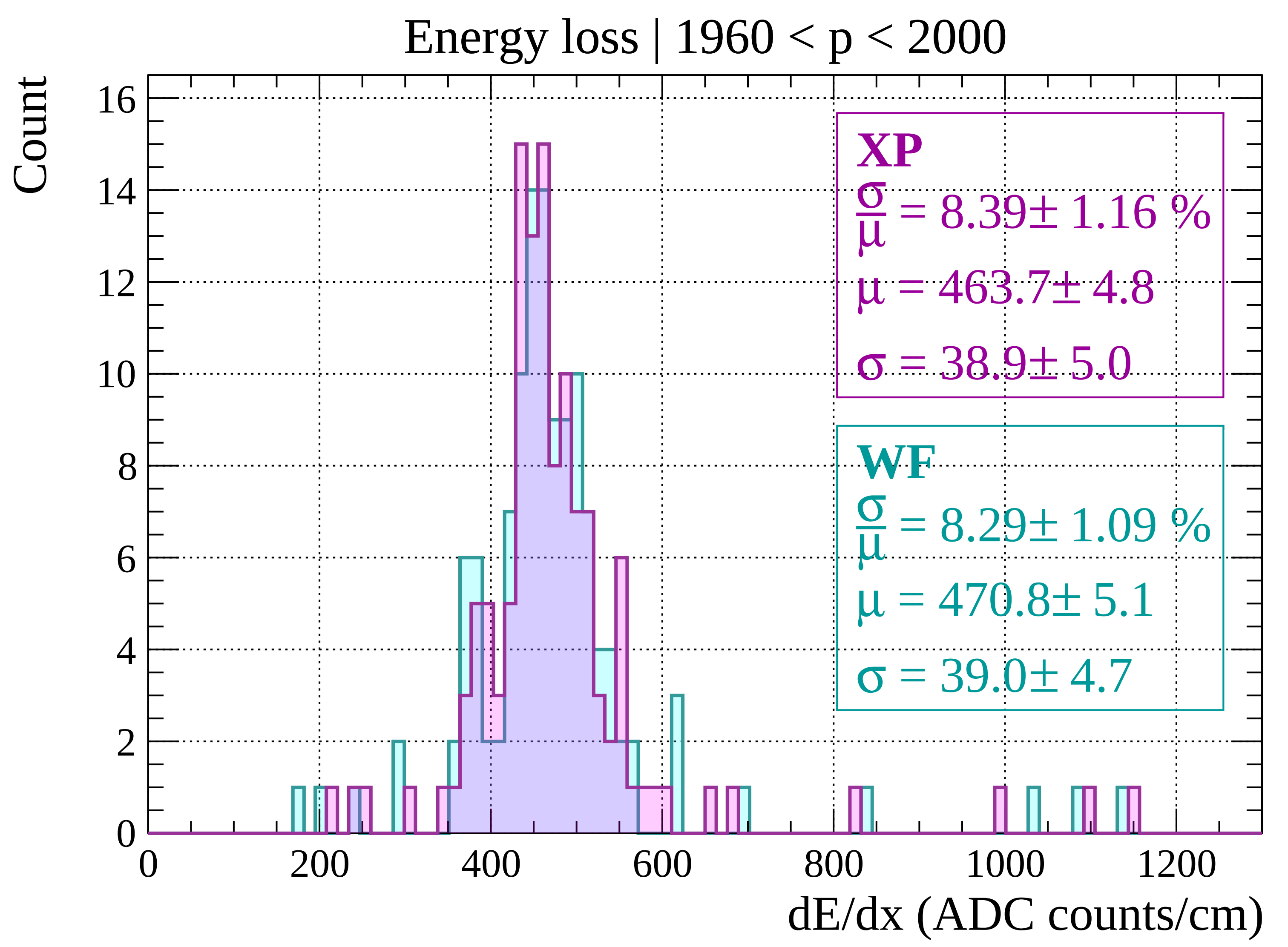
Energy loss | 1800 < p < 1840
Count
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.71± 1.76 %
μ = 462.0± 6.9
σ = 58.7± 7.2
WF
$\frac{\sigma}{\mu}$ = 14.30± 2.03 %
μ = 457.5± 8.4
σ = 65.4± 8.1


Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
XP
$\frac{\sigma}{\mu}$ = 8.60± 0.85 %
μ = 461.4± 4.1
σ = 39.7± 3.5
WF
$\frac{\sigma}{\mu}$ = 7.95± 1.03 %
μ = 471.3± 4.3
σ = 37.5± 4.5

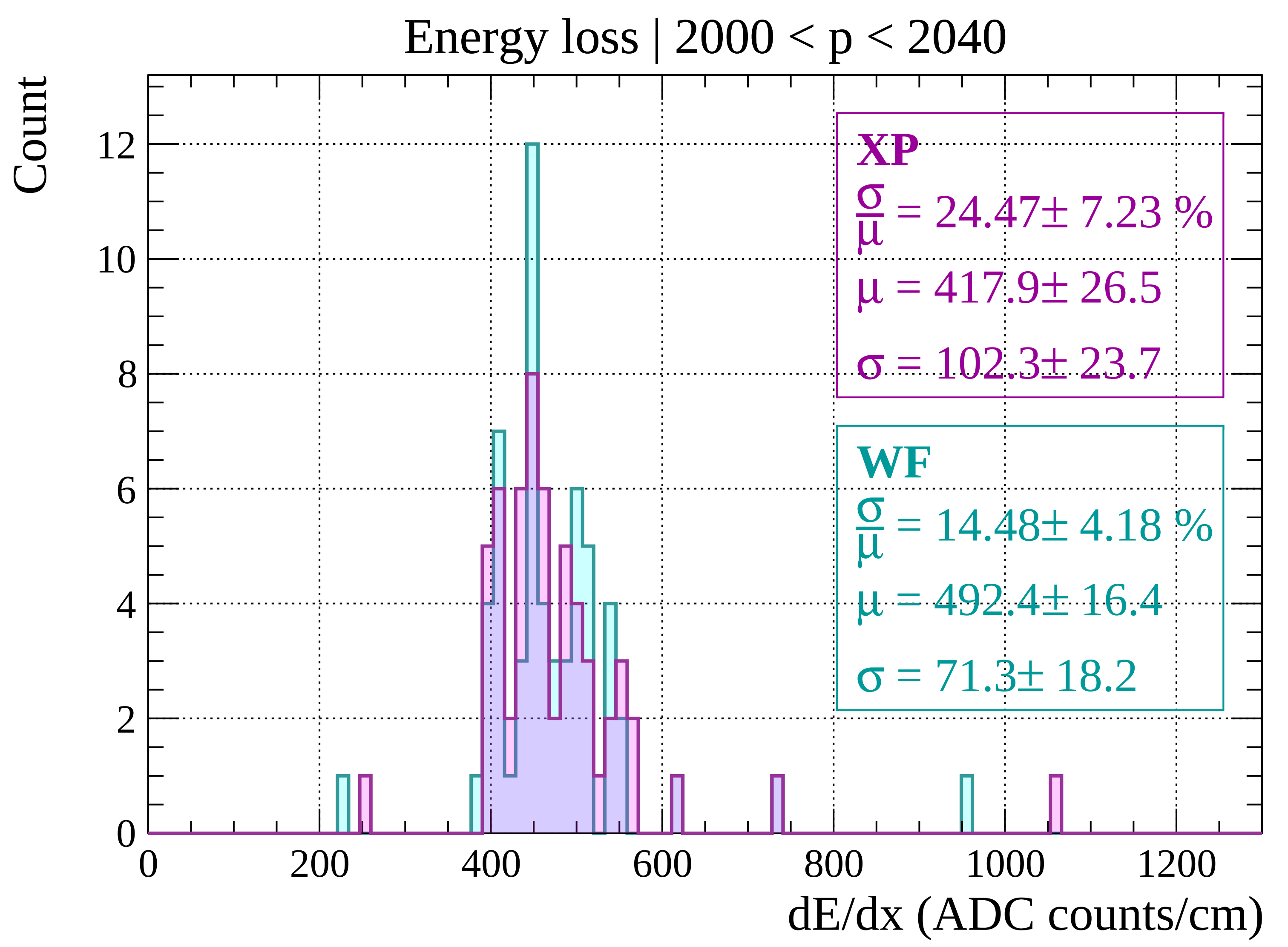


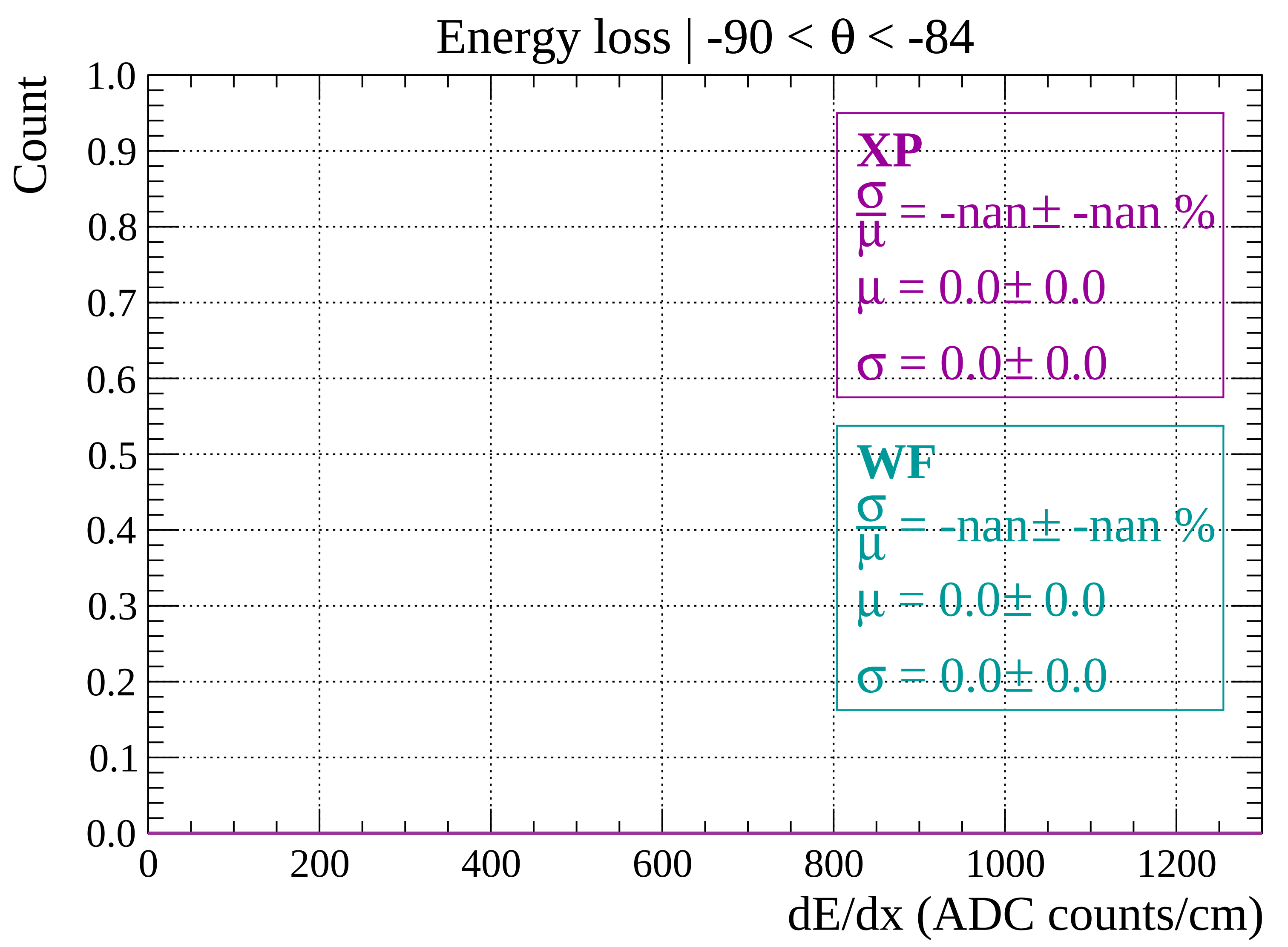
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
XP
$\frac{\sigma}{\mu}$ = 12.74± 1.72 %
μ = 452.4± 7.7
σ = 57.6± 6.8
WF
$\frac{\sigma}{\mu}$ = 9.82± 1.03 %
μ = 466.7± 5.2
σ = 45.8± 4.3


Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.99± 1.42 %
μ = 473.0± 6.4
σ = 56.7± 6.0
WF
$\frac{\sigma}{\mu}$ = 10.11± 1.54 %
μ = 468.4± 5.8
σ = 47.3± 6.6

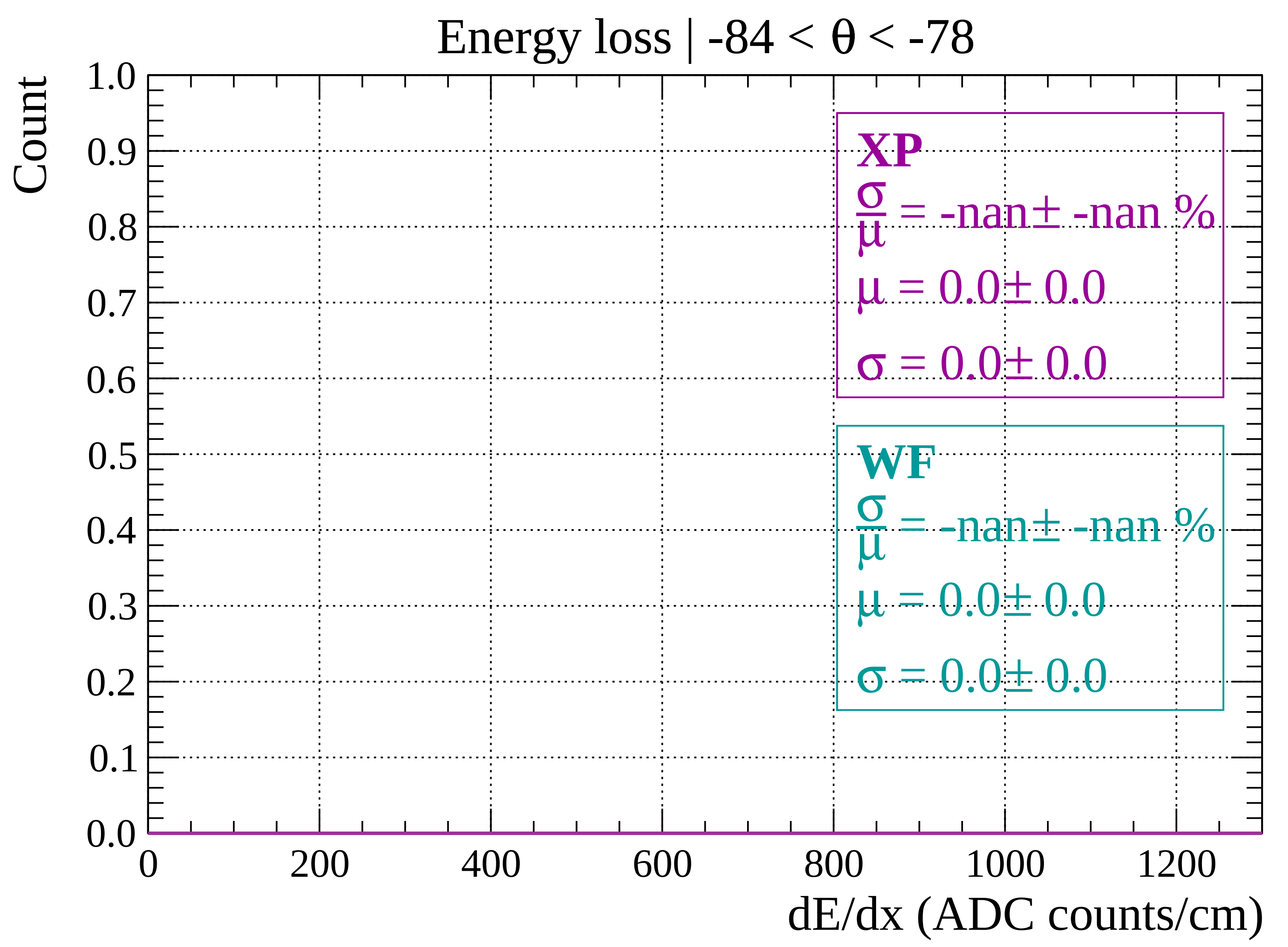


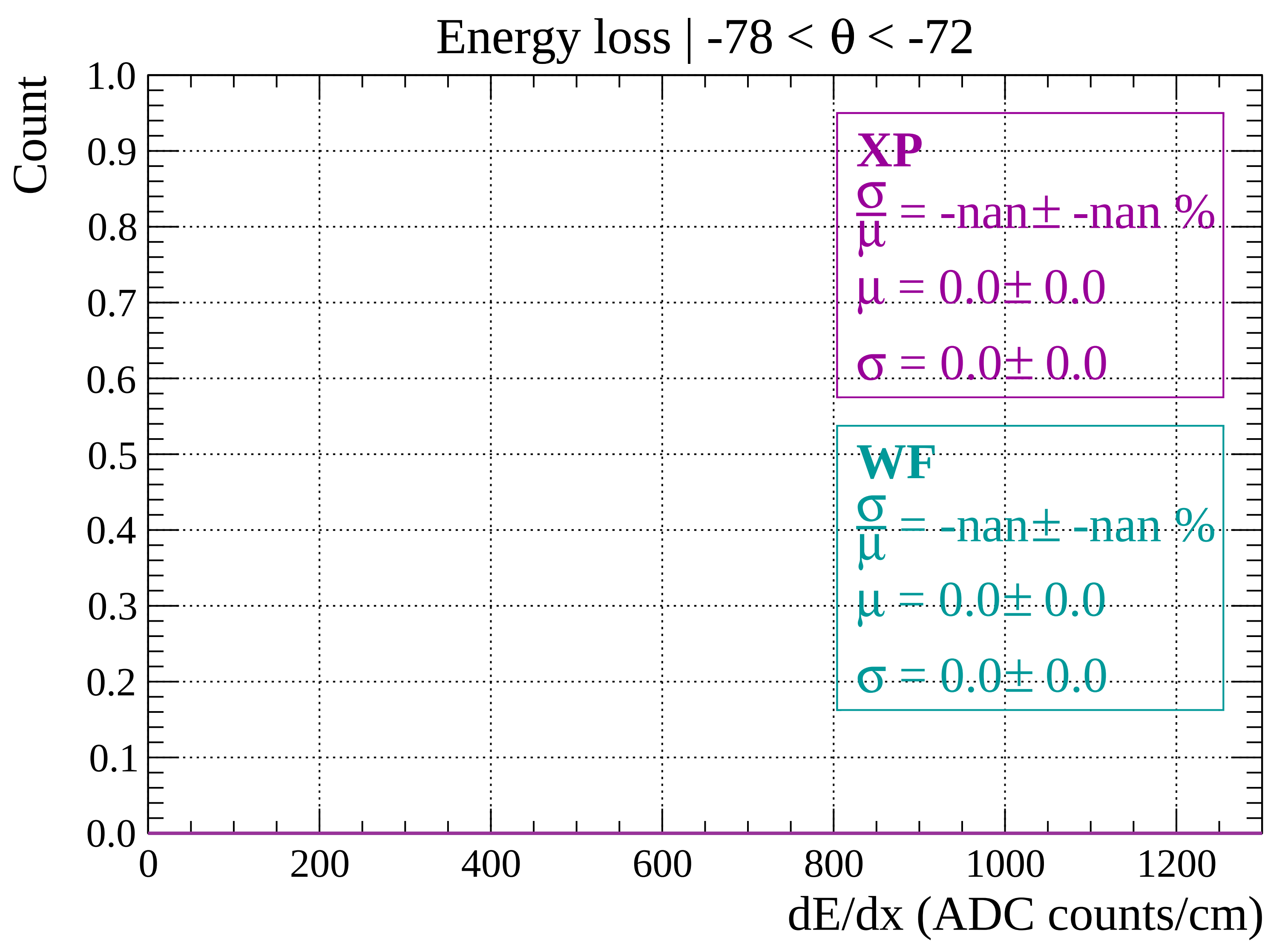
Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 8.39± 1.16 %
μ = 463.7± 4.8
σ = 38.9± 5.0
WF
σ/μ = 8.29± 1.09 %
μ = 470.8± 5.1
σ = 39.0± 4.7


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
σ/μ = 24.47± 7.23 %
μ = 417.9± 26.5
σ = 102.3± 23.7
WF
σ/μ = 14.48± 4.18 %
μ = 492.4± 16.4
σ = 71.3± 18.2

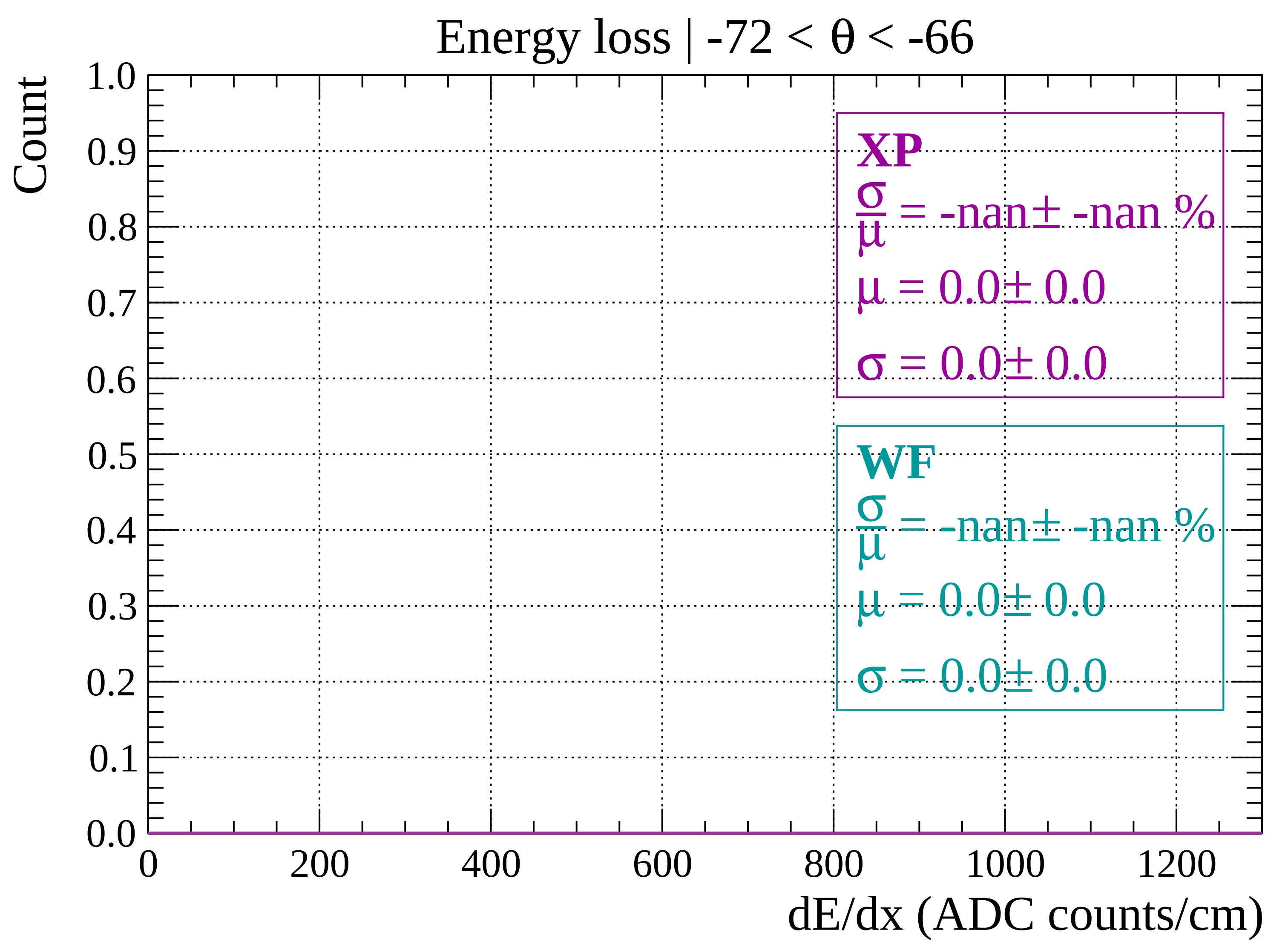


Energy loss | -90 < θ < -84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

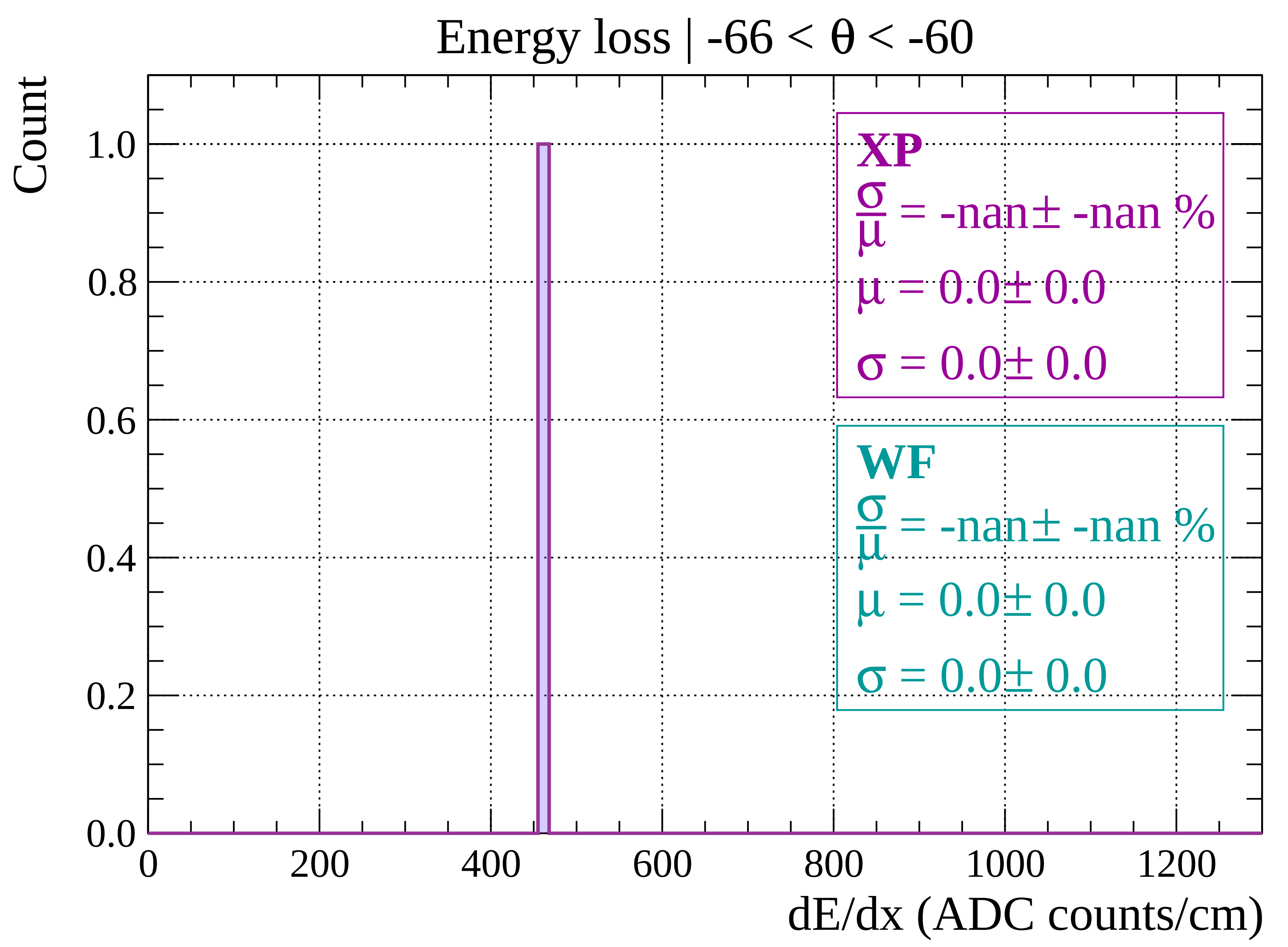

Energy loss | -84 < θ < -78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



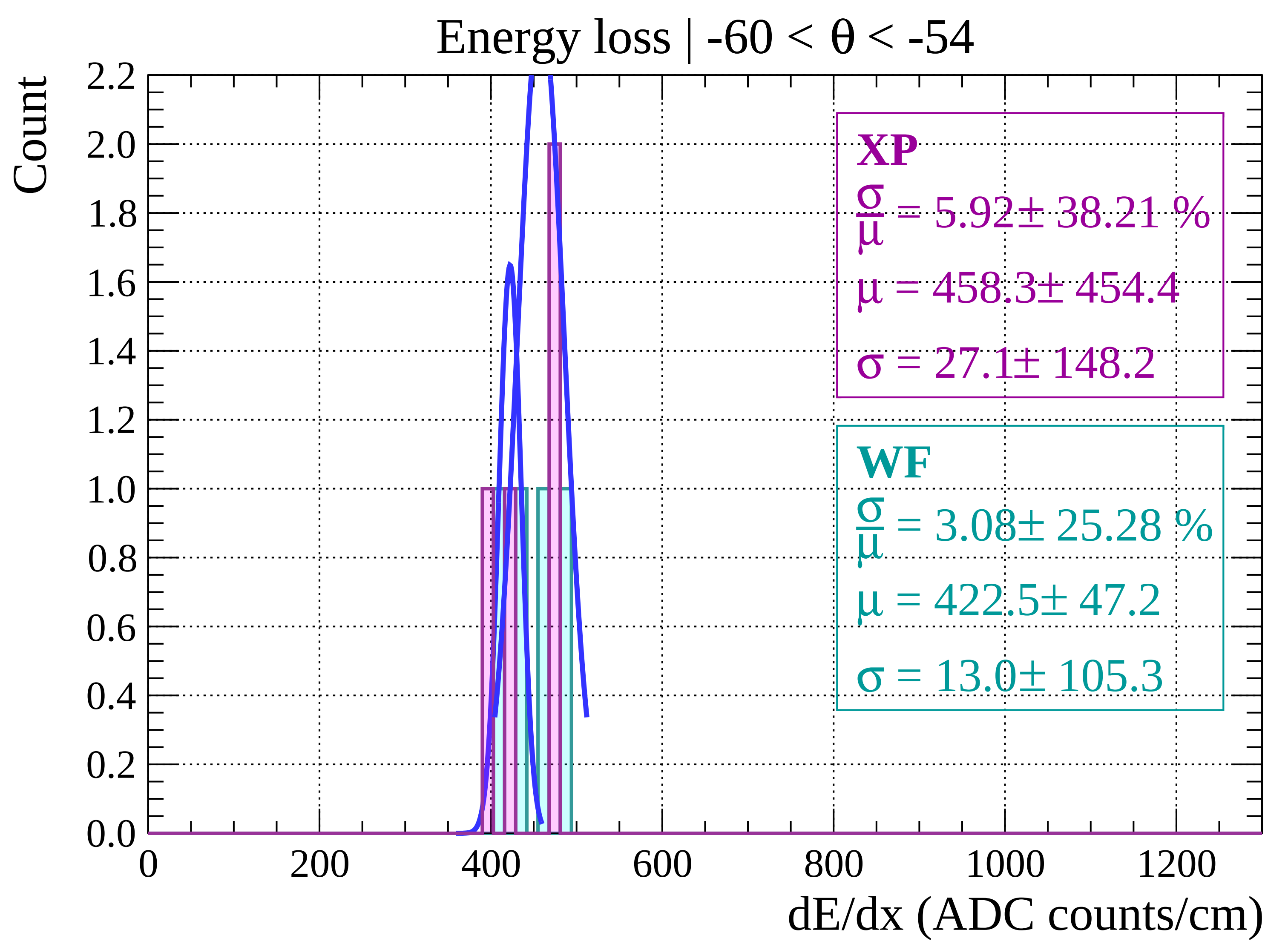
Energy loss | -78 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



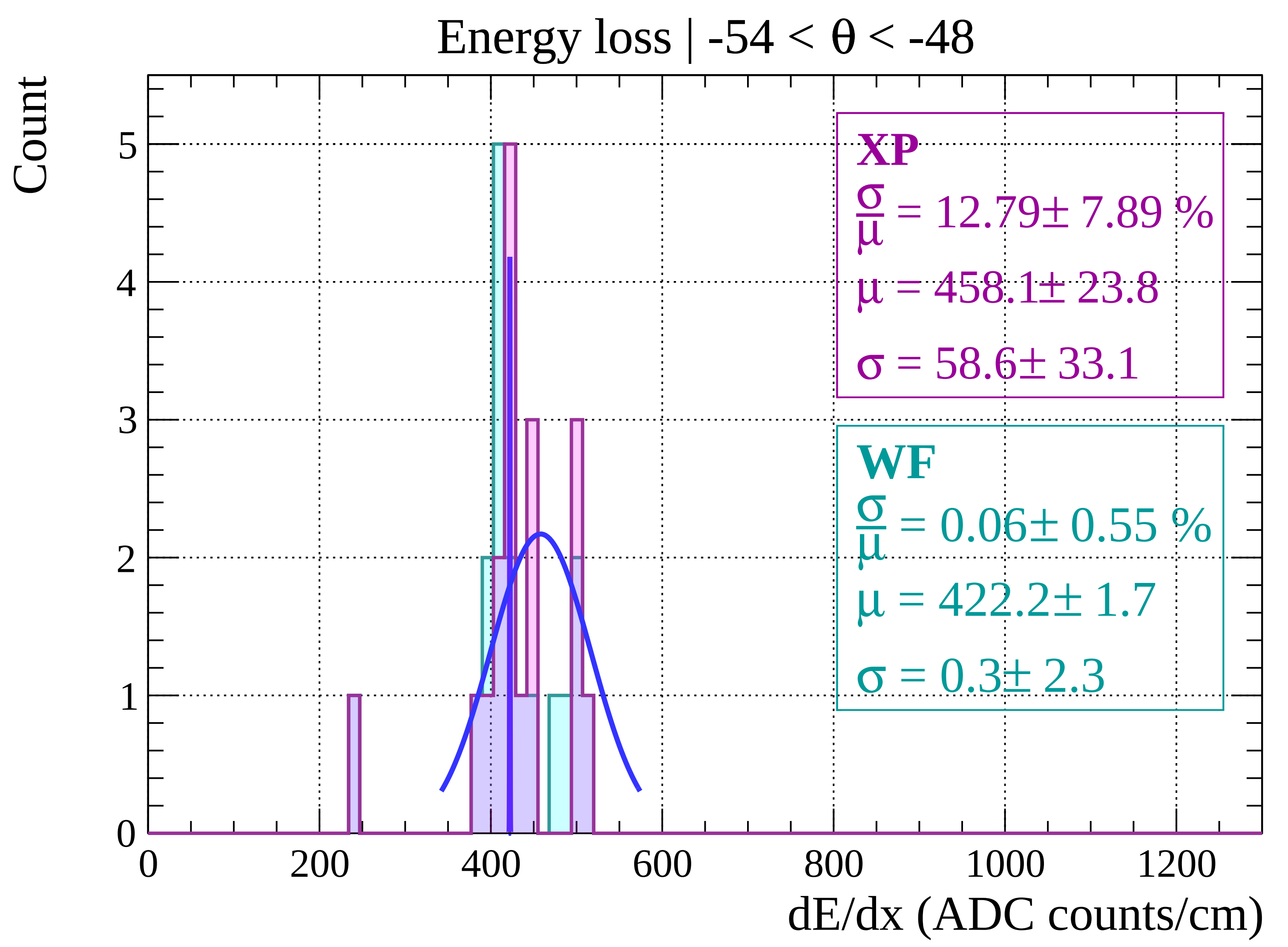
Energy loss | -72 < θ < -66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



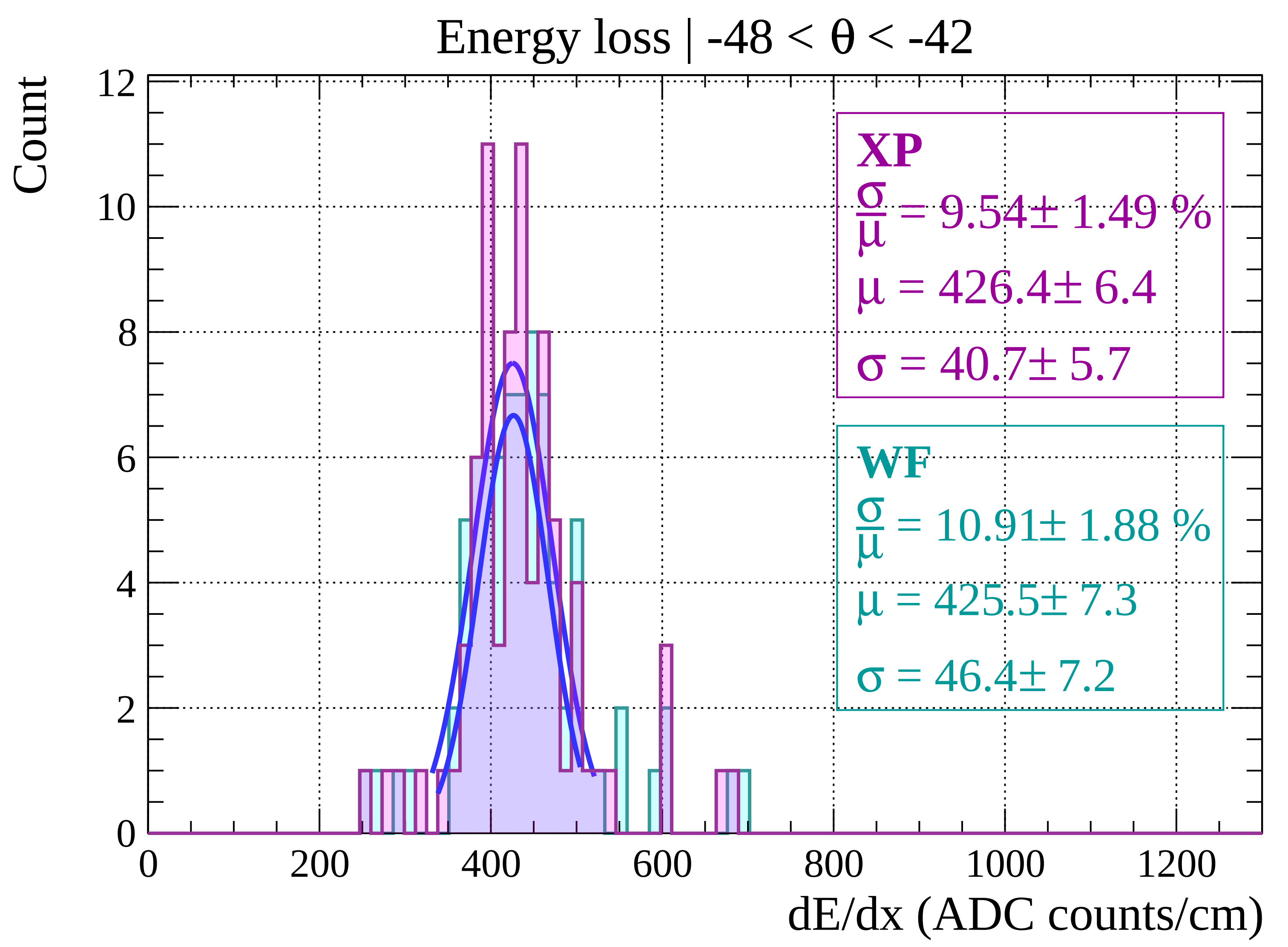
Energy loss | -66 < θ < -60
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



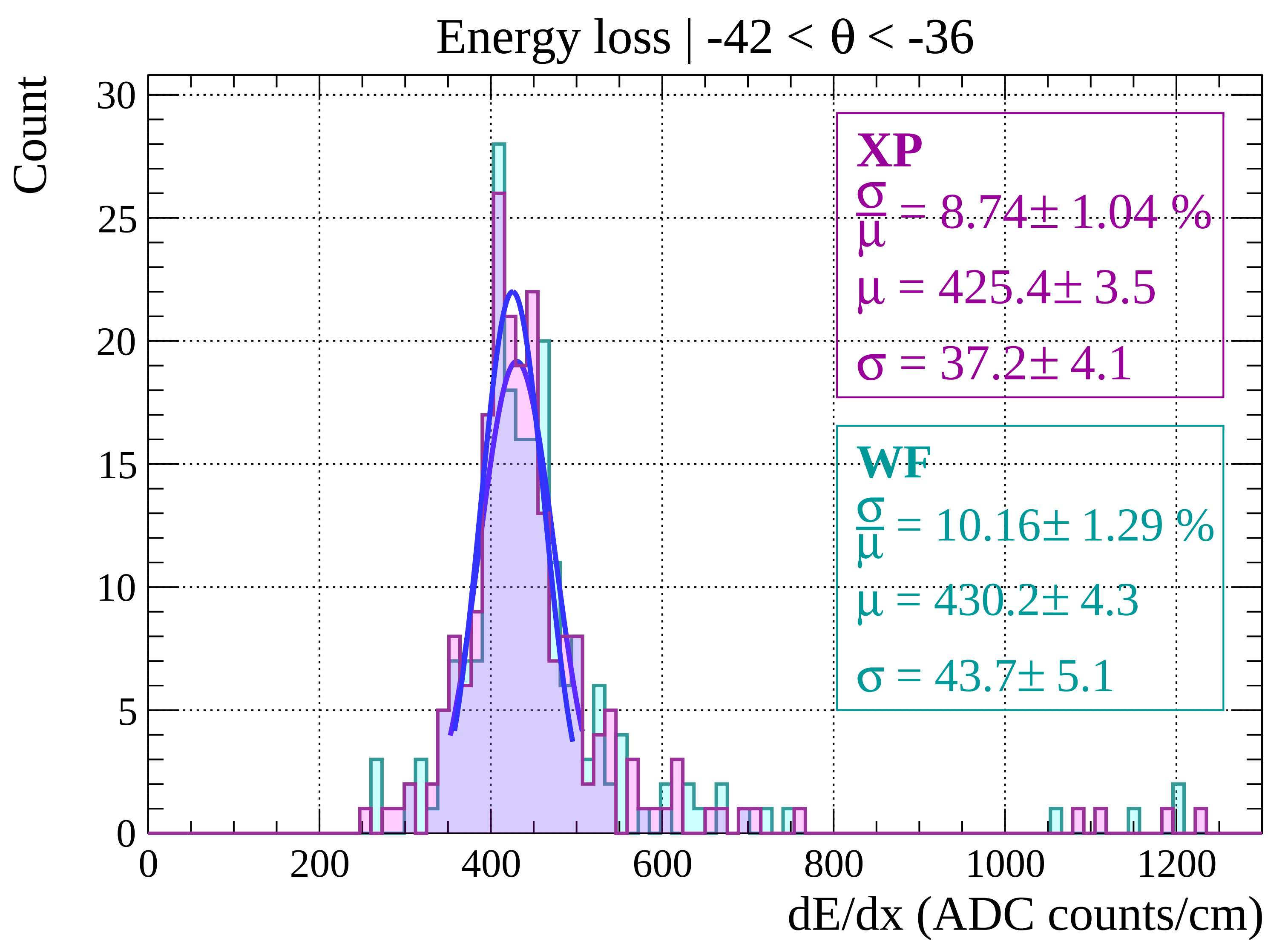
Energy loss | -60 < θ < -54
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.92± 38.21 %
μ = 458.3± 454.4
σ = 27.1± 148.2
WF
$\frac{\sigma}{\mu}$ = 3.08± 25.28 %
μ = 422.5± 47.2
σ = 13.0± 105.3

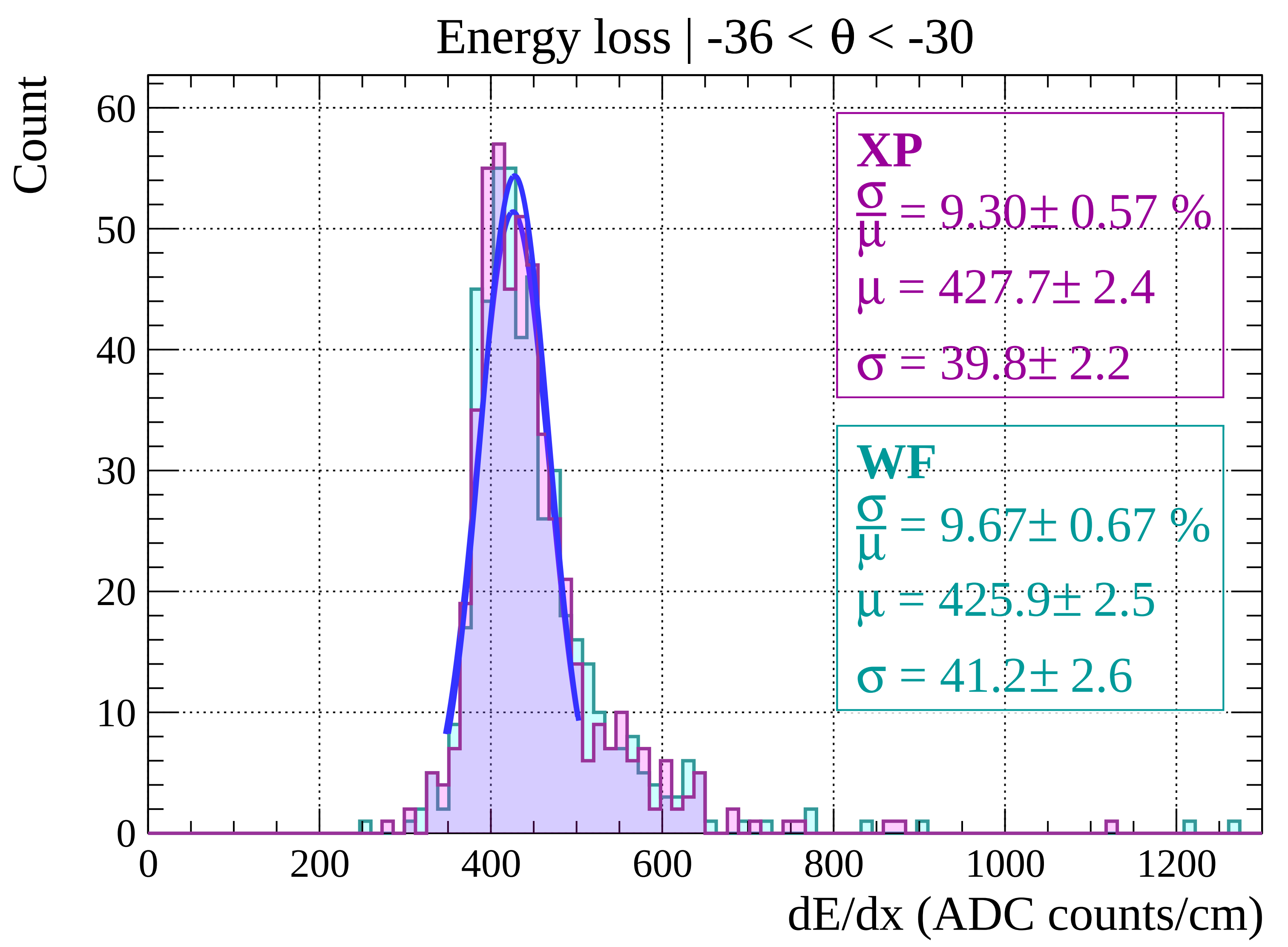


Energy loss | -54 < θ < -48
Count
5
4
3
2
1
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.79± 7.89 %
μ = 458.1± 23.8
σ = 58.6± 33.1
WF
σ/μ = 0.06± 0.55 %
μ = 422.2± 1.7
σ = 0.3± 2.3

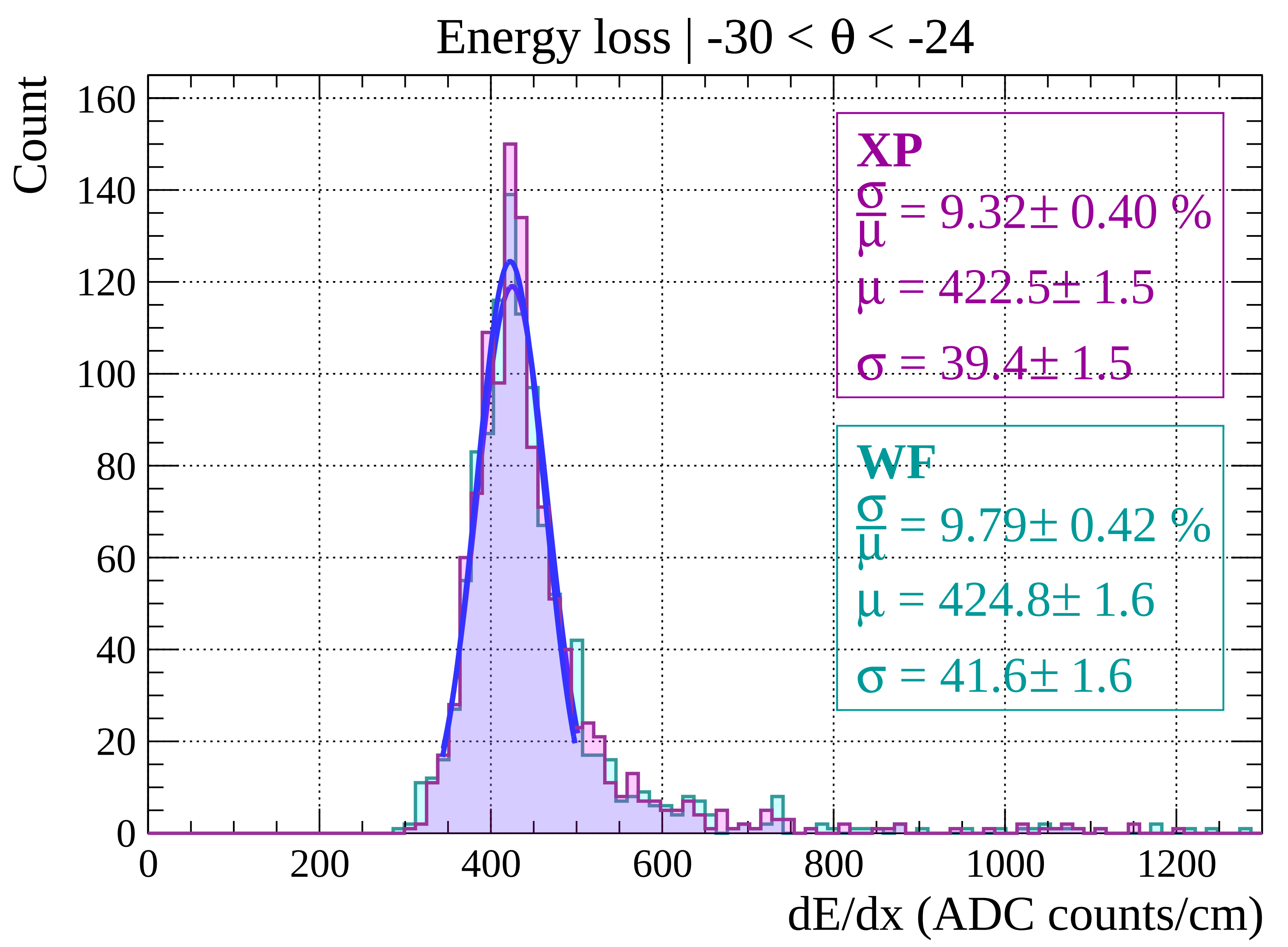


Energy loss | -48 < θ < -42
Count
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.54± 1.49 %
μ = 426.4± 6.4
σ = 40.7± 5.7
WF
$\frac{\sigma}{\mu}$ = 10.91± 1.88 %
μ = 425.5± 7.3
σ = 46.4± 7.2

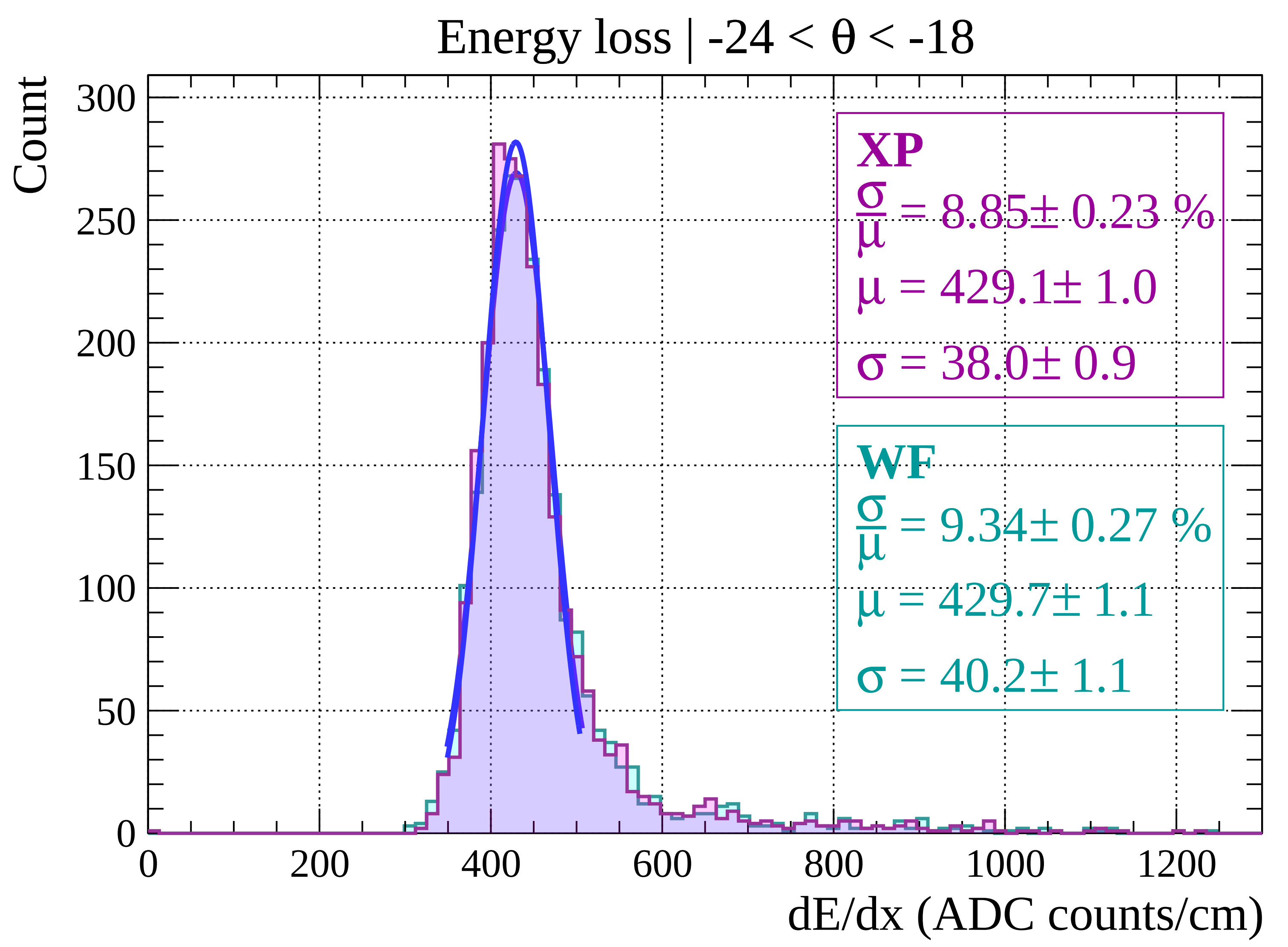

Energy loss | -42 < θ < -36
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.74± 1.04 %
μ = 425.4± 3.5
σ = 37.2± 4.1
WF
σ/μ = 10.16± 1.29 %
μ = 430.2± 4.3
σ = 43.7± 5.1



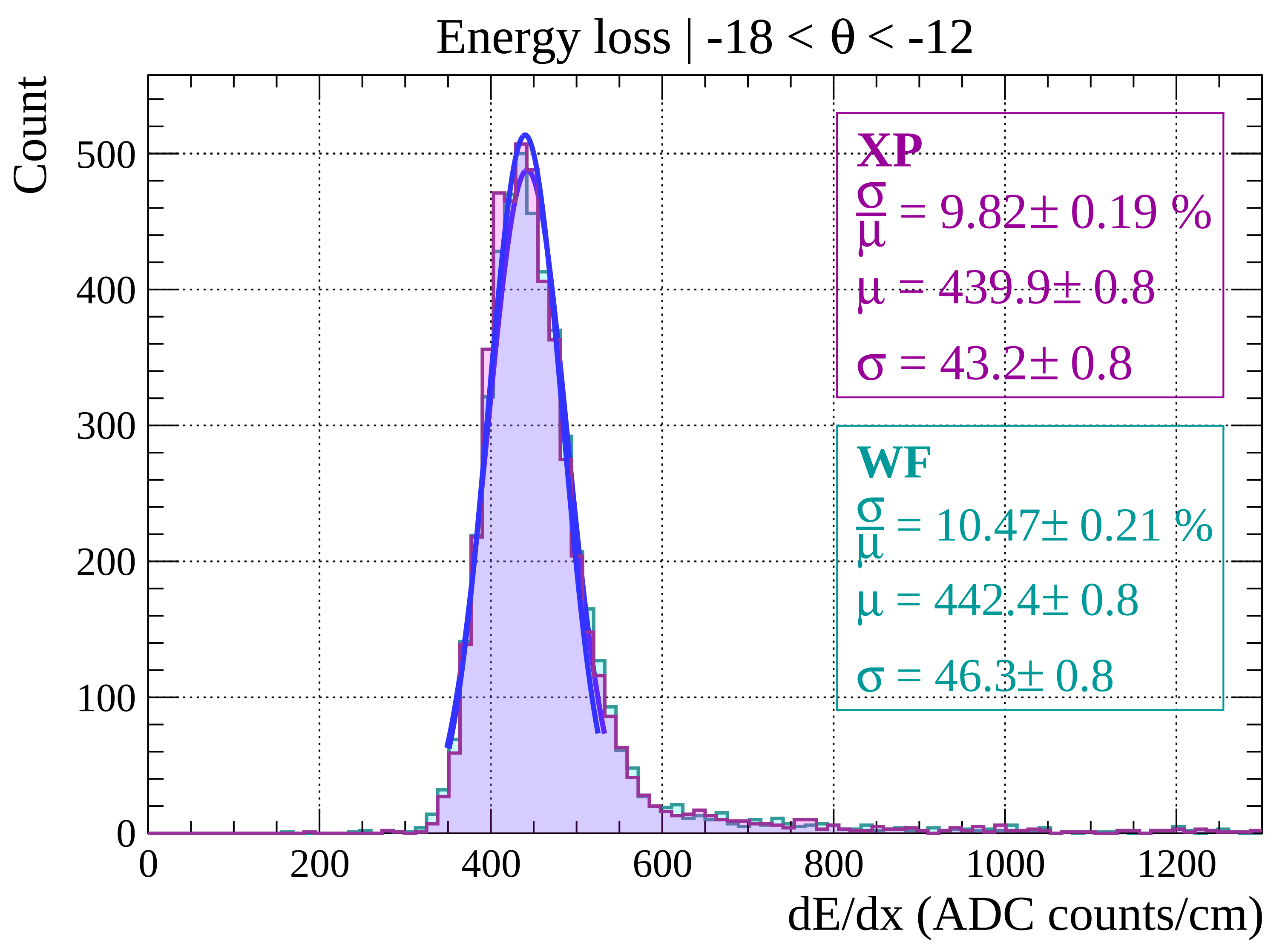
Energy loss | -36 < θ < -30
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 9.30± 0.57 %
μ = 427.7± 2.4
σ = 39.8± 2.2
WF
σ/μ = 9.67± 0.67 %
μ = 425.9± 2.5
σ = 41.2± 2.6



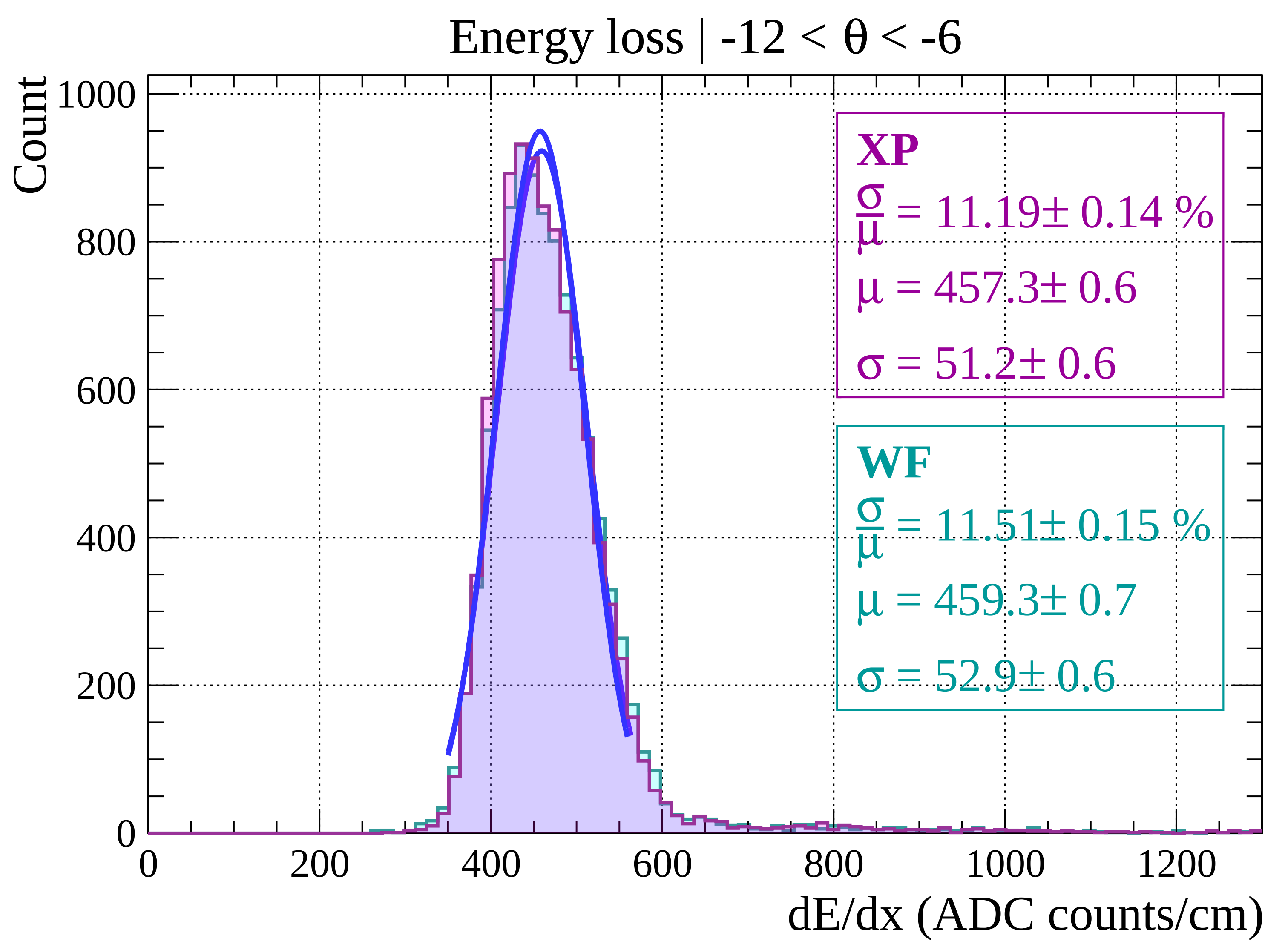
Energy loss | -30 < θ < -24
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 9.32± 0.40 %
μ = 422.5± 1.5
σ = 39.4± 1.5
WF
σ/μ = 9.79± 0.42 %
μ = 424.8± 1.6
σ = 41.6± 1.6



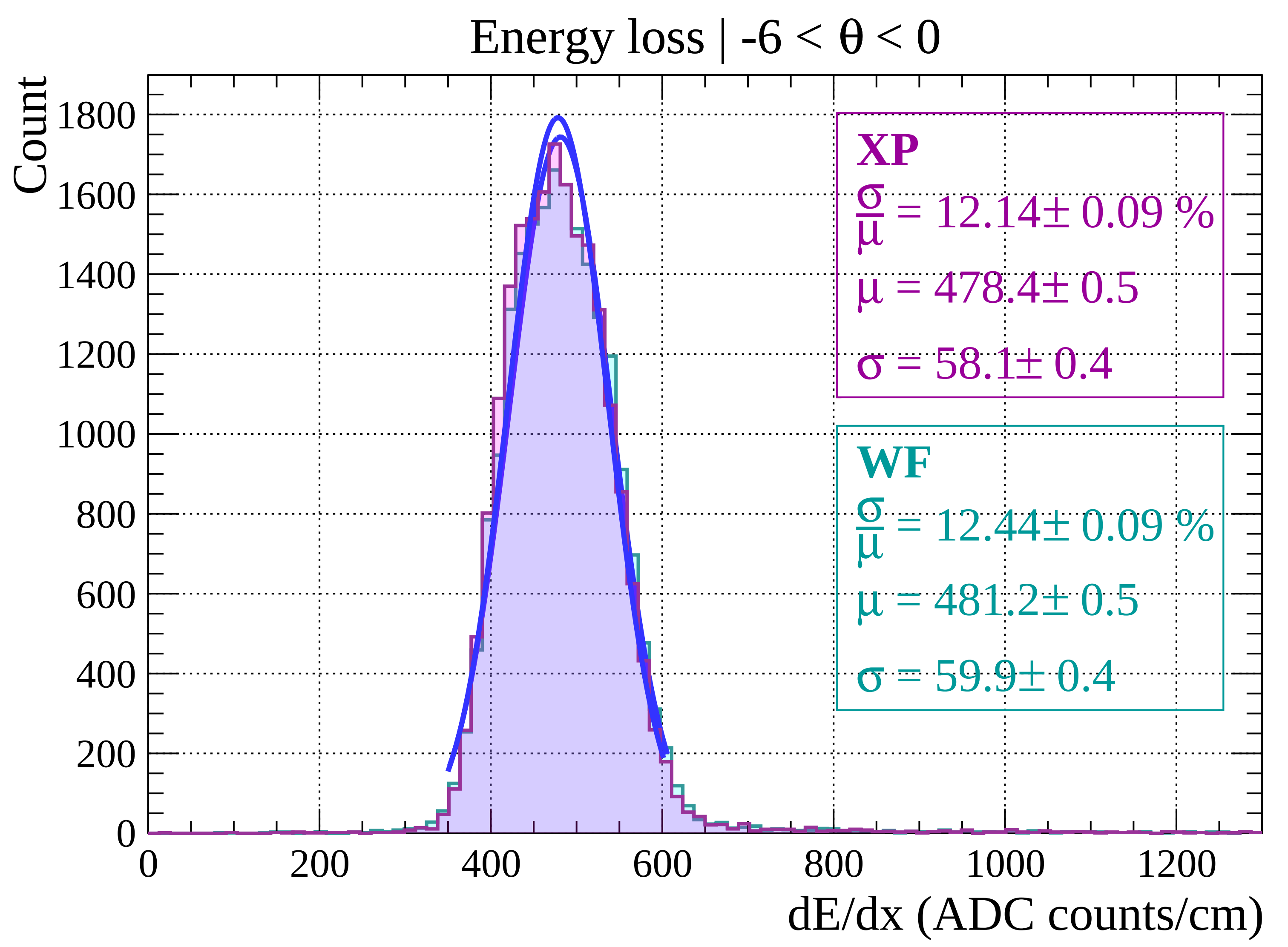
Energy loss | -24 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.85± 0.23 %
μ = 429.1± 1.0
σ = 38.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.34± 0.27 %
μ = 429.7± 1.1
σ = 40.2± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



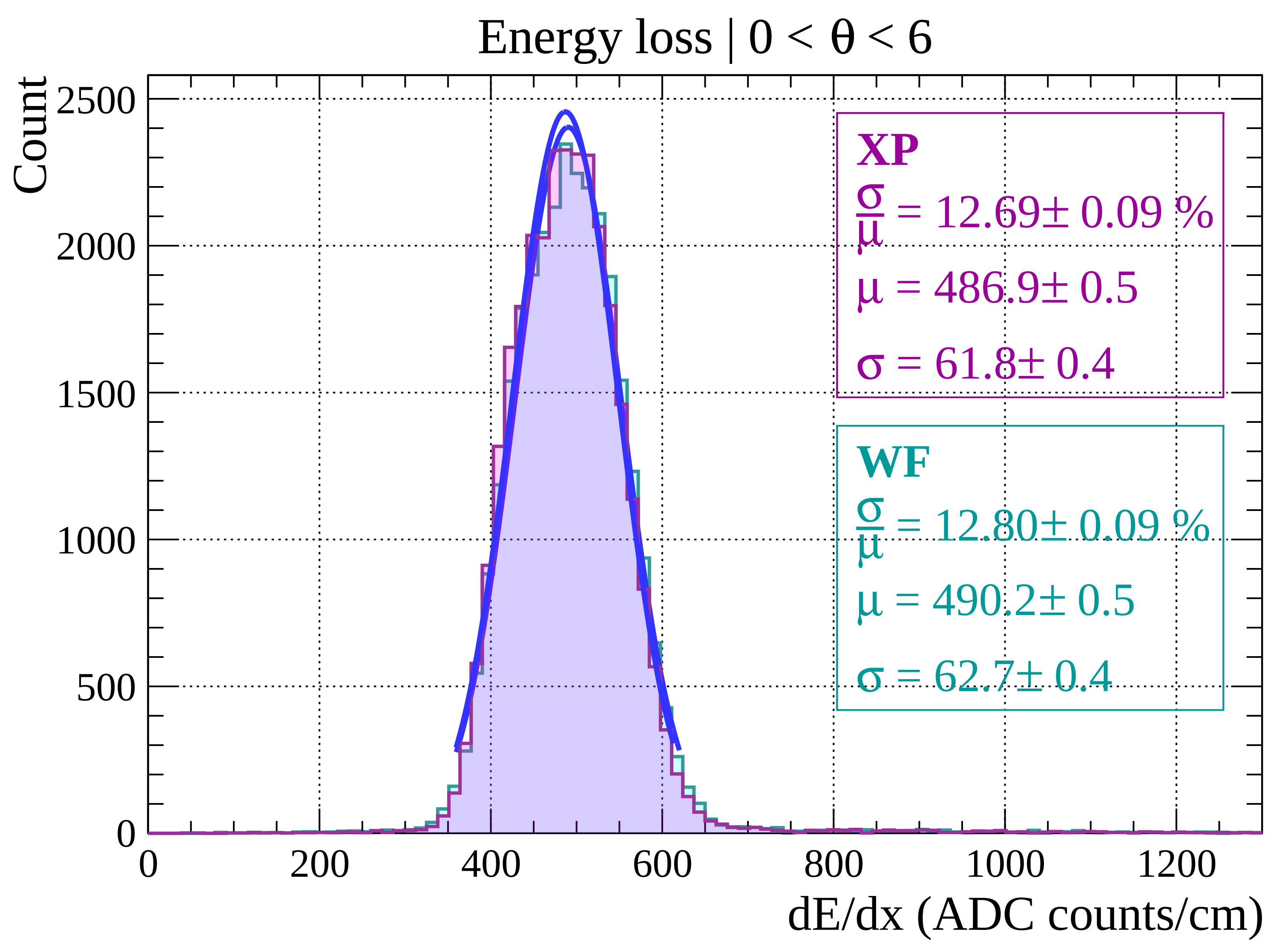
Energy loss | -18 < θ < -12
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.82± 0.19 %
μ = 439.9± 0.8
σ = 43.2± 0.8
WF
σ/μ = 10.47± 0.21 %
μ = 442.4± 0.8
σ = 46.3± 0.8

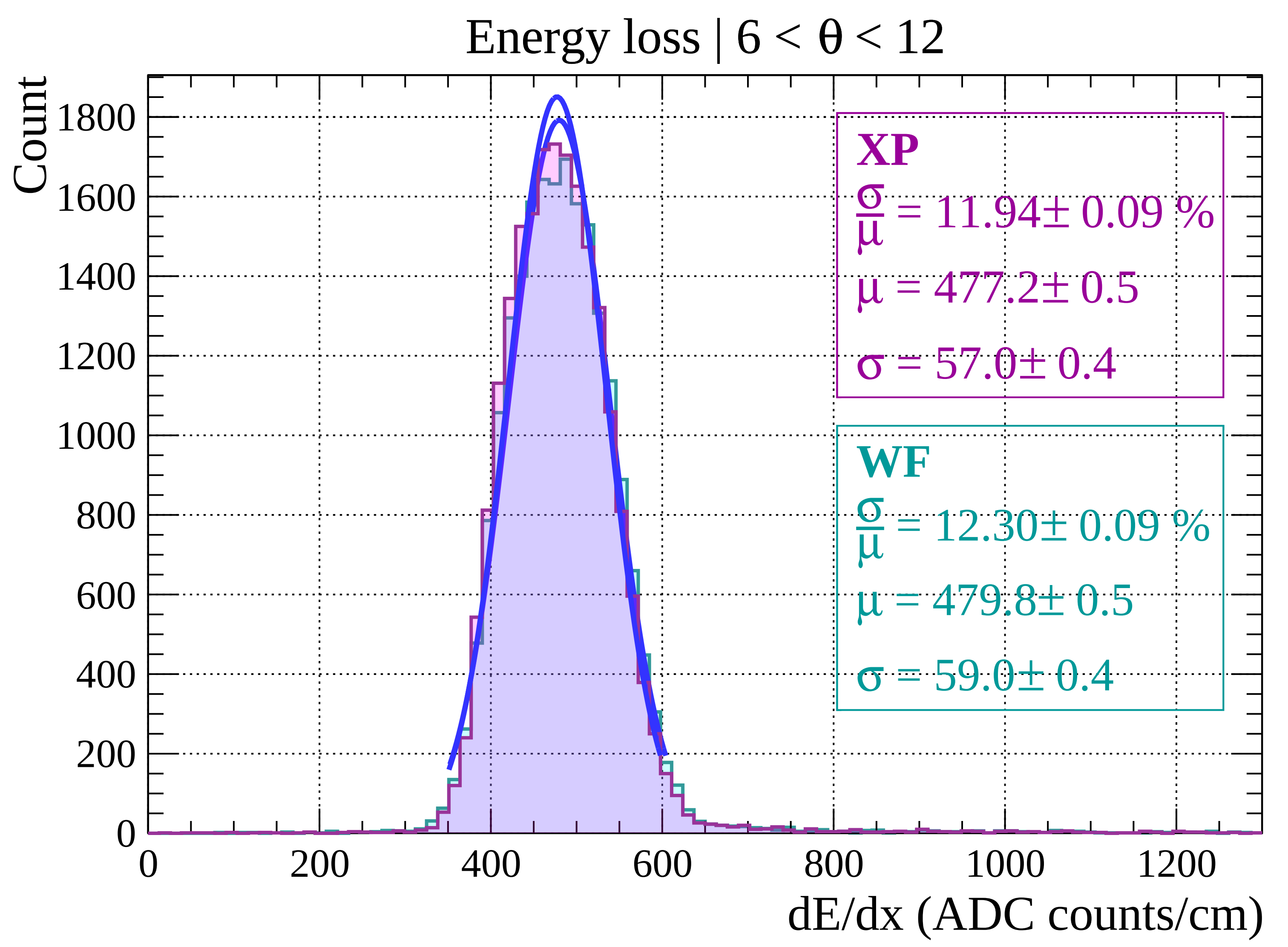


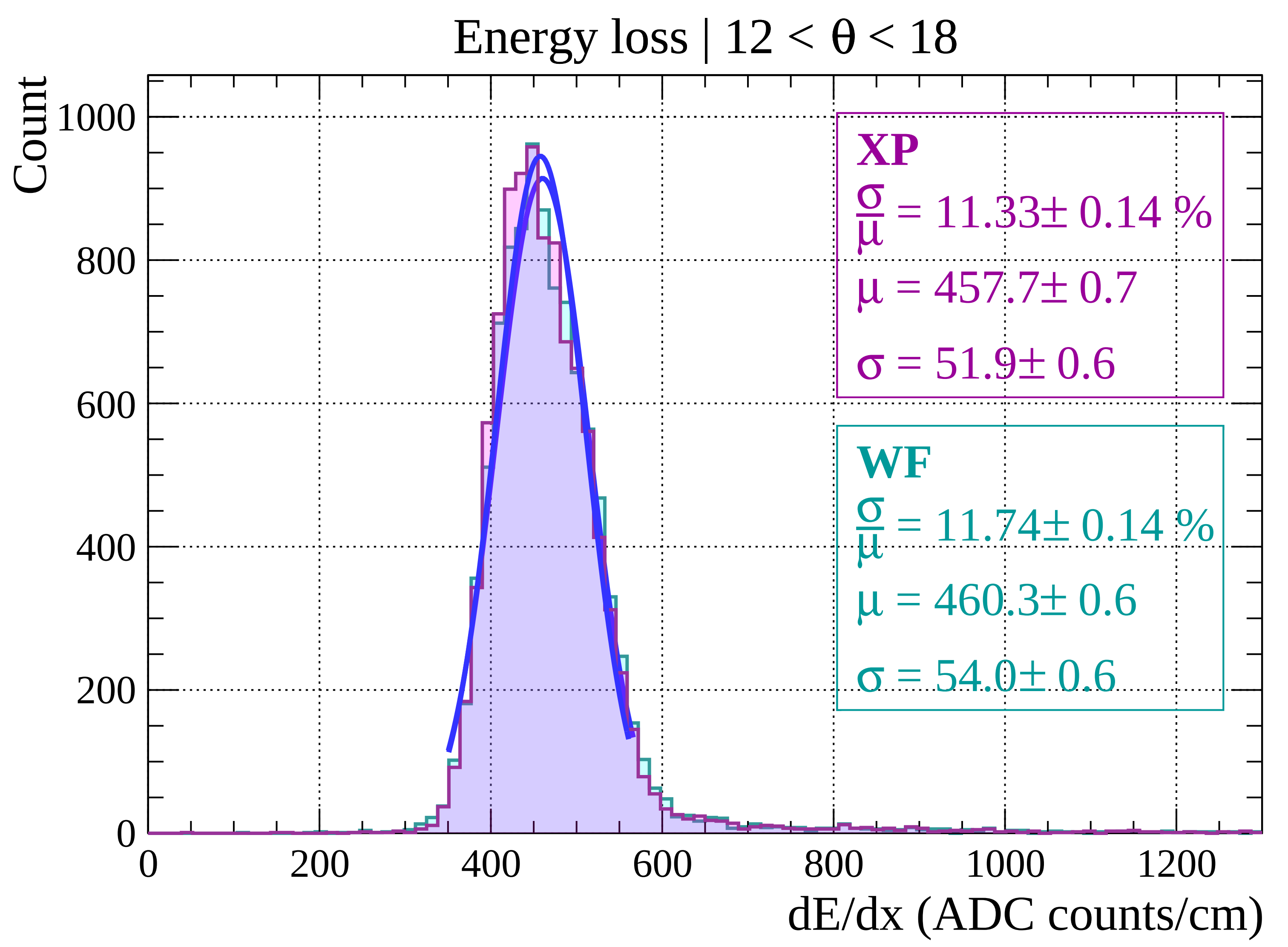
Energy loss | -12 < θ < -6
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.19± 0.14 %
μ = 457.3± 0.6
σ = 51.2± 0.6
WF
σ/μ = 11.51± 0.15 %
μ = 459.3± 0.7
σ = 52.9± 0.6



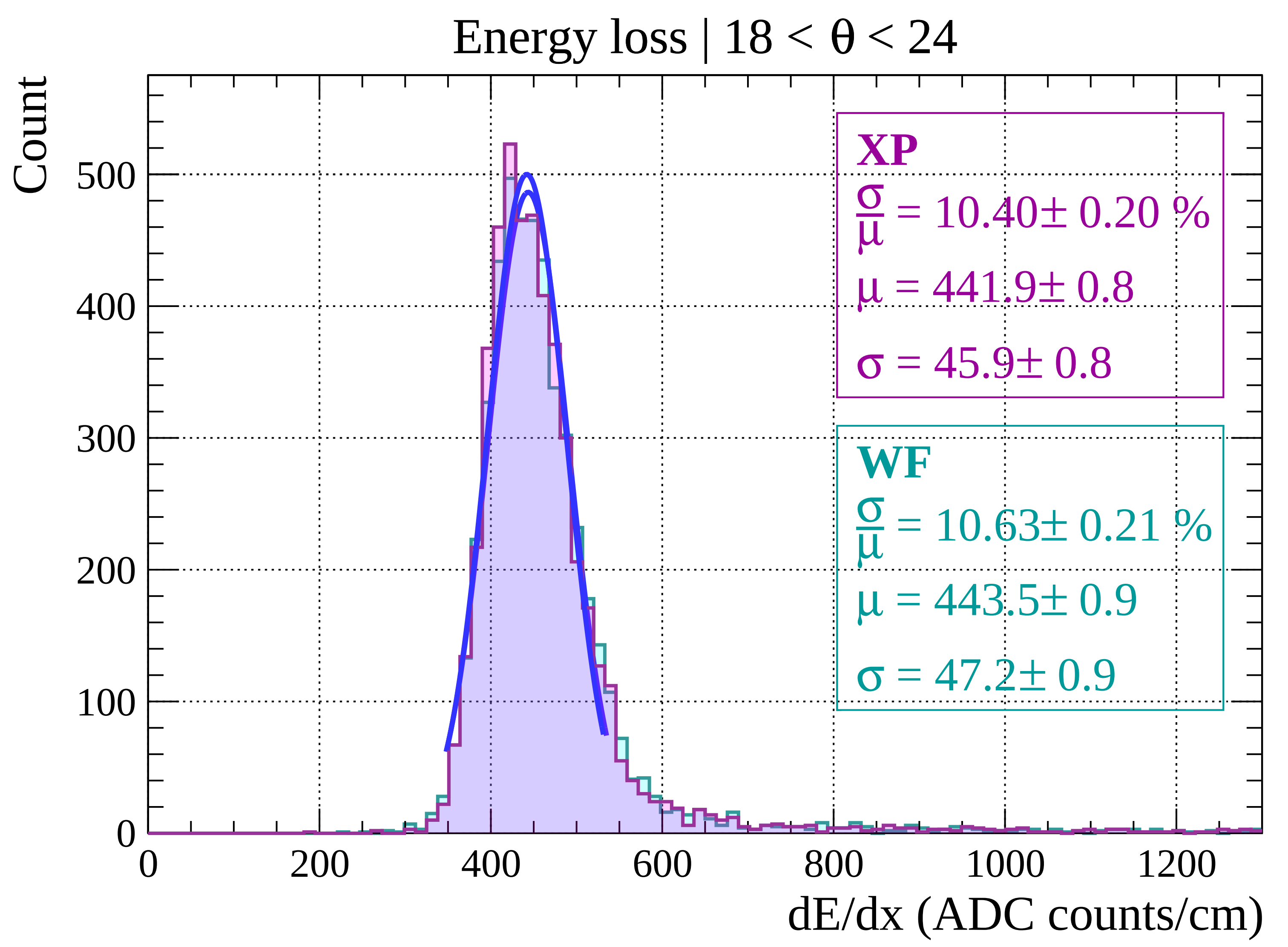
Energy loss | -6 < θ < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
XP
σ/μ = 12.14± 0.09 %
μ = 478.4± 0.5
σ = 58.1± 0.4
WF
σ/μ = 12.44± 0.09 %
μ = 481.2± 0.5
σ = 59.9± 0.4

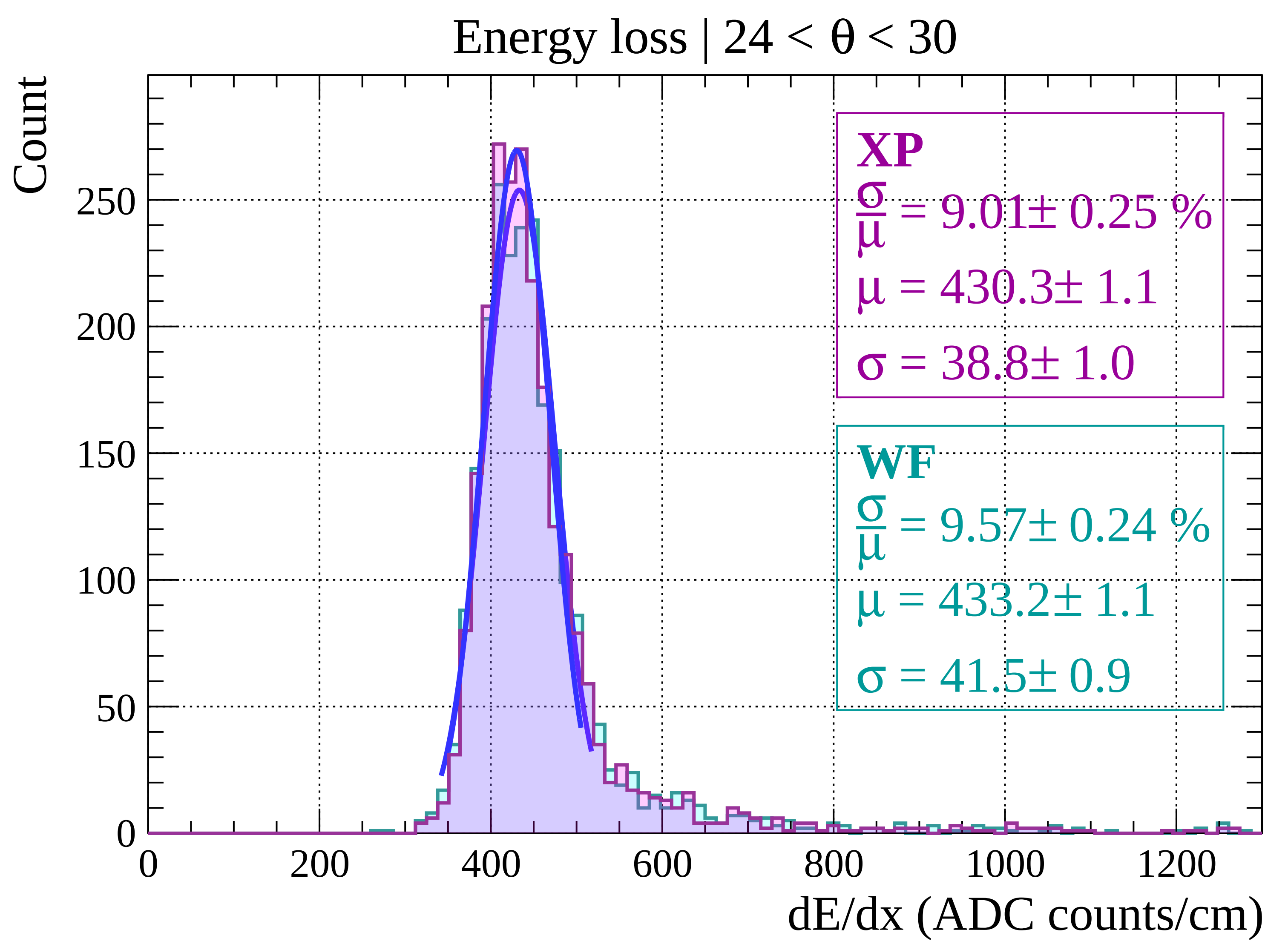

Energy loss | 0 < θ < 6
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.69± 0.09 %
μ = 486.9± 0.5
σ = 61.8± 0.4
WF
σ/μ = 12.80± 0.09 %
μ = 490.2± 0.5
σ = 62.7± 0.4

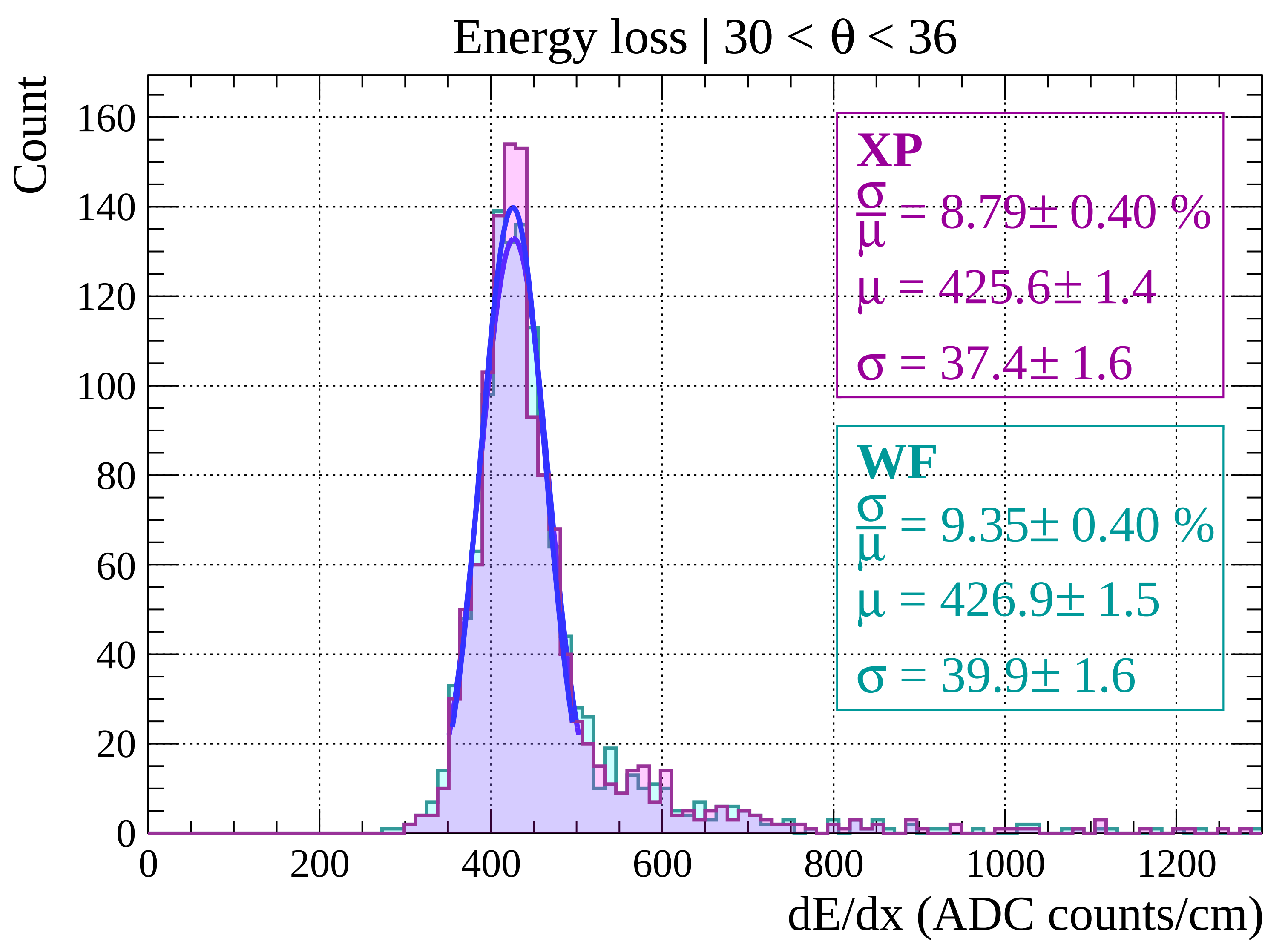

Energy loss | 6 < θ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.94± 0.09 %
μ = 477.2± 0.5
σ = 57.0± 0.4
WF
σ/μ = 12.30± 0.09 %
μ = 479.8± 0.5
σ = 59.0± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200



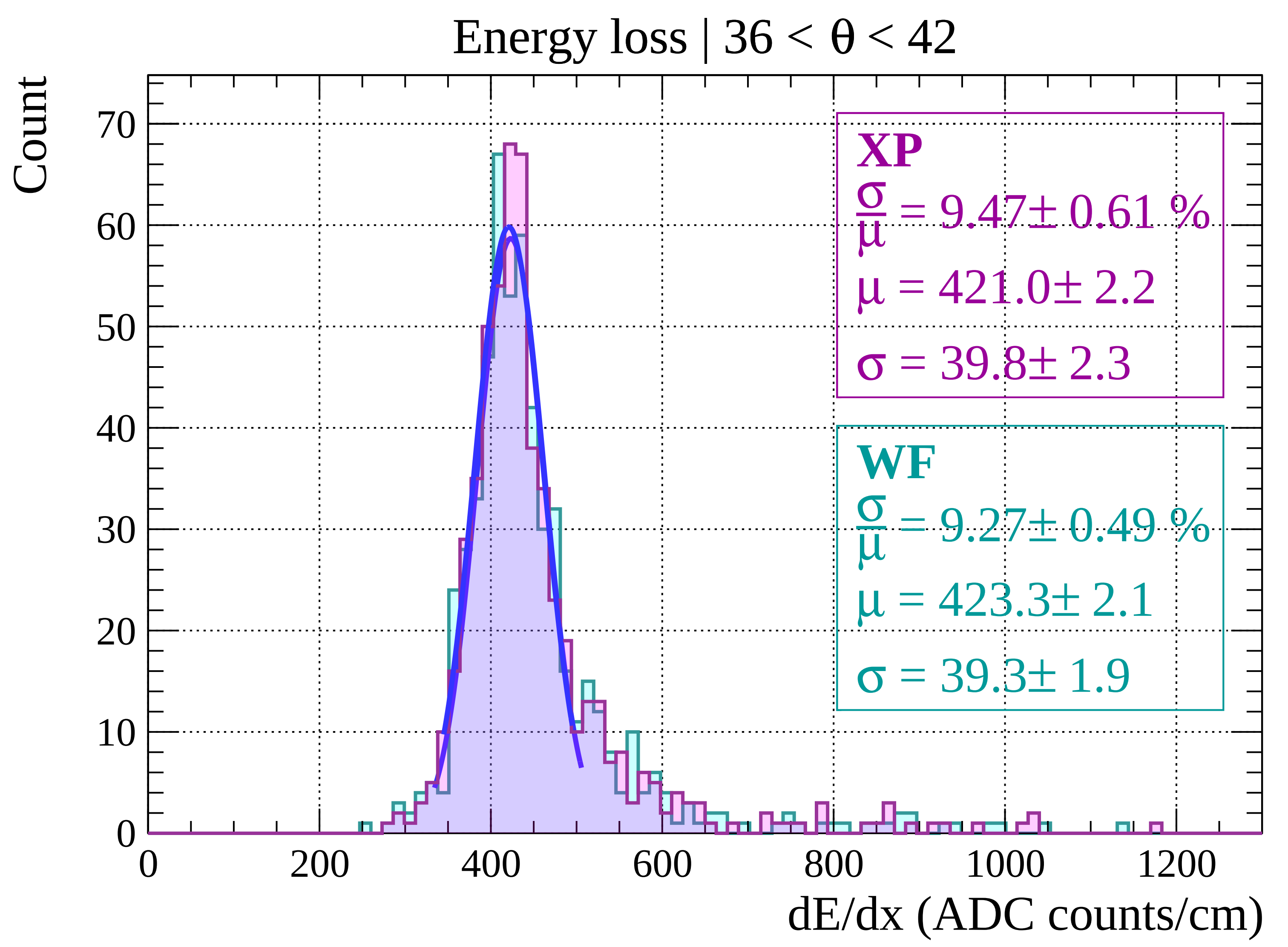
Energy loss | 12 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.33± 0.14 %
μ = 457.7± 0.7
σ = 51.9± 0.6
WF
σ/μ = 11.74± 0.14 %
μ = 460.3± 0.6
σ = 54.0± 0.6


Energy loss | 18 < θ < 24
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.40± 0.20 %
μ = 441.9± 0.8
σ = 45.9± 0.8
WF
σ/μ = 10.63± 0.21 %
μ = 443.5± 0.9
σ = 47.2± 0.9

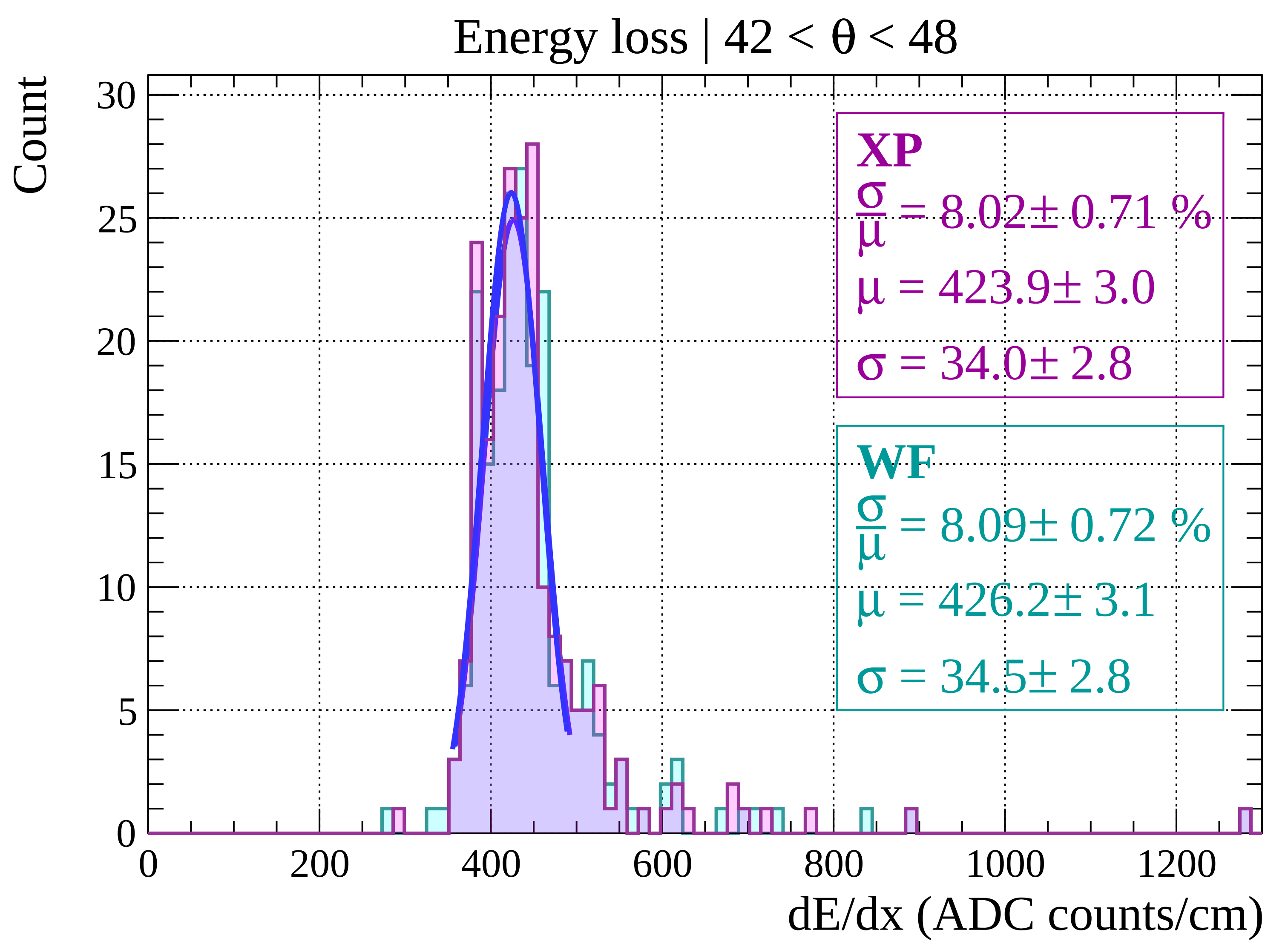

Energy loss | 24 < θ < 30
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.01± 0.25 %
μ = 430.3± 1.1
σ = 38.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.57± 0.24 %
μ = 433.2± 1.1
σ = 41.5± 0.9



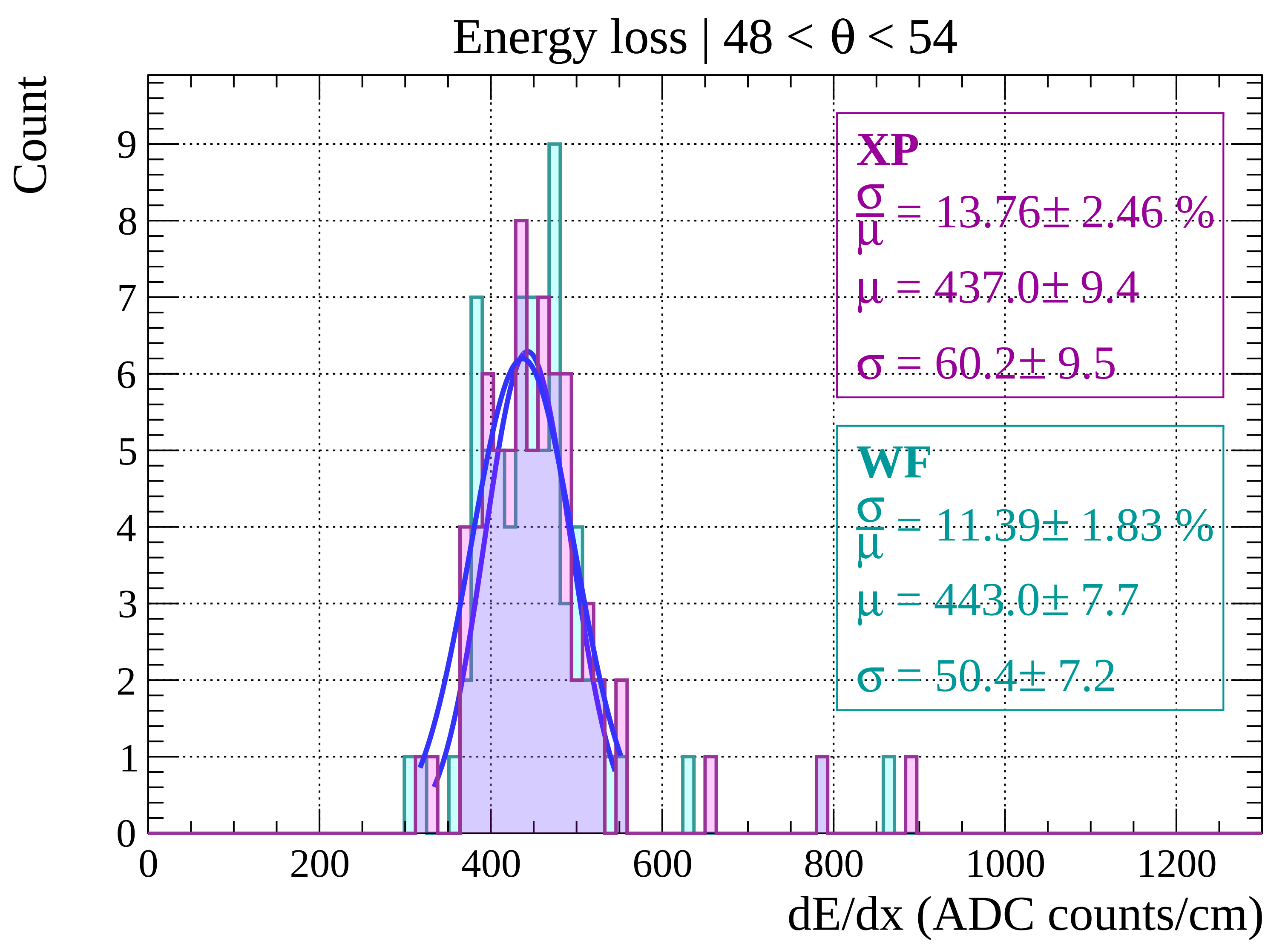
Energy loss | 30 < θ < 36
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.79± 0.40 %
μ = 425.6± 1.4
σ = 37.4± 1.6
WF
σ/μ = 9.35± 0.40 %
μ = 426.9± 1.5
σ = 39.9± 1.6

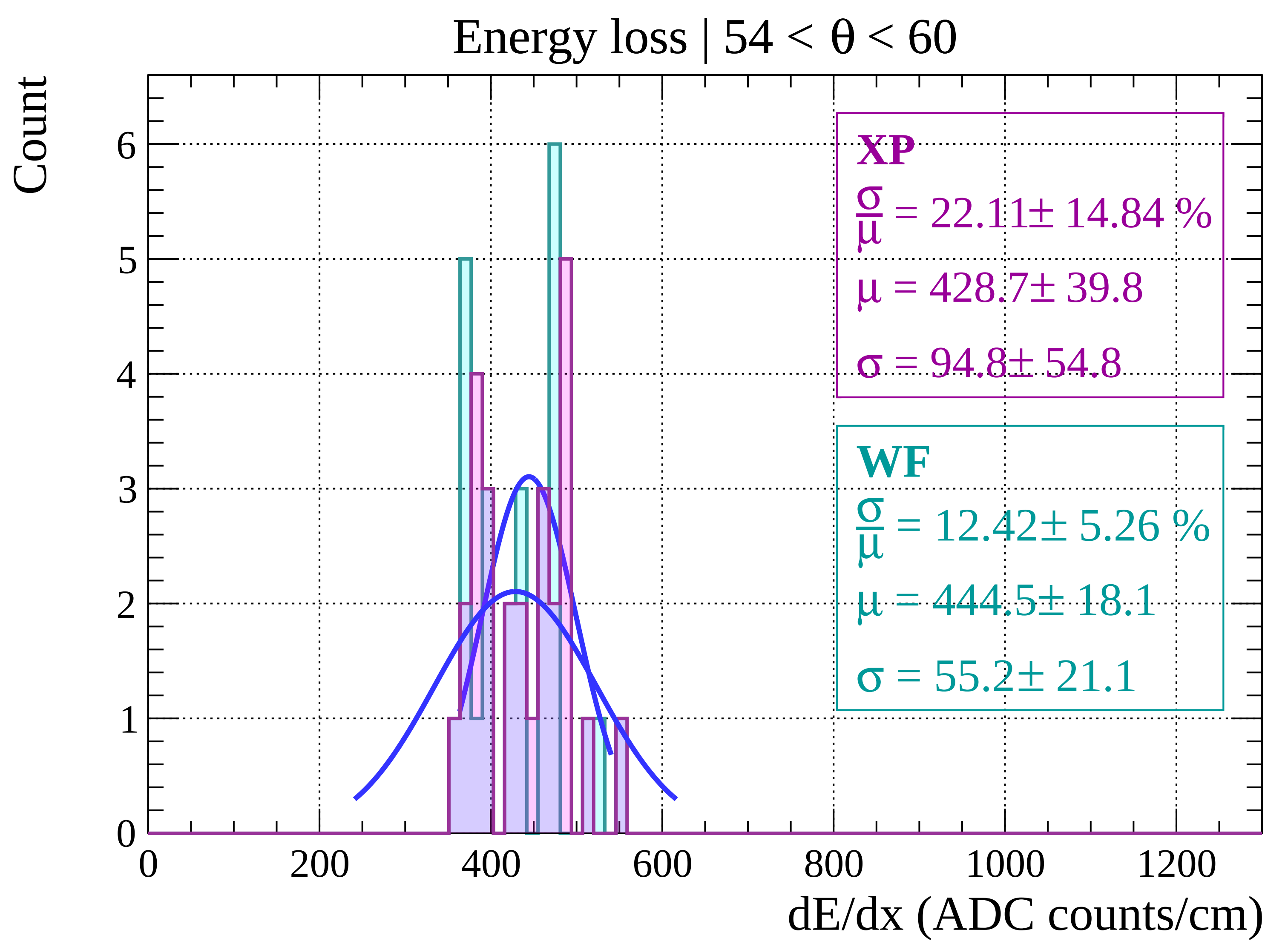


Energy loss | 36 < θ < 42
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.47± 0.61 %
μ = 421.0± 2.2
σ = 39.8± 2.3
WF
σ/μ = 9.27± 0.49 %
μ = 423.3± 2.1
σ = 39.3± 1.9

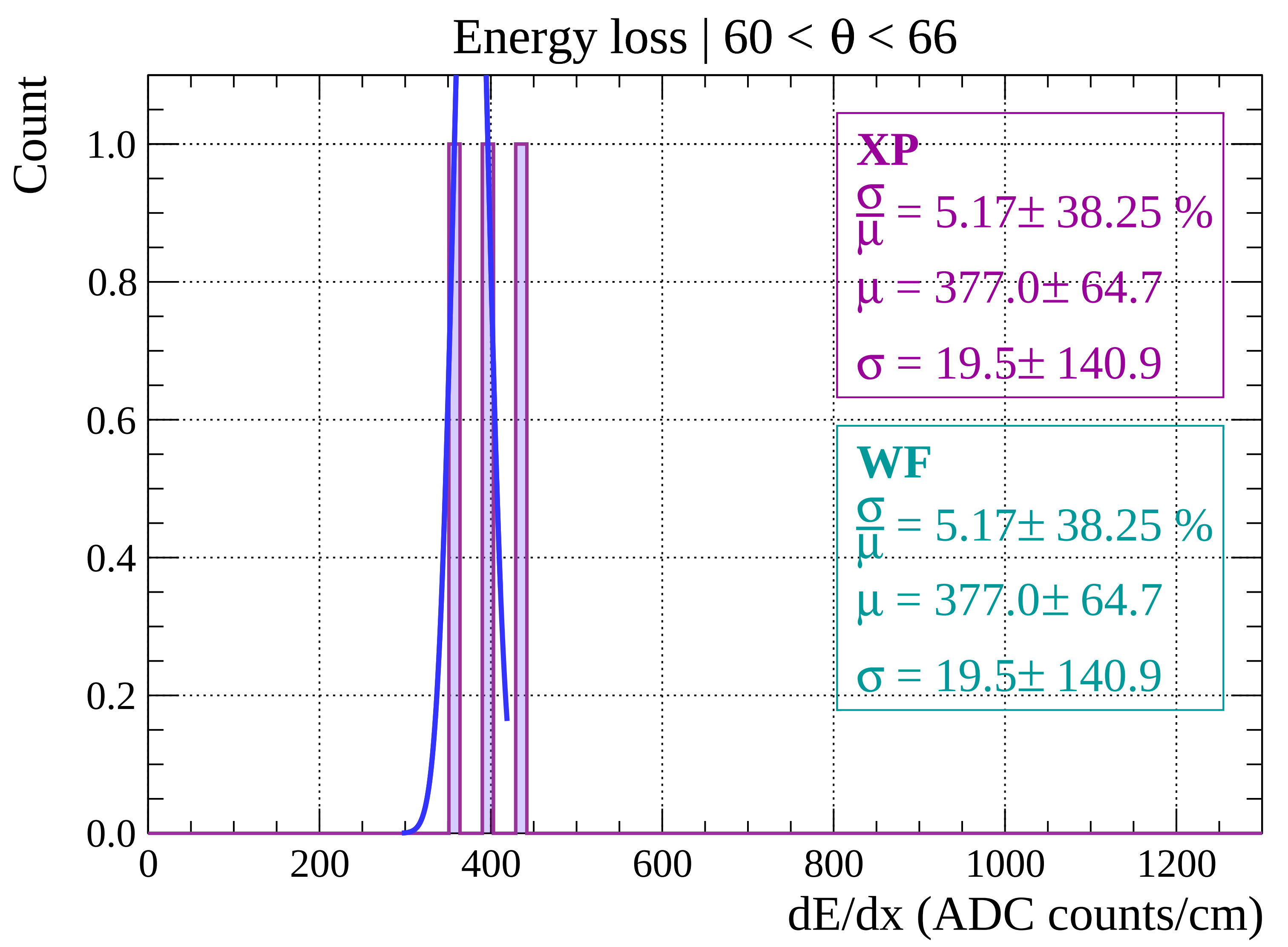


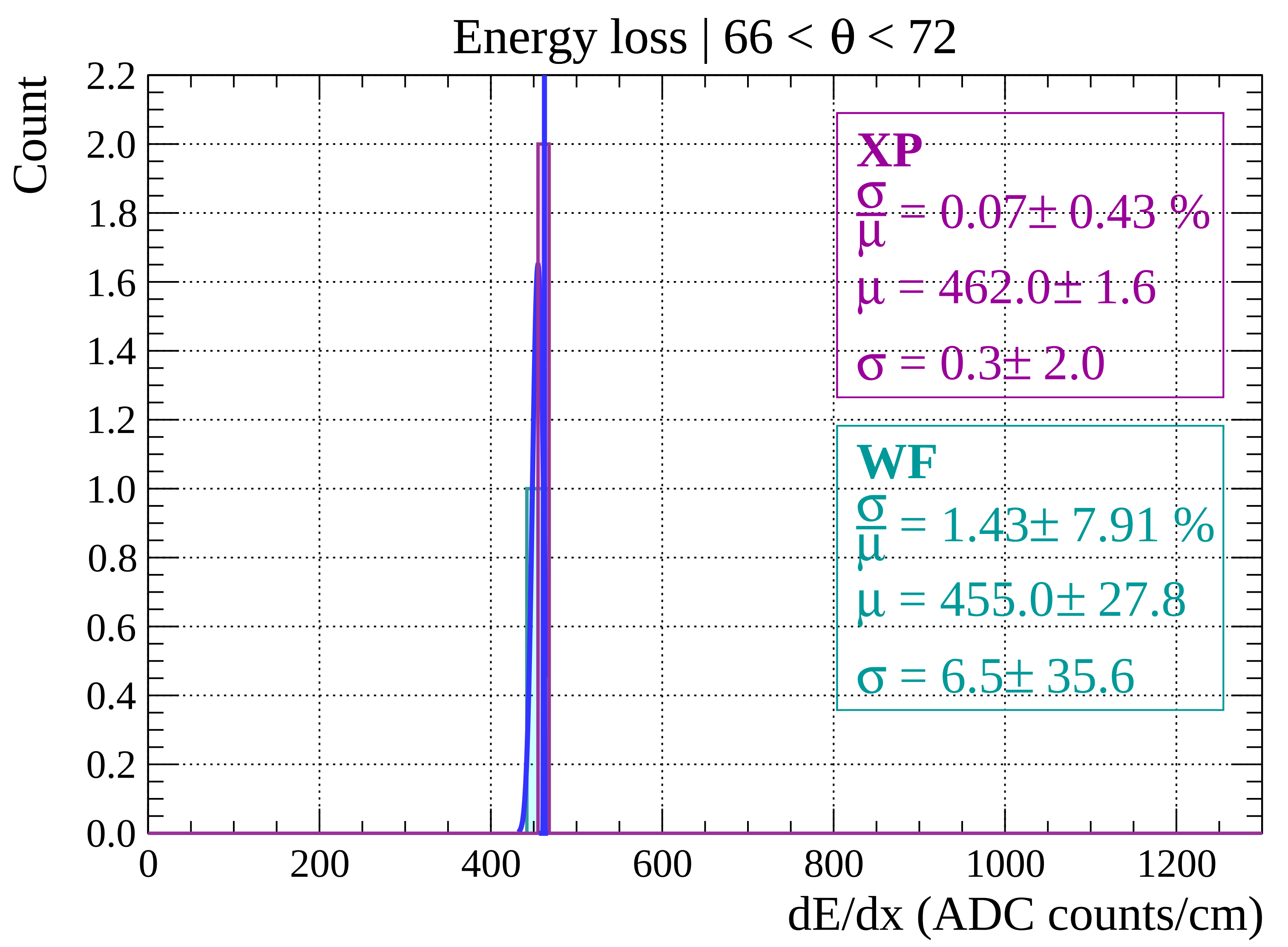
Energy loss | 42 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 8.02± 0.71 %
μ = 423.9± 3.0
σ = 34.0± 2.8
WF
σ/μ = 8.09± 0.72 %
μ = 426.2± 3.1
σ = 34.5± 2.8

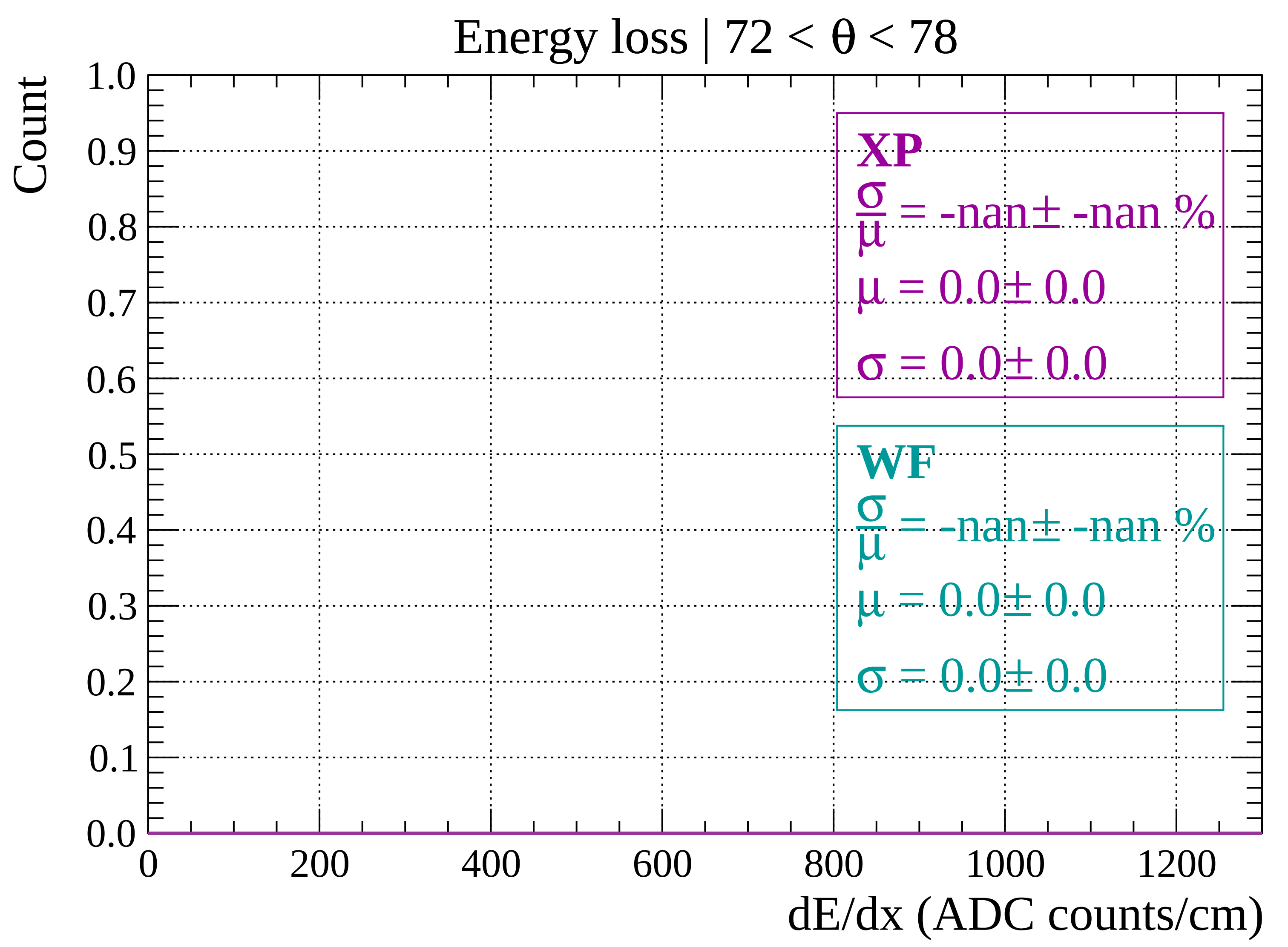

Energy loss | 48 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 13.76± 2.46 %
μ = 437.0± 9.4
σ = 60.2± 9.5
WF
σ/μ = 11.39± 1.83 %
μ = 443.0± 7.7
σ = 50.4± 7.2

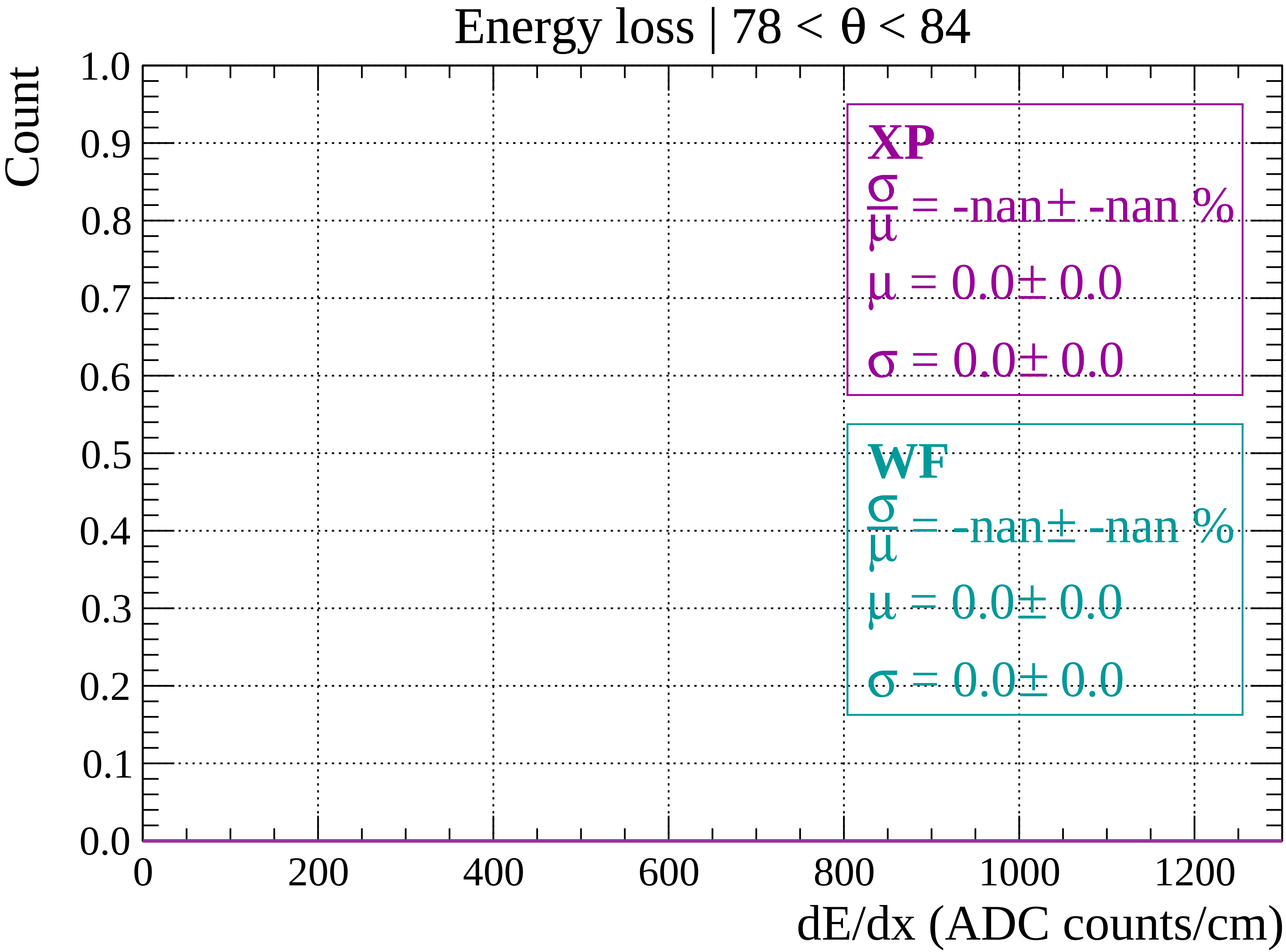


Energy loss | 54 < θ < 60
Count
6
5
4
3
2
1
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 22.11± 14.84 %
μ = 428.7± 39.8
σ = 94.8± 54.8
WF
σ/μ = 12.42± 5.26 %
μ = 444.5± 18.1
σ = 55.2± 21.1

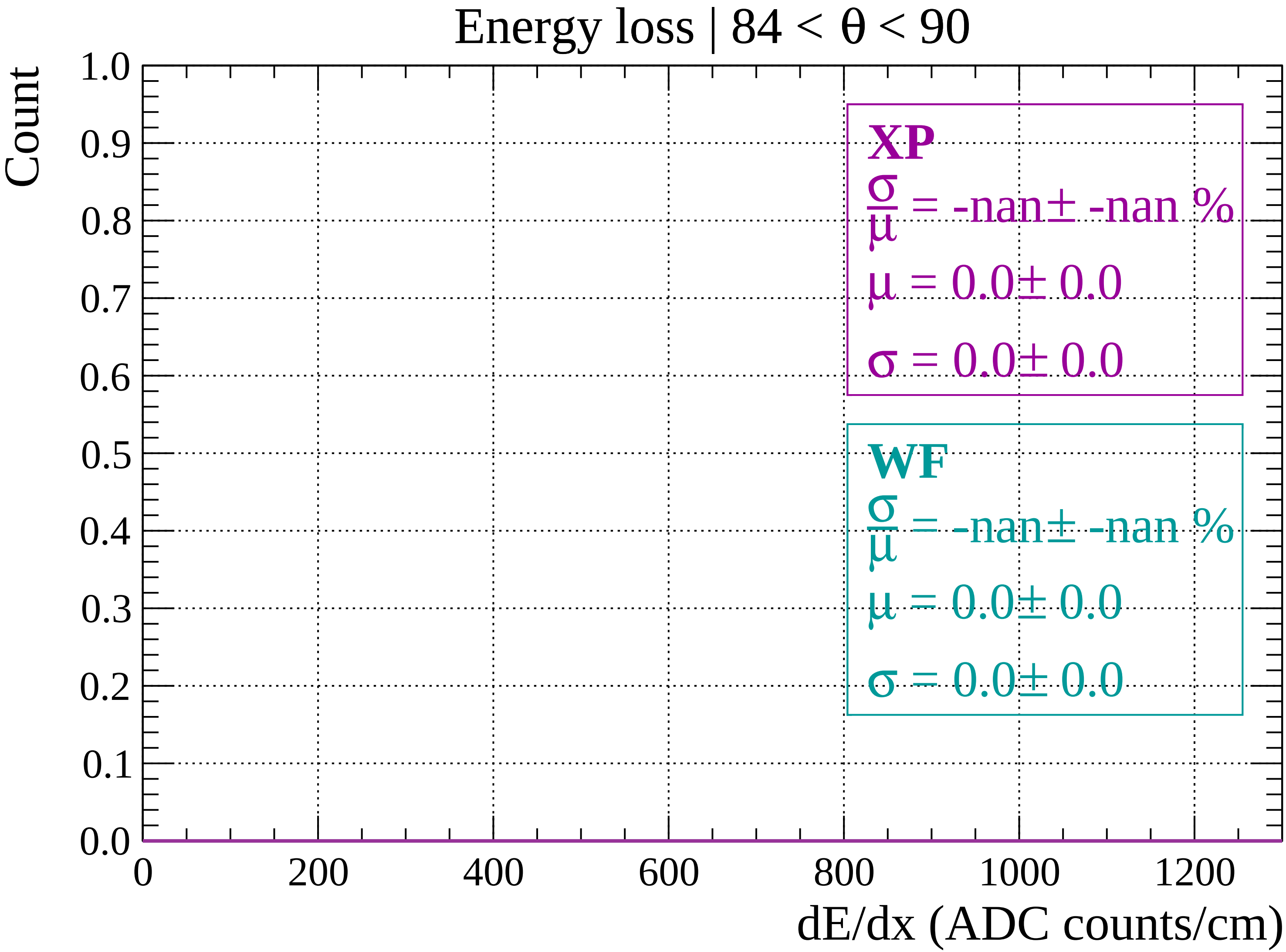

Energy loss | 60 < θ < 66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.17± 38.25 %
μ = 377.0± 64.7
σ = 19.5± 140.9
WF
σ/μ = 5.17± 38.25 %
μ = 377.0± 64.7
σ = 19.5± 140.9

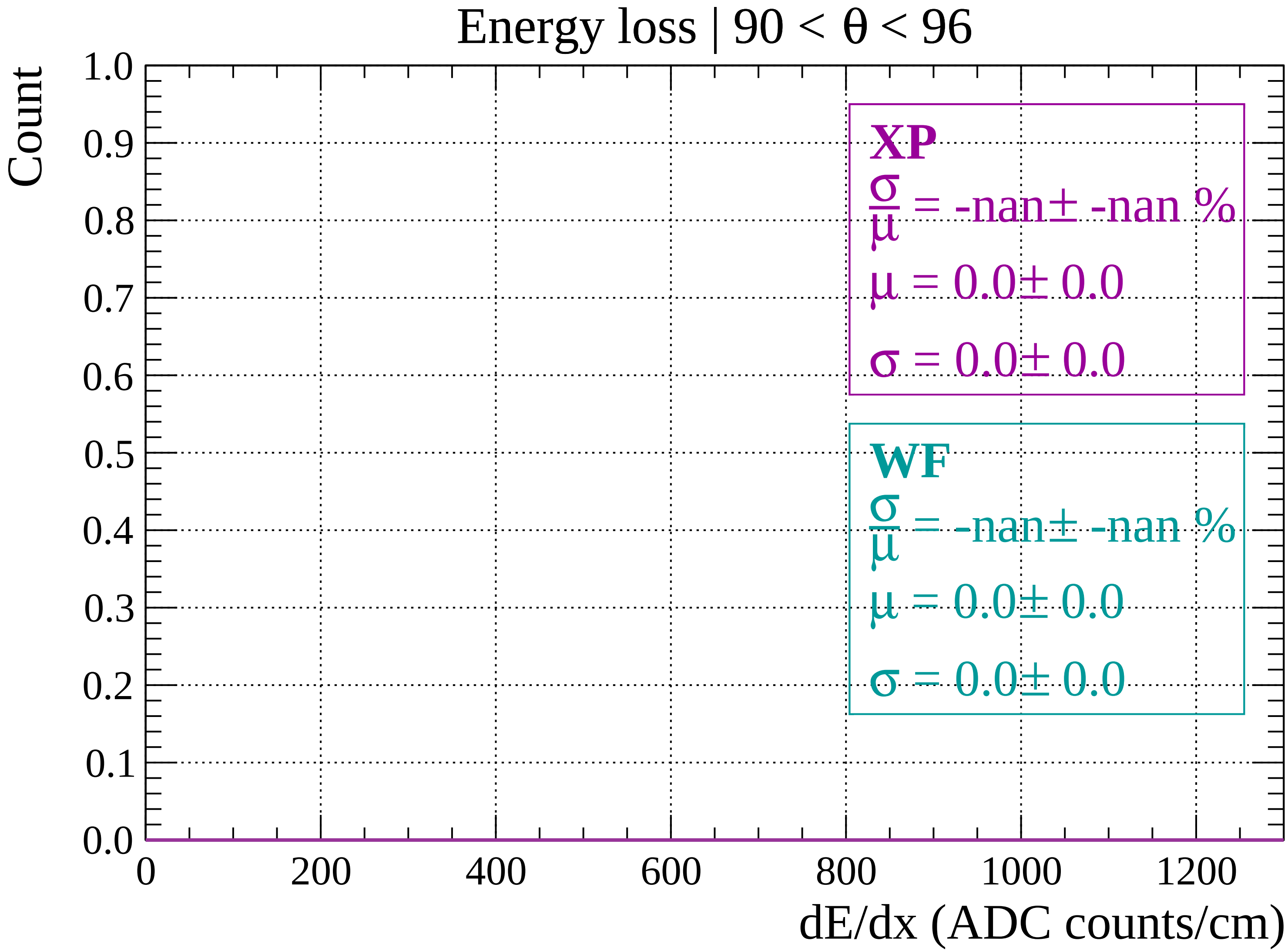

Energy loss | 66 < θ < 72
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 0.07± 0.43 %
μ = 462.0± 1.6
σ = 0.3± 2.0
WF
σ/μ = 1.43± 7.91 %
μ = 455.0± 27.8
σ = 6.5± 35.6


Energy loss | 72 < θ < 78
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

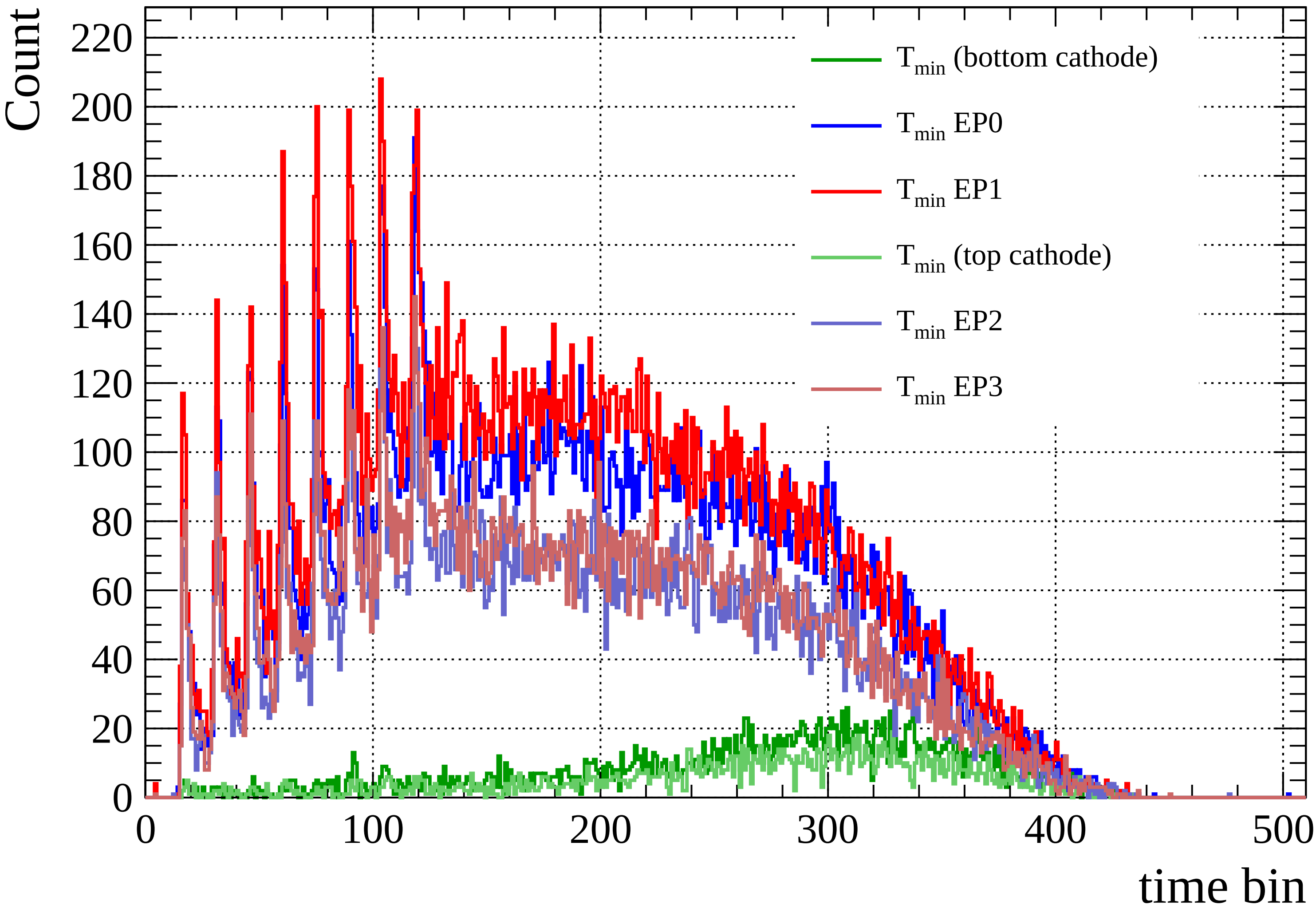


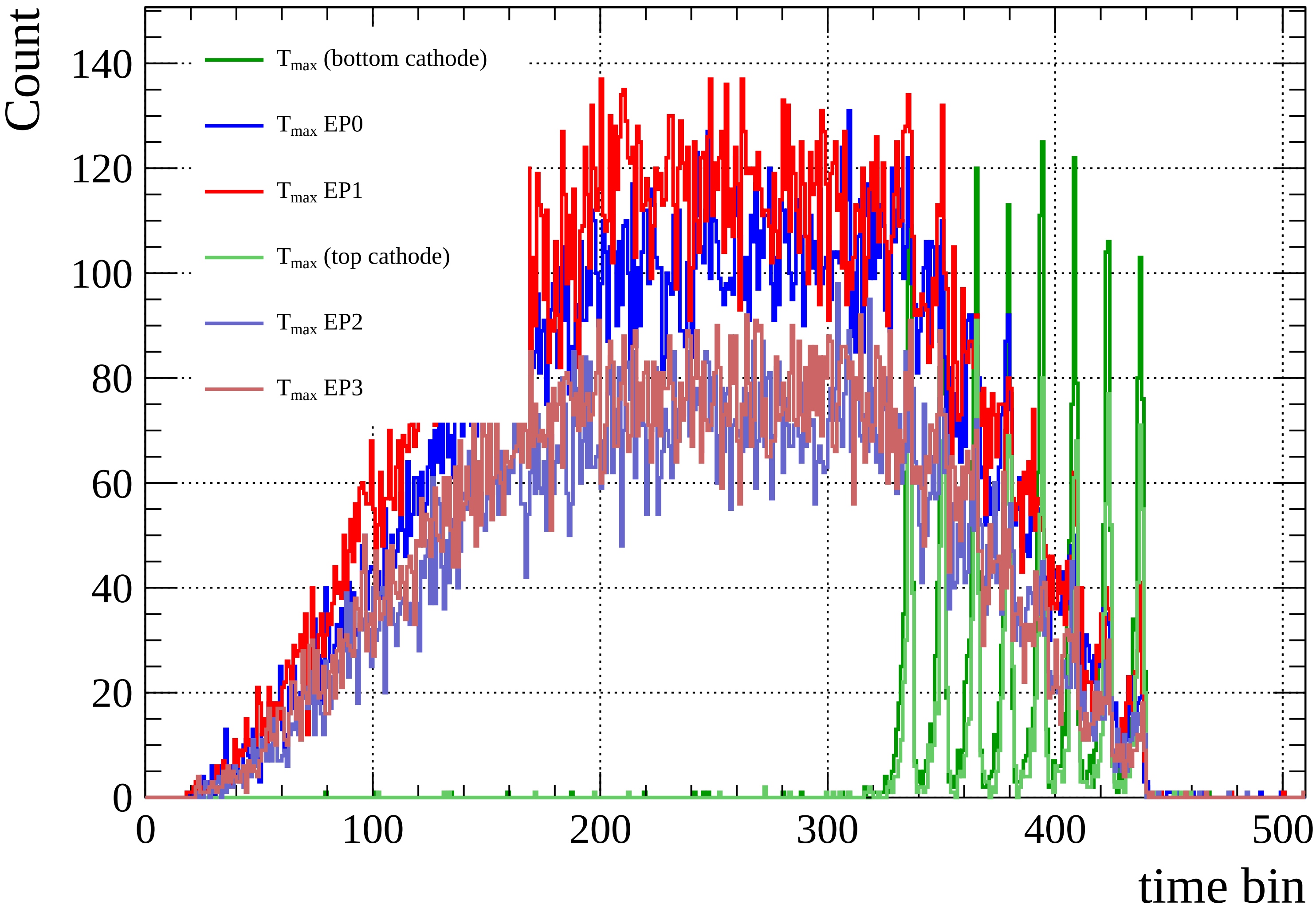
Energy loss | 78 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 96
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
time bin
0
100
200
300
400
500
220
200
180
160
140
120
100
80
60
40
20
0
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
140
120
100
80
60
40
20
0
0
100
200
300
400
500
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3